

Smart Metering Implementation Programme

Great Britain Companion Specification

(GBCS)

Version 4.4

6 November 2025

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Documentation Alignment

1 DLMS Green and Blue Books

This document aligns with the published versions of the Green Book (DLMS UA 1000-2 Ed. 8) and Blue Book (DLMS UA 1000-1 Ed. 12.0). These documents can be obtained from the DLMS User Association: <http://www.dlms.com>.

2 ZigBee Smart Energy Profile

This document aligns with:

ZigBee Smart Energy (ZSE) Profile Specification 1.4;

ZigBee Cluster Library Specification, reference 07-5123-04;

ZigBee OTA Upgrade Cluster Specification, reference 09-5264-23;

ZigBee Specification Revision 22, reference 05-3474-22; and

ZigBee PRO/2007 Layer PICS and Stack Profiles – 08-0006-05.

These documents are available from <http://zigbee.org/About/GBCSPartner.aspx> or by contacting the ZigBee Alliance.

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136		Specifications (ESMETS) requires that Electricity Smart Metering Equipment (ESME) including	
137		variants, meet the requirements described in the Great Britain Companion Specification	
138		(GBCS).	
139		ESME shall be certified by NCSC as compliant with a relevant version of the ‘Commercial	
140		Product Assurance Security Characteristic Smart Metering – Electricity Smart Metering	
141		Equipment’ as identified in the Smart Energy Code Section A.	
142		GSME shall be certified by NCSC as compliant with a relevant version of the ‘Commercial	
143		Product Assurance Security Characteristic Smart Metering – Gas Smart Metering Equipment’	
144		as identified in the Smart Energy Code Section A.	
145		The Communications Hub Technical Specifications (CHTS) requires that Communications	
146		Hubs meet the requirements described in the GBCS.	
147		A Communications Hub shall be certified by NCSC as compliant with a relevant version of the	
148		‘Commercial Product Assurance Security Characteristic Smart Metering - Communications	
149		Hub’ as identified in the Smart Energy Code Section A.	
150		The HAN Connected Auxiliary Load Control Switches (HCALCS) Technical Specifications	
151		(HCALCSTS) requires that HCALCS meet the requirements described in the GBCS.	

152 A HAN Connected Auxiliary Load Control Switch, to meet the requirements of this version of
153 the GBCS, shall be certified by NCSC as compliant with a relevant version of the ‘Commercial
154 Product Assurance Security Characteristic Smart Metering – HAN Connected Auxiliary Load
155 Control Switch’ as identified in the Smart Energy Code Section A.

156 The Prepayment Interface Device (PPMID) Technical Specifications (PPMIDTS) requires that
157 PPMID meet the requirements described in the GBCS.

158 The In Home Display (IHD) Technical Specifications (IHDTs) requires that IHD meet the ZSE
159 requirements described in the GBCS1.

160 The Standalone Auxiliary Proportional Controller (SAPC) Technical Specifications (SAPCTS)
161 requires that SAPC meet the requirements described in the GBCS.

162 SAPC shall be certified by NCSC as compliant with a relevant version of the ‘Commercial
163 Product Assurance Security Characteristic - Standalone Auxiliary Proportional Controller’ as
164 identified in the Smart Energy Code Section A.

165 The versions of GSMETS, ESMETS, PPMIDTS, HCALCSTS, SAPCTS and CHTS to which this
166 version of GBCS is relevant are identified in the Smart Energy Code Section A.

167 GBCS was notified to the European Commission in accordance with the requirements of the
168 Technical Standards and Regulations Directive² laying down a procedure for the provision of
169 information in the field of technical regulations and rules on Information Society services.

170 Structure of the GB Companion Specification (GBCS)

171 Normative Requirements

172 Some Sections of the GBCS are informative and others normative. Unless Sections are
173 marked ‘informative’ in the header, they shall be normative. Subsections of Sections marked
174 informative shall also be informative.

175 For defined terms (those capitalised), please see the Glossary at Section 21. Where terms
176 are in courier new font, they are Abstract Syntax Notation One (ASN.1³) specified structures

1 The only GBCS requirements for IHD are ZSE requirements.

2 GBCS v2.0 was notified (2017/0350/UK) under Article 8 of Directive 98/34/EC of the European Parliament and of the Council (OJ L 204, 21.7.1998, p. 37) as amended by Directive 98/48/EC of the European Parliament and of the Council (OJ L 217, 5.8.1998, p. 18).

Directive 98/34/EC has now been replaced by Directive 2015/1535/EU of the European Parliament and of the Council (OJ L 241, 17.9.2015, p.1), which came into force on 7 October 2015

3 <http://www.itu.int/rec/T-REC-X.680-X.693-201508-I/en>

defined in this document, or in IETF RFC 59124. Definitions of such ASN.1 structures are not repeated in the Glossary.

Structure of the GB Companion Specification (GBCS) and its relationship to other documents – informative

The whole of this Section 2.2 is informative. A number of documents specify what Devices should do and how they should do it, including:

the ‘Technical Specifications’ (GSMETS, ESMETS, SAPCTS IHDTS, HCALCSTS, PPMIDTS and CHTS):

lay out minimum physical requirements and minimum functional capabilities for Devices;

specify that all Devices must use the ZSE protocol specifications; and

specify that Electricity Smart Metering Equipment (ESME) and SAPC must additionally use DLMS COSEM protocol specifications.

International Standards documents, including those which lay out what is required to use ZSE and DLMS COSEM protocols. However, the standards are flexible and could be used in many different ways to implement technically the minimum functional requirements of the Technical Specifications;

the end to end protocol that is defined in the GBCS, which deviates from the standard ZigBee SEP and DLMS COSEM protocols in some instances. Suppliers and the DCC are required to deploy Devices that are certified against those aspects of the GBCS that are fully compliant with the ZigBee and DLMS COSEM protocols. Certification is not required against those aspects of the GBCS where the ZigBee and DLMS COSEM protocols are actively dis-applied or modified. For additional information on areas that would not require certification please see Section 25 for ZigBee SEP, and Section 26 for DLMS COSEM.

GB Smart Metering requires technical interoperability, and so requires a single, consistent, technical implementation of the capabilities laid out in the Technical Specifications across all Devices, in so far as the network communications with Devices are concerned, be those communications over the Smart Metering Home Area Network (SMHAN) or Wide Area Network (WAN). The Devices in scope of this GBCS are:

Electricity Smart Metering Equipment (ESME), including Single Element, Polyphase, Twin Element, Auxiliary Load Control Switch (ALCS), Auxiliary Proportional Controller (APC) and Boost Function variants thereof;

Gas Smart Metering Equipment (GSME);

4 <http://tools.ietf.org/html/rfc5912>

209 Communications Hub Function (CHF) and Communications Hub, Gas Proxy Function (GPF);
210 Standalone Auxiliary Proportional Controller (SAPC), including Auxiliary Load Control Switch
211 (ALCS) and Boost Function variants thereof;
212 Prepayment Interface Device (PPMID) and HAN Connected Auxiliary Load Control Switch
213 (HCALCS); and
214 Type 2 Devices, including In Home Displays (IHDs).
215 The purpose of this GBCS, and related documents, is to specify the single, consistent
216 technical implementation in sufficient detail to achieve operational interoperability of
217 Devices.
218 The Smart Metering technical and security architecture is based on a suite of agreed, open
219 standards, reflecting the UK Government strategy to facilitate the development of third
220 party innovative solutions for consumer devices. These include standards relating to DLMS
221 COSEM, ZSE, ASN.1, NIST cryptography and X.509 related IETF RFCs. The GBCS does not
222 duplicate what is laid out in such standards but rather provides references to them.
223 Scope and Terminology
224 Introduction – informative
225 This Section 3.1 is informative and summarises Section 3.
226 Section 3 introduces key terms used in the GBCS:
227 Messages are how Devices communicate between themselves and with organisations
228 remote from Consumers' Premises. Such Messages are 'end-to-end' and 'unicast' in that:
229 they all identify the sender (e.g. a Supplier) and the intended recipient (e.g. an ESME); and
230 they are all intended for processing by the intended recipient, even though they may pass
231 through intermediate Devices, such as a Communications Hub Function. Most Messages
232 pass through Communications Hub Functions unaltered, save for any 'wrapping' information
233 needed for transport purposes. The only exceptions are where a CHF or GPF is the intended
234 recipient or is the sender (in these cases the Message is processed by the CHF or GPF), or
235 where covered by the Tapping Off Mechanism (Section 10);
236 Messages are one of:
237 a Command to a Device or a corresponding Response;
238 an Alert from a Device; or
239 an information provision transaction (HAN Only Message) solely between Devices;

240 Organisations (such as Suppliers and Network Operators) communicating with Devices are
241 called Remote Parties;

242 Messages to and from Remote Parties are called Remote Party Messages; and

243 Messages solely between Devices are called HAN Only Messages.

244 Section 3 then:

245 explains that the GBCS only covers the Messages needed for the minimum functionality laid
246 out in the Technical Specifications;

247 explains that the GBCS specifies how all such Messages are constructed and related
248 processing performed; and

249 notes that Type 2 Devices (e.g. IHDs) can only send or receive HAN Only Messages.

250 Section 3 also explains some technical terminology and technical conventions used in this
251 GBCS.

252 Scope

253 This Section 3.2 lays out the scope of the GBCS and introduces definitions relied upon in this
254 GBCS.

255 A Message shall be of one the following:

256 a Command;

257 a Response to a Command;

258 an Alert; or

259 an information provision transaction (HAN Only Message).

260 A Message instance shall be an instance of one of the Messages detailed in this GBCS.

261 The Technical Specifications define the minimum functional capabilities required of Devices.

262 Except where those functional capabilities are internal to the Devices or are accessed via the
263 Device's User Interfaces, the minimum functional capabilities shall be invoked by, and / or
264 result in, Messages being passed via the Devices' Network Interfaces.

265 The GBCS is the technical specification, sufficient for the creation by the originator(s) and
266 processing by the target(s), of each Message, where the Message is required in order to
267 implement minimum functionality defined in the Technical Specifications.

268 Specifically, the GBCS details the format, structure and associated processing for each of the
269 Messages required to implement the Technical Specifications' minimum functionality.

270 There are two classifications of Message:

271 HAN Only Message⁵, where both the original sender and ultimate recipient are Devices
272 within the same Smart Metering Home Area Network (SMHAN); and

273 Remote Party Message, where either the original sender or the ultimate recipient is not a
274 Device.

275 A Remote Party Message shall only be of one of the following:

276 a Command;

277 a Response to a Command; or

278 an Alert.

279 Each Remote Party Message shall have a unique Message Code, which shall be as specified
280 in Section 15.

281 Where a Remote Party is known to a Device by way of that Remote Party's Security
282 Credentials being stored on the Device (as specified in Section 4.3.2.5), the Remote Party is
283 referred to as a Known Remote Party (KRP). Otherwise, it is referred to as an Unknown
284 Remote Party (URP).

285 Commands requiring a Response to an Unknown Remote Party shall always be sent to the
286 Device by the Device's Access Control Broker (see Section 4.3.2.5).

287 For clarity, Type 2 Devices shall not be required to support any Remote Party Messages.
288 Thus, provisions in this GBCS in relation to Remote Party Messages shall not apply to Type 2
289 Devices.

290 Remote Parties and Devices are collectively referred to in this GBCS as Smart Metering
291 Entities.

292 Terminology

293 Numbers

294 Numbers within this GBCS are expressed in one of three ways, to avoid potential ambiguity:

295 where a number has no prefix, it is a decimal number (base 10);

296 the 0x prefix is used for hexadecimal numbers (base 16). For example, 0x10 equates to the
297 decimal number 16; and

5 HAN Only Messages are ZigBee commands or response commands. This includes HAN Only Messages passed between Devices using the ZSE TransferData, for example a Command from a PPMID to a GSME.

298 the 0b prefix is for binary numbers (base 2). For example, 0b1010 equates to the decimal
299 number 10.

300 Bit numbering

301 Numbering of bits uses the 'LSB 0' bit numbering scheme, where the least significant bit is
302 referred to as bit 0 and the most significant bit is referred to using the highest bit number.

303 Octets and bytes – informative

304 The term 'octet' is used to refer to units of 8 bits of digital information, to avoid potential
305 ambiguity with the term 'byte', and to align with protocol terminology.

306 An octet string is a sequence of octets, with the first octet shown on the left and the last
307 octet shown on the right, for example 0x12 0xAB 0x34 0xCD. If an octet string is shown
308 formatted as a hexadecimal number, the first octet is shown on the left (most significant)
309 and the last octet is shown on the right (least significant). For example, 0x12AB34CD used in
310 the context of an octet string represents 0x12 0xAB 0x34 0xCD.

311 Tag and MAC – informative

312 In this GBCS:

313 the word 'tag' is always used in the sense it is meant in encoding standards, such as A-XDR6
314 and Distinguished Encoding Rules (DER)7;

315 'tag' is never used to mean Authentication tag, in the cryptographic sense;

316 'MAC' is always used to mean Message Authentication Code, which is a cryptographic
317 checksum on data. Thus, MAC is used instead of Authentication tag; and

318 'MAC' is never used to refer to Medium Access Control, as used in 'MAC address', which is a
319 unique identifier assigned to network interfaces.

320 Concatenation

321 X || Y shall mean the concatenation of the two octet strings X and Y.

322 For example:

323 X = 0xCAFE

324 Y = 0xBEEF

325 X || Y = 0xCAFE BEEF

6 IEC 61334-6

7 <https://www.itu.int/rec/T-REC-X.690/en>

326 Encoding and length of variable length unsigned integers

327 Encoding(X) shall be the encoding of a variable size unsigned integer X as follows:

328 if $0 < X < 128$, then Encoding(X) is a single octet whose value is X; or

329 if $128 \leq X < 256$, then Encoding(X) is an octet string composed of the concatenation $0x81 ||$
330 Y, where Y is one octet in length and has a value equal to X; or

331 if $256 \leq X < 65,536$, then Encoding(X) is an octet string composed of the concatenation $0x82$
332 $||$ Y, where Y is two octets in length and has a value equal to the big endian representation
333 of X; or

334 if $65,536 \leq X < 16,777,216$, then Encoding(X) is an octet string composed of the
335 concatenation $0x83 ||$ Y, where Y is three octets in length and has a value equal to the big
336 endian representation of X.

337 Len(Encoding(X)) shall be the length in octets of Encoding(X).

338 GeneralizedTime

339 The GeneralizedTime ASN.1 type used in this GBCS shall be a UTC Time with a
340 resolution of one second. See section 46 of the ASN.1 specification for format.

341 Octet Endianness – Informative

342 Some fields (or elements) are one octet long, for example an 8 bit integer field.

343 Some fields are made up of more than one octet. For example a 16 bit integer is made up of
344 two octets. There are two ways of expressing these multiple octet elements, called ‘big
345 endian’ or ‘little endian’. Big endian means that the octets are listed from most significant
346 octet first to least significant octet last. Little endian is the reverse. Whether a field is
347 expressed as little or big endian is called endianness.

348 To illustrate, the decimal 500 can be represented as the hexadecimal value 0x01F4. As a 16
349 bit integer, its big endian representation is 0x01 then 0xF4 and its little endian
350 representation is 0xF4 then 0x01.

351 Endianness does not affect single octet elements, nor does it affect the sequence of
352 elements. Thus, it does not affect sequences of single octet elements such as 8 bit integer
353 arrays, byte arrays, octet strings or character strings.

354 The GBCS defaults to ‘big endian’ except for parts where a protocol specifies ‘little endian’
355 sequence.

356 The ZCL specification requires elements to be ‘little endian’ in ZCL commands. This applies to
357 all ZCL and ZSE commands, including those required by the GBCS (see Section 7.4 and

elsewhere in the GBCS). It also applies to the ZCL header and ZCL payload parts of GBZ Payloads, as required by Section 7.2. Note that all other elements of the GBZ Payloads, all DLMS COSEM Payloads, all ASN.1 Security Payloads and all other parts of Messages constructed according to Section 7.2 are 'big endian'. This is because they are defined in the GBCS and not in the ZCL specification. Thus, the GBZ Payload elements labelled 'Profile ID', 'Alert Code', 'Timestamp', 'Extended Header Cluster ID', 'Extended Header GBZ Command Length', 'From Date Time' and 'Length of Ciphered Information' are big endian. Note that the GBZ Payload element 'Ciphered Information' is an octet string and so endianness is irrelevant (since it is a sequence of single octet elements).

Note that GBCS does not use the two octet 'Manufacturer code' field in the ZCL Header. The other three fields in the ZCL Header are single octet elements and so endianness has no effect in relation to ZCL Header.

Security

Introduction – informative

This Section 4.1 is informative and summarises Section 4.

Section 4.2 lays out security provisions that are common across Messages, specifically stating that:

at the application layer, all Messages must have integrity and authenticity protections, Critical Messages must have non-repudiation protections and some parts of Messages must have Confidentiality protections applied to specific data content; and

ZSE protections will be relied upon when Devices within the same Smart Metering Home Area Network (SMHAN) communicate with each other.

Section 4.3 lays out security provisions that are common across Remote Party Messages, specifically:

Identifiers, Counters and Protection Against Replay: lays out requirements in relation to identifiers, counters and their use in Protection Against Replay;

Security Credentials: lays out requirements for all Devices, except for Type 2 Devices, to: have Public-Private Key Pairs, and to make their Public Keys available; and

have Trust Anchor Cells, including those which are storage areas within a Device, capable of holding Public Key Security Credentials for a number of Remote Parties, with the set of Remote Parties being derived from the functionality the Device supports; and

Cryptographic Primitives and their Usage: lays out requirements for Cryptographic Algorithms and their usage, in relation to Remote Party Messages.

Note that the cryptographic protections are intentionally independent of whether a Message Payload is structured according to the ZSE, ASN.1 or DLMS COSEM standards. This means that Suppliers, Network Operators, the Access Control Broker and Other Users who may communicate with Devices need only implement cryptographic requirements in one way, regardless of the type of Device they are communicating with.

The same requirements for security apply regardless of whether a Message is delivered by the Wide Area Network (WAN), SMHAN, Hand Held Terminal (HHT) or local interface. Note that, for Prepayment Top Up, there are a number of different Messages. The content of each particular Message will always be processed in the same way regardless of delivery mechanism.

The following additional points are to be noted:

the governance and structures to ensure uniqueness of identifiers are set out in the Smart Energy Code (SEC) and are outside the scope of the GBCS.

a single Originator Counter can be used for the whole of a Remote Party Organisation (e.g. by that Party counting small enough time intervals). A separate counter per Device is not required;

the Supplementary Originator Counter as specified in Section 4.3.1.4 is required where the corresponding Response has to be cryptographically protected (by way of both Encryption and a MAC), to the Supplementary Remote Party. In all other cases, the Response containing Supplementary Remote Party details is protected back to the Access Control Broker; and

Smart Metering entities make extensive use of a range of Counters as part of the unique identification of Smart Metering Messages. Counters are also a key component used to support Protection Against Replay functionality.

Cryptographic Protections applying to all Messages

Each Message shall have Cryptographic Protections to give assurance to the Message recipient(s) as to:

the Message's integrity; and

the Authenticity of the party or parties creating or augmenting the Message.

The minimum set of such Cryptographic Protections is laid out in this GBCS.

This GBCS lays out the Cryptographic Protections for non-repudiation, where this quality is required for specific Messages, so for Critical Messages.

Where part of a Message is Confidential, that part shall have Cryptographic Protections to ensure both its Confidentiality and its integrity, as detailed in this GBCS.

For HAN Only Messages the Cryptographic Protections required by this GBCS shall be those provided by ZSE.

For clarity, the HAN Only Message Cryptographic Protections require that all Devices shall: be provisioned with the corresponding ZSE related Security Credentials; and be capable of performing the associated cryptographic operations.

Security for Remote Party Messages

This Section 4.3 shall:

apply only to Remote Party Messages;

apply to all Remote Party Messages, regardless of the mechanism (i.e. across the WAN, SMHAN, HHT or User Interface) by which they are delivered to, or received from, the Device in question; and

apply to the processing of Remote Party Messages by Remote Parties and Devices.

Identifiers, Counters and Protection Against Replay

Identifiers

All Smart Metering Entities shall have an Entity Identifier which shall be an octet string of length 8. Each Entity Identifier shall be unique across GB Smart Metering.

Entity Identifiers shall be used in the Business Originator ID and Business Target ID fields of Remote Party Messages as shown in Table 4.3.1.1.

Message Type	Business Originator ID	Business Target ID
Command	Entity Identifier for the Known Remote Party which is requesting execution of this Command	Entity Identifier for the Device that the Remote Party wants to action the Command
Response	The Entity Identifier for the Device. This is always the same as the Business Target ID supplied in the corresponding Command	The Business Originator ID provided in the corresponding Command. For Commands to which the corresponding Response is intended for an Unknown Remote Party, the Business Originator ID in the Command shall always be that of the Access

Message Type	Business Originator ID	Business Target ID
		Control Broker
Alert	The Entity Identifier for the Device	The Entity Identifier for the Known Remote Party to which the Alert is to be addressed. Section 16 of this GBCS specifies which Known Remote Party role each type of Alert shall be addressed to

443 Table 4.3.1.1: Entity Identifiers for Business Originator and Target ID fields

444 For clarity, Commands must contain the correct Entity Identifier value in the Business
 445 Originator ID field of Commands to align to Device Requirements. Specifically, the Business
 446 Originator ID field in a Command:

447 Must contain the Entity Identifier value in the {remotePartyRole,
 448 digitalSignature, management} Trust Anchor Cell, if the Command requires a KRP
 449 signature;

450 Must contain the Entity Identifier value in the {supplier, keyAgreement,
 451 prePaymentTopUp} Trust Anchor Cell, if the Command is a Prepayment Top Up; and

452 Must contain the Entity Identifier value in the {remotePartyRole, keyAgreement,
 453 management} Trust Anchor Cell, if the Command is not a Prepayment Top Up and does
 454 not require a KRP Signature.

455 Originator Counter

456 Except where specified otherwise in the GBCS, a Remote Party Message shall include an
 457 Originator Counter, which shall be octet string of length 8 whose contents shall be set and
 458 read as an unsigned 64-bit integer. Responsibility for generating the Originator Counter shall
 459 be as shown in Table 4.3.1.2.

Message	Responsibility for generating the Originator Counter
Command	The Known Remote Party identified by the Business Originator ID in the Command
Response	The Originator Counter shall have the same value as in the corresponding Command
Alert	The Device generating the Alert

460 Table 4.3.1.2: Responsibility for generating the Originator Counter

Where a Device is required to generate an Originator Counter, the Device shall ensure that the value it generates is strictly numerically greater than any previous Originator Counter value it has placed in any previous Message it has generated, and strictly numerically greater than any Supplementary Originator Counter it has placed in any previous Message it has generated.

Where a Remote Party is required to generate an Originator Counter, the Remote Party shall ensure that:

the value it generates is strictly numerically greater than any previous Originator Counter value it has provided for use in any previous Command to the Device in question;

the 32 least significant bits shall not all have the value 0b0 unless the Command is a Prepayment Top Up Command (see Section 14.3.6 for use of the Originator Counter as the UTRN Counter); and

if the Command is a Prepayment Top Up then the 32 least significant bits shall all have the value 0b0.

Message Identifier

A Message Identifier shall be the concatenation:

Business Originator ID || Business Target ID || CRA Flag || Originator Counter

All Messages shall include a Message Identifier which shall be:

constructed according to the requirements of this Section 4.3.1; and

incorporated in the Message according to the requirements of Section 7.

Additional Counters and Identifiers

The following attributes shall be incorporated in Commands where (1) the Business Originator ID is set to be that of the Access Control Broker and (2) the Message Code is listed in the 'Use Case reference' worksheet of the Mapping Table as '*Supplementary Remote Party Data required*':

Supplementary Remote Party ID, which shall be the Entity Identifier of the Remote Party requesting the creation of the Command by the Access Control Broker; and

Supplementary Remote Party Counter, which shall be an octet string of length 8.

All Responses to such Commands shall incorporate:

the same Supplementary Remote Party ID and Supplementary Remote Party Counter as the Command; and

for those marked as '*Supplementary Originator Counter required in Response*' in the 'Use Case reference' worksheet of the Mapping Table', a Supplementary Originator Counter which shall be generated by the Device, and shall be an octet string of length 8 whose contents shall be set and read as an unsigned 64-bit integer. The Device shall ensure that the value it generates is strictly numerically greater than any previous Originator Counter value it has placed in any previous Message it has generated, and strictly numerically greater than any Supplementary Originator Counter it has placed in any previous Message it has generated.

Protection Against Replay mechanisms

Where a Device supports one or more Remote Party Commands that are marked as requiring 'Protection Against Replay' in the Use Cases, the Device shall implement the requirements detailed in this Section 4.3.1.5.

For each type of Command that a Device supports, and that is marked as 'Protection Against Replay Required' in the Use Case Reference tab of the Mapping Table, the Device shall:

- have the capability to store an Originator Counter value for each Remote Party Role allowed to request execution of that type of Command (the 'Execution Counter'); and

- have all Execution Counters initially set to zero at manufacture.

For clarity, an SAPC is not required to have the capability to store an Execution Counter for a Message Code where the corresponding Use Case:

- is marked as 'optional' in the 'SAPC support' column of the 'Use Case reference tab' in Table 20; and

- is not supported by the SAPC.

Security Credentials

Introduction – informative

A Device shall be able to process four kinds of Security Credential Document:

- its own Security Credential Documents, provided in the form of Device Certificates. Here the Device needs processing to cover (1) generating new Public-Private Key Pairs and so issuing Device Certification Requests, (2) storing its Device Certificates and (3) providing a copy of those Device Certificates on request;

- Security Credential Documents relating to Known Remote Parties, provided in the form of Organisation Certificates. For these, the Device needs to be capable of (1) storing, (2) replacing and (3) providing details of those it holds on request;

Security Credential Documents relating to Unknown Remote Parties, provided in the form of Organisation Certificates. For these, the Device will receive them in a Command so that parts of the Response can be Encrypted. The Device does not need to store such Documents; and

Security Credential Documents relating to Certification Authorities, provided in the form of Certification Authority Certificates. These are processed by the Device only when replacing Remote Parties' Security Credential Documents.

Sections 8 and 13 cover the above functionality.

Section 12 covers requirements related to the structure and content of such Security Credential Documents, where such requirements are relevant to Device processing requirements.

This Section 4.3.2 covers requirements for the storage of such Security Credentials on Devices and their usage in verifying cryptographic protections on Commands the Device receives.

Security Credential Documents

A Security Credential Document shall be either:

a Device Certificate; or

a Remote Party's Organisation Certificate; or

a Certification Authority Certificate.

Device Certificate

A Device Certificate shall relate to only one Device and shall meet the requirements specified at Section 12. A Device Certificate shall either be used for Key Agreement or Digital Signing but not both. Device Certificates shall only be issued by Authorised Public Key Infrastructure (APKI) issuing Certification Authority. Where Security Credentials relating to a Device are incorporated in a Message, the Security Credentials shall be incorporated in the Message in the form of the Device Certificate.

Remote Party's Certificate

A Remote Party Certificate shall be one of that Remote Party's Organisation Certificates and so shall relate to only one Remote Party and shall meet the requirements specified at Section 12. As per Section 12, except where remotePartyRole = root a Remote Party Certificate shall either be used for Key Agreement or Digital Signing but not both. Remote Party Certificates shall only be issued by APKI authorised issuing Certification Authorities. Where Security Credentials relating to a Remote Party are incorporated in a Message, the

557 Security Credentials shall be incorporated in the Message in the form of the Remote Party's
558 Certificate.

559 Certification Authority Certificate

560 A Certification Authority Certificate shall relate to only one Certification Authority and shall
561 meet the requirements specified at Section 12. A Certification Authority Certificate shall
562 only be used by a Device for verifying Digital Signatures on Certificates. Where Security
563 Credentials relating to a Certification Authority are incorporated in a Message, the Security
564 Credentials shall be incorporated in the Message in the form of the Certification Authority's
565 Certificate.

566 Device Security Credentials

567 Where a Device is of deviceType that is gSME, eSME8,
568 communicationsHubCommunicationsHubFunction, or
569 communicationsHubGasProxyFunction, that Device shall have the capacity to store and
570 use securely four private keys:

571 for Key Agreement, a Current Private Key and a Pending Private Key; and

572 for Digital Signing, a Current Private Key and a Pending Private Key.

573 Where a Device is of deviceType that is type1HANConnectedAuxiliaryLoadControlSwitch or
574 type1PrepaymentInterfaceDevice, that Device shall have the capacity to store and use
575 securely two private keys:

576 for Key Agreement, a Current Private Key; and

577 for Digital Signing, a Current Private Key.

578 These stores shall be referred to as Private Key Cells.

579 Wherever one of a Device's Private Keys is required to be used by a GBCS Cryptographic
580 Protection process, only the relevant Current Private Key shall be used. A Device shall not
581 use any Pending Private Key in any GBCS Cryptographic Protection.

582 Where a Device holds a Private Key that is to be used for Key Agreement, the corresponding
583 Public-Private Key Pair shall have been generated according to the section B.4.1 of *FIPS 186-*
584 *49* using the 'ECC Key Pair Generation Using Extra Random Bits' method.

8 So including SAPC as per Section 4.3.2.5.

9 <https://csrc.nist.gov/publications/detail/fips/186/4/final>

585 Where a Device holds a Private Key that is to be used for Digital Signing, the corresponding
586 Key Pair shall have been generated according to the section B.4.1 of *FIPS 186-4* using the
587 ‘ECC Key Pair Generation Using Extra Random Bits’ method.

588 Where a Device supports the processing of Remote Party Messages, the Device shall:

589 have two Trust Anchor Cells to store two Device Certificates relating to itself, with one Trust
590 Anchor Cell for storing Device Certificates where keyUsage = keyAgreement and one for
591 Device Certificates where keyUsage = digitalSignature;

592 where those two Trust Anchor Cells are populated, ensure the Device Certificates have the
593 following attributes:

594 both Device Certificates meet the requirements specified at Section 13;

595 both Device Certificates’ hwSerialNum fields have a value the same as the Devices’ Entity
596 Identifier; and

597 each Device Certificate’s keyUsage field has the same value as the Trust Anchor Cell in which
598 it is placed.

599 Remote Party Security Credentials

600 A Device shall only action a Remote Party Command where:

601 the Known Remote Party identified by the Command has, according to the Security
602 Credentials held on the Device, a Remote Party Role which, according to the Mapping Table
603 for the Message Code in question, is allowed to request execution of the Command; and

604 the Cryptographic Protections in the Command instance received by the Device have been
605 verified, in line with the requirements for a Command with the Message Code in question.

606 To enable this, Security Credentials relating to the Remote Parties in question:

607 shall be held in Trust Anchor Cells on the Device; and

608 shall act as the corresponding Trust Anchors.

609 Required Trust Anchor Cells and related Device requirements

610 The Trust Anchor Cells specified in Table 4.3.2.5 by TrustAnchorCellIdentifier are those
611 required on each deviceType. Additionally:

10 <https://csrc.nist.gov/publications/detail/fips/186/4/final>

612 a GSME shall have a Trust Anchor Cell capable of storing Key Agreement Security
613 Credentials for a PPMID; and

614 a PPMID shall have a Trust Anchor Cell capable of storing Key Agreement Security
615 Credentials for a GSME.

616 The types of Device and the corresponding value of deviceType shall be defined in ASN.1
617 notation by:

```
618 DeviceType ::= INTEGER {
```

619 gSME (0),

620 eSME (1),

621 communicationsHubCommunicationsHubFunction (2),

622 CommunicationsHubGasProxyFunction (3),

623 type1HANConnectedAuxiliaryLoadControlSwitch (4),

```
624 type1PrepaymentInterfaceDevice (5),
```

625 type2 (6)

626 }

627 An SAPC shall have a deviceType of eSME (1) throughout this GBCS.

628 Every Device shall:

629 have storage allocated capable of holding Security Credentials as required by Table 4.3.2.5
630 for its Device type; and

631 have all the Trust Anchor Cells, specified in Table 4.3.2.5 as being required for its Device type,
632 populated with Security Credentials that comply with the requirements of this GBCS.

633 Critically, root, recovery and accessControlBroker Trust Anchor Cells shall be populated with
634 valid credentials for each of those three Remote Parties.

				Type of Device (✓ = is required; empty = is not required)					
				ESME /SAPC	GSME	CH (CHF)	CH (GPF11)	HCALCS	PPMID
deviceType value(s)				1	0	2	3	4	5
TrustAnchorCellIdentifier									
No	remotePartyRole	keyUsage	cellUsage						
1	root	keyCertSign	management	✓	✓	✓	✓	✓	✓
2	recovery	digitalSignature	management	✓	✓	✓	✓	✓	✓
3	supplier	digitalSignature	management	✓	✓		✓	✓	
4	supplier	keyAgreement	management	✓	✓		✓		
5	supplier	keyAgreement	prePaymentTopUp	✓ ¹²	✓				
6	networkOperator	digitalSignature	management	✓			✓		
7	networkOperator	keyAgreement	management	✓			✓		
8	accessControlBroker	digitalSignature	management			✓			✓
9	accessControlBroker	keyAgreement	management	✓	✓	✓	✓	✓	✓
10	transitionalCoS	digitalSignature	management	✓	✓		✓	✓	
11	wanProvider	digitalSignature	management			✓			
12	loadController	digitalSignature	management	✓					
13	loadController	keyAgreement	management	✓					

635 Table 4.3.2.5: Requirements for Trust Anchor Cells by Device Type

636 For clarity, the GPF and CHF shall each have their own set of Trust Anchor Cells.

11 Supplier and Network Operator credentials on the Communications Hub (Gas Proxy Function) relate to the supply of gas only. These Trust Anchor Cells on a Communications Hub are still required and valid where there is no GSME connected to the SMHAN, but the stores should be populated with Access Control Broker certificates (so ensuring the Gas Proxy Function functionality, apart from Update Security Credentials, is inoperable)

12 This Trust Anchor Cell is required on an SAPC, recognising that, in line with SMETS, the SAPC may or may not support prepayment related functionality.

637 A specific Trust Anchor Cell shall be identified in this GBCS using the notation
 638 {remotePartyRole, keyUsage, cellUsage}. For example {supplier, digitalSignature,
 639 management} shall refer to the Trust Anchor Cell that holds the Device's Supplier Digital
 640 Signing Security Credentials, so including the Supplier's:

641 Entity Identifier;
 642 Remote Party Role; and
 643 Digital Signing Public Key.

644 Where a Device supports the processing of Remote Party Messages, that Device:
 645 shall support the processing of the Update Security Credentials Commands required by
 646 Section 13; and

647 shall not allow execution of any Remote Party Command other than an Update Security
 648 Credentials Command, Update Load Controller Security Credentials Command or Provide
 649 Security Credentials Command, nor issue any Remote Party Alerts, in relation to a Remote
 650 Party Role where the Remote Party Role stored in a Trust Anchor Cell is different than that of
 651 the Trust Anchor Cell itself.

652 When verifying a Cryptographic Protection applied to a Command instance it receives, a
 653 Device shall use the Remote Party Security Credentials that it holds at the time of Command
 654 processing.

655 Devices shall only be capable of replacing Remote Party Security Credentials on receipt of an
 656 Update Security Credentials Command specified in this GBCS.

657 What is the Public Key in each Trust Anchor Cell to be used for – informative

TrustAnchorCellIdentifier			Usage of the Public Key in the Trust Anchor Cell
remotePartyRole	keyUsage	cellUsage	
Root	keyCertSign	management	Used only in Certification Path Validation to check that Certification Authority Certificates and Certificates related to change of root credentials were validly issued
recovery	digitalSignature	management	Used only to verify recovery's signature on Update Security Credentials Commands addressed to the Device
supplier	digitalSignature	management	Used to verify the supplier's signature on Critical Commands the supplier has addressed to the Device
supplier	keyAgreement	management	Used in applying MACs to Alerts and Responses addressed to the supplier, where they are not Critical.

TrustAnchorCellIdentifier			Usage of the Public Key in the Trust Anchor Cell
remotePartyRole	keyUsage	cellUsage	
			Used in encrypting data in Alerts and Responses addressed to the supplier
supplier	keyAgreement	prePaymentTopUp	Used to check the supplier MAC on Prepayment Top Up Commands. The supplier can decide whether this is the same key as the Key Agreement key used for other purposes
networkOperator	digitalSignature	management	Used to check the signature of the networkOperator on Critical Commands the networkOperator has sent to the Device. This only equates to Update Security Credentials Commands
networkOperator	keyAgreement	management	Used in applying MACs to Alerts and Responses addressed to the networkOperator, where they are not Critical. Used in encrypting data in Responses addressed to the networkOperator
accessControlBroker	digitalSignature	management	Used to verify the accessControlBroker's signature on Commands addressed to the Device
accessControlBroker	keyAgreement	management	Used in checking the accessControlBroker MAC on Commands received and to calculate the MAC for Responses addressed to the accessControlBroker
transitionalCoS	digitalSignature	management	Used only to check transitionalCoS's signature on Update Security Credentials Commands received by the Device
wanProvider	digitalSignature	management	Used by the Communications Hub (CHF) to verify the wanProvider's signature on Critical Commands addressed to the Communications Hub (CHF)
loadController	digitalSignature	management	Used to check the signature of the loadController on Critical Commands the loadController has sent to the Device
loadController	keyAgreement	management	Used in encrypting data in Alerts and Responses addressed to the loadController

658 Table 4.3.2.6: Use of Public Keys in each Trust Anchor Cell

659 Mapping a Command to the Remote Party Security Credentials to be used in verifying the
660 Command's cryptographic protections

Except for the Security Credentials related Commands (see Section 13), a Device shall apply the requirements of this Section 4.3.2.7 to identify which of the Remote Party Public Keys that it holds are to be used to verify the cryptographic protections on a Command.

Message Authentication Codes

Where a Command is a Prepayment Top Up Command, the supplier MAC in that Command shall be verified using the Public Key in Trust Anchor Cell {**remotePartyRole** *supplier*, **keyUsage** *keyAgreement*, **cellUsage** *prePaymentTopUp*}, along with the Device's Key Agreement Private Key.

All other MACs in Commands shall be verified using the Public Key in Trust Anchor Cell {**remotePartyRole** *accessControlBroker*, **keyUsage** *keyAgreement*, **cellUsage** *management*}, along with the Device's Key Agreement Private Key.

Signature

Where a Command has a Digital Signature, the Device shall identify the Remote Party Role(s) which can legitimately sign the Command according to the message code identified in the Mapping Table.

If there is only one Remote Party Role so identified, then the signature shall be verified using the Public Key in Trust Anchor Cell {**remotePartyRole** *(the identified remote party role)*, **keyUsage** *digitalSignature*, **cellUsage** *management*}.

If there is more than one Remote Party Role so identified, the Device shall use the Business Originator ID in the Command to identify the Trust Anchor Cell(s) where:

keyUsage = *digitalSignature*;

cellUsage = *management*; and

existingSubjectUniqueID = the Business Originator ID in the Command

If there is only one Trust Anchor Cell so identified, then the signature shall be verified using the Public Key in that Trust Anchor Cell.

If there is more than one Trust Anchor Cell so identified the Device shall attempt to verify the Digital Signature using each Trust Anchor Cell identified. These attempts shall be according to the following precedence, and attempts to verify shall cease when a signature verification succeeds:

supplier;

wanProvider;

networkOperator;

693 accessControlBroker.

694 For clarity, other Remote Party Roles on Devices are limited to Commands related to Security
695 Credentials and so cannot have Trust Anchor Cells identified according to this Section
696 4.3.2.7.2.

697 Certification Path Validation

698 Access Control Broker requirements

699 Before it calculates the Access Control Broker to Device MAC (ACB-SMD MAC) in line with
700 Section 6.2.3, the Access Control Broker shall undertake Certification Revocation List (CRL)
701 Validation for any Organisation Certificate in a Command:

702 either by using the algorithm specified in IETF RFC 5280¹³ section 6.3; or

703 by using functionality equivalent to the external behaviour resulting from that algorithm.

704 Only if the CRL Validation is successful shall the Access Control Broker calculate the ACB-
705 SMD MAC. For clarity, the Access Control Broker shall never send a Message to a Device
706 which contains any Certificate that has failed CRL Validation.

707 Device requirements

708 The requirements in this Section 4.3.2.8.2 shall apply only to Use Cases CS02b (Update
709 Security Credentials) and CS02g (Update Load Controller Security Credentials).

710 Where a Device has successfully completed all required Command Authenticity and Integrity
711 checks on a Command, of a type covered by Use Cases CS02b and CS02g, the Device shall
712 undertake either:

713 Certification Path Validation, including time checks; or

714 Certification Path Validation, excluding time checks.

715 If the Device does not have Reliable Time (as defined in Use Cases GCS28 and ECS70 Set
716 Clock) it shall always undertake Certification Path Validation, excluding time checks.

717 Otherwise the validation to be undertaken shall be determined by the contents of the
718 Remote Party Command instance. For clarity, Device types which are not required to have a
719 clock, shall always undertake Certification Path Validation, excluding time checks.

720 The Device shall undertake Certification Path Validation, including time checks:

721 either by using the algorithm specified in IETF RFC 5280 section 6.1; or

722 by using functionality equivalent to the external behaviour resulting from that algorithm.

13 <http://datatracker.ietf.org/doc/rfc5280/>

723 The Device shall undertake Certification Path Validation, excluding time checks:
 724 either by using the algorithm specified in IETF RFC 5280 section 6.1 but not applying the
 725 check at 6.1.3 (a) (2) ('the certificate validity period includes the current time'); or
 726 by using functionality equivalent to the external behaviour resulting from that algorithm
 727 where not applying the check that 'the certificate validity period includes the current time'.
 728 For clarity, Certification Path Validation shall only apply to replacement Certificates in the
 729 Command, not to an existing Certificate in a Trust Anchor cell.
 730 The 'trust anchor' information (with the meaning in IETF RFC 5280) shall be in the root
 731 Security Credentials held on the Device.
 732 If the Device's Certificate Path Validation does not confirm the required certification path
 733 validity, then the Device shall undertake no further processing of the Command, except for
 734 the issuance of a Message containing an `executionOutcome` (with its Section 13.3.4.6
 735 meaning) notifying that the Command was unsuccessful in applying at least some of the
 736 changes in the Command.

737 DLMS Client and Server

738 The Access Control Broker shall perform the role of DLMS COSEM client in relation to the
 739 DLMS COSEM Application Associations, and the Device shall perform the role of DLMS
 740 COSEM server.

741 Cryptographic primitives and their usage

742 In relation to any Remote Party Message, Smart Metering Entities shall:

743 use SHA-256, as specified in *FIPS 180-4*¹⁴, as the Hash function;
 744 use the AES-128 cipher, as specified in *FIPS 197*¹⁵, as the block cipher primitive;
 745 use the Galois Counter Mode (GCM) mode of operation as specified in *NIST Special
 746 Publication 800-38D*¹⁶ ;
 747 use the GMAC technique, based on the use of AES-128, for the calculation of Message
 748 Authentication Codes (MACs), as specified in *NIST Special Publication 800-38D* (see above);

14 <http://csrc.nist.gov/publications/fips/fips180-4/fips-180-4.pdf>

15 <http://csrc.nist.gov/publications/fips/fips197/fips-197.pdf>

16 <http://csrc.nist.gov/publications/nistpubs/800-38D/SP-800-38D.pdf>

use, as the Digital Signature technique, ECDSA (as specified in *FIPS PUB 186-4*) in combination with the curve P-256 (as specified in *FIPS PUB 186-4* at section D1.2.3) and SHA-256 as the Hash function. Within Messages, Signatures shall be in the Plain Format;

use, to calculate the Shared Secret Z, the Static Unified Model, C(0e, 2s, ECC CDH) Key Agreement technique (as specified in *NIST Special Publication 800-56Ar2*¹⁷ save for the requirement to zeroize the Shared Secret) with:

the Single-step Key Derivation Function (KDF) based on SHA-256, as specified in *NIST Special Publication 800-56Ar2*; and

the P-256 curve for the elliptic curve operations.

Resulting DerivedKeyingMaterial (with its meaning in *NIST Special Publication 800-56Ar2*) shall only ever be used in relation to one Message instance. Any Shared Secret that is not 'zeroized' shall be stored and used with the same security protections as Private Keys.

Scope of Cryptographic Protections

The fields that shall always contribute to MAC and Digital Signature are detailed in Section 7.2. Fields that vary across Messages are specified in Section 6, and in the relevant Use Cases. For clarity, a Message instance may transit through multiple Smart Metering Entities before delivery to its target Device, and more than one Smart Metering Entity may be required to apply a Cryptographic Protection to that Message instance. Thus, the scope of protection can only be across fields in the Message instance as constructed at the point the protection is applied.

Where a Message has multiple Cryptographic Protections, the order in which the Smart Metering Entities apply these Cryptographic Protections is specified in this GBCS.

A Device verifying the Cryptographic Protections in such Messages shall undertake such verifications in the reverse sequence to that in which the Cryptographic Protections were applied. This order is also specified in this GBCS.

ECDSA per message secret number

When generating a Digital Signature, the Smart Metering Entity shall calculate the DSA Per-Message Secret Number 'k' with respect to ECDSA (with the meaning in section 6.3 of *FIPS 186-4*) to be the SHA-256 hash of the concatenation of:

the parts of the Message to be signed, as defined in Section 7.2.7; and

the Private Key that the Smart Metering Entity will use in the Digital Signature generation.

¹⁷ <http://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-56Ar2.pdf>

If the value of k so calculated is zero or greater than $n - 1$, or results in an ' r ' or ' s ' value of 0, where r and s have the meanings in *FIPS 186-4*, then a new value for k shall be calculated to be the SHA-256 hash of the concatenation of:

the parts of the Message to be signed, as defined in Section 7.2.7;

the Private Key that the Smart Metering Entity will use in the Digital Signature generation; and

0x00.

The addition of 0x00 to the concatenation shall be repeated until a value of k is generated that does not result in k being zero or greater than $n - 1$, or an ' r ' or ' s ' value of 0.

As an alternative to the above, a Remote Party may choose to derive k as defined in section 6.3 of *FIPS 186-4*, using either of the methods defined in section B.5 of *FIPS 186-4*.

Calculating unique Shared Secret Keys for a Remote Party Message Instance

Where a Smart Metering Entity executes the KDF in relation to a Message instance, the *OtherInfo* field, with the meaning in *NIST Special Publication 800-56Ar2*, shall be populated using the value of information provided in, or to be placed in, the originator-system-title, recipient-system-title and transaction-id fields of the Grouping Header, as per the requirements of Section 7.2.7.

The *OtherInfo* shall be in the Concatenation Format as defined in section 5.8.1.2.1 of NIST Special Publication 800-56Ar2 and shall be the concatenation:

AlgorithmID || value of originator-system-title || length of transaction-id || value of transaction-id || value of recipient-system-title

where:

AlgorithmID is that for AES-GCM-128 and so has a value 0x60857406080300, as specified by section 9.2.3.4.6.5 of the Green Book; and

length of transaction-id has the value 0x09.

Calculating the Initialization Vector for GCM and GMAC

In relation to Remote Party Messages, Smart Metering Entities shall use a 96 bit Initialization Vector (IV) for the GCM and GMAC algorithms as defined in *NIST Special Publication 800-38D*. The IV shall be the concatenation:

FixedField || *InvocationField*

where:

811 *FixedField* shall always have the same value as the Business Originator ID in the Grouping
812 Header part of the Message being processed (see Section 7.2.7); and

813 *InvocationField* = 0x00000000.

814 The DLMS COSEM Authentication Key (AK), as defined in the Green Book, shall be a zero
815 length string.

816 Other input parameters to MAC and Encryption / Decryption operations – informative

817 Other input parameters for MAC, Encryption and Decryption are not specified in this Section
818 4.3.3 because they vary dependent on a number of factors. These other input parameters
819 are listed in tables of the same format as Table 4.3.3.4.1 and their values are specified in
820 each part of the GBCS where such an operation is specified.

821 The template for such tables is the Table 4.3.3.4.1. Please note that this table does not
822 contain any values as it is a template only.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key		
Public Key Agreement Key		
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:		

823 Table 4.3.3.4.1: Template for other input parameters

824 Size of MAC

825 The bit length of the MAC shall be 96 except for the MAC contained in the
826 WrappedApexContingencyKey extension within root Certificates, where the bit length of the
827 MAC shall be 128.

828 Remote Party Message construction, protection and verification – informative

829 Much of the content, processing and structure of Remote Party Messages is common across
830 multiple Messages. The GBCS lays out such common requirements. This is to allow Use

831 Cases to detail only those requirements that are specific to the Message(s) covered by that
832 Use Case.

833 Common Message Structures – informative

834 Parts of the structure and content of Remote Party Messages are common across multiple
835 Remote Party Messages. These common parts of the structure and content are laid out in
836 Section 7 of this GBCS. Section 7 also lays out specific requirements for DLMS COSEM and
837 ZSE compliance for Devices compliant with this GBCS.

838 Note that Remote Party Messages in this GBCS are all constructed using aggregation
839 structures. The GBCS does not allow for more granular message structures (e.g. for DLMS
840 COSEM, individual set, get or action messages).

841 Common Encryption and Decryption approach – informative

842 The content and processing of fields in relation to Confidentiality shall be common across all
843 parts of Messages requiring such protections. Where specified in a Use Case, a Remote
844 Party Message may contain one or more encrypted parts. For such requirements, the
845 corresponding Authenticated Encryption and Authenticated Decryption shall always be
846 undertaken using the approach laid out in Section 8.

847 Note that the GBCS does not require Encryption of the whole of a Message.

848 Message Categories – informative

849 The content and processing of fields related to integrity, authenticity and non-repudiation
850 varies according to whether:

851 the Message is a Command, Response or Alert; and

852 the Message is a Critical Message or not.

853 This leads to groupings which are referred to as Message Categories. Message Categories
854 are structured in a hierarchical way, with the more generally applicable categories being at
855 the tiers of the hierarchy with lower numbers. A category which is derived from another
856 category (i.e. in a tier with a higher number) is called a subordinate Message Category. A
857 category from which another category is derived (i.e. in a tier with a lower number) is called
858 a superordinate Message Category. Figure 5.3 summarises the hierarchy.

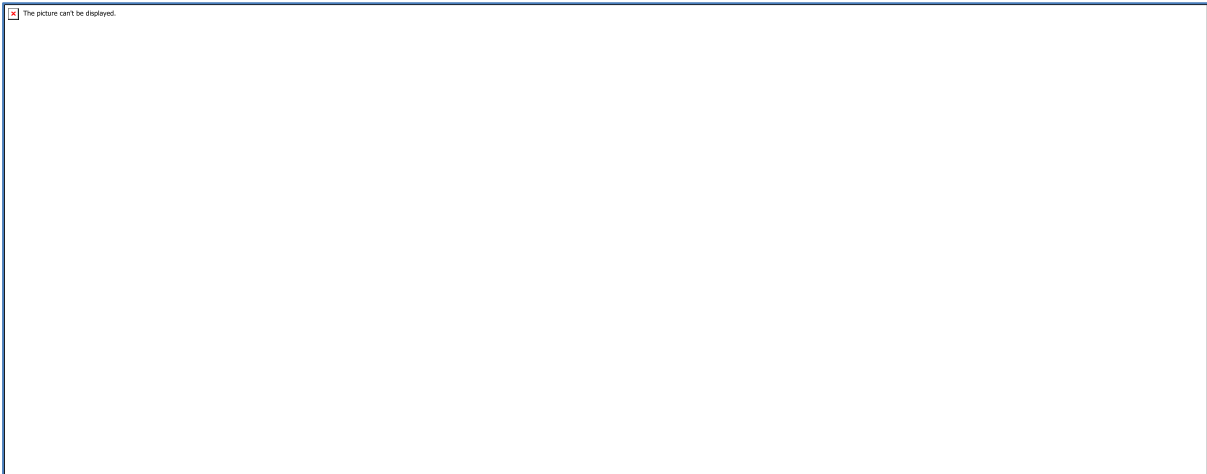


Figure 5.3: Message Categories

Note that the 'Command' part of the hierarchy covers requirements for both the Command and the corresponding Response. Except in certain error cases (e.g. cryptographic processing failure), a Command always leads to a Response.

Section 6 is structured according to the hierarchy at Figure 5.3.

Common Message Processing steps – informative

A common set of stages for Remote Party Message processing is used in this GBCS and the Use Cases, except for Variant Messages¹⁸. Variant Messages include Security Credentials and Prepayment Top Up related Messages.

The common set of stages for Commands is shown in Table 5.4a.

Name of Stage	Summary of the stage	Responsible Smart Metering Entity
Command Construction	The Command is fully populated, apart from cryptographic fields	N/A The entity undertaking this phase is not known to the Device
Command Cryptographic Protection I	This stage is only needed where a Remote Party, other than the Access Control Broker, is required to add Cryptographic Protection to the Command. So for digital signing of Critical Commands only	Known Remote Party
Command Cryptographic	The Access Control Broker adds its Cryptographic Protection to the	Access Control Broker

¹⁸ See Mapping Table for identification of Variant Messages

Name of Stage	Summary of the stage	Responsible Smart Metering Entity
Protection II	Message. This is by way of the ACB adding a MAC	
Command Authenticity and Integrity Verification	The Device undertakes the range of checks needed, including those to ensure authenticity of the sender and integrity of the Message. This includes checking the Identifiers and Counter in the Command and verifying the Access Control Broker's MAC	Device

870 Table 5.4a: Common stages for Commands

871 That common set of stages for Responses is shown in Table 5.4b.

Name of Stage	Summary of the stage	Responsible Smart Metering Entity
1. Response Construction	The Response is fully populated by the Device, apart from cryptographic fields	Device
Response Cryptographic Protection	The Device adds the required Cryptographic Protection to the Response	Device
Response Recipient Verification	The Remote Party (Parties) can undertake the range of checks, including those to ensure authenticity of the sender and integrity of the Message	Remote Party named in the Response

872 Table 5.4b: Common stages for Responses

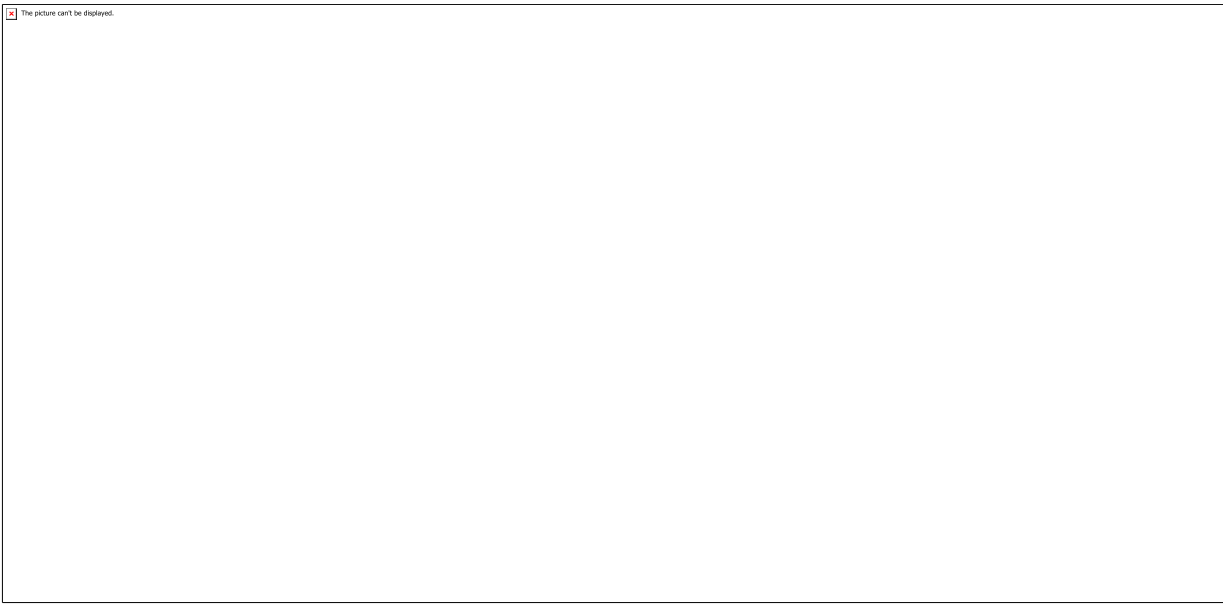
873 That common set of stages for Alerts is shown in Table 5.4c.

Name of Stage	Summary of the stage	Responsible Smart Metering Entity
1. Alert Construction	The Alert is fully populated by the Device, apart from cryptographic fields	Device
Alert Cryptographic Protection	The Device adds the required cryptographic fields to the Alert	Device

Name of Stage	Summary of the stage	Responsible Smart Metering Entity
Alert Recipient Verification	The Remote Party (Parties) can undertake the range of checks, including those to ensure the authenticity of the sender and integrity of the Message	Remote Party named in the Alert

874 Table 5.4c: Common stages for Alerts

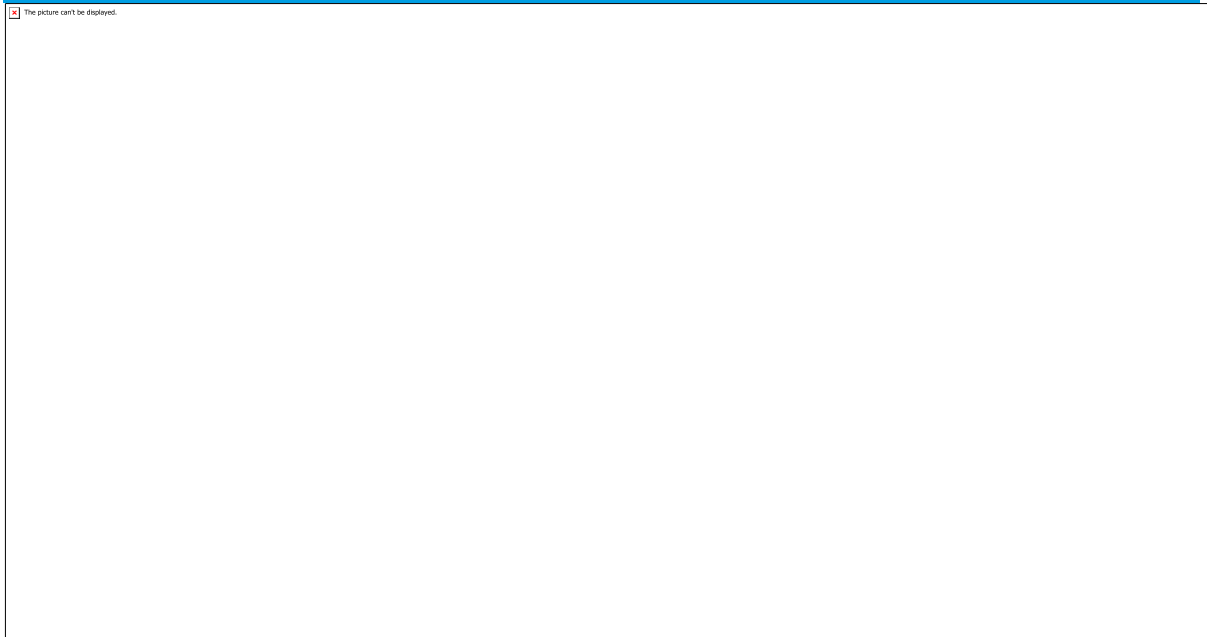
875 The generic processing applied to Commands and their Responses (in relation to integrity,
876 authenticity and non-repudiation) in a Message Category is summarised in Table 5.4d.



877

878 Table 5.4d: Generic Command and Response processing

879 The generic processing applied to Alerts in a Message Category is summarised in Table 5.4e.



880

881 Table 5.4e: Generic Alert processing

882 Message Categories

883 Requirements for the content and processing of fields in Remote Party Messages:

884 related to integrity, Authenticity and non-repudiation; and

885 common across groups of Remote Party Messages;

886 are laid out in this Section 6. Such groupings of Remote Party Messages are referred to as

887 Message Categories.

888 Commands sent by a PPMID to a GSME and Responses to such Commands have

889 requirements similar to Message Categories, and common requirements for this group of

890 Messages are also laid out in this Section 6.

891 Introduction – informative

892 Please see the Mapping Table for the mapping of Use Cases to the Message Categories in

893 this Section 6.

894 For clarity, this Section 6 does not detail requirements for ZigBee commands or messages

895 that may result from a Device processing Remote Party Messages.

896 Message Category SME.C

897 Definitions

898 The superordinate Message Category for SME.C is SME.

For a Message to be of Message Category SME.C it shall be a Command to a Device which is a Remote Party Message, or a Command from a PPMID to a GSME, or a Response to such Commands.

All SME.C Commands and any corresponding Response shall comply with the requirements of this Section 6.2 which covers:

generation of a MAC by the Access Control Broker / PPMID and verification of that MAC by the Device; and

validation by the Device of the Message Identifier.

Processing Stages

The processing of each SME.C Command shall have the stages set out in Table 6.2.2a.

Stage	Responsible Smart Metering Entity
1. Command Construction	The entity undertaking this phase is not known to the Device
Command Cryptographic Protection I	Known Remote Party
Command Cryptographic Protection II	Access Control Broker / PPMID
Command Authenticity and Integrity Verification	Device

Table 6.2.2a: SME.C Command Processing Stages

For a Command, should any of the checks required in the Command Authenticity and Integrity Verification step fail, the Device shall take the steps laid out in Section 6.2.4.2. Otherwise the stages of processing set out in Table 6.2.2b shall be undertaken.

Stage	Responsible Smart Metering Entity
1. Response Construction	Device
Response Cryptographic Protection	Device
Response Recipient Verification	Remote Party named in the Response, or the PPMID named in the Response

Table 6.2.2b: SME.C Response Processing Stages

Processing stages defined in the superordinate Message Category

There are no processing stages defined in the superordinate Message Category (SME).

Processing stages defined in subordinate Message Categories

- 917 There are no requirements for the following processing stages as they are wholly defined in
 918 subordinate Message Categories:
- 919 Command Construction;
- 920 Command Cryptographic Protection I;
- 921 Response Construction;
- 922 Response Cryptographic Protection; and
- 923 Response Recipient Verification.
- 924 Command Cryptographic Protection II
- 925 Requirements in this Section 6.2.3 for Command Cryptographic Protection II shall apply to
 926 Message Category SME.C and all subordinate categories.
- 927 For Remote Party Commands, the Access Control Broker shall calculate the Access Control
 928 Broker to Device Message Authentication Code (ACB-SMD MAC) using the parameters in
 929 Table 6.2.3a.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Access Control Broker's	
Public Key Agreement Key	Device's	As identified by the Business Target ID in Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	Where a KRP Signature is present: 0x110000000000 Grouping Header Command Payload 0x40 KRP Signature	

Input Parameter	Value	Note
	Where a KRP Signature is not present: 0x110000000000 Grouping Header Command Payload 0x00	

930 Table 6.2.3a: Calculation of Access Control Broker to Device MAC

931 The ACB-SMD MAC for incorporation in the Command shall only be calculated once all fields
932 of the Command are populated, as per requirements for the Command Construction and
933 Command Cryptographic Protection I stages for the Message in question.

934 For HAN Only Commands from the PPMID to a GSME, the PPMID shall calculate the PPMID
935 to GSME Message Authentication Code (PPMID-GSME MAC) using the parameters in Table
936 6.2.3b.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	PPMID's	
Public Key Agreement Key	GSME's	As held by the PPMID in the GSME Trust Anchor Cell
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Command Payload 0x00	

937 Table 6.2.3b: Calculation of PPMID-GSME MAC

938 The PPMID-GSME MAC for incorporation in the Command shall only be calculated once all
939 fields of the Command are populated, as per requirements for the Command Construction
940 and Command Cryptographic Protection I stages for the Message in question.

941 Command Authenticity and Integrity Verification

Requirements in this Section 6.2.4 shall apply to Message Category SME.C and all subordinate categories.

Checks to be undertaken

The Device shall undertake the checks in Section 6.2.4.1.1 before any other checks in this Section 6.2.4.1, and shall undertake the other checks in the sequence set out in this Section 6.2.4.1, except, where relevant, as specified in Sections 13.5.4 and 13.7.4.2.2, before undertaking any other processing of the Command.

Message Identifier Validation

The Device shall verify that:

1. the Business Target ID in the Command has the same value as the Device's Entity Identifier;

the Message Code is for a Message that the Device is capable of processing, according to the associated Use Case. For an SAPC:

1. this test shall be applied as if the SAPC were Single Element Electricity Metering Equipment (with its SMETS meaning), and so ESME as defined in Part A of Section 5 of SMETS;

2. this test shall succeed if the Use Case states that an ESME is required to support it, regardless of whether the SAPC does support the corresponding Use Case; and

3. where the SAPC does not support the corresponding Use Case, the SAPC shall not undertake any of the subsequent checks in this Section 6.2.4.1.1;

the Business Originator ID in the Command has the same value as the Entity Identifier held by the Device within a Trust Anchor Cell, where the Smart Metering Entity associated with that Trust Anchor Cell is allowed to request execution of a Command of this type, as specified by the Message Code in the Command and the Mapping Table ('Use Case reference' worksheet Message Code columns); and

the contents of the Message Payload conform to the GBCS in that:

where the Use Case, as identified by its Message Code, specifies a DLMS COSEM Payload:

the DLMS COSEM Payload complies with the Green Book section 9.5 ASN.1 definition of *@CosemPDU.XDLMS-PDU.access-request*; and

the presence and order of *@CosemPDU.XDLMS-PDU.access-request.access-request-body.access-request-specification* is as per the corresponding Message Template, excluding

974 the order of content within any occurrences of the *Selective-Access-Descriptor* structure;
 975 and

976 all parameters within the *@CosemPDU.XDLMS-PDU.access-request* that identify a script
 977 table or a script within a script table are within the range of values specified in the
 978 corresponding Use Case;

979 where the Use Case, as identified by its Message Code, specifies a GBZ Payload:

980 the combination of values in the Extended Header Cluster ID and ZCL Header Command ID
 981 fields in each GBZ Use Case Specific Component are explicitly required in the corresponding
 982 Message Template; and

983 for Critical Commands, the GBZ Use Case Specific Component, as identified by Extended
 984 Header Cluster ID and ZCL Header Command ID fields, are in the order defined in the
 985 corresponding Message Template;

986 where the Use Case, as identified by Message Code, specifies an ASN.1 Security Payload, the
 987 Payload conforms to the requirements of the corresponding Use Case.

988 ACB-SMD MAC Verification

989 To verify the ACB-SMD MAC in Remote Party Commands, the Device shall calculate a MAC
 990 using the parameters in Table 6.2.4.1.2 and ensure the MAC so calculated has the same
 991 value as the ACB-SMD MAC.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	
Public Key Agreement Key	Access Control Broker's	As held by the Device in the Trust Anchor Cell {accessControlBroker, keyAgreement, management}
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated	Where a KRP Signature is	

Input Parameter	Value	Note
Data shall be the concatenation:	present: 0x110000000000 Grouping Header Command Payload 0x40 KRP Signature Where a KRP Signature is not present: 0x110000000000 Grouping Header Command Payload 0x00	

992 Table 6.2.4.1.2: MAC calculation for ACB-SMD MAC verification

993 PPMID-GSME MAC Verification

994 To verify the PPMID-GSME MAC in HAN Only Commands from a PPMID to a GSME, the
 995 Device shall calculate a MAC using the parameters in Table 6.2.4.1.3 and ensure the MAC so
 996 calculated has the same value as the PPMID-GSME MAC.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	GSME's	
Public Key Agreement Key	PPMID's	As held by the GSME in the PPMID Trust Anchor Cell.
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Command Payload 0x00	

997 Table 6.2.4.1.3: MAC calculation for PPMID-GSME MAC verification

998 Processing based on the outcome of checks

999 If:

1000 the Device is an SAPC; and

1001 all of the checks required in Section 6.2.4.1 have succeeded; and

1002 the Device does not support processing of Commands with the Message Code in this

1003 Command

1004 then the Device shall:

1005 generate an entry in the Event Log with Alert / Event Code of 0x8F85;

1006 discard the Command without execution and without sending a Response; and

1007 send a 'Command not supported by Device' Alert, constructed as required by Section 7.2.9.1.

1008 If the Message requires 'Protection Against Replay' according to the corresponding Use

1009 Case, the Device shall, unless it is an SAPC that does not support processing of Commands

1010 with the Message Code in this Command, ensure that the Originator Counter in the

1011 Command has a value that is greater than the value held by the Device for this type of

1012 Command in the corresponding Execution Counter.

1013 Where this check or any of the other prior required checks for this type of Command have

1014 failed, the Device shall:

1015 generate an entry in the Security Log recording failed Authentication;

1016 discard the Command without execution and without sending a Response; and

1017 send an Alert notifying the failed Authentication, constructed as specified in Section 6.7,

1018 populated with the relevant Alert Code from Section 16 (the one of 0x8F1E, 0x8F30 or

1019 0x8F3D that is required), to the Known Remote Party specified in Section 16. If the Device is

1020 an ESME or a CHF, the Alert Payload shall be a DLMS COSEM Alert Payload. Otherwise, the

1021 Alert Payload shall be a GBZ Alert Payload.

1022 Where all of the checks required to be undertaken by the Device have succeeded, the Device

1023 shall:

1024 if the Message requires 'Protection Against Replay' according to the corresponding Use Case,

1025 update the Execution Counter for a Command with the Message Code contained within the

1026 Message from the Remote Party Role identified by the Message, to the value of the

1027 Originator Counter in the Command; and

1028 where the Command contains one or more activation times, set the corresponding

1029 activation times stored on the Device to the relevant values detailed in Section 9.2.2.4, and

1030 process the Command and produce a Response.

1031 Message Category SME.C.C

1032 Definitions

1033 The superordinate Message Category for SME.C.C is SME.C.

1034 For a Message to be of Message Category SME.C.C it shall be:

1035 a subordinate Message Category of Message Category SME.C;

1036 from or to a Remote Party; and

1037 a Critical Message.

1038 A Device shall only be capable of processing the Critical Commands laid out in the GBCS.

1039 All SME.C.C Commands and any corresponding Response shall comply both with the

1040 requirements for SME.C Messages and with the requirements of this Section 6.3 which

1041 covers:

1042 Digital Signing of the Command by the Known Remote Party;

1043 verification of the Digital Signature in the Command by the Device;

1044 Digital Signing of the Response by the Device; and

1045 verification of the Digital Signature in the Response by the Known Remote Party.

1046 Processing stages

1047 Processing stages defined in the superordinate Message Category

1048 There are no requirements additional to those of the superordinate Message Category

1049 (SME.C) for the Command Cryptographic Protection II stage.

1050 Processing stages defined in subordinate categories

1051 There are no requirements for the following processing stages as they are wholly defined in

1052 subordinate categories:

1053 Command Construction; and

1054 Response Construction.

1055 Command Cryptographic Protection I

1056 Requirements in this Section 6.3.3 shall apply to Message Category SME.C.C and all

1057 subordinate categories.

1058 The Remote Party originating the Command shall generate a Known Remote Party Signature

1059 (KRP Signature) for the Command.

1060 The KRP Signature, for incorporation in the Command, shall only be generated once all fields
1061 of the Command Payload and Grouping Header are populated as per the requirements for
1062 the Command Construction stage, for the Message in question.

1063 The KRP Signature shall be calculated across those fields of Grouping Header specified in
1064 Section 7.2.7 and all fields of the Command Payload, as specified in Section 7.2.7.

1065 The Remote Party shall use its Private Digital Signing Key to generate the KRP Signature.

1066 Command Authenticity and Integrity Verification

1067 Requirements in this Section 6.3.4 shall apply to Message Category SME.C.C and all
1068 subordinate categories.

1069 The Device shall undertake the checks set out in this Section 6.3.4:

1070 only after all checks in Section 6.2.4.1 have been successfully completed; and

1071 before undertaking any other processing of the Command.

1072 The Device shall use the Command Payload, Grouping Header and the Public Digital Signing
1073 Key of the Remote Party identified by the checks in Section 4.3.2.7.2 for Digital Signature
1074 verification of the KRP Signature.

1075 The actions laid out in Section 6.2.4.2 shall then apply, as required by the success or failure
1076 of the Digital Signature verification.

1077 Response Cryptographic Protection

1078 The Device creating the Response shall generate a Device Signature (SMD Signature) for the
1079 Response.

1080 The SMD Signature, for incorporation in the Response, shall only be generated once all fields
1081 of the Response Payload and Grouping Header are populated, as per requirements for the
1082 Response Construction stage, for the Message in question.

1083 The SMD Signature shall be calculated across those fields of Grouping Header specified in
1084 Section 7.2.7 and all fields of the Response Payload, as specified in Section 7.2.7.

1085 The Device shall use its Private Digital Signing Key to generate the SMD Signature.

1086 Response Recipient Verification

1087 A Remote Party may verify the SMD Signature in the Response by using the Response body
1088 and the Public Digital Signing Key for the Device identified in the Response.

1089 Message Category SME.C.NC

1090 Definitions

1091 For a Message to be of Message Category SME.C.NC, it shall be:
1092 a subordinate Message Category of Message Category SME.C;
1093 from or to a Remote Party; and
1094 not a Critical Message.

1095 All SME.C.NC Commands and any corresponding Response shall comply both with the
1096 requirements for SME.C Messages and with the requirements of this Section 6.4 which
1097 covers:
1098 generation by the Device of a MAC for the Response; and
1099 verification of that MAC by the intended recipient of the Response.

1100 Processing stages
1101 Processing stages defined in the superordinate Message Category
1102 There are no requirements additional to those of the superordinate Message Category
1103 (SME.C) for the Command Cryptographic Protection II processing stage.

1104 Processing stages defined in subordinate categories
1105 There are no requirements for the following processing stages as they are wholly defined in
1106 subordinate categories:
1107 Command Construction; and
1108 Response Construction.

1109 Command Cryptographic Protection I
1110 There are no additional requirements at the Command Cryptographic Protection I stage
1111 applicable to all Messages of Message Category SME.C.NC and any subordinate Message
1112 Category.

1113 Command Authenticity and Integrity Verification
1114 There are no additional requirements at the Command Authenticity and Integrity
1115 Verification stage applicable to all Messages of Message Category SME.C.NC and any
1116 subordinate Message Category.

1117 Response Cryptographic Protection
1118 Requirements in this Section 6.4.5 shall apply to Message Category SME.C.NC and all
1119 subordinate categories.

- 1120 The Device shall calculate the Device to Known Remote Party MAC (SMD-KRP MAC) using
 1121 the parameters in Table 6.4.5.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	
Public Key Agreement Key	Known Remote Party's	As held by the Device in the relevant Trust Anchor Cell {remotePartyRole, keyAgreement, management}. The relevant Cell will contain Business Originator ID as specified in Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Response Payload 0x00	

- 1122 Table 6.4.5: Calculation of Device to Known Remote Party MAC

- 1123 The SMD-KRP MAC for incorporation in the Response shall only be calculated once all fields
 1124 of the Response, except for the SMD-KRP MAC itself, are populated as per requirements for
 1125 the Response Construction stage, for the Message in question.

- 1126 Response Recipient Verification

- 1127 Requirements in this Section 6.4.6 shall apply to Message Category SME.C.NC and all
 1128 subordinate categories.

- 1129 The Remote Party, as identified by the Business Target ID in the Response, may validate the
 1130 SMD-KRP MAC in the Response by calculating a MAC using the parameters in Table 6.4.6 and
 1131 comparing the MAC to the SMD-KRP MAC.

Input Parameter	Value	Note
-----------------	-------	------

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Known Remote Party's	
Public Key Agreement Key	Device's	As identified by the Business Originator ID in Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Response Payload 0x00	

1132 Table 6.4.6: MAC calculation for SMD-KRP MAC validation

1133 Message Category SME.C.PPMID-GSME

1134 Definitions

1135 For a Message to be of Message Category SME.C.PPMID-GSME, it shall be:

1136 a subordinate Message Category of Message Category SME.C; and

1137 a Message between a PPMID and a GSME.

1138 All SME.C.PPMID-GSME Commands and any corresponding Response shall comply both with
1139 the requirements for SME.C Messages and with the requirements of this Section 6.4 which
1140 covers:

1141 generation by the Device of a MAC for the Response; and

1142 verification of that MAC by the intended recipient of the Response.

1143 Processing stages

1144 Processing stages defined in the superordinate Message Category

1145 There are no requirements additional to those of the superordinate Message Category
1146 (SME.C) for the Command Cryptographic Protection II processing stage.

1147 Processing stages defined in subordinate categories

- 1148 There are no requirements for the following processing stages as they are wholly defined in
 1149 subordinate categories:
- 1150 Command Construction; and
- 1151 Response Construction.
- 1152 Command Cryptographic Protection I
- 1153 There are no additional requirements at the Command Cryptographic Protection I stage
 1154 applicable to all Messages of Message Category SME.C.PPMID-GSME and any subordinate
 1155 Message Category.
- 1156 Command Authenticity and Integrity Verification
- 1157 There are no additional requirements at the Command Authenticity and Integrity
 1158 Verification stage applicable to all Messages of Message Category SME.C.PPMID-GSME and
 1159 any subordinate Message Category.
- 1160 Response Cryptographic Protection
- 1161 Requirements in this Section 6.5.5 shall apply to Message Category SME.C.PPMID-GSME and
 1162 all subordinate categories.
- 1163 The GSME shall calculate the GSME to PPMID MAC (GSME-PPMID MAC) using the
 1164 parameters in Table 6.5.5.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	
Public Key Agreement Key	PPMID's	As held by the GSME in the PPMID Trust Anchor Cell
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Response Payload 0x00	

1165 Table 6.5.5: Calculation of GSME-PPMID MAC

1166 The GSME-PPMID MAC for incorporation in the Response shall only be calculated once all
 1167 fields of the Response, except for the GSME-PPMID MAC itself, are populated as per
 1168 requirements for the Response Construction stage, for the Message in question.

1169 Response Recipient Verification

1170 Requirements in this Section 6.5.6 shall apply to Message Category SME.C.PPMID-GSME and
 1171 all subordinate categories.

1172 The PPMID, as identified by the Business Target ID in the Response, shall validate the GSME-
 1173 PPMID MAC in the Response by calculating a MAC using the parameters in Table 6.5.6 and
 1174 comparing the MAC to the GSME-PPMID MAC.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	PPMID's	
Public Key Agreement Key	GSME's	As held by the PPMID in the GSME Trust Anchor Cell
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation :	0x110000000000 Grouping Header Response Payload 0x00	

1175 Table 6.5.6: MAC calculation for GSME-PPMID MAC validation

1176 Message Category SME.A

1177 Definitions

1178 The superordinate Message Category for SME.A is SME.

1179 For a Message to be of Message Category SME.A it shall be an Alert from a Device which is
 1180 addressed to a Remote Party.

1181 There are no common requirements that shall be applied to all Messages of Message
 1182 Category SME.A.

1183 Processing Stages

1184 The processing of each SME.A Alert shall have the stages set out in Table 6.6.2:

Stage	Responsible Smart Metering Entity
1. Alert Construction	Device
Alert Cryptographic Protection	Device
Alert Recipient Verification	Remote Party named in the Alert

1185 Table 6.6.2: SME.A Processing Stages

1186 Processing stages defined in the superordinate Message Category

1187 There are no processing stages defined in the superordinate Message Category (SME).

1188 Processing stages defined in subordinate categories

1189 There are no requirements for the following processing stages as they are wholly defined in
1190 subordinate categories:

1191 Alert Construction;

1192 Alert Cryptographic Protection; and

1193 Alert Recipient Verification.

1194 Message Category SME.A.C

1195 Definitions

1196 For a message to be categorised as Message Category SME.A.C, it shall be:

1197 a subordinate Message Category of Message Category SME.A; and

1198 a Critical Message.

1199 All SME.A.C Messages shall comply both with the requirements for SME.A Messages and
1200 with the requirements of this Section 6.7 which covers:

1201 Digital Signing of the Alert by the Device; and

1202 Verification of the Digital Signature in the Alert by the Remote Party.

1203 Processing stages

1204 Processing stages defined in the superordinate Message Category

1205 There are no processing stages defined in the superordinate Message Category (SME.A).

1206 Processing stages defined in subordinate categories

1207 There are no requirements for the Alert Construction processing stage as they are wholly
1208 defined in subordinate categories.

1209 Alert Cryptographic Protection

1210 Requirements in this Section 6.7.3 shall apply to Message Category SME.A.C and all
1211 subordinate categories.

1212 The Device creating the Alert shall generate a Device Signature (SMD Signature) for the
1213 Alert.

1214 The SMD Signature, for incorporation in the Alert, shall only be generated once all fields of
1215 the Alert Payload and Grouping Header are populated, as per requirements for the Alert
1216 Construction stage for the Message in question.

1217 The SMD Signature shall be calculated across those fields of Grouping Header and all fields
1218 of the Alert Payload, both as specified in Section 7.2.7.

1219 The Device shall use its Private Digital Signing Key to generate the SMD Signature.

1220 Alert Recipient Verification

1221 Requirements in this Section 6.7.4 shall apply to Message Category SME.A.C and all
1222 subordinate categories.

1223 A Remote Party may verify the SMD Signature in the Alert by using the Alert Payload,
1224 Grouping Header and the Public Digital Signing Key for the Device, as identified in the Alert.

1225 Message Category SME.A.NC

1226 Definitions

1227 For a Message to be of Message Category SME.A.NC it shall be:

1228 a subordinate Message Category of Message Category SME.A; and

1229 not a Critical Message.

1230 All SME.A.NC Messages shall comply both with the requirements for SME.A Messages and
1231 with the requirements of this Section 6.8 which covers:

1232 generation by the Device of a MAC for the Alert and validation of that MAC by the intended
1233 recipient of the Alert.

1234 Processing stages

1235 Processing stages defined in the superordinate Message Category

1236 There are no processing stages defined in the superordinate Message Category (SME.A).

- 1237 Processing stages defined in subordinate categories
- 1238 There are no requirements for the Alert Construction processing stage as they are wholly
1239 defined in subordinate categories.
- 1240 Alert Cryptographic Protection
- 1241 Requirements in this Section 6.8.3 shall apply to Message Category SME.A.NC and all
1242 subordinate categories.
- 1243 The Device shall calculate the Device to Known Remote Party MAC (SMD-KRP MAC) using
1244 the parameters in Table 6.8.3.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	
Public Key Agreement Key	Known Remote Party's	As held by the Device in the relevant Trust Anchor Cell {remotePartyRole, keyAgreement, management}. The relevant Trust Anchor Cell will contain Business Target ID as specified in Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Alert Payload 0x00	

- 1245 Table 6.8.3: Calculation of the Device to Known Remote Party MAC
- 1246 The SMD-KRP MAC for incorporation in the Alert shall only be calculated once all fields of
1247 the Alert, except for the SMD-KRP MAC itself, are populated as per requirements for the
1248 Alert Construction stage, for the Message in question.

1249 As stated in Table 6.8.3, the Public Key Agreement Key used must be the 'Known Remote
 1250 Party's' and must be that 'As held by the Device in the relevant Trust Anchor Cell
 1251 {remotePartyRole, keyAgreement, management}'.

1252 Alert Recipient Verification

1253 Requirements in this Section 6.8.4 shall apply to Message Category SME.A.NC and all
 1254 subordinate categories.

1255 The Remote Party, as identified by the Business Target ID in the Alert, may validate the SMD-
 1256 KRP MAC in the Alert by calculating a MAC using the parameters in Table 6.8.4 and
 1257 comparing the MAC to the SMD-KRP MAC.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Known Remote Party's	
Public Key Agreement Key	Device's	As identified by the Business Originator ID in Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header Alert Payload 0x00	

1258 Table 6.8.4: MAC calculation for SMD-KRP MAC verification

1259 Message structure and DLMS COSEM / ZSE / ASN.1 requirements

1260 Introduction – informative

1261 This Section 7:

1262 defines the structure of Remote Party Messages containing DLMS COSEM, ASN.1 and GBZ
 1263 Payloads. A GBZ Payload is a Payload containing one or more ZigBee messages;

1264 defines the structure of Messages between a PPMID and a GSME on the same SMHAN; and

1265 lays out specific requirements for DLMS COSEM and ZigBee compliance to which Devices
1266 shall adhere.

1267 Note that Remote Party Messages all use an aggregation structure which allows for multiple,
1268 protocol-specific instructions within the same Message. The aggregation structures are used
1269 for all Messages, are based on DLMS COSEM access service, general signing service and
1270 general ciphering service formats, and provide protections across all types of Message
1271 payload (be they DLMS COSEM, ZSE or security related).

1272 The GBCS does not provide more granular Message structures (e.g. for DLMS COSEM,
1273 individual set, get or action messages).

1274 The Technical Specifications require that the Critical Commands mandated by them (and so
1275 those defined in the GBCS) are the only Critical commands allowed. Devices may implement
1276 additional non Critical features only.

1277 It should be noted that:

1278 SMETS only requires DLMS COSEM certification on the ESME and the SAPC;

1279 any action that the Known Remote Party takes to remedy a failure will need to factor in that
1280 some of the instructions succeeded and others did not;

1281 in ASN.1 notation, the signature field in the general-signing service is a variable length
1282 OCTET STRING. When encoded, this means that the length of the signature needs to be
1283 incorporated before the actual signature value. The length is either 64 (0x40) if a signature
1284 is present or 0 (0x00) if signature is not present;

1285 these requirements are to ensure that all Devices behave consistently and in the way
1286 required by originating Remote Party requests, including in error states; and

1287 the WAN Provider may read CHF Operational Data and CHF Configuration Data, with their
1288 CHTS meanings, using mechanisms other than those defined in this GBCS.

1289 Remote Party Message construction – general

1290 This Section 7.2 shall apply to Messages which are of a Message Category that is not
1291 ‘Variant’. For Messages of a Message Category that is ‘Variant’, this Section 7.2 shall only
1292 apply where explicitly stated in the Use Case, with the exception of Section 7.2.11 which
1293 shall apply to all Messages.

1294 Except for elements detailed as being defined in the ZSE or ZCL specifications, the octet
1295 strings constructed in compliance with this Section 7 shall be in ‘big endian’ order according
1296 to IETF RFC 170019. Elements detailed in this Section 7 as being defined in the ZSE or ZCL

19 <http://tools.ietf.org/html/rfc1700>

1297 specifications, shall be serialised into the corresponding parts of octet strings as defined in
1298 the corresponding ZSE or ZCL specification.

1299 Commands

1300 Whether a Command requires a KRP Signature is specified in the corresponding Message
1301 Category requirements in Section 6.

1302 Where a KRP Signature is required, a Remote Party Command received by a Device shall be
1303 the concatenation:

1304 MAC Header || Grouping Header || Command Payload || 0x40 || KRP Signature || ACB-
1305 SMD MAC

1306 Where a KRP Signature is not required, a Remote Party Command received by a Device shall
1307 be the concatenation:

1308 MAC Header || Grouping Header || Command Payload || 0x00 || ACB-SMD MAC

1309 A HAN Only Command from a PPMID to a GSME shall be the concatenation:

1310 MAC Header || Grouping Header || Command Payload || 0x00 || PPMID-GSME MAC

1311 Responses

1312 Whether a Response requires an SMD Signature is specified in the corresponding Message
1313 Category requirements in Section 6.

1314 Where a SMD Signature is required, a Remote Party Response shall be the concatenation:

1315 Grouping Header || Response Payload || 0x40 || SMD Signature

1316 Where a SMD Signature is not required, a Remote Party Response shall be the
1317 concatenation:

1318 MAC Header || Grouping Header || Response Payload || 0x00 || SMD-KRP MAC

1319 A HAN Only Response from a GSME to a PPMID shall be the concatenation:

1320 MAC Header || Grouping Header || Response Payload || 0x00 || GSME-PPMID MAC

1321 Alerts

1322 Whether an Alert requires an SMD Signature is specified in the corresponding Message
1323 Category requirements in Section 6.

1324 Where a SMD Signature is required, a Remote Party Alert shall be the concatenation:

1325 Grouping Header || Alert Payload || 0x40 || SMD Signature

1326 Where a SMD Signature is not required, a Remote Party Alert shall be the concatenation:

-
- 1327 MAC Header || Grouping Header || Alert Payload || 0x00 || SMD-KRP MAC
- 1328 Payload sequence and Break On Error
- 1329 All Message Payloads - Command Payloads, Response Payloads and Alert Payloads - shall:
- 1330 only be constructed in the sequence specified in the corresponding Use Case;
- 1331 only be processed in the sequence specified in the corresponding Use Case; and
- 1332 be processed by a recipient Device on a Break On Error basis.
- 1333 Where a Command Payload contains multiple instructions, processing of further instructions
- 1334 shall cease at the point any one instruction fails. In line with the DLMS COSEM Access
- 1335 Services requirements, a Response shall contain one result for each instruction in the
- 1336 Command, except where the instruction in the Command is a ZSE *GetDayProfiles* or
- 1337 *GetWeekProfiles* command where one or more ZSE commands may be produced in the
- 1338 Response in line with the ZSE specification. The corresponding result in the Response
- 1339 Payload shall detail that instruction's success or failure. The Response Payload shall
- 1340 explicitly detail a result for each of the subsequent instructions that were not attempted.
- 1341 The results in the Response shall be in the same order as the instructions in the Command.
- 1342 The specific result codes shall be as specified in the relevant ZSE / DLMS COSEM document
- 1343 or in this GBCS where standard-based error codes do not exist. Where execution of
- 1344 instructions was not attempted due to the Break On Error requirement, the response shall
- 1345 return:
- 1346 for DLMS instructions, a Data-Access-Result / Action-Result of other-reason;
- 1347 for each such ZCL / ZSE instruction, a Default Response command where the ZCL payload
- 1348 status field has a value of FAILURE (0x01), ZCL payload *Command ID* is set to 0xFF and the
- 1349 ZCL header Frame Control field is set to 0x00 or 0x08, with the Direction Sub-field being 0b0
- 1350 or 0b1 in line with the ZigBee specifications.
- 1351 A ZSE command returning a status of 'NOT_FOUND' shall not be treated as a failure.
- 1352 Message construction – MAC Header
- 1353 The required components of the MAC Header shall be populated with the values as per
- 1354 Table 7.2.5.20

MAC Header

²⁰ See Green Book.

No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	General-Ciphering	0xDD	1	DLMS COSEM APDU tag for General-Ciphering (221 in decimal)
	transaction-id	0x00	1	A value for this element is not needed so the length field is 0x00
	originator-system-title	0x00	1	A value for this element is not needed so the length field is 0x00
	recipient-system-title	0x00	1	A value for this element is not needed so the length field is 0x00
	date-time	0x00	1	A value for this element is not needed so the length field is 0x00
	other-information	0x00	1	A value for this element is not needed so the length field is 0x00
	key-info	0x00	1	Key-info values are not present so encoded as 0x00
	ciphered-service			
	Length	Encoding(X)	Len(Encoding(X))	X shall be the length in octets of the subsequent parts of the Message after this Length value. This includes the security header, the DLMS APDU being protected and the MAC
	security header			
	security control byte (SC)	0x11	1	Bits 3..0 are security suite which is 0b0001 since Security Suite 1 is required Bit 4 is set to 0b1 since Authentication of the APDU is required

MAC Header				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
				Bit 5 is set to 0b0 since the whole of an APDU is never encrypted Bit 6 is set to 0b0 since messages with MACs are unicast Bit 7 is set to 0b0 as per the Green Book
	invocation counter (IC)	0x00000000	4	IC is always zero as specified in Section 8.4

1355 Table 7.2.5: Required components of MAC Header

1356 Additional Authenticated Data (AAD) for the MAC calculation – informative

1357 Terms in italics in this Section 7.2.6 shall have the meanings as specified in Green Book.

1358 The Green Book requires that the AAD used as input to the MAC calculation is the
 1359 concatenation of:

1360 SC || AK || transaction-id || originator-system-title || recipient-system-title || date-time || other-
 1361 information || information to be protected

1362 The Green Book also requires that, for the elements contributing to AAD, the lengths of
 1363 certain octet string fields are included, as well as the values of all fields. The Green Book
 1364 defines octet strings within the general-ciphering service where lengths have to be included
 1365 as:

1366 *transaction-id* OCTET STRING,

1367 *originator-system-title* OCTET STRING,

1368 *recipient-system-title* OCTET STRING,

1369 *date-time* OCTET STRING,

1370 *other-information* OCTET STRING,

1371 As stated in Table 7.2.5, in GBCS-compliant APDUs:

1372 SC takes the value 0x11; and

the following octet strings in the general-ciphering service shall have zero length (so a length octet of 0x00) and so have no value:

transaction-id,

originator-system-title,

recipient-system-title,

date-time,

other-information.

As required by Section 4.3.3.4, AK is a zero length octet string.

Thus, the AAD to be used in MAC calculations that protect APDUs is the concatenation:

0x110000000000 || information to be protected

Message construction – Grouping Header

The following shall be the required components of the Grouping Header and shall be populated with the values as per Table 7.2.7.

Where a Signature is required in a message, it shall be calculated using only those attributes marked 'Yes' in the 'Input to the ECDSA calculation' column of Table 7.2.7, in the sequence they appear in the table.

Thus, a KRP Signature or SMD Signature shall be calculated across the concatenation:

CRA Flag || Originator Counter || Business Originator ID || Business Target ID || date-time (if present) || Message Code || Supplementary Remote Party ID (if present) || Supplementary Remote Party Counter (if present) || Supplementary Originator Counter (if present) || Supplementary Remote Party Key Agreement Certificate (if present) || (information to be protected)

where (information to be protected) shall be:

the Command Payload in a Command;

the Response Payload in a Response; or

the Alert Payload in an Alert.

Grouping Header				
Input to the ECDSA calculation	DLMS COSEM Message Elements	Contents	Length (octets)	Note

Grouping Header				
Input to the ECDSA calculation	DLMS COSEM Message Elements	Contents	Length (octets)	Note
No	General-Signing	0xDF	1	DLMS COSEM APDU tag for General-Signing (223 in decimal)
	transaction-id			
Yes	Length	0x09	1	Length of Originator Counter plus 1
Yes	Value	CRA Flag Originator Counter	9	CRA Flag shall be: 0x01 for Commands 0x02 for Responses 0x03 for Alerts
	originator-system-title			
Yes	Length	0x08	1	Length of Entity Identifier
Yes	Value	Business Originator ID	8	
	recipient-system-title			
Yes	Length	0x08	1	Length of Entity Identifier
Yes	Value	Business Target ID	8	
	date-time			
Yes	Length	0x00 where no date / time is required in this Message	1	Where date-time is not required for a Message, it shall be a 0 octet string as per the DLMS specification

Grouping Header				
Input to the ECDSA calculation	DLMS COSEM Message Elements	Contents	Length (octets)	Note
		0x0C where a date / time field is required		Where date-time is required for a Message, it shall be a 12 octet string as per the DLMS specification. See 'date-timestamp in response' column, 'Use Case reference' tab in Mapping Table
Yes	value	Either empty or a 12 character octet-string containing the date-time stamp for this Response	0 or 12	If (1) this is a Response from a GPF in one of Use Cases GCS13a, GCS13b, GCS13c, GCS14 or GCS60a, and (2) the GPF has a valid value stored for the <i>ReadingSnapshotTime</i> reported by the GSME, then that date-time shall be used to construct the value in this field, with bit 2 of clock status being set to 0b1 to show 'different clock base', since the GSME's clock is a different timing source than the GPF's normal timing source (the CH Clock). Otherwise, where a value is required by the Use Case, the field shall

Grouping Header				
Input to the ECDSA calculation	DLMS COSEM Message Elements	Contents	Length (octets)	Note
				be populated using the date-time in the Device's clock, and bit 2 set to 0b0 (since the time source is the Device's Clock).
	other-information			
Yes	Length	Encoding(X)	variable Len(Encoding(X))	X is length of other information octet string. X is variable
Yes	Value	Message Code Supplementary Remote Party ID Supplementary Remote Party Counter Supplementary Originator Counter Supplementary Remote Party Key Agreement Certificate	variable	The Message Code shall always be present In an Alert, Supplementary Remote Party ID shall be present, if it is required by Section 16 In a Command or Response, the Supplementary Remote Party ID, Supplementary Remote Party Counter and Supplementary Originator Counter, shall be present or not in line with the requirements of Section 4.3.1.4 Supplementary Remote Party Key Agreement

Grouping Header				
Input to the ECDSA calculation	DLMS COSEM Message Elements	Contents	Length (octets)	Note
				Certificate shall only be present where (1) this is a Command, (2) the Response to it should contain encrypted attributes and (3) the Supplementary Remote Party ID is for a Remote Party which does not already have a Key Agreement Public Key on the Device. It may only be present in Commands marked as allowing it in the column 'Key Agreement Certificate Potentially in Command?' of the Use Case reference tab of the Mapping Table
	Content			
Yes	length	Encoding(X)	Len(Encoding(X))	X is the length in octets of the Message Payload

1399 Table 7.2.7: Required components of Grouping Header

1400 Message construction – ASN.1 Security Payloads

1401 For Messages containing ASN.1 Security Payloads, the Payloads shall be constructed as
 1402 detailed in the Use Case for that Message Code (as defined by the Mapping Table).

1403 Message construction – DLMS COSEM Payloads

1404 For Messages containing DLMS COSEM payloads (as defined by the Message Code and Use
 1405 Cases in this GBCS):

1406 any Command Payload shall comply with the requirements of Table 7.2.9a and the
1407 associated Use Case;

1408 any Response Payload shall comply with the requirements of Table 7.2.9b and the associated
1409 Use Case; and

1410 any Alert Payload shall comply with the requirements of Table 7.2.9c and the associated Use
1411 Case.

DLMS COSEM Payloads – Commands				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	access-request	0xD9	1	DLMS COSEM APDU tag for Access Request (217 in decimal)
	long-invoke-id-and-priority	0x20 Least significant 24 bits of Originator Counter	4	Construction explained in rows below detailing bit (31..0) usage
	(bits 0-23) invoke-id	Least significant 24 bits of Originator Counter		
	(bits 24 -27) reserved	0b0000		Fixed value
	(bit 28) self-descriptive	0b0		Not-Self-Descriptive
	(bit 29) processing-option	0b1		Break on Error
	(bit 30) service-class	0b0		Unconfirmed
	(bit 31) priority	0b0		Normal
	date-time	0x00	1	A value for this element is not present so the length field is 0x00
	access-request-body			
	access-request-specification			
1	SEQUENCE OF	Use Case	1	The total number of gets, sets and actions in the Use Case (means that

DLMS COSEM Payloads – Commands				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
		<i>specific</i>		<i>there will be less than 128 in total). This content is specified in each DLMS COSEM Use Case</i>
2	<i>Use Case Specific Content</i>	<i>Use Case specific</i>		<i>The list of Gets, Sets and Actions specific to the Use Case. This content is specified in each DLMS COSEM Use Case</i>
	access-request-list-of-data			
	list-of-data			
3	SEQUENCE OF Data	<i>Use Case specific</i>	1	<i>The total number of attributes in the list-of-data in the Use Case (means that there will be less than 128 in total). This content is specified in each DLMS COSEM Use Case</i>
4	<i>Use Case Specific Content</i>	<i>Use Case specific</i>	<i>Use Case set</i>	<i>Values of the attributes required by the Use Case. This content is specified in each DLMS COSEM Use Case</i>

1412 Table 7.2.9a: Required components of Command Payload

1413 Elements marked in Table 7.2.9a as Use Case specific shall be populated according to the

1414 Use Case for the Message Code (see Section 19).

DLMS COSEM Payloads – Responses				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	access-response	0xDA	1	DLMS COSEM APDU tag for Access Response (218 in decimal)
	long-invoke-id-and-priority	0x20 Least significant 24 bits of	4	

DLMS COSEM Payloads – Responses				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
		Originator Counter		
	date-time	0x00	1	A value for this element is not needed so the length field is 0x00
	access-response-body			
	access-request - specification OPTIONAL	0x00	1	Not present so false (0x00)
	access-response-list-of-data			
	list-of-data			
5	SEQUENCE OF Data	Use Case specific	1	The total number of attributes in the Response in the Use Case. This content is specified in each DLMS COSEM Use Case
6	Use Case Specific Content	Use Case specific	Use Case set	Values of the attributes required by the Use Case. This content is specified in each DLMS COSEM Use Case
	access-response-specification			
7	SEQUENCE OF CHOICE	Use Case specific	1	The total number of responses, including the 1 here and those in the Use Case
8	Use Case Specific Content	Use Case specific	Use Case set	Fields stating the result of each Gets, Sets and Actions specific to the Use Case.

1415 Table 7.2.9b: Required components of Response Payload

1416 Elements marked in Table 7.2.9b as Use Case specific shall be populated according to the
 1417 Use Case for the Message Code (see Section 19).

DLMS COSEM Payloads – Alerts				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	data-notification	0x0F	1	DLMS COSEM APDU tag for data-notification (15 in decimal)
	long-invoke-id-and-priority	0x20 least significant 24 bits of Originator Counter	4	
	date-time	0x00	1	A value for this element is not needed so the length field is 0x00
	notification-body			
	Structure	0x02	1	
1	SEQUENCE OF Data	0x02 unless there is Use Case specific data additional	1	The majority of Alerts do not contain any additional data. For Alerts without additional data, there is no corresponding Use Case (since there is no Use Case specific content). Where an Alert does contain additional content, it has a specific Use Case. The additional content is specified in each such Use Case. In such cases, this field shall contain the total number of Data in the Use Case sequence plus the one in this template
	Data			
	Tag	0x12	1	Tag for LONG UNSIGNED
	Value	Alert Code	2	The Alert Code for this Alert, shall be as defined in Section 16

DLMS COSEM Payloads – Alerts				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	Data			
	Tag	0x09	1	Tag for octet-string
	Length	0x0C	1	Twelve characters long as DLMS date times are octet-string(12)
	Value	Time Stamp	12	The time stamp for this Alert, shall be as defined in Section 16
2	<i>Use Case Specific Additional Content</i>	<i>Use Case specific</i>	<i>Use Case</i>	<i>See Note at row 1, which means that, for most Alerts, there will be no Use Case specific content.</i>

1418 Table 7.2.9c: Required components of Alert Payload

1419 Elements marked in Table 7.2.9c as Use Case specific shall be populated according to the Use
 1420 Case for the Message Code.

1421 DLMS COSEM Alert population

1422 Devices shall construct Alerts with the following Message Codes according to the
 1423 requirements of this Section 7.2.9.1:

Use Case Name	ECS100 Command not supported by Device
<u>Facet</u>	<u>Value</u>
Grouping	Remote Party Message
Message Type	Alert
Summary	The SAPC shall create and send this Alert when it receives a Command which, in line with SMETS 9.1, it does not support but which would be supported if the SAPC were an ESME
Message Type Category	SME.A.C (see Section 6.7)
Payload	DLMS COSEM (see Table 7.2.9c)
Grouping Header Fields:	

1.	Message Code	0x0120
2.	Business Target ID	The Business Originator ID from the corresponding field in the Command
3.	Supplementary Remote Party ID	This field shall be included in the Alert if Supplementary Remote Party ID is present in the Command, and it shall take the same value as in the Command
4.	Supplementary Remote Party Counter	This field shall be included in the Alert if Supplementary Remote Party Counter is present in the Command, and it shall take the same value as in the Command
Payload Fields:		
5.	Alert Code	0x8F85
6.	Use Case Specific Additional Content	The concatenation: 0x09 0x0A Message Code Originator Counter where the last two fields have the values from the corresponding part of the Command. Note that 0x09 is the DLMS COSEM tag for octet-string and 0x0A is the length of the concatenation Message Code Originator Counter

1424 Table 7.2.9.1a: Use Case - ECS100 Command not supported by Device

1425

Use Case Name	ECS101 Limit APC [n] Level Command processed
<u>Facet</u>	<u>Value</u>
Grouping	Remote Party Message
Message Type	Alert
SMETS Reference(s)	The Device shall create and send this Alert as required by the SMETS sections detailed in Table 16.2 for this Alert Code
Message Type Category	SME.A.C (see Section 6.7)
Payload	DLMS COSEM (see Table 7.2.9c)
Grouping Header Fields:	
7.	Message Code
	0x0121

8.	Business Target ID	The Supplier's Entity Identifier
9.	Supplementary Remote Party ID	This field shall be absent
Payload Fields:		
10.	Alert Code	0x8F86
11.	Use Case Specific Additional Content	See the Message Template for 'ECS101 Limit APC [n] Level Command processed' in Section 18.2

1426 Table 7.2.9.1b: Use Case - ECS101 Limit APC [n] Level Command processed

1427

Use Case Name	ECS102 Limit APC [n] Level ended or cancelled
<u>Facet</u>	<u>Value</u>
Grouping	Remote Party Message
Message Type	Alert
SMETS Reference(s)	The Device shall create and send this Alert as required by the SMETS sections detailed in Table 16.2 for this Alert Code
Message Type Category	SME.A.C (see Section 6.7).
Payload	DLMS COSEM (see Table 7.2.9c)
Grouping Header Fields:	
12. Message Code	0x0122
13. Business Target ID	The Supplier’s Entity Identifier
14. Supplementary Remote Party ID	This field shall be absent
Payload Fields:	
15. Alert Code	0x8F87
16. Use Case Specific Additional Content	See the Message Template for ‘ECS102 Limit APC [n] Level ended or cancelled’ in Section 18.2

1428 Table 7.2.9.1c: Use Case - ECS102 Limit APC [n] Level ended or cancelled

1429

Use Case Name	ECS200 Operational Update
<u>Facet</u>	<u>Value</u>
Grouping	Remote Party Message
Message Type	Alert
SMETS Reference(s)	The Device may create and send this Alert when it wishes to notify either the Supplier, Network Operator or Load Controller (or any two of them) of some change in operational status. For clarity, GBCS does not specify the value of the utf8-string in 'Use Case Specific Additional Content'. However, the utf8-string shall be limited in length so that the total size of the Alert created does not exceed the Maximum PDU Size
Message Type Category	SME.A.C (see Section 6.7)
Payload	DLMS COSEM (see Table 7.2.9c)
Grouping Header Fields:	
17. Message Code	0x0123
18. Business Target ID	The Entity Identifier of either the Supplier, Network Operator or Load Controller
19. Supplementary Remote Party ID	If included, the Entity Identifier of either the Supplier, Network Operator or Load Controller
Payload Fields:	
20. Alert Code	0x8F88
21. Use Case Specific Additional Content	The concatenation: 0x0C Encoding (Len('value')) 'value' where 'value' is the content of the UTF-8 string the Device wishes to send, which shall comply with Section 7.2.9.2. Note that 0x0C is the DLMS COSEM tag for utf8-string

1430 Table 7.2.9.1d: Use Case – ECS200 Operational Update

1431 JavaScript Object Notation (JSON) in DLMS COSEM utf8-strings

1432 Where a Use Case specifies that the value of a DLMS COSEM utf8-string must comply with

1433 this Section 7.2.9.2, then the value of the utf8-string shall:

1434 comply with the requirements of IETF RFC825921;

1435 where it is part of a Remote Party Response, be limited in length so that the
1436 Response does not exceed 63 times 1,149 octets, so that the Response can be sent in at
1437 most 63 GBT Messages;

1438 where it is part of a Remote Party Alert, be limited in length so that the Alert does
1439 not exceed Maximum PDU Size octets;

1440 where it is to indicate that the Device does not hold information, be the string:

1441 {}

1442 This includes the case where the Device is required to report information for
1443 Auxiliary Controller [n], but there is currently no Auxiliary Controller associated with
1444 Auxiliary Controller [n], where those terms have their SMETS meaning; and

1445 where the Device is reporting Auxiliary Controller [n] State and there is currently an
1446 Auxiliary Controller associated with Auxiliary Controller [n], be the string:

1447 {"outputState": *p*, "inputState": *q other-data* }

1448 where:

1449 *p* is an integer value between 0 and 100 inclusive specifying the Auxiliary Controller [n]'s
1450 output state (which shall correspond to the commanded output state value in Section
1451 7.3.6.1);

1452 *q* is an integer value between 0 and 100 inclusive specifying the Auxiliary Controller [n]'s
1453 input state (which shall correspond to the commanded input state value in Section 7.3.6.1);
1454 and

1455 *other-data* is any other UTF-822 encoded information the Device includes, subject to the
1456 value of the whole string conforming to IETF RFC8259. For clarity, *other-data* may be the
1457 empty string.

1458 Message construction – GBZ Payloads

1459 A GBZ Payload shall be a Payload containing one or more ZSE / ZCL commands. For clarity,
1460 this includes Payloads in HAN Only Commands between a PPMID and a GSME.

1461 For Messages containing GBZ Payloads (as defined by the Mapping Table):

21 <https://tools.ietf.org/html/rfc8259>

22 <https://tools.ietf.org/html/rfc3629>

- 1462 any Command Payload shall comply with the requirements of Table 7.2.10a and the
1463 associated Use Case;
- 1464 any Response Payload shall comply with the requirements of Table 7.2.10b and the
1465 associated Use Case; and
- 1466 any Alert Payload shall comply with the requirements of Table 7.2.10c and the associated
1467 Use Case.
- 1468 Each GBZ Use Case Specific Component shall comply with:
- 1469 1. Table 7.2.10d if the ZSE / ZCL command within it is not encrypted; or
- 1470 2. Table 7.2.10e if the ZSE / ZCL command within it is encrypted.

GBZ Payloads – Commands				
No	Message Elements	Contents	Length (octets)	Note
	Profile ID	0x0109	2	ZSE
1	Total number of GBZ Use Case Specific Component(s)	See 'Note' column	1	This octet is to be interpreted as an 8 bit unsigned integer specifying the total number of GBZ Use Case Specific Component(s)
2	GBZ Use Case Specific Component(s)	Use Case specific		See Tables 7.2.10d and 7.2.10e

1471 Table 7.2.10a: Required components of GBZ Command Payload

1472 Elements marked in Table 7.2.10a as Use Case specific shall be populated according to the
1473 Use Case for the Message Code (see Section 15).

GBZ Payloads – Response				
No	Message Elements	Contents	Length (octets)	Note
	Profile ID	0x0109	2	ZSE
1	Total number of GBZ Use Case Specific Component(s)	See 'Note' column	1	This octet is to be interpreted as an 8 bit unsigned integer specifying the total number of GBZ Use Case Specific Component(s) in this Message

GBZ Payloads – Response				
No	Message Elements	Contents	Length (octets)	Note
2	<i>GBZ Use Case Specific Component(s)</i>	<i>Use Case specific</i>		<i>See Tables 7.2.10d and 7.2.10e</i>

1474 Table 7.2.10b: Required components of GBZ Response Payload

1475 Elements marked in Table 7.2.10b as Use Case specific shall be populated according to the
 1476 Use Case for the Message Code (see Section 15).

GBZ Payloads – Alerts				
No	Message Elements	Contents	Length (octets)	Note
	Profile ID	0x0109	2	ZSE
1	<i>Total number of GBZ Use Case Specific Component(s)</i>	<i>See 'Note' column</i>	1	<i>0x00 if there is no Use Case Specific Component or 0x01 if there is</i>
	Alert Code	<i>See 'Note' column</i>	2	The Alert Code for this Alert as defined in Section 16
	Timestamp	UTCTime	4	The <i>UTCTime</i> , with its ZCL meaning, at which this Alert was created
2	<i>GBZ Use Case Specific Component(s)</i>	<i>Use Case specific</i>		<i>For Alert Code 0x8F0A, see GCS53 and Table 7.2.10e. For Alert Codes 0x8F1C and 0x8F72, see Section 11.2.6. For Alert Codes 0x8F66 and 0x8F67, see Section 9.2.2.6. For Alert Code 0x81A0, see Section 16.4</i>

1477 Table 7.2.10c: Required components of GBZ Alert Payload

1478 Elements marked in Table 7.2.10c as Use Case specific shall be populated according to the
 1479 Use Case for the Message Code (see Section 15).

GBZ Use Case Specific Component without encrypted content

No	Message Elements	Contents	Length (octets)	Note
	Extended Header Control Field	0x00, 0x10, 0x11 or 0x01	1	Most significant nibble: 0x0 if 'From Date Time' not present; or 0x1 if 'From Date Time' present Least significant nibble: 0x0 (if not the last GBZ Use Case Specific Component in this Message); or 0x1 (if the last GBZ Use Case Specific Component in this Message)
	Extended Header Cluster ID	See 'Note' column	2	The Cluster ID of the ZSE / ZCL command contained in this GBZ Use Case Specific Component
	Extended Header GBZ Command Length	See 'Note' column	2	These two octets shall be interpreted as a 16 bit unsigned integer specifying the total length in octets of the remainder of this GBZ Component (so excluding this and prior fields)
	From Date Time	See 'Note' column	4	The earliest date-time of any entry that can be returned in the response, specified as a ZigBee UTCTime
	ZCL header	Use Case specific	3	These fields shall have the meaning specified in the ZSE / ZCL Specifications except for the Transaction Sequence Number. The Transaction Sequence Number shall be set to 0 for the first request-style ZSE / ZCL command in the Message and shall be incremented by one for every subsequent request-style ZSE / ZCL command frame in the

GBZ Use Case Specific Component without encrypted content				
No	Message Elements	Contents	Length (octets)	Note
				Message. The corresponding response-style ZSE / ZCL command frame shall copy the Transaction Sequence Number from the request-style ZSE / ZCL command frame
	ZCL payload	Use Case specific	Variable	These fields shall have the meaning specified in the ZSE / ZCL specifications, except where the GBCS specifies otherwise

1480 Table 7.2.10d: Required components of GBZ Use Case Specific Component without
 1481 encrypted content

1482

GBZ Use Case Specific Component with encrypted content				
No	Message Elements	Contents	Length (octets)	Note
	Extended Header Control Field	0x02 or 0x03	1	0x02 (if not the last GBZ Use Case Specific Component in this Message); or 0x03 (if the last GBZ Use Case Specific Component in this Message)
	Extended Header Cluster ID	See 'Note' column	2	The Cluster ID of the ZCL Command contained in this GBZ Use Case Specific Component
	Extended Header GBZ Command Length	See 'Note' column	2	These two octets shall be interpreted as a 16 bit unsigned integer specifying the total length in octets of the remainder of this GBZ Component (so excluding this and prior fields but including the 2 octets of Additional Header)

GBZ Use Case Specific Component with encrypted content				
No	Message Elements	Contents	Length (octets)	Note
	Additional Header Control	0x00	1	Reserved for future extensibility
	Additional Header Frame Counter	See 'Note' column	1	This octet is to be interpreted as an 8 bit unsigned integer. Its value shall be 0x00 for the first GBZ Use Case Specific Component with encrypted content in a Message. The value shall increase by one in each subsequent GBZ Use Case Specific Component with encrypted content in a Message
	ZCL header	See 'Note' column	3	These fields shall have the meaning specified in the ZigBee Cluster Library
	Length of Ciphred Information	See 'Note' column	2	These two octets shall be interpreted as a 16 bit unsigned integer specifying the total length in octets of the Ciphred Information
	Ciphred Information	See 'Note' column	Variable	See Section 8.4

1483 Table 7.2.10e: Required components of GBZ Use Case Specific Component with encrypted
 1484 content

1485 Transfer of Large Remote Party Messages²³

1486 All Devices which are not Type 2 Devices shall be capable of supporting the General Block
 1487 Transfer (GBT) requirements of this Section 7.2.11.

1488 GBT Terminology and Parameters

1489 A GBT Message shall be an APDU constructed and processed as defined by this Section
 1490 7.2.11.

²³ Terms defined within this Section are only used within this Section, and therefore not included in the Glossary (Section 21).

1491 A GBT Message Series shall be the set of GBT Messages needed to exchange one complete
 1492 Remote Party Message between a GBT Initiator and a GBT Recipient.

1493 For a Remote Party Command sent using a GBT Message Series, the GBT Initiator shall be the
 1494 Access Control Broker, or the HHT for Remote Party Commands delivered via the HHT and
 1495 the GBT Recipient shall be the target Device.

1496 For a Remote Party Response or a Remote Party Alert sent using a GBT Message Series, the
 1497 GBT Initiator shall be the sending Device and the GBT Recipient shall be the Access Control
 1498 Broker and the HHT where such Remote Party Messages are also delivered to the HHT in line
 1499 with the requirements of Section 10.5.3.2.

1500 A GBT Third Party shall be the Remote Party identified by:
 1501 in a Remote Party Command, the value in the Business Originator ID field; and
 1502 in a Remote Party Response or a Remote Party Alert, the value in the Business Target ID field.

1503 GBT Streaming Window shall be the number of GBT Messages the GBT Initiator sends
 1504 without receipt of a GBT Message (Acknowledgement), or since receipt of the most recent
 1505 GBT Message (Acknowledgement).

1506 GBT Streaming Window shall be:
 1507 63 where the GBT Message Series carries a Remote Party Response;
 1508 6 where the GBT Message Series carries a Remote Party Command; or
 1509 the number of GBT Messages the sender wishes to be resent in response to a GBT Message
 1510 (Request Block Resend).

1511 Maximum PDU Size shall be 1200 octets.

1512 For clarity, recovery of lost blocks may take place before the end of the streaming window.

1513 Remote Party Message size

1514 Where a Remote Party Message exceeds the Maximum PDU Size, the GBT Initiator and the
 1515 GBT Responder shall exchange the Remote Party Message in a GBT Message Series.

1516 Where a Remote Party Message does not exceed the Maximum PDU Size, the GBT Initiator
 1517 and the GBT Responder may exchange the Remote Party Message in a GBT Message Series.

1518 GBT Message Structure

1519 A GBT Message shall, if it is a GBT Message (Acknowledgement) or a GBT Message (Request
 1520 Block Resend), be the concatenation:

1521 Message Routing Header || GBT Header

1522 A GBT Message shall, if it is neither a GBT Message (Acknowledgement) nor a GBT Message
1523 (Request Block Resend), be the concatenation:

1524 Message Routing Header || GBT Header || GBT Block Data

1525 where:

1526 Message Routing Header shall be structured and populated according to Table 7.2.11.5.

1527 Note that Message Routing Header (1) uniquely identifies the GBT Message Series, (2)
1528 identifies whether the GBT Message is being sent to or from the Device and (3)
1529 unambiguously ties all GBT Messages in the GBT Message Series to the single Remote Party
1530 Message being exchanged;

1531 GBT Header shall be structured and populated according to Table 7.2.11.6; and

1532 GBT Block Data shall be the part of the Remote Party Message, constructed as per Section
1533 7.2.11.4, being carried in this GBT Message.

1534 GBT Message processing

1535 The GBT Initiator shall, once the Remote Party Message is fully constructed and all
1536 cryptographic protections are applied, slice the octet string produced so that:

- 1537 1. GBT Block Data with GBT Initiator Block Number of 1 is the 1149 most significant octets
1538 of the Remote Party Message, or all of the octets if the size of the Remote Party
1539 Message is less than 1149 octets;
- 1540 2. GBT Block Data with GBT Initiator Block Number of 2 is the next 1149 most significant
1541 octets of the Remote Party Message, or all of the octets if the size of the remaining
1542 octets in Remote Party Message is less than 1149 octets; and
- 1543 3. remaining GBT Block Data are created by repeating Step 2, each time incrementing GBT
1544 Initiator Block Number by 1, until there are no remaining octets in the Remote Party
1545 Message.

1546 The GBT Recipient shall not undertake any processing, in the sense of Section 6, of the
1547 Remote Party Message carried in a GBT Message Series until it has received:

1548 a GBT Message in this GBT Message Series where the 'last-block' field contains 0b1 (meaning
1549 last block); and

1550 all GBT Messages in this GBT Message Series with 'block-number' fields less than the 'block-
1551 number' field in the last block. Where the GBT Recipient has not received all such GBT
1552 Messages, it shall send a GBT Message (Request Block Resend) for each missing block-
1553 number. Where the GBT Recipient is a Device, it may discard all blocks in a GBT Message

1554 Series if it has received no response to a GBT Message (Request Block Resend) after 60
1555 minutes.

1556 For clarity, the GBT Recipient may, where it is a Device, immediately discard, without any
1557 further processing, any GBT Message where:

1558 the Message Code in the Message Routing Header is for a Command, which requires
1559 Protection Against Replay; and

1560 the Device's Execution Counter, for the Business Originator ID and Message Code specified in
1561 the Message Routing Header, is equal to or greater than the Originator Counter in the
1562 Message Routing Header

1563 When a GBT Message Series carries a Response and that GBT Message Series consists of
1564 more than 63 GBT Messages, the GBT Recipient shall, if it wishes to receive all parts of the
1565 GBT Message Series, send a GBT Message (Acknowledgement) when it has received each
1566 complete set of 63 sequential GBT Messages. Note, when the Device sending the Response
1567 receives such a GBT Message (Acknowledgement), it is able to begin sending the next
1568 sequence of GBT Messages in the GBT Message Series.

1569 GBT Recipient Block Number shall be set to 0x0001 in the first GBT Message sent by the GBT
1570 Recipient. It shall be incremented by 1 in each subsequent GBT Message it sends.

1571 GBT Initiator Block Number Ack shall be the highest of:

1572 0x0000; and

1573 the highest block-number in any GBT Message the GBT Initiator has received in this GBT
1574 Message Series.

1575 GBT Recipient Block Number Ack shall:

1576 in a GBT Message (Acknowledgement), be the highest block-number in any GBT Message the
1577 GBT Recipient has received in this GBT Message Series; and

1578 in a GBT Message (Request Block Resend), the value of block-number up to which the GBT
1579 Recipient has received all the prior numbered GBT Messages in this GBT Message Series.

1580 Where the GBT Initiator is a Device, the Device shall be able to resend any GBT Message
1581 within a GBT Message Series, for a minimum period from when it sends the first GBT
1582 Message in that series, to whichever is the sooner of:

1583 it receiving an authenticated GBT Message (Acknowledgement) where the GBT Recipient
1584 Block Number Ack contains a value equal to the highest value of GBT Initiator Block Number
1585 in this GBT Message Series; or

1586 24 hours later.

1587 For clarity, Devices shall discard malformed GBT messages without further processing in the
 1588 sense of Section 6. For example, the following GBT Messages would be malformed:

1589 one with 'block-number' equal to zero (this does not comply with the GBCS); and

1590 one delivered to a recipient where 'block-number-ack' is greater than the greatest 'block-
 1591 number' in any GBT Message the recipient has sent in that GBT Message Series.

1592 Where the GBT Recipient is an ESME or an SAPC, the Device shall not send a GBT Message
 1593 (Request Block Resend), in relation to a GBT Message Series, until and unless 30 seconds
 1594 have elapsed since receipt of the most recent GBT Message in that GBT Message Series. If 30
 1595 minutes elapse without the Device receiving corresponding GBT Message(s) in response, the
 1596 Device may discard the GBT Messages it has received in this GBT Message Series.

1597 Message Routing Header

Message Routing Header				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	general-ciphering	0xDD	1	Tag used is the same as for a normal DLMS General-Ciphering header
	transaction-id			
	Length	0x09	1	Length of Originator Counter
	Value	See 'Note' column	9	Shall be populated with the corresponding field from the Grouping Header in the Remote Party Message that is being carried in this GBT Message Series
	originator-system-title			
	Length	0x08	1	Length of Entity Identifier
	Value	Business Originator ID	8	If the GBT Message is sent from the Device, the value shall be the Entity Identifier of the Device

Message Routing Header				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
				If the GBT Message is sent to the Device, the value shall be the Entity Identifier of the GBT Third Party
	recipient-system-title			
	Length	0x08	1	Length of Entity Identifier
	Value	Business Target ID	8	If the GBT Message is sent to the Device, the value shall be the Entity Identifier of the Device If the GBT Message is sent from the Device, the value shall be the Entity Identifier of the GBT Third Party
	date-time	0x00	1	A value for this element is not needed so the length field is 0x00
	other-information			
	Length	0x02	1	Length of Message Code
	Value	Message Code	2	Shall be populated with the corresponding field from the Grouping Header in the Remote Party Message that is being carried in this GBT Message Series
	key-info	0x00	1	key-info values are not present so encoded as 0x00
	ciphered-service			
	Length	Encoding(X)	Len(Encoding(X))	X shall be the length in octets of the subsequent parts of the GBT

Message Routing Header				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
				Message after this length value
	security header			
	security control byte (SC)	0x01	1	Specifies that no MAC field is present at the end of the APDU
	invocation counter (IC)	0x00000000	4	IC is always zero

1598 Table 7.2.11.5: Message Routing Header

1599 GBT Header

GBT Header				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	general-block-transfer	0xE0	1	DLMS COSEM APDU tag for General-Block-Transfer
	block-control			
	last-block (bit 7)	See 'Note' column	1/8	0b0 if not the last GBT Message in this GBT Message Series the sender has to send, or 0b1 if this is the last GBT Message in this GBT Message Series the sender has to send
	streaming (bit 6)	See 'Note' column	1/8	0b1 if the sender does not require a GBT Message in response, or 0b0 if the sender does require a GBT Message in response
	window (bits 0 - 5)	See 'Note' column	6/8	The value of GBT Streaming Window as required by Section 7.2.11.1

GBT Header				
No	DLMS COSEM Message Elements	Contents	Length (octets)	Note
	block-number	See 'Note' column	2	GBT Initiator Block Number, if this GBS Message is sent by the GBT Initiator GBT Recipient Block Number, if this GBS Message is sent by the GBT Recipient
	block-number-ack	See 'Note' column	2	GBT Initiator Block Number Ack, if this GBS Message is sent by the GBT Initiator GBT Recipient Block Number Ack, if this GBS Message is sent by the GBT Recipient
	block-data			
	Length	Encoding(X)	Len(Encoding(X))	X is the length in octets of the following parts of this APDU

1600 Table 7.2.11.6: GBT Header

1601 Illustrations – informative

1602 GBT allows for the transport of Messages where the Message is greater than the Maximum
 1603 PDU Size. A number of Use Cases can result in this larger Message size, either as a
 1604 Command or a Response. There are no Alert Use Cases that result in the larger Message size.

1605 GBT does not change any part of the Remote Party Message content that is being
 1606 transported.

1607 Example 1: A small Command with small Response – e.g. read MPAN.

1608 Without GBT, the Command is:

1609 MAC Header || Grouping Header || read MPAN Command Payload || 0x00 || ACB-SMD
 1610 MAC

1611 and the Response is:

1612 MAC Header || Grouping Header || read MPAN Response Payload || 0x00 || SMD-KRP
 1613 MAC

1614 GBT can be applied to this Use Case. The Command becomes:

1615 Message Routing Header || GBT Header || MAC Header || Grouping Header || read MPAN
 1616 Command Payload || 0x00 || ACB-SMD MAC

1617 and the Response becomes:

1618 Message Routing Header || GBT Header || MAC Header || Grouping Header || read MPAN
 1619 Response Payload || 0x00 || SMD-KRP MAC

1620 Example 2: A large Command with small Response – e.g. set tariff on an ESME where the
 1621 tariff is complex.

1622 Without GBT, the Command is:

1623 MAC Header || Grouping Header || set Tariff Command Payload || 0x40 || KRP Signature
 1624 || ACB-SMD MAC

1625 For the purposes of example, assume this is divided into three blocks, so block-numbers 1, 2
 1626 and 3. Actual set tariff Commands will vary from this number of blocks.

1627 The Command is transmitted as the GBT Message Series:

1628 Message Routing Header || GBT Header || block 1

1629 Message Routing Header || GBT Header || block 2

1630 Message Routing Header || GBT Header || block 3

1631 This is reconstructed at the ESME, by concatenating blocks 1, 2 and 3 to give:

1632 MAC Header || Grouping Header || set Tariff Command Payload || 0x40 || KRP Signature
 1633 || ACB-SMD MAC

1634 The Response could have the following structure:

1635 Message Routing Header || GBT Header || Grouping Header || Response Payload || 0x40
 1636 || SMD Signature

1637 It will be smaller than the Maximum PDU Size and so can be sent as a single APDU without
 1638 any use of GBT. In this case it would have the following structure:

1639 Grouping Header || Response Payload || 0x40 || SMD Signature

1640 Example 3: small Command with large Response – e.g. read half-hourly profile (Export)

1641 The Command is:

1642 MAC Header || Grouping Header || read half-hourly profile (Export) Command Payload ||
 1643 0x00 || ACB-SMD MAC

1644 It will be smaller than the Maximum PDU Size and so can be sent as a single APDU without
1645 any use of GBT.

1646 The ESME Response is:

1647 MAC Header || Grouping Header || read half-hourly profile (Export) Response Payload ||
1648 0x00 || SMD-KRP MAC

1649 Assuming that there are 75 blocks to send the GBT Message Series would, if no retries are
1650 needed, be as follows:

1651 The ESME will send:

1652 Message Routing Header || GBT Header || Block 1

1653 ...

1654 Message Routing Header || GBT Header || Block 63

1655 and wait for acknowledgement.

1656 The Access Control Broker will construct and send that acknowledgement whose structure
1657 is:

1658 Message Routing Header || GBT Header

1659 When this acknowledgement is received by the ESME, the ESME will send:

1660 Message Routing Header || GBT Header || Block 64

1661 ...

1662 Message Routing Header || GBT Header || Block 75

1663 When the whole Message is received by the Access Control Broker, the Response can then
1664 be reconstructed:

1665 MAC Header || Grouping Header || read half-hourly profile (Export) Response Payload ||
1666 0x00 || SMD-KRP MAC

1667 Once the response has been reconstructed, the MAC can be checked.

1668 Device Requirements – DLMS COSEM

1669 Introduction – informative

1670 The DLMS COSEM server in the ESME and SAPC (and CHF where a DLMS COSEM server is
1671 present) responds to requests for information, and also provides Alerts in response to events
1672 within the Device (e.g. push data at the end of billing period; Alert in the event of a tamper;

1673 disable supply when prepayment credit expires). To achieve this, a level of configuration is
1674 needed to ensure that the behaviour of the Device is as expected.

1675 The Technical Specifications require that the Critical Commands mandated by them (and so
1676 those defined in the GBCS) are the only Critical commands allowed. Devices may implement
1677 additional non Critical features only.

1678 The Technical Specifications only require DLMS COSEM certification on the ESME and the
1679 SAPC.DLMS COSEM objects (or functionality equivalent to them) are required to deliver the
1680 ESME / SAPC functionality defined in the Use Cases in a consistent way but should not be
1681 accessible via the Device's HAN interface (i.e. it is internal functionality).

1682 General Requirements

1683 Constant values specified in Table 7.3.8a shall be fixed before operation and shall be
1684 immutable save via a firmware upgrade. This is to ensure consistent functioning and guard
1685 against potential attacks. Except where explicitly required by this Section 7.3, a Device shall
1686 not expose any part of any DLMS COSEM object, either for the writing of an attribute or for
1687 the invocation of a method that could, if used, constitute a Critical action.

1688 For Devices which are not ESME, SAPC or CHF (so where deviceType <> 1 or 2), the GBCS
1689 does not require the implementation of any DLMS COSEM objects.

1690 All Devices which are ESME (so where deviceType = 1):

1691 shall implement all of the DLMS COSEM objects, attributes and methods detailed in 'SMETS
1692 Required Objects' tab of the table in Section 20, and expose the specified attributes and
1693 methods over its network interface; and

1694 shall have the constant values set for the DLMS COSEM attributes specified as requiring
1695 constant values in Table 7.3.8a, and shall ensure that such values cannot be amended, save
1696 via activation of new firmware.

1697 All Devices which are SAPC (so where deviceType = 1):

1698 shall implement all of the DLMS COSEM objects, attributes and methods detailed in the
1699 'SMETS Required Objects' tab of Table 20 which are:

1700 marked 'mandatory' in the 'SAPC DLMS COSEM support' column; or

1701 marked 'optional' in the 'SAPC support' column and are required for Use Cases that the
1702 SAPC does support,

1703 and shall expose such specified attributes and methods over its network interface;
1704 and

1705 shall, where the SAPC supports the corresponding SMETS functionality, have the constant
1706 values set for the DLMS COSEM attributes specified as requiring constant values in Table
1707 7.3.8a, and shall ensure that such values cannot be amended, save via activation of new
1708 firmware.

1709 Application Associations

1710 An ESME, SAPC or CHF shall communicate using pre-established Application Associations
1711 (AA). These shall be set at manufacture, and the Device shall reject all subsequent attempts
1712 to open or release Application Associations.

1713 An ESME, SAPC or CHF shall support the Application Associations in Table 7.3.8c. An ESME,
1714 SAPC or CHF shall not support any additional Application Associations.

1715 The Application Associations in Table 7.3.8c shall limit access to DLMS features by
1716 configuring the object_list attribute to reflect the access granted to the role in 'SMETS
1717 Required Objects' tab of the Mapping Table in Section 20. For SAPC, the object_list attribute
1718 settings shall limit access in line with the 'SMETS required objects' tab of Table 20 to only
1719 those those rows which are either:

1720 marked 'mandatory' in the 'SAPC DLMS COSEM support' column; or

1721 marked 'optional' in the 'SAPC DLMS COSEM support' column and are required for Use
1722 Cases that the SAPC does support.

1723 Any other methods and attributes of any class shall be made inaccessible by listing them in
1724 the object_list attribute such that there is no access.

1725 The Public AA shall only expose:

1726 the SAP Assignment object; and

1727 the DLMS COSEM Logical Device name object and object_list with no objects listed other
1728 than the Association Logical Name (LN) Object (with its Blue Book meaning) and the SAP
1729 Assignment Object.

1730 When a Message is received by the Device, the Message shall be validated against the AA
1731 based on the Business Originator ID and the Message Code within the Grouping Header.

1732 Other attributes in the Association LN Objects and the Security Setup Objects shall be set at
1733 manufacture in accordance with Tables 7.3.8d and 7.3.8e.

1734 The 'SAP Assignment' object shall be configured at manufacture in accordance with Table
1735 7.3.8f. The method associated with the 'SAP Assignment' object shall not be accessible to
1736 any Application Association.

1737 Interface Classes and Objects

-
- 1738 Devices shall support the version of Interface Classes shown as current in the Blue Book.
- 1739 An ESME and an SAPC shall support the 'Class 9000' as detailed in Section 22 of this GBCS.
- 1740 Unless explicitly required in a predetermined script or the SMETS 'Required Objects' tab of
 1741 the Mapping Table, Class 3 objects shall not have a reset method that is accessible external
 1742 to the Device.
- 1743 Unless otherwise stated, Generic Profile objects with a non-zero attribute 4 shall capture the
 1744 first entry at midnight UTC.
- 1745 Values normally negotiated when an AA is established
- 1746 Conformance Block Contents
- 1747 The conformance block shall be set according to Table 7.3.8g.
- 1748 Other Items.
- 1749 Other items for pre-establishing the Application Associations and other communication
 1750 parameters shall be implemented as detailed in Table 7.3.8h.
- 1751 Security Setup Objects
- 1752 Security Setup Objects shall be limited to those listed in Table 7.3.8c.
- 1753 Manufacturer specific attributes and methods for these objects shall not be accessible
 1754 external to the Device.
- 1755 The methods of the Security Setup objects shall not be accessible external to the Device. The
 1756 attributes of the Security Setup objects shall be as specified in Table 7.3.8e.
- 1757 Note that Security Credentials are updated as specified in Section 13.
- 1758 Scripts for operation of the ESME / SAPC
- 1759 Scripts required for operation of the Device shall be as listed in Table 7.3.8b.
- 1760 The Device shall ensure that the script table objects shall be read only. The Device shall
 1761 ensure that a script table object entries shall only be executable by the corresponding
 1762 Application Association specified in Table 7.3.8b.
- 1763 The Device shall ensure that a script table object's entries shall only be executable from an
 1764 activity calendar, scheduler, or single action scheduler controlled by the corresponding
 1765 Application Association application in Table 7.3.8b.
- 1766 Auxiliary Controller related scripts
- 1767 All italicised terms in this Section 7.3.6.1 shall have their DLMS COSEM meanings.

1768 As required by DLMS COSEM, the *execute* method (*method 1*) of a *script table* object
1769 (*class_id* = 9) takes a *long-unsigned* value, so two octet value, as its parameter. To have an
1770 effect, the *value* of that *long-unsigned* parameter has to match one of the *script_identifiers* in
1771 that object's *scripts* attribute (*so attribute 2*). Similar requirements apply to the
1772 *script_selector* values in *schedule* objects (*class_id* = 10).

1773 This Section 7.3.6.1 specifies the *script_identifier* values that SAPC and ESME shall support
1774 (and so the values such Devices shall accept in associated Use Cases) for *script table* objects
1775 with OBIS codes 0-0:10.0.103.255 and 0-0:10.0.107.255, and the functionality that such
1776 Devices shall implement when each such *script_identifier* in these two objects is executed.

1777 SAPC and ESME shall support all *script_identifiers* in objects 0-0:10.0.103.255 and 0-
1778 0:10.0.107.255 where:

1779 the most significant nibble of the most significant octet has a value of either 0x1, indicating
1780 output to the controlled load, or 0x2, indicating input;

1781 the least significant nibble of the most significant octet has a value of either 0x1, 0x2, 0x3,
1782 0x4 or 0x5, indicating 'n' in relation to Auxiliary Controller [n];

1783 for 0-0:10.0.103.255, the most significant bit of the least significant octet has a value of 0b0
1784 or 0b1;

1785 for 0-0:10.0.107.255, the most significant bit of the least significant octet has a value of 0b0;
1786 and

1787 the seven least significant bits of the least significant octet have a value between 0b00000000
1788 and 0b1100100 inclusive, so between 0 and 100.

1789 SAPC and ESME shall undertake the following processing based on the value of
1790 *script_identifier*, when a *script* in these *script tables* is invoked:

1791 where the most significant octet has a value of 0x1n, the Device shall only undertake any
1792 further script invocation processing where there is currently an Auxiliary Controller
1793 associated with Auxiliary Controller [n] and shall only undertake processing in relation to
1794 that Auxiliary Controller;

1795 where the most significant octet has a value of 0x2n, the Device shall only undertake any
1796 further script invocation processing where there is currently an Auxiliary Proportional
1797 Controller associated with Auxiliary Controller [n] and shall only undertake processing in
1798 relation to that Auxiliary Proportional Controller;

1799 where the script invocation is being requested as part of an 'ECS47e Limit APC [n] Level'
1800 Command's execution, the Device shall only undertake any further script invocation

1801 processing where the currently associated Auxiliary Controller is an Auxiliary Proportional
1802 Controller;

1803 for 0-0:10.0.103.255, where the most significant bit of the least significant octet has a value
1804 of 0b1, the Device shall only take script invocation action where the *script* execution is being
1805 requested by the Auxiliary Controller Calendar (with its SMETS meaning) and so the *schedule*
1806 object with OBIS code 0-0:12.0.2.255;

1807 where the most significant bit of the least significant octet has a value of 0b0, the Device
1808 shall only take script invocation action where the *script* execution is being requested as part
1809 of an 'ECS47a Set Auxiliary Controller [n] State' or 'ECS47e Limit APC [n] Level' Command's
1810 execution (so where the value of the *data* parameter in the *execute* method / the
1811 controllerNumberAndCommandedState field meets this requirement);

1812 where the preceding conditions are all met, the Device shall undertake processing for the
1813 commanded state of the Auxiliary Controller [n] according to the SMETS requirements, using
1814 the following interpretation of the seven least significant bits of the least significant octet
1815 and, where relevant, the most significant nibble of the most significant octet:

1816 for an Auxiliary Proportional Controller (with its SMETS meaning), the value of the seven
1817 least significant bits of the least significant octet shall be interpreted as a percentage
1818 between 0% (so 0b0000000) and 100% (so 0b1100100), which shall relate:

1819 to the output state, if the most significant nibble of the most significant octet is 0x1; or
1820 to the input state, if the most significant nibble of the most significant octet is 0x2;

1821 for an ALCS, the value of 0b1100100 in the seven least significant bits of the least significant
1822 octet shall be interpreted as meaning closure of the switch (so allowing energy to flow) and
1823 any other value shall be interpreted as meaning opening of the switch (so not allowing
1824 energy to flow); and

1825 for HCALCS, the value of 0b1100100 in the seven least significant bits of the least significant
1826 octet shall be interpreted as meaning closure of the switch (so allowing energy to flow) and
1827 so requiring the 'Duty Cycle' parameter in any resulting Load Control Event command to be
1828 0x64; any other value shall be interpreted as meaning opening of the switch (so not allowing
1829 energy to flow) and so requiring the 'Duty Cycle' parameter in any resulting Load Control
1830 Event command to be 0x00.

1831 Where that processing leads to a change in the commanded state of Auxiliary Controller [n],
1832 the Device shall update the Auxiliary Controller [n] State's commanded state (so the value of
1833 either '*p*' or '*q*' in Section 7.2.9.2 and, where the Alerts are required by SMETS, the value of
1834 'resultingLevel' in each of Use Cases 'ECS101 Limit APC [n] Level Command processed' and
1835 'ECS102 Limit APC [n] Level ended or cancelled') to reflect the current commanded state:

1836 where Auxiliary Controller [n] is an ALCS or HCALCS and its current commanded state is
 1837 closure of the switch, 'p' and 'q' shall have the value 100. If its current commanded state is
 1838 opening of the switch, 'p' and 'q' shall have the value 0; and

1839 where Auxiliary Controller [n] is an Auxiliary Proportional Controller, 'p' shall have a value
 1840 between 0 and 100 inclusive, reflecting the percentage to which its commanded output
 1841 state is currently set, 'q' shall have a value between 0 and 100 inclusive, reflecting the
 1842 percentage to which its commanded input state is currently set.

1843 Correspondingly, as part of the step 4 checks in Section 6.2.4.1.1, Devices shall, for:

1844 *script_selector* values in Use Case 'ECS46d Set Auxiliary Controller Calendar' Commands; and,

1845 *data* values in the *execute method* of Use Case 'ECS47a Set Auxiliary Controller [n] State' /
 1846 'ECS47e Limit APC [n] Level' Commands,

1847 apply the following validation to each of those values:

1848 the most significant nibble of the most significant octet has a value of 0x1 or 0x2;

1849 the least significant nibble of the most significant octet has a value of either 0x1, 0x2, 0x3,
 1850 0x4 or 0x5;

1851 the seven least significant bits of the least significant octet has a value between 0b0000000
 1852 and 0b1100100 inclusive, so between 0 and 100; and

1853 for Use Case 'ECS46d Set Auxiliary Controller Calendar', the most significant bit of the least
 1854 significant octet is 0b1; and

1855 for Use Cases 'ECS47a Set Auxiliary Controller [n] State' and 'ECS47e Limit APC [n] Level', the
 1856 most significant bit of the least significant octet is 0b0.

1857 When creating an entry in the Auxiliary Controller Event Log (with its SMETS meaning) which
 1858 relates to the execution of a script in objects 0-0:10.0.103.255 and 0-0:10.0.107.255, the
 1859 Device shall record the *script_identifier* value in the
 1860 *entry_auxiliaryControllerLogEntry.switchNumberAndAction* field of that log entry.

1861 Pricing matrices, scripts and registers

1862 Where an SAPC supports the related SMETS functionality, statements and requirements in
 1863 this Section 7.3.7 shall apply to the SAPC as if it were an ESME.

1864 Summary of approach – informative

1865 As required by SMETS, an ESME has:

1866 a 1 * 48 matrix of primary element TOU (Time Of Use) consumption registers, and an
 1867 associated 1 * 48 matrix of consumption based prices;

1868 a 4 block by 8 time band matrix of primary element 'TOU with Blocks' consumption registers,
 1869 and an associated 4 * 8 matrix of consumption based prices;

1870 a tariff switching table, which specifies time based switching between primary consumption
 1871 registers and so between different prices. (Switching between blocks is based on
 1872 consumption passing thresholds and not on time);

1873 for Twin Element ESME only, a 1 * 4 matrix of secondary element TOU consumption registers,
 1874 and an associated 1 * 4 matrix of consumption based prices; and

1875 for Twin Element ESME only, a secondary tariff switching table, which specifies time based
 1876 switching between registers and so between different consumption based prices.

1877 There needs to be a clear mapping, at the level of encoded instructions in Commands to the
 1878 ESME, between the switching table entries (identified by Script Selector), Consumption
 1879 Registers and consumption based prices (identified by Index). The next Section details that
 1880 mapping for the encoded DLMS COSEM elements used by the ESME.

1881 Note that the mapping of Script Selector is included for information in Table 7.3.7.2, but the
 1882 requirements for that mapping are in Section 7.3.8.

1883 ESME requirements

1884 An ESME shall:

1885 in calculating the cost of Consumption, for a Consumption Register specified in Table 7.3.7.2
 1886 apply the charge_per_unit in the charge_table_element identified by corresponding Index
 1887 specified in Table 7.3.7.2 of the unit_charge_active attribute of the corresponding Charge
 1888 Object specified in Table 7.3.7.2; and

1889 reject any instruction to set the unit_charge_passive attribute of a Charge Object in Table
 1890 7.3.7.2 that does not include all of the charge_table_elements specified in Table 7.3.7.2 for
 1891 that Charge Object or contains charge_table_elements with values of index that are not in
 1892 Table 7.3.7.2 for that Charge Object.

Description	Script Selector (long unsigned)	Consumption Register	Charge Object	Index (octet-string(1))	Notes
TOU(1)	0x0001	1-0:1.8.1.255	0-0:19.20.0.255	0x01	
TOU(2)	0x0002	1-0:1.8.2.255	0-0:19.20.0.255	0x02	
TOU(3)	0x0003	1-0:1.8.3.255	0-0:19.20.0.255	0x03	
TOU(4)	0x0004	1-0:1.8.4.255	0-0:19.20.0.255	0x04	
TOU(5)	0x0005	1-0:1.8.5.255	0-0:19.20.0.255	0x05	

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Description	Script Selector (long unsigned)	Consumption Register	Charge Object	Index (octet-string(1))	Notes
TOU(6)	0x0006	1-0:1.8.6.255	0-0:19.20.0.255	0x06	
TOU(7)	0x0007	1-0:1.8.7.255	0-0:19.20.0.255	0x07	
TOU(8)	0x0008	1-0:1.8.8.255	0-0:19.20.0.255	0x08	
TOU(9)	0x0009	1-0:1.8.9.255	0-0:19.20.0.255	0x09	
TOU(10)	0x000A	1-0:1.8.10.255	0-0:19.20.0.255	0x0A	
TOU(11)	0x000B	1-0:1.8.11.255	0-0:19.20.0.255	0x0B	
TOU(12)	0x000C	1-0:1.8.12.255	0-0:19.20.0.255	0x0C	
TOU(13)	0x000D	1-0:1.8.13.255	0-0:19.20.0.255	0x0D	
TOU(14)	0x000E	1-0:1.8.14.255	0-0:19.20.0.255	0x0E	
TOU(15)	0x000F	1-0:1.8.15.255	0-0:19.20.0.255	0x0F	
TOU(16)	0x0010	1-0:1.8.16.255	0-0:19.20.0.255	0x10	
TOU(17)	0x0011	1-0:1.8.17.255	0-0:19.20.0.255	0x11	
TOU(18)	0x0012	1-0:1.8.18.255	0-0:19.20.0.255	0x12	
TOU(19)	0x0013	1-0:1.8.19.255	0-0:19.20.0.255	0x13	
TOU(20)	0x0014	1-0:1.8.20.255	0-0:19.20.0.255	0x14	
TOU(21)	0x0015	1-0:1.8.21.255	0-0:19.20.0.255	0x15	
TOU(22)	0x0016	1-0:1.8.22.255	0-0:19.20.0.255	0x16	
TOU(23)	0x0017	1-0:1.8.23.255	0-0:19.20.0.255	0x17	
TOU(24)	0x0018	1-0:1.8.24.255	0-0:19.20.0.255	0x18	
TOU(25)	0x0019	1-0:1.8.25.255	0-0:19.20.0.255	0x19	
TOU(26)	0x001A	1-0:1.8.26.255	0-0:19.20.0.255	0x1A	
TOU(27)	0x001B	1-0:1.8.27.255	0-0:19.20.0.255	0x1B	
TOU(28)	0x001C	1-0:1.8.28.255	0-0:19.20.0.255	0x1C	
TOU(29)	0x001D	1-0:1.8.29.255	0-0:19.20.0.255	0x1D	
TOU(30)	0x001E	1-0:1.8.30.255	0-0:19.20.0.255	0x1E	
TOU(31)	0x001F	1-0:1.8.31.255	0-0:19.20.0.255	0x1F	
TOU(32)	0x0020	1-0:1.8.32.255	0-0:19.20.0.255	0x20	
TOU(33)	0x0021	1-0:1.8.33.255	0-0:19.20.0.255	0x21	
TOU(34)	0x0022	1-0:1.8.34.255	0-0:19.20.0.255	0x22	

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Description	Script Selector (long unsigned)	Consumption Register	Charge Object	Index (octet-string(1))	Notes
TOU(35)	0x0023	1-0:1.8.35.255	0-0:19.20.0.255	0x23	
TOU(36)	0x0024	1-0:1.8.36.255	0-0:19.20.0.255	0x24	
TOU(37)	0x0025	1-0:1.8.37.255	0-0:19.20.0.255	0x25	
TOU(38)	0x0026	1-0:1.8.38.255	0-0:19.20.0.255	0x26	
TOU(39)	0x0027	1-0:1.8.39.255	0-0:19.20.0.255	0x27	
TOU(40)	0x0028	1-0:1.8.40.255	0-0:19.20.0.255	0x28	
TOU(41)	0x0029	1-0:1.8.41.255	0-0:19.20.0.255	0x29	
TOU(42)	0x002A	1-0:1.8.42.255	0-0:19.20.0.255	0x2A	
TOU(43)	0x002B	1-0:1.8.43.255	0-0:19.20.0.255	0x2B	
TOU(44)	0x002C	1-0:1.8.44.255	0-0:19.20.0.255	0x2C	
TOU(45)	0x002D	1-0:1.8.45.255	0-0:19.20.0.255	0x2D	
TOU(46)	0x002E	1-0:1.8.46.255	0-0:19.20.0.255	0x2E	
TOU(47)	0x002F	1-0:1.8.47.255	0-0:19.20.0.255	0x2F	
TOU(48)	0x0030	1-0:1.8.48.255	0-0:19.20.0.255	0x30	
Block(1)TOU(1)	0x0065	1-1:1.8.1.255	0-0:19.20.0.255	0xA1	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(1)TOU(2)	0x0066	1-1:1.8.2.255	0-0:19.20.0.255	0xA2	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(1)TOU(3)	0x0067	1-1:1.8.3.255	0-0:19.20.0.255	0xA3	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(1)TOU(4)	0x0068	1-1:1.8.4.255	0-0:19.20.0.255	0xA4	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(1)TOU(5)	0x0069	1-1:1.8.5.255	0-0:19.20.0.255	0xA5	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(1)TOU(6)	0x006A	1-1:1.8.6.255	0-0:19.20.0.255	0xA6	Which block is activated by this Script Selector will depend on

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Description	Script Selector (long unsigned)	Consumption Register	Charge Object	Index (octet-string(1))	Notes
					ESME consumption since last block reset
Block(1)TOU(7)	0x006B	1-1:1.8.7.255	0-0:19.20.0.255	0xA7	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(1)TOU(8)	0x006C	1-1:1.8.8.255	0-0:19.20.0.255	0xA8	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(1)	0x006D	1-2:1.8.1.255	0-0:19.20.0.255	0xB1	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(2)	0x006E	1-2:1.8.2.255	0-0:19.20.0.255	0xB2	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(3)	0x006F	1-2:1.8.3.255	0-0:19.20.0.255	0xB3	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(4)	0x0070	1-2:1.8.4.255	0-0:19.20.0.255	0xB4	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(5)	0x0071	1-2:1.8.5.255	0-0:19.20.0.255	0xB5	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(6)	0x0072	1-2:1.8.6.255	0-0:19.20.0.255	0xB6	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(7)	0x0073	1-2:1.8.7.255	0-0:19.20.0.255	0xB7	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(2)TOU(8)	0x0074	1-2:1.8.8.255	0-0:19.20.0.255	0xB8	Which block is activated by this Script Selector will depend on ESME consumption since last block reset

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Description	Script Selector (long unsigned)	Consumption Register	Charge Object	Index (octet-string(1))	Notes
Block(3)TOU(1)	0x0075	1-3:1.8.1.255	0-0:19.20.0.255	0xC1	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(2)	0x0076	1-3:1.8.2.255	0-0:19.20.0.255	0xC2	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(3)	0x0077	1-3:1.8.3.255	0-0:19.20.0.255	0xC3	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(4)	0x0078	1-3:1.8.4.255	0-0:19.20.0.255	0xC4	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(5)	0x0079	1-3:1.8.5.255	0-0:19.20.0.255	0xC5	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(6)	0x007A	1-3:1.8.6.255	0-0:19.20.0.255	0xC6	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(7)	0x007B	1-3:1.8.7.255	0-0:19.20.0.255	0xC7	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(3)TOU(8)	0x007C	1-3:1.8.8.255	0-0:19.20.0.255	0xC8	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(1)	0x007D	1-4:1.8.1.255	0-0:19.20.0.255	0xD1	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(2)	0x007E	1-4:1.8.2.255	0-0:19.20.0.255	0xD2	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(3)	0x007F	1-4:1.8.3.255	0-0:19.20.0.255	0xD3	Which block is activated by this Script Selector will depend on

Description	Script Selector (long unsigned)	Consumption Register	Charge Object	Index (octet-string(1))	Notes
					ESME consumption since last block reset
Block(4)TOU(4)	0x0080	1-4:1.8.4.255	0-0:19.20.0.255	0xD4	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(5)	0x0081	1-4:1.8.5.255	0-0:19.20.0.255	0xD5	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(6)	0x0082	1-4:1.8.6.255	0-0:19.20.0.255	0xD6	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(7)	0x0083	1-4:1.8.7.255	0-0:19.20.0.255	0xD7	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
Block(4)TOU(8)	0x0084	1-4:1.8.8.255	0-0:19.20.0.255	0xD8	Which block is activated by this Script Selector will depend on ESME consumption since last block reset
TOU(1) (Secondary Element)	0x00C9	1-20:1.8.1.255	0-0:19.20.5.255	0x01	Only present on Twin Element ESME
TOU(2) (Secondary Element)	0x00CA	1-20:1.8.2.255	0-0:19.20.5.255	0x02	Only present on Twin Element ESME
TOU(3) (Secondary Element)	0x00CB	1-20:1.8.3.255	0-0:19.20.5.255	0x03	Only present on Twin Element ESME
TOU(4) (Secondary Element)	0x00CC	1-20:1.8.4.255	0-0:19.20.5.255	0x04	Only present on Twin Element ESME

1893 Table 7.3.7.2: ESME requirements for pricing matrices, scripts and registers

1894 DLMS Device Requirements Tables

1895 Table 7.3.8a: Objects tab in embedded file

1896 Table 7.3.8b: Scripts tab in embedded file

1897 Table 7.3.8c: Application Associations tab in embedded file

1898 Table 7.3.8d: Association LN Object Content tab in embedded file

- 1899 Table 7.3.8e: Security Setup Object Content tab in embedded file
- 1900 Table 7.3.8f: SAP Assignment Object content tab in embedded file
- 1901 Table 7.3.8g: Conformance Content tab in embedded file
- 1902 Table 7.3.8h: End to End Communications tab in embedded file



- 1903
- 1904 ESME / SAPC accounts, credits and charges – informative
- 1905 Where an SAPC supports the related SMETS functionality, the statements in this Section
- 1906 7.3.9 apply to the SAPC as if it were an ESME.
- 1907 As detailed in the Mapping Table, an ESME shall have two *Account* objects (*Class ID 111*)
- 1908 which shall be used in both Credit and Prepayment Modes:
- 1909 a single ‘active’ *Account* object (*OBIS code 0-0:19.0.0.255*) which can be read in Use Cases
- 1910 but which is never written to directly; and
- 1911 a single ‘passive’ *Account* object (*OBIS code 0-1:19.0.0.255*) which can be written to in Use
- 1912 Cases but which is never read.
- 1913 Both relate to import energy.
- 1914 The activation attribute and method of the ‘passive’ object leads to its *static* values being
- 1915 copied from the passive to the active object, rather than the passive becoming active (see
- 1916 Section 9.2.2.7).
- 1917 The ‘Set Payment Mode to Credit’ and ‘Set Payment Mode to Prepayment’ Use Cases are
- 1918 used to trigger such activation of the ‘passive’ object. The SMETS attributes Suspend Debt
- 1919 Emergency and Suspend Debt Disabled are implemented as part of these objects and so are
- 1920 set in these Use Cases. If an ESME is in Prepayment Mode and the value of either Suspend
- 1921 Debt attribute is to be changed, this can be achieved by sending a Set Payment Mode to
- 1922 Prepayment Command containing the new values.
- 1923 ESME / SAPC requirements for using Special Days objects
- 1924 When applying Blue Book special days related requirements to a Calendar / Scheduler object listed in Table
- 1925 7.3.10, an ESME and an SAPC, where it supports the related SMETS functionality, shall use the
- 1926 corresponding Specials Days Object in Table 7.3.10.

Calendar / Scheduler object

Special Days Object to
be used

SMETS Reference	Class ID	OBIS	OBIS
TariffSwitchingTable(SpecialDays)	20	0-0:13.0.0.255	0-0:11.0.0.255
TariffSwitchingTable(SecondaryElement)(SpecialDays)	20	0-0:13.0.1.255	0-0:11.0.1.255
Non-DisablementCalendar(SpecialDays)	10	0-0:12.0.1.255	0-0:11.0.2.255
AuxiliaryControllerCalendar(SpecialDays)	10	0-0:12.0.2.255	0-0:11.0.3.255

1927 Table 7.3.10: Special Days Object

1928 Device requirements – ZSE

1929 This Section 7.4 details the ZigBee clusters, attributes and commands that shall be
 1930 supported by Devices in their interactions with other Devices on the same HAN, including
 1931 whether the support is as a ZSE client or a server. Note, this Section does not detail the ZCL
 1932 / ZSE commands that Devices will need to process as part of processing Remote Party
 1933 Commands, or Commands sent by a PPMID to a GSME. Such requirements are detailed in
 1934 Sections 18 and 19.

1935 Only Devices capable of operating at Sub-GHz shall be required to support the requirements
 1936 in rows of Table 7.4 where the cell in the column labelled ‘Sub GHz capable Devices only?’
 1937 contains ‘Yes’.

1938 For clarity and as required by ZSE, all Devices shall support the Key Establishment Cluster as
 1939 both Client and Server.

1940 A GSME shall implement a ZSE *Metering Device* and shall implement all the *clusters*,
 1941 *commands*, *attribute sets* and *attributes* in Table 7.4 where column A is ‘GSME: Metering
 1942 Device’.

1943 A GPF shall implement a ZSE *Metering Device* and shall implement all the *clusters*,
 1944 *commands*, *attribute sets* and *attributes* in Table 7.4 where column A is ‘GPF: Metering
 1945 Device (Gas Mirror Endpoint)’.

1946 A GPF shall implement a ZSE *Energy Services Interface* and shall implement all the *clusters*,
 1947 *commands*, *attribute sets* and *attributes* in Table 7.4 where column A is ‘GPF: Energy
 1948 Services Interface (Gas ESI Endpoint)’.

1949 A CHF shall implement a ZSE *Remote Communications Device* and shall implement all the
 1950 *clusters*, *commands*, *attribute sets* and *attributes* in Table 7.4 where column A is ‘CHF:
 1951 Remote Communications Device (Remote Communications Endpoint)’.

1952 An SAPC shall implement a ZSE *Energy Services Interface* and shall implement all the *clusters*,
 1953 *commands*, *attribute sets* and *attributes* in Table 7.4 where column A is ‘SAPC: Energy
 1954 Services Interface (Electricity ESI Endpoint)’.

-
- 1955 Where it supports the corresponding SMETS functionality, an SAPC shall implement the
 1956 *clusters, commands, attribute sets and attributes* in Table 7.4 where column A is ‘SAPC
 1957 optional: Energy Services Interface (Electricity ESI Endpoint)’.
- 1958 Additionally, an SAPC may support other *clusters, commands, attribute sets and attributes* in
 1959 Table 7.4 where column A is ‘ESME: Energy Services Interface (Electricity ESI Endpoint)’.
- 1960 An ESME which is not a Twin Element ESME shall implement a *ZSE Energy Services Interface*
 1961 and shall implement all the *clusters, commands, attribute sets and attributes* in Table 7.4
 1962 where column A is ‘ESME: Energy Services Interface (Electricity ESI Endpoint)’.
- 1963 An ESME which is a Twin Element ESME shall implement three *ZSE Energy Services*
 1964 *Interfaces*:
- 1965 1. the first which shall implement all the clusters, commands, attribute sets and
 1966 attributes in Table 7.4 where column A is ‘ESME: Energy Services Interface (Twin
 1967 ESME aggregate ESI Endpoint)’;
 - 1968 2. the second which, in relation to the primary measuring element, shall implement all
 1969 the clusters, commands, attribute sets and attributes in Table 7.4 where column A is
 1970 ‘ESME: Energy Services Interface (Twin ESME primary/secondary ESI Endpoint)’; and
 - 1971 3. the third which, in relation to the secondary measuring element, shall implement all
 1972 the clusters, commands, attribute sets and attributes in Table 7.4 where column A is
 1973 ‘ESME: Energy Services Interface (Twin ESME primary/secondary ESI Endpoint)’.
- 1974 A PPMID shall implement a *ZSE In-Home Display*, shall implement all the *clusters, commands,*
 1975 *attribute sets and attributes* in Table 7.4 where column A is ‘PPMID: In-Home Display’, and
 1976 shall support the other clusters, attributes and commands necessary to meet the SMETS
 1977 requirements.
- 1978 An HCALCS shall implement a *ZSE Load Control Device* and shall implement all the *clusters,*
 1979 *commands, attribute sets and attributes* in Table 7.4 where column A is ‘HCALCS: Load
 1980 Control Device’.
- 1981 An HHT shall implement a *ZSE Remote Communications Device* and shall implement all the
 1982 *clusters, commands, attribute sets and attributes* in Table 7.4 where column A is ‘HHT:
 1983 Remote Communications Device’.
- 1984 An IHD shall implement all the clusters, commands, attribute sets and attributes in Table 7.4
 1985 where column A is ‘IHD: In-Home Display’ and shall support the other clusters, attributes
 1986 and commands necessary to meet the SMETS requirements.

- 1987 Where a row in Table 7.4 is required for a Device, that Device shall support the cluster,
 1988 attribute or command specified in that row as client or server, as specified in column C
 1989 (labelled 'Client / Server').
- 1990 Support for *clusters, commands, attribute sets and attributes* shall be as defined in columns
 1991 B ('Cluster'), D ('Command'), E ('Attribute Set') and F ('Attribute').
- 1992 Note that the other columns in Table 7.4 are informative and for requirements traceability
 1993 only.
- 1994 Except where explicitly required by this Section 7.4 or by Section 19.3, a Device shall not
 1995 execute any ZSE command, be that in a GBZ Command Payload or provided as a native ZSE
 1996 command, that could, if executed, constitute a Critical action. For clarity, a Device shall not
 1997 execute a ZSE *Publish Change of Supplier* command if bits 11-12 of the *Provider Change*
 1998 *Control* parameter (*Meter Contactor State*) of that command has any value other than 0b11
 1999 (*Supply UNCHANGED*).
- 2000 In relation to the *calendar cluster's* use for exchanging information about the Auxiliary
 2001 Controller Calendar (with its SMETS meaning) between Devices:
- 2002 the *Calendar Type* value of 0x04 (named *Auxillary Load Switch Calendar*) shall be used, to
 2003 differentiate from the values of 0x00 (named *Delivered Calendar*) and 0x03 (named *Friendly*
 2004 *Credit Calendar*) for the Tariff Switching Table and Non-Disablement Calendar, respectively
 2005 (with their SMETS meanings); and
- 2006 the *Auxiliary Load Switch State* parameters in *PublishDayProfile* commands shall be set so
 2007 that:
- 2008 *bit0* to *bit4* have values corresponding to Auxiliary Controller [1] to Auxiliary Controller [5]
 2009 respectively, with each such bit being 0b1 where the commanded output state is 100 for the
 2010 switching instruction in question (so the value of 'p' with its Section 7.3.6.1 meaning), or 0b0
 2011 otherwise; and
- 2012 *bit5* to *bit7* have the value 0b0.
- 2013 For clarity, this means that settings related to commanded input states are not
 2014 shared, and commanded output settings of 99 or less on an Auxiliary Proportional
 2015 Controller (with its SMETS meaning) are all represented as 0b0 to other HAN Devices.



GBCS v4.4 Table
7.4.xlsx

2016

2017 Table 7.4: Device Requirements

2018 Encryption of Attributes in Remote Party Messages

2019 In some Use Cases, some attributes are marked as Encrypted.

2020 This Section 8 lays out requirements as to how such Encryption and related Decryption shall
2021 be undertaken.

2022 Approach – informative

2023 Since ZSE and DLMS have differing data types to represent the same attribute of SMETS
2024 information, there are some differences in the format of the data that is encrypted. These
2025 differences are laid out in this Section 8. However, Encryption and Decryption use the same
2026 cryptographic AES GCM primitives in the same way in all cases, regardless of protocol. The
2027 usage is the same as that to generate MACs for Remote Party Message protection, and
2028 therefore as per the AES GCM approach laid out in the Green Book.

2029 Encryption of SMETS attributes is required when:

2030 the Supplier reads the amounts held in Time Debt Register [1..2] and Payment Debt Register.
2031 Each of these is a single integer value;

2032 the Supplier reads the values held in the Active Import Register or Secondary Active Import
2033 Register or Consumption Register. Each of these is a single integer value;

2034 the Supplier reads the values held in the Tariff Block Counter Matrix, Tariff TOU Register
2035 Matrix or Tariff TOU Block Register Matrix;

2036 a Known Party or an Unknown Party reads one or more entries from a Log (with each entry
2037 in the specific log having a Log specific structure), specifically:

2038 the current or previous Supplier reads the Billing Data, the Daily Read Log (excluding the
2039 export related parts), or the Prepayment Daily Read Log. Note that a previous Supplier is an
2040 Unknown Remote Party as far as the meter is concerned;

2041 the Supplier, Network Operator, or an Unknown Remote Party reads the Daily Consumption
2042 Log or the Profile Data Log (Consumption parts);

2043 the Network Operator reads the Network Data Log;

2044 a Device sends an Alert containing a single entry from the Billing Data Log (excluding the
2045 export related parts).

2046 Common requirements

2047 All Encryption shall be Authenticated Encryption which:

2048 shall use the cryptographic primitives, input value structures and cryptographic material
2049 specified in Section 4; and

2050 shall, for Key Agreement, use the Key Agreement key pair of the Device and the Remote
2051 Party which is accessing the data item.

2052 A Device shall, where it stores a data item listed in the Mapping Table as Encrypted, only
2053 provide that data in a Remote Party Message in Encrypted form.

2054 Where the Encrypted data item is within a Log, a Command requesting that data shall always
2055 have 'from' and 'to' date-times specified.

2056 Where all the octets in the 'from' date-time are 0x00 (excluding the least significant 3 bytes
2057 in Blue Book octet string formatted date-times), the Device shall interpret the 'from' field as
2058 meaning from the oldest in the Log.

2059 Where all the octets in the 'to' date-time are 0xFF (excluding the least significant 3 bytes in
2060 Blue Book octet string formatted date-times), the Device shall interpret the 'to' field as
2061 meaning to the newest in the Log.

2062 Where the Encrypted data item in the Mapping Table is not in a Log, a Command requesting
2063 that data shall never have 'from' or 'to' date-times specified.

2064 Key Derivation Inputs

2065 Where a Remote Party Message (1) contains Encrypted data items and (2) contains a
2066 Supplementary Remote Party ID, then the Encryption Remote Party shall be that identified
2067 by the Supplementary Remote Party ID. Otherwise, the Encryption Remote Party shall be
2068 the Remote Party identified in the Grouping Header of the Message.

2069 If the Message is to include a Supplementary Originator Counter generated by the Device
2070 (see Section 4.3.1.4), then the Encryption Originator Counter shall be the Supplementary
2071 Originator Counter. Otherwise the Encryption Originator Counter shall be the Originator
2072 Counter with the value in the Grouping Header of the Message.

2073 In relation to the Key Derivation Function requirements at Section 4.3.3, fields shall be
2074 populated as follows:

2075 'value of transaction-id' shall be the concatenation 0x04 || Encryption Originator Counter.
2076 Note 0x04 ensures this value is not used in any other Key Derivation Function invocation
2077 save that related to this Encryption / Decryption; and

2078 for Encrypted data items in Responses and Alerts, 'value of recipient-system-title' shall be
2079 Encryption Remote Party and 'value of originator-system-title' shall be the Device's Entity
2080 Identifier.

2081 AAD, Plaintext and Ciphertext

2082 The Plaintext shall be set to the structure and content of the data item(s) as they would have
 2083 been exposed on the Device's HAN interface, if access to them were not constrained to be
 2084 via Encrypted form by this Section 8.4.

2085 AAD shall be set to security control byte (SC) which shall have the value of 0x31 (see Section
 2086 8.4.1).

2087 The Invocation Counter (IC) shall have a value of 0x00000000.

2088 The Authenticated Encryption MAC (AE MAC) shall be the MAC produced by applying
 2089 Authenticated Encryption to AAD and Plaintext, as defined in *NIST Special Publication 800-*
 2090 *38D*, with the values specified in this Section 8.4.

2091 Authenticated Encryption (AE) Ciphertext shall be the Ciphertext produced by applying
 2092 Authenticated Encryption to Plaintext, with the values specified in this Section 8.4.

2093 Ciphred Information shall be the concatenation:

2094 SC || IC || AE Ciphertext || AE MAC

2095 Meaning of SC – informative

2096 The SC is set to 0x31 to reflect the following:

2097 Bits 3..0 are security suite which is 0b0001 since Security Suite 1 is required;

2098 Bit 4 is set to 0b1 since Authentication of the data is required;

2099 Bit 5 is set to 0b1 since Encryption of the data is required;

2100 Bit 6 is set to 0b0 since Messages containing the encrypted data are unicast; and

2101 Bit 7 is set to 0b0 as per the Green Book.

2102 Access to sensitive data – COSEM attribute access

2103 Access to sensitive data items shall be via the Data Protection class, as specified in the Blue
 2104 Book. The required OBIS codes and associated details for each attribute shall be as specified
 2105 in the 'SMETS required objects' tab in the Mapping Table.

2106 For clarity and in line with the Blue Book, the first octet of the plaintext input, constructed
 2107 by the Device when using the Data Protection class, shall be 0x02 (the tag for structure
 2108 according to Section 9.5 of the Green Book) and the next octet shall be the number of
 2109 captured DLMS COSEM attributes in the Use Case.

2110 The Device shall only allow read access to attributes listed as Encrypted in the Mapping
 2111 Table using the `get_protected_attributes(data)` method of the Data Protection class and not
 2112 allow access to any other methods of such objects.

2113 Values of the Data Protection class attributes

2114 The values of attributes 1, 3, 4, 5 and 6 of an object of the Data Protection class shall be set
2115 on a Device at manufacture, and those values shall not be capable of amendment except by
2116 firmware upgrade. For each object of the Data Protection class:

2117 protection_object_list (attribute 3) shall be an array containing object_definition(s). Within
2118 an object_definition: class_id, logical_name, attribute_index, data_index, restriction_type
2119 and restriction_value shall take values as per Table 7.3.8a;

2120 the value of protection_parameters_get (attribute 4) shall be a single entry array containing
2121 one protection_parameters_element, which shall have the values specified in Table 8.5.1;

2122 the value of protection_parameters_set (attribute 5) shall be an array containing zero
2123 entries; and

2124 the value of required_protection (attribute 6) shall be 0b01100000 (0x60) where the object
2125 exposes the get_protected_attributes(data) method, since Authenticated Encryption is
2126 required on the output of the method.

2127

Attribute	Type	Value
protection_type	Enum	(2) authentication and encryption
protection_options	Structure	
transaction_id	octet-string	Empty string
originator_system_title	octet-string	Empty string
recipient_system_title	octet-string	Empty string
other_information	octet-string	Empty string
key_info	Structure	
key_info_type:	Enum	(2) agreed_key
key_info_options	CHOICE	agreed_key_options
agreed_key_info_options	Structure	
key_parameters	octet-string	0x02 (meaning C(0e, 2s ECC CDH))
key_ciphersed_data	octet-string	An octet string of length zero

2128 Table 8.5.1: Values of protection_parameters_element

2129 Parameters of the get_protected_attributes method

2130 The get_protected_attributes_request parameter of the get_protected_attributes method
 2131 shall:

2132 be populated in the Command to the Device according to Table 8.5.2a; and

2133 be verified by the Device receiving the Command according to Table 8.5.2a.

2134 The protection_parameters part of the get_protected_attributes_response returned by the
 2135 get_protected_attributes method shall be populated by the Device according to Table 8.5.2b.

2136 The value of protected_attributes part of the protected_attributes_response_data returned
 2137 by the get_protected_attributes method shall be populated by the Device with Ciphred
 2138 Information, calculated as per the requirements of Section 8.2. The tag for
 2139 protected_attributes shall be 'octet-string' (0x09) and the length shall be the length of
 2140 Ciphred Information.

Field	Value	Device Validation	Note
get_protected_attributes_request			
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x02	Must have this value	2 elements in the structure
object_list			The first element in the get_protected_attributes_request structure
Tag	0x01	Must have this value	Meaning 'array'
Length	See 'Note' column	Must have this value	Number of entries in the array
object_definition			Entry in the object_list array. May be more than 1 occurrence
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x05	Must have this value	5 elements in the structure
class_id			
Tag	0x12	Must have this value	Meaning 'long-unsigned'
Value	See 'Note'	Must be the same as the	The class_id of the object which is the source of the

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Field	Value	Device Validation	Note
	column	class_id in attribute 3 of the Data Protection object being accessed	Encrypted data
logical_name			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x06	Must have this value	Logical_name is always 6 octets long
Value	See 'Note' column	Must be the same as the logical_name in attribute 3 of the Data Protection object being accessed	The logical_name of the object which is the source of the Encrypted data
attribute_index			
Tag	0x0F	Must have this value	Meaning 'integer'
Value	See 'Note' column	Must be the same as the attribute_index in attribute 3 of the Data Protection object being accessed	The attribute_index of the object which is the source of the Encrypted data
data_index			
Tag	0x12	Must have this value	Meaning 'long-unsigned'
Value	0x0000	Must have this value	Meaning the whole attribute is captured or set
Restriction			
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x02	Must have this	2 elements in the structure

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Field	Value	Device Validation	Note
		value	
EITHER		Must be present if this invocation is not to access a Log as defined in Section 8.2	If this is not to access a Log as defined in Section 8.2
restriction_type			
Tag	0x16	Must have this value	Meaning 'enum'
Value	0x00	Must have this value	Meaning 'none'
restriction_value			Assumes that the CHOICE does not need encoding since the value of 'restriction_type' defines the CHOICE [Note, there are no tags in the Blue Book for this CHOICE]
Tag	0x00	Must have this value	Meaning 'null-data'
OR		Must be present if this invocation is to access a Log as defined in Section 8.2	If this is to access a Log as defined in Section 8.2
restriction_type			
Tag	0x16	Must have this value	Meaning 'enum'
Value	0x01	Must have this value	Meaning 'restriction by date'
restriction_value			Assumes that the CHOICE does not need encoding since the value of 'restriction_type' defines the CHOICE [Note, there are no tags in the Blue Book for this CHOICE]
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x02	Must have this value	2 elements in the structure
from_date			In the date-time format of the Blue Book
Tag	0x09	Must have this value	Meaning 'octet-string'

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Field	Value	Device Validation	Note
Length	0x0C	Must have this value	Date-time is always 12 octets long
Value	See 'Note' column		Log entries with a date-time stamp prior to this date-time shall not be returned.
to_date			In the date-time format of the Blue Book
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x0C	Must have this value	Date-time is always 12 octets long
Value	See 'Note' column		Log entries with a date-time stamp after this date-time shall not be returned.
protection_parameters			The second element in the get_protection_attributes_request structure
Tag	0x01	Must have this value	Meaning 'array'
Length	0x01	Must have this value	1 entry in the array
protection_parameters_element			The 1 entry in the protection_parameters array
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x02	Must have this value	2 elements in the structure
protection_type			The first element in the protection_parameters_element
Tag	0x16	Must have this value	Meaning 'enum'
Value	0x02	Must have this value	Meaning 'authentication and encryption'
protection_options			The second element in the protection_parameters_element
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x05	Must have this value	5 elements in the structure

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Field	Value	Device Validation	Note
transaction_id			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x09	Must have this value	transaction_id is always 9 octets in length
Value	See 'Note' column		The concatenation 0x04 the Originator Counter value part of the transaction_id in the Grouping Header of this Command
originator_system_title			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x08	Must have this value	Entity Identifier is always 8 octets in length
Value	See 'Note' column		The Entity Identifier of the Encryption Remote Party
recipient_system_title			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x08	Must have this value	Entity Identifier is always 8 octets in length
Value	See 'Note' column	Must be the Device's Entity Identifier	The Entity Identifier of the Device
other_information			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x00	Must have this value	Zero length since this string is empty
key_info			
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x02	Must have this value	2 elements in the structure
key_info_type:			

Field	Value	Device Validation	Note
Tag	0x16	Must have this value	Meaning 'enum'
Value	0x02	Must have this value	Meaning 'agreed_key'
key_info_options		CHOICE	Assumes that the CHOICE does not need encoding since the value of 'restriction_type' defines the CHOICE [Note, there are no tags in the Blue Book for this CHOICE]
agreed_key_info_options			
Tag	0x02	Must have this value	Meaning 'structure'
Length	0x02	Must have this value	2 elements in the structure
key_parameters			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x01	Must have this value	Length fixed by the Blue Book
Value	0x02	Must have this value	Meaning 'C(0e, 2s ECC CDH)'
key_ciphared_data			
Tag	0x09	Must have this value	Meaning 'octet-string'
Length	0x00	Must have this value	Zero length since this string is empty

2141 Table 8.5.2a: values of get_protected_attributes_request

2142

Field	Value	Validation	Note
protection_parameters			
Tag	0x01	Must have this value	Meaning 'array'
length	0x01	Must have this value	1 entry in the array
protection_parameters_element			The 1 entry in the protection_parameters array
tag	0x02	Must have this value	Meaning 'structure'

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Field	Value	Validation	Note
length	0x02	Must have this value	2 elements in the structure
protection_type			The first element in the protection_parameters_element
tag	0x16	Must have this value	Meaning 'enum'
value	0x02	Must have this value	Meaning 'authentication and encryption'
protection_options			The second element in the protection_parameters_element
tag	0x02	Must have this value	Meaning 'structure'
length	0x05	Must have this value	5 elements in the structure
transaction_id			
tag	0x09	Must have this value	Meaning 'octet-string'
length	0x09	Must have this value	transaction_id in is always 9 octets in length
value	See note		The concatenation 0x04 Encryption Originator Counter
originator_system_title			
tag	0x09	Must have this value	Meaning 'octet-string'
length	0x08	Must have this value	Entity Identifier is always 8 octets in length
value	See note		The Entity Identifier of the Device
recipient_system_title			
tag	0x09	Must have this value	Meaning 'octet-string'
length	0x08	Must have this value	Entity Identifier is always 8 octets in length
value	See note		The Entity Identifier of the Encryption Remote Party
other_information			
tag	0x09	Must have this value	Meaning 'octet-string'
length	0x00	Must have this value	Zero length since this string is empty
key_info		Structure	
tag	0x02	Must have this value	Meaning 'structure'

Field	Value	Validation	Note
length	0x02	Must have this value	2 elements in the structure
key_info_type:			
tag	0x16	Must have this value	Meaning 'enum'
value	0x02	Must have this value	Meaning 'agreed_key'
key_info_options		CHOICE	Assumes that the CHOICE does not need encoding since the value of 'restriction_type' defines the CHOICE [Note, there are no tags in the Blue Book for this CHOICE]
agreed_key_info_options			
tag	0x02	Must have this value	Meaning 'structure'
length	0x02	Must have this value	2 elements in the structure
key_parameters			
tag	0x09	Must have this value	Meaning 'octet-string'
length	0x01	Must have this value	Length fixed by the Blue Book
value	0x02	Must have this value	Meaning 'C(0e, 2s ECC CDH)'
key_ciphred_data			
tag	0x09	Must have this value	Meaning 'octet-string'
length	0x00	Must have this value	Zero length since this string is empty

2143 Table 8.5.2b: values of protection_parameters

2144 Billing Data Log Alert – DLMS COSEM

2145 'Use Case Specific Additional Content' within the Billing Data Log Alert shall be populated
2146 according to the Message Template for the Billing Data Log Alert in Section 18.2.

2147 Access to sensitive data – ZSE attribute access

2148 Ciphred Information shall be used to populate each 'GBZ Use Case Specific Component
2149 with encrypted content' as specified in Section 7.2.10.

2150 Time Synchronisation and Future Dated Remote Party Messages

2151 This Section 9 details how time synchronisation shall operate, and how future dated Remote
2152 Party Messages shall be processed by Devices. The latter applies only where a Command is
2153 specified in 'Use Case reference' tab in the Mapping Table, as 'Capable of future dated
2154 invocation'.

2155 Note that all references in the GBCS to time shall be to UTC date-time unless explicitly stated
2156 otherwise.

2157 Time synchronisation

2158 Introduction – informative

2159 SMETS requires that SAPC, ESME and GSME have clocks and that, under normal operating
2160 circumstances, the time on those clocks is accurate to within 10 seconds.

2161 CHTS requires that Communications Hubs have clocks and that, under normal operating
2162 circumstances, the time on those clocks is accurate to within 10 seconds.

2163 Critical functionality on Communications Hubs can function predictably without reliance on
2164 time. Time setting mechanisms on Communications Hubs therefore are not constrained or
2165 specified in the GBCS. However, under normal operating circumstances, a Communications
2166 Hub will provide the time reference for all dependent Devices on the HAN.

2167 Significant parts of SAPC, ESME and GSME functionality are time-dependent for their correct
2168 and predictable functioning. This includes Critical functionality which can only be controlled
2169 by the Device's Supplier with responsibility for that Device. Thus, time must be accurate in
2170 terms of alignment with the time set by the Supplier on the Device. However, the accuracy
2171 requirements measured in seconds are smaller than end-to-end network latency for delivery
2172 of Commands to Devices.

2173 This leads to a time synchronisation approach for ESME and SAPC as specified in Section
2174 9.1.4, and GSME as specified in Section 9.1.7.

2175 That approach is:

2176 for the Supplier to send a Set Clock Command with the Supplier's current time and a future
2177 time (reflecting a time tolerance) in the Command; and

2178 if, when the Device receives the Command, the Communications Hub's time is within
2179 tolerance of the Supplier's time, the Device aligns itself to the Communications Hub's time
2180 and treats its time as Reliable. Otherwise the Device treats its time as Unreliable.

2181 The time synchronisation for a GSME follows the same principles but tolerance needs to
2182 differ because a GSME is 'sleepy'. 'Sleepy' means that its SMHAN radio will not be active
2183 most of the time and therefore the tolerance provided by the Supplier needs to reflect the
2184 extended latency.

2185 Common Requirements – Set Clock

2186 Supplier Current Time shall be the Supplier's time at the point the Supplier sends a Set Clock
2187 Command.

2188 GSME and ESME shall maintain a record of its Time Status, which, for clarity, is not the same
2189 as the ZSE *TimeStatus* attribute in the Remote Communications Endpoint. Time Status shall
2190 have one of the values in Table 9.1.2.

Value	Meaning
Invalid	The Device has no meaningful time
Unreliable	The Device has a meaningful time but that time may not be accurate and needs to be affirmed / reaffirmed by the Supplier
Reliable	The Device has a meaningful time and that time has been affirmed by its Supplier

2191 Table 9.1.2: Time Status

2192 Device Requirements relating to the ZCL Time Cluster and its usage

2193 Requirements in this Section 9.1.3 shall apply to an SAPC as if it were an ESME.

2194 All italicised terms in this Section 9.1.3 shall have the meanings defined in the *Time Cluster*
2195 specification within the ZigBee Cluster Library Specification.

2196 In relation to the ZCL *Time Cluster* in the Remote Communications Endpoint, a
2197 Communications Hub shall:

2198 set the *Time* attribute to the *UTCTime* provided to it via its WAN interface, whenever such
2199 time information is available to it via its WAN interface and maintain its internal clock to the
2200 tolerance required by the CHTS;

2201 set the *Time* attribute to 0xFFFFFFFF whenever it cannot maintain its time via its WAN
2202 interface to the tolerance required by the CHTS due to resetting its internal clock;

2203 whenever its Time attribute has the value 0xFFFFFFFF, store the value 0xFFFFFFFF in any
2204 unsigned 32 bit integer format time stamps it records and store the value
2205 0xFFFFFFFFFFFFFFFF8000FF in any DLMS COSEM octet-string(12) format time stamps it
2206 records; and

2207 always have *TimeStatus* attributes set as:

2208 Attribute Bit Number 0 (Master) equal to 0b1 (master clock);

2209 Attribute Bit Number 2 (MasterZoneDst) equal to 0b0 (not master for Time Zone and DST);
2210 and

2211 Attribute Bit Number 3 (Superseding) equal to 0b1 (time synchronisation can supersede).

2212 In relation to any ZigBee Time Cluster on the ESME, the ESME shall always have *TimeStatus*
2213 attributes set as:

2214 Attribute Bit Number 0 (Master) equal to 0b0 (not master clock); and

2215 Attribute Bit Number 3 (Superseding) equal to 0b0 (time synchronisation should not be
 2216 superseded).

2217 At power on of the clock, an ESME or GSME shall:

2218 set its Time Status as 'Invalid';

2219 attempt to synchronise time, using the Communications Hub's Time Cluster; and

2220 where a valid *Time* (so not 0xFFFFFFFF) is provided by the Communications Hub before any
 2221 Set Clock Command is received, set its time to the value of *Time* provided and set its Time
 2222 Status to 'Unreliable'.

2223 ESME and GSME shall attempt to synchronise time, using the Communications Hub's Remote
 2224 Communications Endpoint Time Cluster, once every 24 hour period in line with the SMETS
 2225 requirement. ESME and GSME shall undertake the following processing dependent on the
 2226 outcome of each attempted synchronisation:

2227 if a time of 0xFFFFFFFF is provided or if no time is received the Device shall retry the
 2228 synchronisation after an elapsed period of 30 minutes, for a minimum of the lesser of three
 2229 retries, or a retry resulting in a valid Time (so not 0xFFFFFFFF) being provided. If no valid
 2230 Time has been provided after three retries, the Device shall set its Time Status to
 2231 'Unreliable';

2232 if the time provided by the Communications Hub differs from the Device's time by more
 2233 than 10 seconds, then the Device shall:

2234 set its Time Status to 'Unreliable'; and

2235 if this results in a change to Time Status, the Device shall construct and issue an Alert with
 2236 Alert Code 0x8F0C, meaning that its time would have been shifted by more than 10 seconds.
 2237 If the Device is a GSME, the Alert Payload shall be a GBZ Alert Payload. If the Device is an
 2238 ESME, the Alert Payload shall be a DLMS COSEM Alert Payload.

2239 if the time provided by the Communications Hub differs from the Device's time by 10
 2240 seconds or less, then the Device shall adjust its time to the Communications Hub's time.

2241 ECS70 Set Clock on ESME

2242 Requirements in this Section 9.1.4 and in Use Case 'ECS70 Set Clock on ESME' shall apply to
 2243 an SAPC as if it were an ESME.

2244 This Use Case covers the setting of the Clock by the Supplier on an ESME.

Cross Reference	Value
Grouping	Remote Party Message

Message Type	Command and Response
Message Type Category	SME.C.C
Capable of future dated invocation?	No
Protection Against Replay Required?	Yes
Service Reference	6.11
Valid Target Device(s)	ESME
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	DLMS COSEM

2245 Table 9.1.4: Use Case Cross References for ECS70 Set Clock on ESME

2246 Pre-conditions

2247 None.

2248 Detailed Steps

2249 The Command Payload shall be constructed as per Table 9.1.4.2a.

Class	OBIS Code	Attribute or Method?	Attribute / Method no.	Set, Get or Action	Attribute/Method name and Blue Book ref.	DLMS COSEM data types	Value (for Sets or Actions)	Notes
8	0-0:1.0.0.255	A	9	Set	clock_base	Enum	5	5 shall mean radio controlled which shall be interpreted as controlled via the Communications Hub Time Cluster that is available over the ESME's HAN interface

Class	OBIS Code	Attribute or Method?	Attribute / Method no.	Set, Get or Action	Attribute/Method name and Blue Book ref.	DLMS COSEM data types	Value (for Sets or Actions)	Notes
8	0-0:1.0.0.255	M	5	Action	preset_adjusting_time	structure{ preset_time: octet-string, validity_interval_start: octet-string, validity_interval_end: octet-string}	{'not specified', Supplier Current Time, Supplier Current Time + Tolerance Period}	'not specified' in preset_time shall be the value 0xFFFFFFFFFFFFFFFF8000FF as required by the Blue Book. All times shall be formatted as octet-string according to section 4.1.6.1 of the Blue Book
8	0-0:1.0.0.255	M	4	Action	adjust_to_preset_time	Integer	0	
8	0-0:1.0.0.255	A	2	Get	Time	Octet-string	Null	This get means that the resulting Time of the ESME shall be provided in the Response
8	0-0:1.0.0.255	A	4	Get	Status	Unsigned	Null	This get means that the resulting Time Status of the ESME shall be provided in the Response

2250 Table 9.1.4.2a: Construction of Command Payload

2251 In this Section 9.1.4.2, the object with OBIS code 0-0:1.0.0.255 shall be referred to as the
2252 *Clock* and italicised terms shall have their Blue / Green Book meaning.

2253 On receipt of the Command, the ESME shall undertake processing in the following sequence:

- 2254 1. the ESME shall undertake the 'Command Authenticity and Integrity Verification'
2255 processing required for Commands of type SME.C.C. If that fails, processing shall cease;
- 2256 2. the ESME shall process the instructions in the *access-request-body* of the Command as
2257 follows:

2258 when attribute 9 of the *Clock* (*clock_base*) is set to '*radio controlled*' (5), the ESME shall
2259 request the value of the Communications Hub Time from the Communications Hub via its
2260 ZigBee radio. If a time of 0xFFFFFFFF is provided or if no time is received:

- 2261 1. if the current Time Status is set to 'Reliable', the ESME shall set bit 1 of
2262 attribute 4 (so setting *status* of Clock to be '*doubtful value*' (Unreliable)); or
- 2263 2. if the current Time Status is not set to 'Reliable', the ESME shall not change
2264 status.

2265 the *preset_adjusting_time* method of the Clock shall be executed. Note this is to set
 2266 parameters for the *adjust_to_present_time* method;

2267 the *adjust_to_present_time* method of the Clock shall be executed as follows:

2268 3. if the Communications Hub Time returned lies between
 2269 *validity_interval_start* and *validity_interval_end*, then:

2270 1. ESME *time* shall be updated to match Communications Hub
 2271 Time;

2272 2. the ESME shall unset bit 0 of attribute 4 (so setting *status* of
 2273 the *Clock* not to be an '*invalid value*'); and

2274 3. the ESME shall unset bit 1 of attribute 4 (so setting *status* of
 2275 the *Clock* not to be a '*doubtful value*').

2276 4. if the Communications Hub Time returned lies before
 2277 *validity_interval_start* (Supplier Current Time) or after
 2278 *validity_interval_end* (Supplier Current Time + Tolerance)) and (bit 0 of
 2279 attribute 4 of the *Clock* is unset), then:

2280 1. time shall remain unchanged, since time is outside the
 2281 *validity_interval*; and

2282 2. the ESME shall set bit 1 of attribute 4 and unset bit 0 of
 2283 attribute 4 (so setting *status* of *Clock* to be a '*doubtful value*'
 2284 (Unreliable)).

2285 the *get request* on the *time* and *status* attributes of the *Clock* shall be executed.

2286 5. the ESME shall undertake the 'Response Construction' and 'Response Cryptographic'
 2287 processing required for a Response of type SME.C.C.

2288 On receipt of the Response, the recipient may undertake the 'Response Recipient Processing'
 2289 for Responses of type SME.C.C.

2290 The meaning of result attributes is as defined in the Green Book.

2291 The meaning of the unsigned integer returned by the *get request* on attribute 4 of the Clock
 2292 (*status*) is as per Table 9.1.4.2b.

Values in attribute 4 of the <i>Clock</i> object	Time Status Meaning
Bit 0 is set	Invalid
Bit 0 is unset and Bit 1 is set	Unreliable

Bit 0 is unset and Bit 1 is unset	Reliable
-----------------------------------	----------

2293 Table 9.1.4.2b: Meaning of unsigned integer

2294 Time related object on ESME

2295 Requirements in this Section 9.1.5 shall apply to an SAPC as if it were an ESME.

2296 Italicised terms in this Section shall have their Blue Book meaning.

2297 An ESME shall have a *Data* object with *OBIS* code 0-0:94.44.100.255 where attribute 2 of
2298 that object:

2299 shall be a double-long-unsigned value;

2300 shall have a value set by the ESME to the number of seconds between 0 hours 0 minutes 0
2301 seconds on 1st January 2000 UTC and the value of UTC time specified by attribute 2 of the
2302 *Clock* object with *OBIS* code 0-0:1.0.0.255;

2303 shall be the value recorded by the ESME in attribute 2 of any *Profile generic* object *entry* as
2304 the date-time stamp, at the time the *entry* is added;

2305 shall be the format recorded by the ESME in attribute 2 of any *Profile generic* object *entry* in
2306 other date-time fields.

2307 Correspondingly:

2308 the '*from_value*' and '*to_value*' fields in the *selective access* structure, which are required by
2309 the GBCS when accessing attribute 2 of any *Profile generic* object directly, shall be *double-*
2310 *long-unsigned* attributes containing a date-time specified in seconds since 0 hours 0 minutes
2311 0 seconds on 1st January 2000 UTC; and

2312 the *restricting_object* field in the *selective access* structure shall be set with values of
2313 class_id = 1; logical_name = 0-0:94.44.100.255; attribute_index = 2 and data_index = 0.

2314 The Blue Book requires that, for a Data Protection class object, *restriction_by_date* access
2315 has *from_date* and *to_date* specified as octet-string. Thus, where a Use Case requires that
2316 the contents of attribute 2 of a *Profile generic* object are returned in Encrypted form (and so
2317 accessed via a Data Protection object):

2318 the *from_date* and *to_date* fields in the Command shall be octet-strings formatted as per
2319 section 4.1.6.1 of the Blue Book; and

2320 the ESME shall undertake the conversion necessary to equate these values to '*from_value*'
2321 and '*to_value*' equivalents in accessing attribute 2 of any *Profile generic* object.

2322 Start of Time and End of Time values

- 2323 Where a date-time is specified as a 32 bit long unsigned integer:
- 2324 the Start of Time shall mean the value 0x00000000; and
- 2325 the End of Time shall mean the value 0xFFFFFFFF.
- 2326 When a date-time is specified in the DLMS COSEM octet-string(12) format:
- 2327 the Start of Time shall mean the value 0x000000000000000000008000FF; and
- 2328 the End of Time shall mean the value 0xFFFFFFFFFFFFFFFF8000FF’.
- 2329 GCS28 Set Clock on GSME
- 2330 This Use Case covers the setting of time by the Supplier on a GSME.

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	SME.C.C
Capable of future dated invocation?	No
Protection Against Replay Required?	Yes
Service Reference	6.11
Valid Target Device(s)	GSME
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

2331 Table 9.1.7: Use Case Cross References for GCS28 Set Clock on GSME

2332 Pre-conditions

2333 None.

2334 Construction of Command

2335 Set Clock Command Payloads shall be constructed according to the requirements of Section
2336 9.1.7.4 and populated as specified in Table 9.1.7.2.

2337 MAC Header, Grouping Header, KRP Signature and ACB-SMD MAC shall be populated as
2338 required for a Command of the SME.C.C Message Category.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value
@SetTime.CommandPayload	SEQUENCE		
validityIntervalStart	GeneralizedTime	The earliest time the Communications Hub can provide if the Command is to set Reliable Time	Mandatory
validityIntervalEnd	GeneralizedTime	The latest time the Communications Hub can provide if the Command is to set Reliable Time	Mandatory

2339 Table 9.1.7.2: @SetTime.CommandPayload population

2340 Device processing of Command and Response handling

2341 The GSME receiving a Set Clock Command shall undertake processing steps in the sequence
2342 defined in this Section 9.1.7.3.

2343 The GSME shall:

2344 1. undertake Command Authenticity and Integrity Verification as required for a Command
2345 of the SME.C.C Message Category;

2346 2. request the now current Communications Hub Time. If the Communications Hub
2347 cannot supply a valid time, it shall provide 0xFFFFFFFF. If this is sent, or no response is
2348 received, the GSME shall:

2349 1. if its current Time Status is set to 'Reliable', set Time Status to 'Unreliable', set
2350 deviceTimeStatus to unreliable, populate deviceTime with its current
2351 Time and process from step 5; or

2352 if its current Time Status is not set to 'Reliable', set deviceTimeStatus to be Time
2353 Status, populate deviceTime with its current Time and process from step 5.

2354 2. if ((the Communications Hub Time < validityIntervalStart) or (Communications Hub Time
2355 > validityIntervalEnd)), set Time Status to 'Unreliable', and set deviceTimeStatus to

2356 unreliable, leave its Time unchanged, populate deviceTime with its current Time and
 2357 process from step 5;

2358 3. set its time to the Communications Hub Time, populate deviceTime with the
 2359 corresponding value, and set deviceTimeStatus to reliable;

2360 4. populate the Response Payload according to the requirements of Section 9.1.7.4 using
 2361 the deviceTimeStatus and deviceTime values produced by the processing in this Section
 2362 9.1.7.3;

2363 5. construct Grouping Header and apply the Response Cryptographic Protection required
 2364 for a Response of the SME.C.C Message Category; and

2365 6. send the Response.

2366 On receipt of the Response, the recipient may undertake the 'Response Recipient Processing'
 2367 for Responses of type SME.C.C.

2368 Set Clock Command and Response Payloads – structure **definition**

2369 Each instance of @SetTime.CommandPayload and of @SetTime.ResponsePayload shall be
 2370 an octet string containing the DER encoding of the populated structure defined in this
 2371 Section 9.1.7.4 which specifies the structure in ASN.1 notation.

2372 SetTime DEFINITIONS ::= BEGIN

2373

2374

2375 CommandPayload ::= SEQUENCE

2376 {

2377 -- specify the period within which the Communications Hub's time must lie

2378 -- if this Command is successfully to set time

2379 validityIntervalStart GeneralizedTime,

2380 validityIntervalEnd GeneralizedTime

2381

2382 }

2383

2384 ResponsePayload ::= SEQUENCE

```

2385
2386 {
2387     -- Specify the Device's now current time
2388     deviceTime                                GeneralizedTime,
2389
2390     -- Specify the Device's now current Time Status
2391     deviceTimeStatus                        DeviceTimeStatus
2392 }
2393
2394 DeviceTimeStatus ::= INTEGER
2395 {
2396     reliable                                (0),
2397     invalid                                (1),
2398     unreliable                            (2)
2399 }
2400
2401 END
2402 Future Dated Remote Party Messages
2403 Future Dated Commands for the Reading of Data Items – informative
2404 Where future dated execution of a Command to read data items is supported in a Use Case,
2405 this is achieved by setting values in a schedule stored on the Device. In such cases, the
2406 sequence of Messages is as follows:
2407 on receipt of a Command to update a schedule, the Device should attempt to Authenticate
2408 then execute the Command. The Device should then create a corresponding Response
2409 either indicating the schedule has been set or providing failure reasons; and
2410 when each trigger time in the schedule is reached (according to the Clock on the Device),
2411 the Device undertakes the required processing then creates and sends an Alert. One initial
2412 Command to set a schedule may generate many such Alerts.

```

-
- 2413 In such circumstances, the Command and Response are specified in one Use Case, and the
2414 Alert is specified in a different Use Case.
- 2415 Such future dated reading can be cancelled by sending a Command which resets the
2416 schedule values.
- 2417 The only example of such a schedule is the Billing Calendar. The Alerts generated are Billing
2418 Calendar Alerts.
- 2419 Future Dated Commands for the Writing of Attributes
- 2420 Introduction – informative
- 2421 Only Commands marked ‘Capable of future dated invocation?’ in the Mapping Table can be
2422 future dated. Such Commands allow a data item or group of data items to be changed at a
2423 date-time in the future.
- 2424 Where a data item or group of data items on a Device are capable of future dated updates,
2425 this is achieved by the Device having:
- 2426 a ‘current’ and a ‘next’ version of the group of data items in question;
- 2427 a data item for recording the date / time at which the ‘next’ version should be made
2428 ‘current’; and
- 2429 a method to set ‘current’ values equal to ‘next’ values.
- 2430 This is the meaning of data items with ‘Current’ and ‘Next’ post fixes in the ‘SMETS / CHTS
2431 Attribute / method’ column of the Mapping Table.
- 2432 In such cases, the sequence of Messages to effect a future dated update would be:
- 2433 1. a Command would be sent to the Device instructing that the data items in question
2434 should be stored in the ‘next’ data items and the corresponding activation date /
2435 time should be set (so overwriting previous ‘next’ values and any previous activation
2436 date / time);
 - 2437 2. only if the Command is Authenticated, would the Device attempt to execute the
2438 instructions in Command. Execution for such Commands means writing to ‘next’
2439 values and setting activation date-times. If the activation date-times are in the past,
2440 the Device will also attempt to make the ‘next’ values ‘current’;
 - 2441 3. the Device would then create a Response. This Response would either confirm that
2442 the ‘next’ values and the activation date / time values have been set (and so any
2443 previous future dated command with this Message Code has been over written), or
2444 would provide failure reasons. The ‘current’ values would be unaffected assuming
2445 the activation date / times are in the future; and
-

2446 4. when an activation date / time is reached (according to the clock on the Device), the
 2447 Device would attempt to make the corresponding 'next' values 'current'. The Device
 2448 would then send an Alert detailing success or failure.

2449 Like all other Commands, future dated Commands cannot be modified once accepted by the
 2450 Device. However, the time activated processing can be stopped from happening by sending
 2451 a new Command of the same Message Type. This is because the new Command over writes
 2452 the values from the old Command.

2453 For example:

2454 a 'cancellation' can be effected by sending a new Command where the activation date / time
 2455 in the new Command has a value that means 'never' to the Device; and

2456 a 'modification' can be effected by sending a new Command where the activation date-time
 2457 and / or the 'next' values in the new Command are different than the old one.

2458 Commands that are marked 'Capable of future dated invocation?' in the Mapping Table can
 2459 also be invoked immediately, as specified in Section 9.2.2.4.

2460 When there is a change of Supplier on a Device which is (1) after a future dated change is
 2461 stored but (2) before it is activated, the processing at Section 13.3.5.10 will be undertaken at
 2462 the point of update of Security Credentials. This ensures future dated commands from the
 2463 old Supplier will not be actioned by the Device.

2464 Date-times in future datable commands

2465 Where a Command contains more than one activation date-time field, the values in all
 2466 activation date-times in an instance of that Command shall be the same, except for ZSE
 2467 Command Payloads where a field contains 0xFFFFFFFF or 0xFFFFFFF, then all activation
 2468 date-times shall be either 0xFFFFFFFF or 0xFFFFFFF. Devices shall reject Commands not
 2469 complying with this requirement. A future datable Command shall be future dated when it
 2470 contains an activation date-time that is not one of the values in Section 9.2.2.4.

2471 Effect on prior Commands of the same Message Code

2472 On receipt of an Authenticated future datable Command, the Device shall overwrite all parts
 2473 of any previously sent future dated Command of the same Message Code and, if the
 2474 activation date-times for the instructions in the Command are in the past, the Device shall
 2475 execute the instructions immediately.

2476 Using a future dated Command to write Attributes immediately

2477 Where a Command is marked 'Capable of future dated invocation?' in the Mapping Table,
 2478 instructions within the Command shall be executed immediately after Authentication by the
 2479 Device when:

2480 for DLMS COSEM Commands Payloads, activation date-time(s) have the value
2481 0x000000000000000008000FF;

2482 for ZSE Command Payloads, the activation date-time(s) have the value 0x00000000; and
2483 for ASN.1 Command Payloads, the activation date-time is not present.

2484 Cancellation of future dated Commands for the writing of Attributes

2485 Where a Command is marked ‘Capable of future dated invocation?’ in the Mapping Table,
2486 instructions within the Command shall never be executed by the Device when:

2487 for DLMS COSEM Commands Payloads, activation date-time(s) have the value
2488 0xFFFFFFFFFFFFFFFF8000FF;

2489 for ZSE Command Payloads, the activation date-time(s) have the value 0xFFFFFFFF for any
2490 ZSE command other than PublishCalendar, PublishSpecialDays, PublishBlockThresholds,
2491 PublishPriceMatrix;

2492 for ZSE Command Payloads, the activation date-time(s) have the value 0xFFFFFFF0 for the
2493 ZSE command PublishCalendar, PublishSpecialDays, PublishBlockThresholds,
2494 PublishPriceMatrix; and

2495 for ASN.1 commands, the activation date-time has a value of 99991231235959Z.

2496 For clarity, sending such a Command has the effect of ‘cancelling’ any previously sent future
2497 datable Command of the same Message Code that the Device has not already executed.

2498 Reactions to Future Dated Commands

2499 Subject to Command Authenticity and Integrity Verification as detailed in Section 6, where a
2500 Command is future dated, at time the Command is received, the Device shall send a
2501 Response to the Command:

2502 where activation date-times are in the past or the instructions detail immediate execution as
2503 per Section 9.2.2.4, the Command shall be executed immediately:

2504 for a CS02b Update Security Credentials Command, the processing shall be as required by
2505 Section 13.3.4.3;

2506 for a CS02g Update Load Controller Security Credentials Command, the processing shall be
2507 as required by Section 13.10.3.4;

2508 for a CS06 Activate Firmware, the processing shall be as required by Section 11.5.2.2; and
2509 for all other Commands, the Response shall detail the outcome of the Command’s execution,
2510 and no Alert shall be generated;

2511 where activation date-times are in the future, that Response shall detail the success or
 2512 otherwise of storing the details in the Command; and

2513 if the activation date-times are in the future and the Command's details were successfully
 2514 stored, the Device shall, at the time the future activation date-time is reached, process each
 2515 of the instructions, which contain an activation date-time, as specified in the Command in
 2516 the sequence specified in that Command and then generate an Alert with an Alert Payload,
 2517 for each instruction, of the same type as the Payload type of the Command and an Alert
 2518 Code of 0x8F66 for successful execution and 0x8F67 for failed execution. Thus:

2519 an ASN.1 Command Payload shall lead to an ASN.1 Alert Payload, which shall be as defined
 2520 in Sections 11 and 13;

2521 a DLMS COSEM Command Payload shall lead to DLMS COSEM Alert Payload(s), which shall
 2522 be as defined in Table 7.2.9c, where the Use Case specific additional content contains the
 2523 concatenation 0x09 || 0x13 || Message Code || Originator Counter || cosem-attribute-
 2524 descriptor from the corresponding part of the Command Payload. Note that 0x09 is the
 2525 DLMS COSEM tag for octet-string and 0x13 is the length of the concatenation Message Code
 2526 || Originator Counter || cosem-attribute-descriptor; or

2527 a GBZ Command Payload shall lead to GBZ Alert Payload(s), shall be as defined in Table
 2528 7.2.10c, where the Use Case specific additional content contains the concatenation 0x0E ||
 2529 Message Code || Originator Counter || Extended Header Cluster ID || Frame control ||
 2530 Command identifier from the corresponding part of the Command payload.

2531 ESME requirements for activation of future datable Commands

2532 Requirements in this Section 9.2.2.7 shall apply to an SAPC as if it were an ESME, where
 2533 either:

2534 the requirements relate to the Auxiliary Control Calendar, with its SMETS meaning; or

2535 the requirements relate to SMETS functionality which the SAPC supports.

2536 When either (1) a Command successfully sets the 'passive' *Account* object's
 2537 *account_activation_time* (attribute ID 13, OBIS code 0-1:19.0.0.255) to
 2538 0x000000000000000000000000000000008000FF or (2) the ESME's clock reaches the value in that attribute or
 2539 (3) the *activate_account* method is invoked, the ESME shall not activate 'passive' *Account*
 2540 object, as detailed in the Blue Book, but shall set the attributes in the 'active' *Account* object
 2541 (OBIS code 0-0:19.0.0.255) as follows:

2542 set the *payment_mode* part of the *account_mode_and_status* attribute (attribute ID 2) to
 2543 the *payment_mode* value in the 'passive' *Account* object; and

2544 set each of the other *static* attributes (as defined in the Blue Book) to the corresponding
2545 value in the 'passive' *Account* object.

2546 When either (1) a Command successfully sets an Activation Date-Time Attribute in Table
2547 9.2.2.7 to 0x00000000000000008000FF or (2) the ESME's clock reaches the value in an
2548 Activation Date-Time Attribute in Table 9.2.2.7, the ESME shall set the corresponding Active
2549 Attribute in Table 9.2.2.7 to the value of the corresponding Passive Attribute in Table 9.2.2.7.

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SMETS Reference	Activation Date-Time Attribute			Passive Attribute			Active Attribute		
	Class ID	OBIS	Attr. ID	Class ID	OBIS	Attr. ID	Class ID	OBIS	Attr. ID
TariffSwitchingTable(SpecialDays)	9000	0-0:94.44.128.29	6	11	0-1:11.0.0.255	2	11	0-0:11.0.0.255	2
TariffThresholdMatrix	9000	0-0:63.1.1.255	6	21	0-0:16.1.12.255	2	21	0-0:16.0.12.255	2
TariffSwitchingTable(SecondaryElement).specialDays	9000	0-0:94.44.128.30	6	11	0-1:11.0.1.255	2	11	0-0:11.0.1.255	2
DisablementThreshold(MeterBalance)	9000	0-0:94.44.128.22	6	9000	0-0:94.44.128.22	4	21	0-0:16.0.1.255	2
DebtRecoveryRateCap	9000	0-0:94.44.128.12	6	9000	0-0:94.44.128.12	4	111	0-0:19.0.0.255	18
DebtRecoveryRateCap(period)	9000	0-0:94.44.128.13	6	9000	0-0:94.44.128.13	4	111	0-0:19.0.0.255	19
EmergencyCreditLimit	9000	0-0:94.44.128.2	6	9000	0-0:94.44.128.2	4	112	0-0:19.10.1.255	9
EmergencyCreditThreshold	9000	0-0:94.44.128.3	6	9000	0-0:94.44.128.3	4	112	0-0:19.10.1.255	10
LowCreditThreshold	9000	0-0:94.44.128.9	6	9000	0-0:94.44.128.9	4	111	0-0:19.0.0.255	16
Non-DisablementCalendar	9000	0-0:94.44.128.28	6	10	0-1:12.0.1.255	2	10	0-0:12.0.1.255	2
LoadLimitPeriod(Timer)	9000	0-0:94.44.128.6	6	9000	0-0:94.44.128.6	4	71	0-0:17.0.0.255	6
LoadLimitPowerThreshold	9000	0-0:94.44.128.7	6	9000	0-0:94.44.128.7	4	71	0-0:17.0.0.255	4
LoadLimitRestorationPeriod(Timer)	9000	0-0:94.44.128.8	6	9000	0-0:94.44.128.8	4	71	0-0:17.0.0.255	7
AuxiliaryControllerCalendar	9000	0-0:94.44.128.26	6	10	0-1:12.0.2.255	2	10	0-0:12.0.2.255	2
Non-DisablementCalendar(SpecialDays)	9000	0-0:94.44.128.31	6	11	0-1:11.0.2.255	2	11	0-0:11.0.2.255	2
AuxiliaryControllerCalendar(SpecialDays)	9000	0-0:94.44.128.35	6	11	0-1:11.0.3.255	2	11	0-0:11.0.3.255	2

2550 Table 9.2.2.7: Values for Active and Passive Attributes for Account objects

2551 ZSE Implementation

2552 Italicised terms in this Section 10 shall have their meaning in the ZCL / ZSE specifications.

2553 Introduction – informative

2554 This Section 10 sets out specific requirements relating to the implementation of ZSE in
2555 Devices:

2556 Tunnels: requirements relating to Devices' support for the *Tunneling Cluster*. This includes
2557 specific differences between GSME, HHT, VWD and other Devices, related to their use of the
2558 *Tunneling Cluster*. Note that all Devices except Type 2 Devices without Virtual WAN
2559 capability shall support the *Tunneling Cluster*, since this is the mechanism by which Remote
2560 Party Messages (and HAN Only Messages between a PPMID and a GSME) are transported
2561 over the HAN;

2562 GSME and GPF interactions (including the Tapping Off Mechanism (TOM)): this includes
2563 requirements relating to the GPF maintaining a copy of GSME data items, where copies are
2564 not supported natively by ZSE mirroring;

2565 GPF structured data items: requirements relating to how structured data items on the GPF
2566 are updated by the GSME and resulting values on the GPF are calculated; and

2567 HHT interactions – requirements relating to HHT connection to the SMHAN, including
2568 specific *Tunneling Cluster* related requirements.

2569 VWD interactions – requirements relating to VWD connection to the SMHAN, including
2570 specific *Tunneling Cluster* related requirements.

2571 Tunnels

2572 Overview – informative

2573 All Remote Party Messages are carried across the SMHAN using the *Tunneling Cluster's*
2574 *TransferData* command.

2575 Type 2 Devices without Virtual WAN capability such as IHDs are not required to send or
2576 receive Remote Party Messages and so are not required to support the *Tunneling Cluster*.

2577 Remote Party Messages to and from the GPF do not cross the SMHAN and so do not use the
2578 *Tunneling Cluster*.

2579 All other types of Device need to be able to send and receive Remote Party Commands over
2580 the SMHAN and so, as specified in Section 10.2.2, shall support the *Tunneling Cluster*.

2581 Section 10.2.2 lays out the associated requirements, across all Devices including those for
2582 the GSME, VWD and HHT.

2583 GSME requirements are different than all other Devices since a GSME is a 'sleepy' Device.
2584 Additional GSME requirements are laid out in Sections 10.2.4 and 10.3.

2585 HHT interactions also have specific requirements due to their function. These specific
2586 requirements are laid out in Section 10.5.

2587 VWD interactions also have specific requirements due to their function. These specific
2588 requirements are laid out in Section 10.9.

2589 A PPMID may be a sleepy device, and therefore may have different requirements to other
2590 devices.

2591 Requirements for the Tunneling Cluster

2592 Remote Party Messages and SME.C.PPMID-GSME Messages shall be transported over the
2593 SMHAN using the *Tunneling Cluster's TransferData* command. Except where a *TransferData*
2594 command is to or from a GSME, the value of the *Data* field's payload in the *TransferData*
2595 command shall be the Remote Party Message or SME.C.PPMID-GSME Message. Where a
2596 *TransferData* command is to or from a GSME, the *Data* field's payload of the *TransferData*
2597 command shall take the values specified in Section 10.2.4.

2598 Devices supporting the *Tunneling Cluster* as a *Server* shall have a
2599 *MaximumIncomingTransferSize* set to 1500 octets, in line with the ZSE default. All Devices
2600 supporting the *Tunneling Cluster* shall use this value in any *RequestTunnelResponse*
2601 command and any *RequestTunnel* command.

2602 Devices shall set the value of the *ManufacturerCode* field in any *RequestTunnel* command to
2603 0xFFFF ('not used').

2604 The *ProtocolID* used for Messages sent or received using the CHF of a Virtual WAN CH as a
2605 *Tunneling Cluster Client* and VWD as a *Tunneling Cluster Server* shall be 0. The CHF shall set
2606 the value of the *ProtocolID* field in the *RequestTunnel* command to 0 and the CHF and VWD
2607 shall associate the *TunnelID* in the *RequestTunnelResponse* command with *ProtocolID* 0.

2608 In all other cases, the *ProtocolID* used for Remote Party Messages shall be 6. Devices shall
2609 set the value of the *ProtocolID* field in any *RequestTunnel* command to 6 and the CHF and
2610 Devices shall associate the *TunnelID* in the *RequestTunnelResponse* command with
2611 *ProtocolID* 6.

2612 Devices shall set the value of the *FlowControlSupport* field in any *RequestTunnel* command
2613 to 'False'.

2614 All Devices except Type 2 Devices without Virtual WAN capability and GPFs shall support the
2615 *Tunneling Cluster*.

-
- 2616 An ESME, an SAPC and an HCALCS shall support the *Tunneling Cluster* as a *Server* using a
2617 protocol with a *ProtocolID* of 6.
- 2618 A GSME, a PPMID and an HHT shall support the *Tunneling Cluster* as a *Client* using a protocol
2619 with a *ProtocolID* of 6.
- 2620 A VWD shall support the *Tunneling Cluster* as a *Client* using a protocol with a *ProtocolID* of 6
2621 and as a *Server* using a protocol with a *ProtocolID* of 0.
- 2622 A CHF shall support the *Tunneling Cluster* as a *Client* and as a *Server*.
- 2623 A GPF shall support mirroring functionality. The *Basic Cluster Physical Environment* attribute
2624 shall, contrary to ZSE, be supported and shall have the value 0x01.
- 2625 When a Device receives a *CloseTunnel* command, the Device shall not close that tunnel
2626 unless the command is sent from the Device which opened the tunnel.
- 2627 Where a Device is required to support the *Tunneling Cluster* as a server, the *tunneling* server
2628 Device shall:
- 2629 when sending a *TransferData* command, use the last *TunnelID* which it holds and reset the
2630 inactivity counter accordingly. If it receives a *TransferDataError* command in response with a
2631 *TransferDataStatus* of 0x00 (No such tunnel), it shall disregard the last *TunnelID* and close
2632 the associated *tunnel*;
- 2633 persist *TunnelID* values, and the associated *tunneling* client Device's short address and
2634 endpoint, including when power is lost; and
- 2635 when receiving a *RequestTunnel* command from a *tunneling* client Device which has an
2636 active *TunnelID* registered with the *tunneling* server Device, reset the corresponding
2637 inactivity timer for that active *TunnelID* and send a *RequestTunnelResponse* command
2638 containing that active *TunnelID*.
- 2639 Where a Device is required to support the *Tunneling Cluster* as a client, the *tunneling* client
2640 Device shall:
- 2641 whenever it sends a *TransferDataError* command with a *TransferDataStatus* of 0x00 ('No
2642 such tunnel') to a *tunneling* server Device, then send a *RequestTunnel* command to that
2643 *tunnelling* server Device; and
- 2644 whenever it receives a *TransferDataError* command with a currently valid *TunnelID* and a
2645 *TransferDataStatus* of 0x00 ('No such tunnel') from a *tunneling* server Device, close any
2646 associated *tunnel* and send a *RequestTunnel* command to that *tunneling* server Device.
- 2647 Tunneling Requirements
-

2648 As soon as possible after a Device, which is required to support the *Tunneling Cluster* as a
 2649 client, has successfully established a shared secret key using *CBKE* with a Device, which is
 2650 required to support the *Tunneling Cluster* as a server, the *tunneling* client Device shall send a
 2651 *RequestTunnel* command to the *tunneling* server Device using a *ProtocolID* of 6 to request a
 2652 *tunnel* association with the Device.

2653 Where such a *tunneling* server Device remains in the *tunneling* client's Device Log, the
 2654 *tunneling* client Device shall send a *RequestTunnel* command to the Device whenever 0xFFFF
 2655 seconds have elapsed since receipt of the most recent *RequestTunnelResponse* command
 2656 from that Device.

2657 Where the CHF receives a *RequestTunnelResponse* command from a Device with a
 2658 *TunnelStatus* of 0x01 (*Busy*), the CHF shall send another *RequestTunnel* command three
 2659 minutes later.

2660 Where the CHF receives a *RequestTunnelResponse* command from a Device with a
 2661 *TunnelStatus* of 0x02 (*No More Tunnel IDs*), the CHF shall send a *CloseTunnel* command for
 2662 any *TunnelID* that may relate to an active tunnel association with that Device and, after
 2663 receiving responses to all such commands, send another *RequestTunnel* command.

2664 GSME

2665 When a GSME has successfully established a shared secret key using *CBKE* with a
 2666 Communications Hub, the GSME shall:

2667 send a request to the *ZigBee Gas ESI Endpoint* requesting the creation of mirrored *Basic*,
 2668 *Metering* and *Prepayment Clusters* using the *RequestMirror* command;

2669 configure, using the *ConfigureMirror* command, the *ZigBee Gas Mirror Endpoint* to use the
 2670 two way mirroring notification scheme '*Predefined Notification Scheme B*'; and

2671 send a *RequestTunnel* command to the CHF to request a tunnel association with the CHF.

2672 In line with ZSE, when a GPF sends a *RequestMirrorResponse* command in response to a
 2673 *RequestMirror* command, the *RequestMirrorResponse* command shall contain the
 2674 *EndPointID* to be used by the GSME regardless of whether the *RequestMirror* created the
 2675 mirror.

2676 A GPF shall only send a *RequestMirrorResponse* containing the *EndPointID* to the Device
 2677 which caused the GPF to create the mirror.

2678 Where a GPF receives a *ConfigureMirror* command to use the two way mirroring notification
 2679 scheme '*Predefined Notification Scheme B*' which has the *Disable Default Response Sub-field*
 2680 in its *Frame Control Field* set to zero, the GPF shall respond with a *Default Response*
 2681 indicating *SUCCESS* if it has a mirror configured to use '*Predefined Notification Scheme B*',
 2682 regardless of whether that was configured by the *ConfigureMirror* command.

2683 Where the Communications Hub has successfully actioned a *ConfigureMirror* command, the
 2684 GPF shall set the *Push All Static Data - Basic Cluster*, *Push All Static Data - Metering Cluster*
 2685 and *Push All Static Data - Prepayment Cluster* flags.

2686 Where a GSME reports a value for the *ManufacturerName* attribute or the *ModelIdentifier*
 2687 attribute, the GPF shall accept that value. For clarity, there are no requirements for the GPF
 2688 to subsequently process or make available any such value.

2689 For clarity, the GSME:

2690 shall not action ZSE / ZCL commands received from the GPF in relation to any of the flags
 2691 within *NotificationFlags2*, *NotificationFlags3* and *NotificationFlags5*;

2692 for *NotificationFlags4*, shall only action ZSE / ZCL commands received from the GPF in
 2693 relation to the flags specified in Table 10.2.2.2a.

Bit Number	Waiting Command
6	<i>Get Prepay Snapshot</i>
7	<i>Get Top Up Log</i>
9	<i>Get Debt Repayment Log</i>

2694 Table 10.2.2.2a: flags in *NotificationFlags4* to be actioned by the GSME

2695 for *FunctionalNotificationFlags*, shall only action ZSE / ZCL commands received from the GPF
 2696 in relation to the flags specified in Table 10.2.2.2b:

Bit Number	Waiting Command
0	<i>New OTA Firmware</i>
1	<i>CBKE Update Request</i>
4	<i>Stay Awake Request HAN</i>
5	<i>Stay Awake Request WAN</i>
6-8	<i>Push Historical Metering Data Attribute Set</i>
9-11	<i>Push Historical Prepayment Data Attribute Set</i>
12	<i>Push All Static Data - Basic Cluster</i>
13	<i>Push All Static Data - Metering Cluster</i>
14	<i>Push All Static Data - Prepayment Cluster</i>
15	<i>NetworkKeyActive</i>
21	<i>Tunnel Message Pending</i>
22	<i>GetSnapshot</i>
23	<i>GetSampledData</i>

25	Energy Scan Pending
26	Channel Change Pending

2697 Table 10.2.2.2b: flags in *FunctionalNotificationFlags* to be actioned by the GSME

2698 shall have access to the *Notification Flags* on the Communications Hub whenever it can

2699 communicate with the Communications Hub; and

2700 shall not provide any metering data to the *ZigBee Gas Mirror Endpoint* until and unless the

2701 GPF's Entity Identifier is recorded in the GSME Device Log.

2702 The GSME shall send a *RequestTunnel* command to the CHF to request a tunnel association

2703 with the CHF whenever it does not have a currently valid tunnel association with the CHF,

2704 and one of the following is true:

2705 the GSME has created an Alert or Response that is to be sent; or

2706 the GSME has ascertained, via the *Tunnel Message Pending* flag, that there is a Command

2707 for it buffered on the Communications Hub.

2708 Where the GSME receives a *RequestTunnelResponse* command from the CHF with a

2709 *TunnelStatus* of 0x01 (*Busy*), the GSME shall send another *RequestTunnel* command the next

2710 time it turns its HAN Interface on.

2711 Where the GSME receives a *RequestTunnelResponse* command from the CHF with a

2712 *TunnelStatus* of 0x02 (*No More Tunnel IDs*), the GSME shall send a *CloseTunnel* command for

2713 any *TunnelID* that may relate to an active tunnel association between it and the CHF and,

2714 after receiving responses to all such commands, send another *RequestTunnel* command.

2715 Immediately following the first successful establishment of a tunnel between the CHF and

2716 the SAPC / ESME / GSME, the SAPC / ESME / GSME shall send an Alert with Alert Code

2717 0x8F69.

2718 MirrorReportAttributeResponse command support

2719 GPF shall:

2720 support *ConfigureMirror* commands where the '*Mirror Notification Reporting*' field is set to

2721 0x01, and treat such a value as a request from the GSME for the GPF to push

2722 *FunctionalNotificationFlags* values using the *MirrorReportAttributeResponse* command;

2723 treat ZSE references to '*when the MirrorReporting attribute is set*' as references to '*when the*

2724 *Mirror Notification Reporting field in the most recently accepted ConfigureMirror command*

2725 *was set to 0x01*'; and

2726 send a *MirrorReportAttributeResponse* command to the GSME, containing the

2727 *FunctionalNotificationFlags* and *NotificationFlags#N* values required for '*Predefined*

2728 *Notification Scheme B*, whenever it receives a *ReportAttributes* command from the GSME
2729 for any '*Attribute Reporting Status*' attribute with a value of '*Attribute Reporting Complete*'.
2730 Note that the GSME may support one such '*Attribute Reporting Status*' attribute in each ZSE
2731 cluster.

2732 Therefore, and for clarity, where a GSME wishes to receive *MirrorReportAttributeResponse*
2733 commands from the GPF, the GSME should:

2734 notify the GPF of that wish by setting the '*Mirror Notification Reporting*' field to 0x01 in
2735 *ConfigureMirror* commands it sends;

2736 support the '*Attribute Reporting Status*' attribute on each of the mirrored clusters, so *Basic*,
2737 *Metering* and *Prepayment*; and

2738 send a *ReportAttributes* command from the GSME for an *Attribute Reporting Status* attribute
2739 with a value of '*Attribute Reporting Complete*', each time it wishes to trigger the GPF to send
2740 a *MirrorReportAttributeResponse* command.

2741 CHF requirements for ESME Transfer Data commands

2742 Requirements in this Section 10.2.2.3 shall apply to an SAPC as if it were an ESME.

2743 When a CHF is sending a Remote Party Message to an ESME in a *TransferData* command, the
2744 CHF shall:

2745 set the *Disable Default Response* sub-field of the *Frame Control* field in the *ZCL Header* of
2746 the *TransferData* command to 0b0;

2747 if it is the first time this Remote Party Message has been sent by the CHF, set a counter (the
2748 '*Message Transmission Counter*') for this Remote Party Message to 1. Otherwise increase
2749 the *Message Transmission Counter* by 1; and

2750 not send any subsequent Remote Party Message to the ESME until either:

2751 it has received a *Default Response* with a *Status Code* of *SUCCESS* in relation to a
2752 *TransferData* command containing the current Remote Party Message; or

2753 it has sent an Alert with Code 0x8F84 ('Failure to Deliver Remote Party Message to ESME') in
2754 relation to the current Remote Party Message.

2755 If the CHF receives a *Default Response* with a *Status Code* of *SUCCESS* to the *TransferData*
2756 command for the current Remote Party Message, the CHF may now send subsequent
2757 Remote Party Messages to the ESME.

2758 If the CHF receives no *Default Response* after 10 seconds, or receives a *Default Response*
2759 with a *Status Code* other than *SUCCESS*, to the *TransferData* command for the current
2760 Remote Party Message, the CHF shall start the Flow Control Replay Timer from 0 seconds.

2761 When the Flow Control Replay Timer reaches 30 seconds:

2762 if the Message Transmission Counter for the current Remote Party Message is less than 3,

2763 the CHF shall retry delivery of the current Remote Party Message, by sending it in a new

2764 *TransferData* command to the ESME; or

2765 if the Message Transmission Counter for the current Remote Party Message is 3, the CHF

2766 shall:

2767 discard the current Remote Party Message; and

2768 log the event in its Event Log and send an Alert with Alert Code 0x8F84 ('Failure to Deliver

2769 Remote Party Message to ESME') and Message Code 0x00D5, in relation to the current

2770 Remote Party Message.

2771 Use Case for Alert: 'Failure to Deliver Remote Party Message to ESME'

2772 Requirements in this Section 10.2.2.4, in all sub-Sections within it and in the Use Case for

2773 Alert: 'Failure to Deliver Remote Party Message to ESME' shall apply to an SAPC as if it were

2774 an ESME.

2775 Description

2776 This Section 10.2.2.4 covers the creation and processing of Alerts resulting from the

2777 requirements of Section 10.2.2.3.

2778 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Alert
Message Type Category	SME.A.C
Capable of future dated invocation?	N/A
Protection Against Replay Required?	N/A
Service Reference	N/A
Valid Initiating Device(s)	CHF
Valid Business Target role(s) for Alert	Access Control Broker
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	DLMS COSEM

2779 Table 10.2.2.4.2: Use Case Cross References for Alert: 'Failure to Deliver Remote Party
2780 Message to ESME'

2781 Construction of Alerts

2782 Grouping Header and SMD Signature shall be populated as required for an Alert of the
2783 SME.A.C Message Category, with the Message Code being 0x00D5 and Alert Code being
2784 0x8F84.

2785 The Business Originator ID field in the Grouping Header shall contain the CHF's Entity
2786 Identifier.

2787 The Business Target ID field in the Grouping Header shall contain the Entity Identifier in the
2788 CHF's {accessControlBroker, digitalSignature, management} Trust
2789 Anchor Cell.

2790 The Use Case Specific Additional Content, with its Section 7.2.9 meaning, shall be an octet-
2791 string containing the value of the Message Identifier from the Remote Party Message that
2792 has not been delivered to the ESME, and so is the subject of this Alert.

2793 Thus, the Use Case Specific Additional Content shall be the following concatenation, with
 2794 values taken from the Remote Party Message that was not delivered:

2795 0x09 || 0x19 || value of Business Originator ID || value of Business Target ID || value of
 2796 CRA Flag || value of Originator Counter

2797 Processing of Alerts

2798 The SMD Signature may be verified by the Access Control Broker as per Section 6.7.4.

2799 Recovery from Device removal from the CHF Device Log then re-addition to that log

2800 GSME recovery of mirroring

2801 Where the GSME has had three consecutive, scheduled wakes without successfully receiving
 2802 a requested value for the GPF's *FunctionalNotificationFlags* attribute, the GSME shall send a
 2803 *RequestMirror* command to the GPF, conforming to the requirements of Section 10.2.2.2. If
 2804 the GSME receives a *RequestMirrorResponse* command in response (so confirming GPF
 2805 communications), the GSME shall send a *ConfigureMirror* command to the GPF, conforming
 2806 with the requirements of Section 10.2.2.2.

2807 For clarity, this section shall apply where the GSME requests the GPF's
 2808 *FunctionalNotificationFlags* attribute value:

2809 either by using a *ReadAttributes* command specifying that attribute; or

2810 by way of a *ReportAttributes* command from the GSME for an '*Attribute Reporting Status*'
 2811 attribute with a value of '*Attribute Reporting Complete*'.

2812 PPMID / Type 2 Device recovery of ZigBee binding triggered updates

2813 A Device may use ZigBee binding requests to receive pushed updates, by way of ZigBee
 2814 commands, from the 'GPF: Metering Device (Gas Mirror Endpoint)' with its Section 7.4
 2815 meaning. If a Device does use this mechanism, the Device may repeat the sending of such
 2816 binding requests where it has not received any resulting pushed ZigBee commands for an
 2817 extended period, although the frequency of such repetition should not be such that it
 2818 materially impacts the performance of the HAN or other Devices.

2819

2820 GSME Tunnel Management – informative

2821 Commands are sent from the Communications Hub via the tunnel to the GSME. Since the
 2822 GSME is a 'sleepy' Device, a mechanism is needed for the GSME to request that Commands
 2823 are sent to it by the CHF.

2824 In common with the transport of all Remote Party or SME.C.PPMID-GSME Messages, the
2825 mechanism used is the *TransferData* command, but *TransferData* commands sent between a
2826 GSME and CHF need to distinguish between when:

2827 the GSME is sending a Message, so an Alert or a Response or a GBT Message containing part
2828 of an Alert / Response;

2829 the GSME is asking the CHF to send it a Command, or a GBT Message containing part of a
2830 Command; and

2831 the CHF is sending the GSME a Command, or a GBT Message containing part of a Command.

2832 To meet this need, the following Sections specify additional structure in the first part of the
2833 *Data* parameter of the *TransferData* commands sent between GSME and CHF. Specifically
2834 the sending Device shall:

2835 where a Message is being sent, set the *Data* parameter payload in a *TransferData* command
2836 to the concatenation:

2837 Tunnel Manager Header || Message

2838 where a Message is not being sent (so when the GSME is requesting that a Message is sent),
2839 set the *Data* parameter payload in a *TransferData* command to the value of Tunnel Manager
2840 Header.

2841 A mechanism is also required to notify the GSME that one or more Commands are available
2842 for retrieval from the CHF.

2843 The ZSE specification has a flag called *Tunnel Message Pending* in the *Functional Flag*
2844 *Notification* definition. This flag is used to notify a GSME that the CHF has a Message
2845 waiting to be transferred to the GSME. The flag is set on the first pending Command and is
2846 reset when all Messages have been transferred to the GSME. The flag is available through
2847 the *ReadAttribute* or *MirrorAttributeResponse* command. The requirements for setting this
2848 flag are specified in Section 10.3.4.

2849 The Tunnel Manager Header identifies three different kinds of *TransferData* command
2850 usage:

2851 GET (the value 0x01): this is used by the GSME to retrieve waiting Message from the CHF;

2852 GET-RESPONSE (the concatenation 0x02 || (Number of Messages remaining for retrieval
2853 after this Message) || (Message addressed to the GSME)): this is used by the CHF to send a
2854 Message to the GSME. It also indicates how many Messages have yet to be retrieved; and

2855 PUT(the value 0x03): this is used by the GSME to send a Message via the CHF.

2856 Where a Command is waiting on the CHF for the GSME to retrieve it, the following sequence
2857 shall apply:

2858 1. the Tunnel Message Pending flag is set on the Communications Hub as detailed in
2859 Section 10.3.4;

2860 2. the GSME turns on its HAN Interface and obtains the value of the Tunnel Message
2861 Pending flag; and

2862 3. If the Tunnel Message Pending flag is set:

2863 1. the GSME sends a *TransferData* command to the CHF with the GET structure in the
2864 Tunnel Manager Header. The Tunnel Manager Header is the only content in the *Data*
2865 field of this *TransferData* command;

2866 the CHF sends a *TransferData* command to the GSME with the GET-RESPONSE structure in
2867 the Tunnel Manager Header and a Message in the remaining part of the *Data* field of the
2868 command. The GET-RESPONSE structure details how many more Messages are available for
2869 retrieval; and

2870 the GET and GET-RESPONSE pattern repeats until all Messages have been transferred or the
2871 GSME decides to stop requesting Messages.

2872 When the GSME wishes to send a Message, the GSME sends a *TransferData* command to
2873 the CHF with the PUT structure in the Tunnel Manager Header and the Message in the
2874 remainder of the *Data* field in the *TransferData* command.

2875 TransferData commands sent between GSME and CHF

2876 When it wishes to send a Message, so an Alert or Response or GBT Message, a GSME shall
2877 send a *TransferData* command to the CHF with the value in the *Data* parameter payload of
2878 the *TransferData* command set to the concatenation:

2879 0x03 || Message

2880 When it wishes to retrieve a Message stored for it on a CHF, a GSME shall send a
2881 *TransferData* command to the CHF with the value in the *Data* field set to 0x01. When the
2882 CHF receives such a *TransferData* command from a GSME, the CHF shall send a *TransferData*
2883 command to the GSME with the value in the *Data* parameter payload set to:

2884 the concatenation

2885 0x02 || (Number of Messages remaining for retrieval after this Message) || (Message
2886 addressed to the GSME)

2887

2888 where it has Messages for the GSME not yet downloaded by the GSME; or

2889 the concatenation 0x02 || 0x00, where it has no Messages for the GSME to retrieve, the
 2890 0x00 representing the number of Messages available to retrieve.

2891 GSME and GPF interactions

2892 Introduction – informative

2893 The GSME is informed that Remote Party or SME.C.PPMID-GSME Commands are available
 2894 for it to retrieve via *Tunnel Message Pending* flag on the GPF.

2895 The GSME should, under normal operating circumstances, retrieve all Commands buffered
 2896 for it when it turns its HAN Interface on. For example, if two Commands are buffered for it,
 2897 the GSME should retrieve both Commands before turning its HAN Interface off.

2898 However, in some circumstances, a GSME may choose not to retrieve all buffered
 2899 Commands in a single session. In such cases, the GSME should retrieve each Command as
 2900 soon as possible after that Command is received by the CHF.

2901 Potential reasons for a GSME failing to retrieve all buffered Commands include:

2902 the GSME battery requires time to recover;

2903 the GSME is entering a ‘low battery’ mode and limiting the use of its radio; or

2904 a radio communications error.

2905 Section 10.3 details actions the CHF may take where Commands, or GBT Messages
 2906 containing parts of Commands, for a GSME are not retrieved by the GSME.

2907 Commands addressed to a GSME must be processed by the GSME and, when successfully
 2908 processed, any changed operational or configuration data must be made available to the GPF.
 2909 The GPF then has updated information to provide to other Devices on the same SMHAN.

2910 In ZSE terms, the GPF incorporates two distinct logical Devices, which are discoverable and
 2911 addressed on different *endpoints*. Section 7 describes which *clusters* reside on which
 2912 *endpoint*.

2913 GSME data residing on the *ZigBee Gas Mirror Endpoint* – informative

2914 The *ZigBee Gas Mirror Endpoint* provides a ‘reflection’ of the data held by the GSME. A
 2915 GSME is typically a battery-powered Device and its HAN Interface is mostly not turned on,
 2916 making it unable to respond to other Devices. The GSME turns its HAN Interface on at
 2917 regular intervals (e.g. 30 minutes) and pushes consumption data to the *ZigBee Gas Mirror*
 2918 *Endpoint*. This provides other Devices on the same SMHAN with access to GSME
 2919 consumption data at any time.

2920 GSME data residing on the *ZigBee Gas ESI Endpoint* – informative

2921 The *ZigBee Gas ESI Endpoint* holds GSME data which is provided by a Remote Party, for
 2922 example pricing. The *ZigBee Gas ESI Endpoint* makes this type of data available to Devices
 2923 on the same SMHAN.

2924 GSME data from a Remote Party is sent to the GSME in a Remote Party Command. Such a
 2925 Command has to be validated by the GSME before any data in it is applied by the GSME. For
 2926 example, a Command to change tariff must be rejected by the GSME if it fails authentication,
 2927 and the data in the Command must not be applied in such circumstances.

2928 If data in a Remote Party Command is accepted by the GSME, a mechanism is needed to
 2929 provide the changed data to the *ZigBee Gas ESI Endpoint*. This is so that the *ZigBee Gas ESI*
 2930 *Endpoint* can then provide that data to other Devices on the same SMHAN.

2931 A mechanism is also needed to deal with a Response not being received from the GSME.
 2932 The lack of a Response may indicate that the GSME and the *ZigBee Gas ESI Endpoint* do not
 2933 contain the same value in one or more data items. If data items on the two are not
 2934 synchronised, Devices on the SMHAN will display incorrect information.

2935 There are several possible reasons why this lack of a Response may arise, not all of which
 2936 mean that data is out of synchronisation:

2937 the Command has failed validation by the GSME and has been discarded;
 2938 the Response has been lost due to a communications error; or
 2939 a software error.

2940 GSME Command retrieval and TOM Requirements

2941 TOM Commands and Responses

2942 A Command shall be a TOM Command if it is a Remote Party Command with one of the
 2943 following Message Codes:

2944 0x006B (GCS01a Set Tariff and Price on GSME);

2945 0x006F (GCS05 Update Prepayment Configurations on GSME) – the GPF shall only process
 2946 the *Calendar* cluster ZSE commands within the Command;

2947 0x0071 (GCS07 Send Message to GSME);

2948 0x0015 (CS11 Clear ZigBee Device Event Log) where the Command is addressed to the
 2949 GSME;

2950 0x007C (GCS23 Set CV and Conversion Factor Value(s) on the GSME);

2951 0x007E (GCS25 Set Billing Calendar on the GSME);

2952 0x00D8 (GCS25a Set Billing Calendar on the GSME – all periodicities);

-
- 2953 0x0088 (GCS44 Write Contact Details on GSME); or
- 2954 0x00A3 (GCS01b Set Price on GSME).
- 2955 A TOM Response shall be a Response to a TOM Command.
- 2956 For clarity, neither a TOM Response nor a TOM Command may contain Encrypted data.
- 2957 Processing of Commands addressed to a GSME
- 2958 The CHF, GPF and GSME shall undertake the processing steps below following receipt of a
- 2959 Remote Party or SME.C.PPMID-GSME Command by the Communications Hub, where that
- 2960 Command is addressed to a GSME on the same SMHAN:
- 2961 1. the CHF shall buffer the Command and instruct the GPF to set the *Tunnel Message*
 - 2962 *Pending* flag to inform the GSME that the Command is awaiting retrieval. If the
 - 2963 Command has been sent as multiple GBT Messages, the GPF *Tunnel Message Pending*
 - 2964 flag shall only be set once all GBT Messages making up the Command have been
 - 2965 received by the Communications Hub. If not all GBT Messages making up a Command
 - 2966 have been received by a Communications Hub within 24 hours of the first GBT Message
 - 2967 in that Command being received, then the CHF may discard the GBT Messages that have
 - 2968 been received for that command;
 - 2969 2. if 24 hours elapse after setting the GPF *Tunnel Message Pending* flag without the
 - 2970 Command being retrieved by the GSME, the CHF may discard the Command. If the CHF
 - 2971 discards a Command in this way, it shall notify the GPF and the GPF shall log the event in
 - 2972 its Event Log and send an Alert with a GBZ Payload containing an Alert Code 0x819D;
 - 2973 1. when the GSME turns its HAN Interface on, it shall read the *Tunnel Message Pending*
 - 2974 flag and retrieve the Command using the *TransferData* command as defined in
 - 2975 Section 10.2.3. Each *TransferData* command received by the GSME shall result in the
 - 2976 GSME sending a *DefaultResponse* command;
 - 2977 2. the CHF shall process the *DefaultResponse* commands it receives to establish when
 - 2978 the Command has successfully been retrieved by the GSME, and shall provide an
 - 2979 indication to the GPF accordingly. The GPF shall, when there are no further
 - 2980 Commands or GBT Messages pending retrieval by the GSME, clear the *Tunnel*
 - 2981 *Message Pending* flag;
 - 2982 3. if a Command is a TOM Command, the CHF shall retain a copy of the Command
 - 2983 contents. For each such Command, the CHF shall start a response timer at the point
 - 2984 where it has received *DefaultResponse* command(s) confirming the GSME has
 - 2985 successfully retrieved the Command;
 - 2986 4. once a Command is successfully retrieved by the GSME, the GSME shall process the
 - 2987 Command in line with the requirements of the GBCS. Note that (1) this processing
-

2988 shall result in the GSME attempting to send a Response to the Command or an Alert
 2989 that it has received an invalid Command and (2) if sending a Response, the Response
 2990 shall, as per the GBCS requirements, detail the success or failure of GSME processing
 2991 for each instruction within the corresponding Command;

2992 5. the GSME shall not, under normal operating conditions, delay sending the Response
 2993 and shall, where possible, send it before turning its HAN Interface off;

2994 6. on receipt of a Response that is a TOM Response, the CH shall inspect the Response
 2995 from the GSME. If the Response indicates successful execution of at least one
 2996 elemental ZCL / ZSE command in the corresponding TOM Command, the CHF shall
 2997 transfer a copy of the corresponding TOM Command contents and the TOM
 2998 Response to the GPF, and shall clear the response timer for the Command;

2999 7. on receipt of a TOM Response and the corresponding TOM Command contents, the
 3000 GPF shall clear any stored copy it has of a TOM Command and then:

3001 if the TOM Command is not future dated, process the elemental ZCL / ZSE commands
 3002 contained within the Command according to the *status* within the Response, updating data
 3003 it holds accordingly. Once processed by the GPF, the GPF shall make any updated data
 3004 available over the WAN and over the HAN to the Devices in the GPF's Device Log;

3005 if the TOM Command is future dated, store a copy of the TOM Command without updating
 3006 any data it makes available over the WAN or HAN;

3007 8. if a Response to a TOM Command has not been received by the Communications
 3008 Hub when the corresponding response timer reaches 6 hours:

3009 the CHF may discard its copy of the TOM Command contents, clear the response timer and
 3010 notify the GPF accordingly; and

3011 on receipt of such a notification, the GPF shall log the event in its Event Log and send an
 3012 Alert with a GBZ Payload containing an Alert Code 0x819E;

3013 9. for clarity, the CHF shall relay all Remote Party Responses received on its HAN
 3014 Interface through the WAN Interface;

3015 10. whenever the CHF receives an Alert detailing activation of a future dated ZCL / ZSE
 3016 command from within a TOM Command (so an Alert with Alert Code 0x8F66 where
 3017 the Message Code in the Alert Payload is 0x006B or 0x00A3), the CHF shall pass a
 3018 copy of that Alert to the GPF;

3019 11. on receipt of such an Alert the GPF shall compare the Originator Counter in the Alert
 3020 Payload with the Originator Counter of any copy of a TOM Command it holds with
 3021 the same Message Code as in the Alert Payload, and:

3022 if the Originator Counters match, the GPF shall update the data it shares over the HAN and
 3023 WAN with the elemental ZCL / ZSE command contained within the TOM Command; or

3024 if the Originator Counters do not match or the GPF does not hold a TOM Command with this
 3025 Message Code, the GPF shall send an Alert with Alert Code 0x819E, as specified by Section
 3026 16.

3027 GPF Structured Data Items

3028 Underlined terms in this Section 10.4 shall have their meaning in the SMETS and / or CHTS.

3029 Introduction – informative

3030 There are GPF requirements to store structured data items which do not have a direct one to
 3031 one mapping in ZSE, or the interpretation may be uncertain. These structured data items
 3032 have to be constructed by the GPF.

3033 Structured Data Items

3034 This Section 10.4.2 details how each structured data item shall be constructed by the GPF.

3035 Daily Read Log

3036 The GSME shall record the Daily Read Log data items at midnight UTC as defined in SMETS.

3037 The GSME shall use the *snapshot cause 'General'* (0x00000001) for the *snapshot* taken.

3038 The GSME shall push the *snapshot* to the GPF using the *PublishSnapshot* command with a
 3039 *SnapshotPayloadType* and a *SnapshotSub-Payload* populated in line with the requirements
 3040 of Use Case 'GCS16a Read GSME Daily Read log(s)'. It is not necessary for the GSME to
 3041 report any attributes which duplicate those contained in the *snapshot*.

3042 The GPF shall populate the relevant attributes upon receipt of the *PublishSnapshot*
 3043 command, providing the command is received between midnight (UTC) and the next
 3044 scheduled wake of the GSME.

3045 The GPF shall store the data contained in the *PublishSnapshot* command in the GPF copy of
 3046 the GSME Daily Read Log.

3047 In the event of a communications outage, the GPF shall retrieve missing *snapshots* using the
 3048 *GetSnapshot* command, with the UTC start time field populated based on the last received
 3049 *snapshot* timestamp, if one has been received.

3050 Prepayment Daily Read Log

3051 If the GSME is operating in prepayment mode it shall record the Prepayment Daily Read Log
 3052 data items at midnight UTC. In ZSE terms, the GSME shall take a *prepayment snapshot* of
 3053 the relevant items. The format and data of the *prepayment snapshot* taken is defined in ZSE.

3054 The GSME shall use the *snapshot cause 'General'* (0x00000001) for the *prepayment*
3055 *snapshot* taken.

3056 The GSME shall push the *prepayment snapshot* to the GPF using the *Publish Prepay*
3057 *Snapshot* command.

3058 The GPF shall populate the relevant attributes upon receipt of the *Publish Prepay Snapshot*
3059 command, providing the command is received between midnight (UTC) and the next
3060 scheduled wake of the GSME.

3061 The GPF shall store the data contained in the *Publish Prepay Snapshot* command in the GPF
3062 copy of the GSME Prepayment Daily Read Log.

3063 In the event of a communications outage, the GPF shall retrieve missing *prepayment*
3064 *snapshots* using the *GetPrepaySnapshot* command (and *GetPrepaySnapshot* notification
3065 flag) with the UTC start time field populated based on the last received *prepayment*
3066 *snapshot* timestamp, if one has been received.

3067 Billing Data Log – informative

3068 SMETS defines Billing Data Log as a log capable of storing the following UTC date and time
3069 stamped entries:

3070 twelve entries comprising Tariff TOU Register Matrix, the Consumption Register and Tariff
3071 Block Counter Matrix;

3072 five entries comprising the value of prepayment credits;

3073 ten entries comprising the value of payment-based debt payments; and

3074 twelve entries comprising Meter Balance, Emergency Credit Balance, Accumulated Debt
3075 Register, Payment Debt Register and Time Debt Registers [1 ... 2].

3076 Requirements for each part are detailed separately in the following Sections.

3077 Billing Data Log – Tariff TOU Register Matrix, the Consumption Register and Tariff Block
3078 Counter Matrix

3079 The GSME shall capture this *snapshot* at the following trigger points:

3080 End of Billing Cycle (snapshot cause 'End of Billing Period');

3081 Change of Payment Mode (snapshot cause 'Change of Meter Mode');

3082 Change of Tariff (snapshot cause 'Change of Tariff Information'); and

3083 as specified in Section 13.3.5.10 (snapshot cause 'Change of Supplier').

3084 When it next turns on its HAN Interface, the GSME shall push this *snapshot* to the GPF using
 3085 the *PublishSnapshot* Command with a SnapshotPayloadType and a SnapshotSub-Payload
 3086 populated in line with the requirements of Use Case 'GCS15b Read GSME Billing Data Log
 3087 (change of mode / tariff triggered)'.

3088 The GPF shall store the data contained in the *PublishSnapshot* command in the GPF copy of
 3089 the GSME Billing data Log.

3090 In the event of a communications outage, the GPF shall retrieve missing *snapshots* using the
 3091 *GetSnapshot* command (and the relevant notification flag) with the UTC start time field
 3092 populated based on the last received *snapshot* timestamp, if one has been received, or
 3093 0x0000 otherwise.

3094 Billing Data Log – value of prepayment credits

3095 Upon completion of processing of a valid Prepayment Top Up, the GSME shall push the
 3096 Prepayment Top Up's details to the GPF using the *PublishTop Up Log* command.

3097 The GPF shall store the data contained in the *Publish Top Up Log* command in the GPF copy
 3098 of the GSME Billing data Log.

3099 If there has been a communications outage, the GPF shall use the *Get Top Up Log* command
 3100 to retrieve all Prepayment Top Ups that may have been processed during the
 3101 communications outage. The GSME shall set the *Date / Time* field of the *Get Top Up Log*
 3102 command to the current UTC time.

3103 Billing Data Log – payment-based debt payments

3104 Upon completion of processing of a valid Prepayment Top Up where the GSME has made a
 3105 debt payment using part of that Prepayment Top Up, the GSME shall push details of that
 3106 debt payment only to the GPF using the *Publish Debt Log* command.

3107 The GPF shall record the details provided in the GPF copy of the GSME Billing Data Log.

3108 In cases of communications outages, the GPF shall request any outstanding payment-based
 3109 debt payments by use of the *GetDebtRepaymentLog* command (and *GetDebtRepaymentLog*
 3110 notification flag) with the Debt Type field set to 0x02 (Debt 3).

3111 Billing Data Log – Meter Balance, Emergency Credit Balance, Accumulated Debt Register,
 3112 Payment Debt Register and Time Debt Registers [1 ... 2]

3113 The GSME shall capture this snapshot at the following trigger points:

3114 End of Billing Cycle (snapshot cause bit 1 set: 'End of Billing Period' , as per PublishSnapshot
 3115 command);

3116 Change of Payment Mode (snapshot cause bit 14 set: 'Change of Meter Mode');

3117 Change of Tariff (snapshot cause including set: bit 3 'Change of Tariff Information'); and
3118 as specified in Section 13.3.5.10 (snapshot cause 'Change of Supplier').

3119 When it next turns on its HAN Interface, the GSME shall push this *snapshot* to the GPF using
3120 the *Publish Prepay Snapshot* command.

3121 The GPF shall store the data contained in the *Publish Prepay Snapshot* command in the GPF
3122 copy of the GSME Billing Data Log.

3123 In the event of a communications outage, the GPF shall retrieve missing *snapshots* using the
3124 *GetPrepaySnapshot* command (and *GetPrepaySnapshot* notification flag) with the UTC start
3125 time field populated based on the last received snapshot timestamp, if one has been
3126 received.

3127 GPF Profile Data Log

3128 The GPF shall create the GPF Profile Data Log from the consumption information pushed by
3129 the GSME each half hour.

3130 The GSME shall, on each half hour, record the following information and push to the GPF:

3131 the *CurrentSummationDelivered* attribute containing total consumption value (with units of
3132 m³);

3133 the *CurrentDayAlternativeConsumptionDelivered* attribute containing total consumption
3134 today (with units of kWh); and

3135 the *CurrentDayCostConsumptionDelivered* attribute containing total cost of consumption
3136 today (with units of Currency Unit);

3137 Upon receipt of the pushed data, the GPF shall calculate the consumption with units of m³
3138 over the previous half hour by subtracting its previously recorded total consumption value
3139 from the total consumption value now sent.

3140 The resulting value shall be stored in the GPF Profile Data Log.

3141 In the event that there are missing values in the GPF Profile Data Log, the GPF shall
3142 interrogate the GSME Profile Data Log using the *GetSampledData (SampleID 0x0000)*
3143 command and the *GetSampledData* notification flag to retrieve missing values.

3144 GPF Daily Gas Consumption Log

3145 The GPF shall create the GPF Daily Gas Consumption Log based on the values pushed from
3146 the GSME. The difference between last total consumption value pushed from the GSME
3147 each UTC day and the last value pushed in the prior UTC day shall be time stamped and
3148 stored in the GPF Daily Gas Consumption Log, so that the values in the log represent
3149 consumption in that UTC day.

3150 In the event of communications outages resulting in the final daily value being missed, the
3151 GPF shall retrieve the values from the GSME Profile Data Log using the *GetSampledData*
3152 (*SampleID 0x0000*) command and *GetSampledData* notification flag.

3153 Historical Attributes

3154 A GSME shall support:

3155 the *Historical Cost Consumption Information* attribute set, measured in Currency Units; and

3156 the *Alternative Historical Consumption* attribute set, measured in kWh.

3157 A GPF shall mirror the attribute sets listed above.

3158 As per Section 10.4.2.8, the GSME shall, on each half hour, record the following information
3159 and push to the GPF:

3160 total consumption value (with units of m³);

3161 total consumption today (with units of kWh); and

3162 total cost of consumption today (with units of Currency Unit);

3163 Using the 'total consumption today' value, the GPF shall update the attributes of the
3164 mirrored *Alternative Historical Consumption* attribute set.

3165 Using the 'total cost of consumption today' value, the GPF shall update the attributes of the
3166 mirrored *Historical Cost Consumption Information* attribute set.

3167 In exception circumstances, the GPF shall request the GSME to push the historical data sets
3168 using the '*Push Historical Metering Data Attribute Set*' and '*Push Historical Prepayment Data*
3169 *Attribute Set*' notification flags. The GSME shall interpret the '*Push Historical Metering Data*
3170 *Attribute Set*' notification flag as requiring it to push the *Alternative Historical Consumption*
3171 attribute set.

3172 Other attributes

3173 The GSME, ESME and the SAPC, where it supports such functionality, shall populate the
3174 *AccumulatedDebt* attribute in line with the SMETS Accumulated Debt requirements, and all
3175 other Devices shall interpret that attribute correspondingly.

3176 The GSME, ESME and the SAPC, where it supports such functionality, shall populate the
3177 *Credit Remaining* attribute in line with the SMETS Meter Balance requirements, and all other
3178 Devices shall interpret that attribute correspondingly. The GSME, ESME and the SAPC, where
3179 it supports such functionality, shall apply functionality related to the *CutOffValue* attribute in
3180 line with this interpretation of *Credit Remaining* and the SMETS requirements for
3181 Disablement Threshold.

3182 The GPF shall calculate the value of the price in any ZCL *PublishPrice* command it creates
3183 using the tariff information it has derived through the TOM and the time from the
3184 Communications Hub's clock.

3185 The ESME and the SAPC shall populate the *AuxSwitchNLabel* attribute in line with the SMETS
3186 Auxiliary Controller [n] Description requirements, and all other Devices shall interpret that
3187 attribute correspondingly.

3188 The *CommodityType* and *MeteringDeviceType* attributes shall be set by devices as follows:

3189 'GPF: Metering Device (Gas Mirror Endpoint)': 128 (Mirrored Gas Metering) –
3190 *MeteringDeviceType* only;

3191 'GSME: Metering Device': 1 (Gas Metering) – *MeteringDeviceType* only;

3192 'GPF: Energy Services Interface (Gas ESI Endpoint)': 1 (Gas Metering) – *CommodityType* only;

3193 'ESME: Energy Services Interface (Electricity ESI Endpoint)' and not a Polyphase ESME: 0
3194 (Electric Metering);

3195 'SAPC: Energy Services Interface (Electricity ESI Endpoint)': 0 (Electric Metering);

3196 'ESME: Energy Services Interface (Electricity ESI Endpoint)' and a Polyphase ESME: 15
3197 (Electric Metering Element/Phase 3);

3198 'ESME: Energy Services Interface (Twin ESME aggregate ESI Endpoint)': 0 (Electric Metering);

3199 'ESME: Energy Services Interface (Twin ESME primary ESI Endpoint)':13 (Electric Metering
3200 Element/Phase 1); and

3201 'ESME: Energy Services Interface (Twin ESME secondary ESI Endpoint)':14 (Electric Metering
3202 Element/Phase 2).

3203 When processing a ZSE *Get Event Log* command or a ZSE *Clear Event Log* command with a
3204 Log ID nibble of 0x6 (GSME Event Log) or 0x7 (GSME Security Log), a GPF shall process the
3205 command using the relevant GSME Proxy Log copy of the GSME Event or Security Log. Other
3206 values of Log ID shall refer to the GPF's own logs.

3207 Where a Device is an ESME which is not a Twin Element ESME, or is an SAPC which supports
3208 such functionality, the Device shall populate the *SiteID* attribute with the 13 most significant
3209 octets being the Import MPAN and the following 13 octets the Export MPAN, where those
3210 terms have their SMETS meanings.

3211 Where an ESME is a Twin Element ESME it shall populate:

3212 the *SiteID* attribute in the 'ESME: Energy Services Interface (Twin ESME aggregate ESI
3213 Endpoint)' with the 13 most significant octets being the Import MPAN on the primary
3214 element and the following 13 octets the Export MPAN; and

3215 the *SiteID* attribute in the 'ESME: Energy Services Interface (Twin ESME secondary ESI
 3216 Endpoint)' with the most significant 13 octets being the Import MPAN on the secondary
 3217 element.

3218 Where a ZCL / ZSE command containing *IssuerEventId* and / or *ProviderID* fields is received
 3219 by a Device as part of a GBZ Remote Party Command, the Device shall undertake no
 3220 processing in relation to those two fields. For clarity, this means the Device shall not use the
 3221 contents of those fields for anti replay purposes.

3222 In ZSE *GetSampledData* and *GetSampledDataResponse* commands:
 3223 the *SampleID* field shall be interpreted as:

3224 0x0000 meaning Profile Data Log;

3225 0x0001 meaning Daily Consumption Log; and

3226 0x0002 meaning Network Data Log;

3227 the *Sample Type* field shall, where the command is to or from an ESME and relates to the
 3228 Profile Data Log, be interpreted as:

3229 0x00 meaning Consumption / Active Energy Imported;

3230 0x01 meaning Active Energy Exported;

3231 0x02 meaning Reactive Energy Imported; and

3232 0x03 meaning Reactive Energy Exported;

3233 each of the integers in the *Samples* field shall, when populated by ESME, be the 24 least
 3234 significant bits of the corresponding value that would be returned when the Profile Data Log
 3235 is read using a Remote Party Command. Thus, the values returned in *Samples* are in whole
 3236 Wh / varh; and

3237 the *SampleRequestInterval* field shall contain 0xFFFF whenever the *SampleID* field is 0x0001.

3238 A GSME shall reject any *PublishPriceMatrix* command that does not contain four *Price* fields.

3239 When processing a *Get Snapshot* or *Get Prepay Snapshot* command, a Device shall return all
 3240 snapshots where the *Snapshot Cause* in the snapshot matches any of the set bits in the
 3241 *Snapshot Cause* parameter of the command. When processing a command where *Issuer*
 3242 *Calendar ID* has the value 0xFFFFFFFF or 0xFFFFFFF, a GPF or GSME shall interpret
 3243 0xFFFFFFFF as meaning the currently in force Tariff Switching Table calendar and 0xFFFFFFF
 3244 as meaning the currently in force Non-Disablement Calendar.

3245 Devices shall support the requirements relevant to their device type detailed in 'Trust Center
 3246 Swap-Out' process section of the ZSE specification. Where a Device would normally use the

3247 CHF's *Keep Alive Cluster* to periodically check communications with the CHF as part of its
3248 '*Trust Center Swap-Out*' support, the Device shall, when first connecting to the HAN:

3249 Check whether the CHF supports the *Keep Alive Cluster*; and

3250 Where the CHF does not support the *Keep Alive Cluster*, use an alternative behaviour to
3251 allow it to detect a failure to communicate with the CHF.

3252 The GSME shall set the *BillDeliveredTrailingDigit* attribute to the same value as
3253 *PriceTrailingDigit* in the *Price* cluster.

3254 In line with the SMETS requirement, the *UnitOfMeasure* parameter in the
3255 *PublishTariffInformation* command, sent to a GSME shall be 0x00 (kWh) as per the Message
3256 Templates for GCS01a and GCS01b, shall apply to the *Block Threshold N* parameter in the
3257 *PublishBlockThresholds* command in such Messages. Contrary to ZSE, the GSME shall
3258 undertake the necessary calculation when comparing these thresholds against the
3259 *CurrentBlockPeriodConsumptionDelivered* attribute (whose unit of measure in line with
3260 SMETS shall be m³).

3261 Contrary to ZSE, the GPF shall accept any valid UTCTime in the value of the *Implementation*
3262 *Date/Time* parameter in a *Publish Change Of Tenancy* command, in a Command complying
3263 with Use Case GCS09.

3264 The GSME shall only action a ZSE *Set Low Credit Warning Level* command it receives at the
3265 specified 'From Date Time' specified with that ZSE command.

3266 In GBZ Remote Party Commands containing the *Issuer Tariff ID* parameter, the value of that
3267 parameter shall always be 0x00000001.

3268 In GBZ Remote Party Commands containing the *Friendly Credit Calendar ID* parameter, the
3269 value of that parameter shall always be 0x00000002.

3270 In responding to GBZ Remote Party Commands requesting information relating to the *Price*
3271 *Cluster*, the GSME & GPF shall provide that information in the GBZ Remote Party Response.

3272 On receipt of a *Credit Adjustment* command with a *Credit Adjustment Type* set to 0x01, the
3273 GSME shall reset the prepayment mode Meter Balance, the Emergency Credit Balance and
3274 the Accumulated Debt Register.

3275 Where a Device receives a GBZ Command containing a ZSE *Change Payment Mode*
3276 command or a ZSE *Get Current Price* command and the Device fails successfully to execute
3277 that ZSE command, the Device shall populate the corresponding GBZ Response with a ZSE
3278 *Default Response* command in the place of either a ZSE *Change Payment Mode Response*
3279 command or a ZSE *Publish Price* command.

3280 For GSME, the value of the *Multiplier* attribute in the *Metering* cluster shall be set to 1 and
3281 the value of the *Divisor* attribute in the *Metering* cluster shall be set to 1000.

3282 Where a ZCL / ZSE command containing a ZCL *Date* field is received by a GSME or by a GPF
3283 as part of a GBZ Remote Party Command, the Device shall always interpret the value 0xFF in
3284 a *Date* sub-field as meaning ‘unused’ and so having the effect of a wildcard. For clarity, this
3285 means the Device shall never interpret the value 0xFF in such *year* sub-fields as meaning the
3286 year 2155.

3287 Where a ZCL / ZSE command containing a ZCL *Date* field is received by a Device from a GPF,
3288 an SAPC or an ESME, the receiving Device shall always interpret the value 0xFF in a *Date* sub-
3289 field as meaning ‘unused’ and so having the effect of a wildcard. For clarity, this means the
3290 Device shall never interpret the value 0xFF in such *year* sub-fields as meaning the year 2155.

3291 ESME, GSME and SAPC, where it supports such functionality, shall set bit 6 of the
3292 Prepayment Alarm Status attribute (*‘Emergency credit is available to be selected’*) to 0b1
3293 only when Emergency Credit is available but not activated (where those terms have their
3294 SMETS meanings). Otherwise the attribute shall have the value 0b0.

3295 For clarity, the GSME shall update *ReadingSnapshotTime* to the UTC time of the GSME’s
3296 clock whenever the GSME updates the *CurrentSummationDelivered* attribute. Where the
3297 GSME does not provide a *ReadingSnapshotTime* attribute value, the GPF shall set the value
3298 of its *ReadingSnapshotTime* attribute to be 0xFFFFFFFF (meaning invalid time in ZigBee), and
3299 shall record that the value was not provided by the GSME.

3300 Where:

3301 A GPF or ESME currently has no meaningful time; and

3302 It has previously received and successfully processed at least one GCS09 or ECS12 Command
3303 (so a Command to restrict access to historic data for other Devices),

3304 the GPF / ESME should not provide, to any other Device, the values for any attributes in the
3305 *Historical Consumption*, *Historical Cost Consumption Information* or *Alternative Historical*
3306 *Consumption Store* attribute sets whose name includes ‘previous’, so those attributes
3307 containing historic data.

3308 For clarity and in line with the requirements of Use Cases GCS15b, GCS15c, GCS16a and
3309 GCS16b and the ZSE requirement that the ‘Latest End Time’ is the ‘latest time of a snapshot
3310 to be returned by a corresponding *Publish Snapshot* command’, when processing a *Get*
3311 *Snapshot* or *Get Prepay Snapshot* command, either from another GPF or as part of a Remote
3312 Party Command, a GPF shall return all snapshots with a timestamp up to and including the
3313 date-time in the *Latest End Time* parameter.

3314 For clarity and in line with the requirements of Use Cases CS10a and CS10b, when
 3315 processing a *Get Event Log* command as part of a Remote Party Command to read either an
 3316 Event Log or a Security Log, a GPF shall return all log entries with a timestamp up to and
 3317 including the date-time in the *End Time* parameter of the *Get Event Log* command.

3318 Sequence of actions where snapshots and attributes are taken

3319 Except where the Technical Specifications explicitly require otherwise, GSME should
 3320 undertake all actions that may affect the contents of a ZigBee snapshot or value of a ZigBee
 3321 attribute, before providing such snapshots or attributes to the GPF.

3322 Hand Held Terminal (HHT) interactions

3323 Introduction – informative

3324 An HHT allows for delivery of Remote Party Messages to and from the SMHAN. This is as an
 3325 alternative delivery route to the Communications Hub’s WAN connection. It is intended for
 3326 one-off configuration of Devices, for example at installation. Hence, there are time outs to
 3327 ensure usage is limited in this way.

3328 This Section 10.5 specifies requirements related to:

3329 how a connection is made between an HHT and a Communications Hub; and

3330 how Remote Party Messages are then to be transferred to and from the HHT.

3331 Establishing a connection between an HHT and a Communications Hub – informative

3332 ZigBee supports an Inter-PAN mechanism which can be used for communication links
 3333 between Devices. This mechanism is used to establish a link key between the HHT and the
 3334 Communications Hub (referred to as the Inter-PAN Link Key (IPLK)), by using ZSE’s *CBKE* (for
 3335 clarity, this IPLK is not used to encrypt any Inter-PAN communication). The Inter-PAN link is
 3336 then used for HHT to Communications Hub communication:

3337 the HHT uses the link to send its Entity Identifier and *Install Code* to the Communications
 3338 Hub as part of a ‘CCS01 Add Device to CHF device log’ Command;

3339 the Communications Hub adds these details to the CHF’s Device Log (so allowing the HHT to
 3340 *join* the SMHAN) and confirms this to the HHT using a ‘CCS01 Add Device to CHF device log’
 3341 Response; and

3342 the HHT then *joins* the SMHAN and so can exchange Remote Party Messages within the
 3343 Communications Hub, and the Communications Hub can relay them to / from the specified
 3344 Device(s) on the HAN. The IPLK is used as the *pre-configured link key* in this *joining* process.

3345 Both the *Inter-PAN communications* and *joining* to the SMHAN use the *CBKE* mechanism that
 3346 is defined in ZSE.

3347 *Inter-PAN* is only to be available for 60 minutes from power on of the Communications Hub.
 3348 So, if needed, *Inter-PAN* can be enabled by power cycling the Communications Hub.

3349 The *Inter-PAN* mechanism defined by ZSE requires the HHT to specify the Communications
 3350 Hub that it wishes to link to. There may be multiple Communications Hubs available to the
 3351 HHT to connect to via *Inter-PAN*.

3352 There are a number of options to provide the HHT with information sufficient to identify
 3353 uniquely the Communications Hub it is to link to, including:

3354 the installer manually reading the GPF's Entity Identifier (which is the IEEE address of the
 3355 Communications Hub's SMHAN radio) printed on the Hub, and confirming / selecting this on
 3356 the HHT; or

3357 the installer using a scanner on the HHT to read the GPF's Entity Identifier.

3358 Two illustrative connection scenarios are provided in the following two Sections

3359 Illustration 1: Installer manually chooses network – informative

- 3360 1. the Communications Hub opens Inter-PAN communication for 60 minutes after
 3361 power on;
- 3362 2. the HHT is powered on;
- 3363 3. the HHT performs an active scan using the Beacon Request mechanism;
- 3364 4. the HHT displays the IEEE addresses returned in the Beacons from all neighbouring
 3365 PAN Coordinators. Note that the GBCS requires the Extended PAN ID initially to be
 3366 set to the Communications Hub's HAN Interface's IEEE address. This is the same as
 3367 the GPF Entity Identifier, which is printed on the Communications Hub in line with
 3368 CHTS. Note that, if the Communications Hub is a replacement for one previously
 3369 installed, the Extended PAN ID will not be its GPF Entity Identifier, and the installer
 3370 will need a different way to establish the correct Extended PAN ID to use;
- 3371 5. the installer (who knows the Consumer's Communications Hub's ZigBee IEEE address
 3372 as the GPF Entity Identifier is printed on the Communications Hub) picks the desired
 3373 IEEE address;
- 3374 6. the HHT initiates Inter-PAN CBKE with the Communications Hub;
- 3375 7. the Communications Hub responds to the Inter-PAN CBKE;
- 3376 8. if Inter-PAN CBKE completes successfully, the HHT sends its Install Code and Entity
 3377 Identifier to the Communications Hub in a CCS01 Command, then
- 3378 9. the Communications Hub adds the HHT to the CHF Device Log and sends a CCS01
 3379 Response to the HHT accordingly, and then

-
- 3380 10. the HHT joins to the SMHAN;
- 3381 11. otherwise, no link is established.
- 3382 Illustration 2: HHT uses barcode scan – informative
- 3383 1. the Communications Hub opens Inter-PAN communication for 60 minutes after
3384 power on;
- 3385 2. the HHT is powered on;
- 3386 3. the HHT optically scans the GPF Entity Identifier printed on the target
3387 Communications Hub;
- 3388 4. the HHT performs an active scan using the Beacon Request mechanism;
- 3389 5. when a Beacon returns an IEEE address equal to the scanned GPF Entity Identifier,
3390 the HHT initiates Inter-PAN communication with the Communications Hub so
3391 identified;
- 3392 6. the Communications Hub responds to the Inter-PAN CBKE;
- 3393 7. if Inter-PAN CBKE completes successfully:
- 3394 8. the HHT sends its Install Code and Entity Identifier to the Communications Hub in a
3395 CCS01 Command, then
- 3396 9. the Communications Hub adds the HHT to the CHF Device Log and sends a CCS01
3397 Response to the HHT accordingly, and then
- 3398 10. the HHT joins to the SMHAN;
- 3399 11. otherwise, no link is established.
- 3400 WAN proxy operation
- 3401 Introduction – informative
- 3402 The HHT has to be capable of holding Remote Party Messages, to which the appropriate
3403 Remote Party Message protection has already been applied, and has to be capable of
3404 exchanging such Messages.
- 3405 The Communications Hub must therefore be able to maintain two interfaces to the WAN;
3406 the real one via the WAN Interface and a 'logical WAN' via the connection to the HHT.
- 3407 WAN Responses
- 3408 The Communications Hub shall send any Responses and Alerts through both its WAN
3409 Interface and the link to the HHT, if present. Whilst this may result in apparent unsolicited

3410 Responses at the Remote Party which have to be dealt with, it ensures the earliest possible
3411 reconciliation of Commands destined for Smart Metering Equipment.

3412 HHT and CHF – Device Requirements

3413 Any Device which provided its Entity Identifier and *Install Code* in a CCS01 Command to the
3414 Communications Hub via Inter-PAN shall be treated by the CHF as being of type HHT. For
3415 clarity, the Device Type specified in any such CCS01 Command shall be disregarded by the
3416 CHF.

3417 As per Section 10.2.2, in interactions between an HHT and a Communications Hub over the
3418 SMHAN:

3419 the HHT shall support the *Tunneling Cluster* as a *Client*; and

3420 the Communications Hub shall support the *Tunneling Cluster* as a *Server*.

3421 The Communications Hub shall only allow *Inter-PAN communications* for 60 minutes from
3422 any power on of the Communications Hub. For clarity, this is the period during which an
3423 HHT can establish a connection, not the period of use of any connection.

3424 At power on, a Communications Hub shall remove any Devices of type HHT from the CHF
3425 Device Log.

3426 The Communications Hub shall prior to installation, set *nwkExtendedPANId* to be the Entity
3427 Identifier of the GPF, which is always the Communications Hub's IEEE address for its HAN
3428 Interface.

3429 HHT and CHF – establishing communications

3430 Prior to being able to exchange Messages, the HHT and Communications Hub shall
3431 undertake the following steps:

3432 1. the HHT shall identify the Communications Hub and initiate the CBKE process using
3433 Inter-PAN communications, as specified in this Section 10.5.3.4;

3434 the Communications Hub shall not respond to any such communications if more than 60
3435 minutes has elapsed since the Communications Hub's most recent power on, or if there is a
3436 Device of type HHT already in the CHF's Device Log. Otherwise, the Communications Hub
3437 shall respond to the CBKE request;

3438 if CBKE does not succeed, processing shall cease. Otherwise, the Communications Hub and
3439 HHT shall each store the link key established through CBKE (referred to as the Inter-PAN Link
3440 Key (IPLK)) along with a record of the other Device's Entity Identifier, which is its IEEE
3441 address. Then processing shall continue from step 4;

the HHT shall send a 'CCS01 Add Device to CHF device log' Command, populated with the HHT's Entity Identifier and Install Code, to the CHF using the mechanism defined in Section 10.5.3.5;

where all parts of the Command are successfully received, the CHF shall first validate that the Entity Identifier in the CCS01 Command matches the Entity Identifier it stored against the IPLK at step 3. If it does not match, processing shall cease. Otherwise the CHF shall process the Command according to the requirements of the 'CCS01 Add Device to CHF device log' Use Case, and shall send any resulting 'CCS01 Add Device to CHF device log' Response to the HHT, via Inter-PAN using the mechanism defined in Section 10.5.3.5. For clarity the Install Code in the CCS01 Command shall not be used in the subsequent ZigBee joining. Where the CHF sends such a Response, it shall start a timer. When that timer reaches 0xFFFF seconds, the CHF shall remove the HHT from its Device Log, remove the HHT from the SMHAN and close any open tunnels to the HHT;

if the 'CCS01 Add Device to CHF device log' Response from the CHF to the HHT states that the HHT has been added to the CHF's Device Log, the HHT may attempt to join the SMHAN. The Devices shall use the IPLK stored at step 3 as the pre-configured link key to secure the joining process. If the joining is successful:

1. the HHT shall send a *RequestTunnel* command to the CHF, with contents as per Section 10.2.2;

the CHF shall process the *RequestTunnel* command and send a *RequestTunnelResponse* command in response; and

if *TunnelStatus* in the *RequestTunnelResponse* command is not 0x00 ('success'), processing by the HHT shall cease. Otherwise the HHT and CHF may now exchange Messages using the *TransferData* command with the *TunnelID* associated with *ProtocolID* 6.

Note that steps 1 to 5 above use *Inter-PAN communications*; the remaining step 6 uses the standard ZigBee SMHAN communications.

Once the HHT has *joined* the SMHAN, any Messages received by the CHF from the HHT in the *Data* parameter payload of a *TransferData* command with the *TunnelID* associated with *ProtocolID* 6, shall be forwarded to the relevant Device on the SMHAN as if they were received via the Communications Hub's WAN Interface. If any such Message contains a Business Target ID which is not in the CHF Device Log and is not the CHF ID or the GPF ID, the Communications Hub may silently discard the Message.

Whilst the HHT is in the CHF's Device Log and *joined* to the SMHAN, any Responses received by the CHF from any SMHAN Device shall be provided to the HHT using the *TransferData*

3477 command with the *TunnelID* associated with *ProtocolID* 6. Such Responses shall also be sent
3478 over the Communications Hub's WAN Interface, if available.

3479 Note that the requirements of Section 7.2.11(Transfer of Large Remote Party Messages)
3480 apply to Messages exchanged once the HHT has joined the SMHAN. Section 7.2.11
3481 requirements do not apply to the CCS01 Command and Response exchanged using the Inter-
3482 PAN mechanism.

3483 Once the HHT usage on the SMHAN is complete, the HHT should send a *CloseTunnel*
3484 command to the Communications Hub. On receipt of such a *CloseTunnel* command from an
3485 HHT, the Communications Hub shall process that command as per the ZSE specification and
3486 shall:

3487 remove the HHT from its Device Log; and

3488 remove the HHT from the SMHAN.

3489 Population of the Inter-PAN ZigBee Frame

3490 A Device may send a Remote Party Message to another Device on the same SM HAN using
3491 the ZSE Inter-PAN Frame. In doing so, the sending Device shall:

3492 Take the 'n' most significant octets of the Remote Party Message to create an 'Inter-PAN
3493 Message Fragment (IPMF)', where 'n' is the maximum number such that the
3494 aMaxPHYPacketSize (with its IEEE 802.15.4 2003 specification meaning) would not be
3495 exceeded in the resulting message over the SM HAN. This IPMF shall have an index of 0x00;

3496 If there are remaining octets in the Remote Party Message, take the next 'n' most significant
3497 octets to create IPMF with an index of one greater than the previous IPMF;

3498 Repeat step 2 until there are no remaining octets in the Remote Party Message; and

3499 Set the 'Total Number of IPMF' to be the number of IPMF so created.

3500 A Device receiving a Remote Party Message by this mechanism shall respond to each IPMF
3501 received with an 'IPMF Response' constructed according to this Section10.5.3.5. The
3502 receiving Device shall re-construct the Remote Party Message in a way that is aligned with
3503 the splitting mechanism defined in this Section10.5.3.5.

3504 If a response to an Inter-PAN ZigBee Frame is required by this Section 10.5.3.5 but the
3505 receiving Device has not received such a Response after 60 seconds, the receiving Device
3506 may treat the transmission of the underlying CCS01 Message as having failed.

3507 When a sending Device is populating the Inter-PAN ZigBee Frame to communicate with a
3508 receiving Device using Inter-PAN, the sending Device shall:

- 3509 1. Populate the Inter-PAN ZigBee NWK Header according to the requirements of Table
3510 10.5.3.5a;
- 3511 2. Populate, in the Inter-PAN ZigBee APS Header, the fields specified in Table 10.5.3.5b,
3512 according to the requirements of Table 10.5.3.5b; and
- 3513 3. Populate the Inter-PAN ZigBee Payload:
- 3514 1. According to the requirements of Table 10.5.3.5c when the Device is sending
3515 an IPMF in this Frame; or
- 3516 2. According to the requirements of Table 10.5.3.5d when the Device is sending
3517 an IPMF Response in this Frame.

Element	Contents	Length (bits)	Note
NWK frame control	See remaining rows in this table.	16 in total	Components as per the following rows in this table
- Frame type (bits 0 -1)	0b11	2	Inter-PAN NWK Frame
- Protocol version (bits 2-5)	0b0010	4	ZigBee Pro
- Remaining sub-fields (bits 6-15)	0b0000000000	10	Unused

3518 Table 10.5.3.5a: Inter-PAN ZigBee NWK Header (two octets total length)

3519

Element	Contents	Length (bits)	Note
APS frame control	See next six rows of this table	8 in total	Components as per the following six rows in this table
- Frame type (bits 0 -1)	0b11	2	Inter-PAN NWK Frame

Element	Contents	Length (bits)	Note
- Delivery Mode (bits 2-3)	0b00	2	Unicast
- Reserved (bit 4)	0b0	1	Unused
- Security (bit 5)	0b0	1	No security
- ACK request (bit 6)	0b0	1	No ACK requested
- Extended Header Present (bit 7)	0b0	1	No Extended Header Present
Cluster identifier	0xFFFF	16	Manufacturer specific
Profile identifier	0x0109	16	ZSE

3520 Table 10.5.3.5b: Inter-PAN ZigBee APS Header (five octets total length)

3521 Note, the Inter-PAN ZigBee Payloads specified in this Section 10.5.3.5 do not contain a ZCL
 3522 Header.

3523

Element	Contents	Length (bits)	Note
Control byte	0x00	8	Indicating that this frame contains an IPMF
Index of this IPMF	See Note column	8	Shall be interpreted as an unsigned 8 bit integer with a value of 0x00 for the first IPMF, and each subsequent IPMF having a value one higher than the previous A Device shall only send IPMFs in index order, starting at index 0x00 and shall not send a subsequent IPMF unless it has received a Frame containing an IPMF Response for the previous IPMF with a response status of 0x00

Element	Contents	Length (bits)	Note
Total number of IPMF	Total number of IPMF	8	Shall be interpreted as an unsigned 8 bit integer. This number shall be calculated by the sending Device and the receiving Device shall not process, in the sense of Section 6, the Remote Party Message until it has successfully received IPMFs for all index values for 0x00 to (Total number of IPMF minus one)
Length of IPMF	Number of octets in the IPMF		Shall be interpreted as an unsigned 8 bit integer
IPMF	The octets making up this IPMF.	Variable	Shall be derived from the Remote Party Message according to this Section 10.5.3.5

3524 Table 10.5.3.5c: Inter-PAN ZigBee Payload contents when sending an IPMF (variable total
 3525 length)

3526

Element	Contents	Length (bits)	Note
Control byte	0x80	8	Indicating that this frame contains an IPMF Response
Index of the IPMF	Index of the IPMF to which this IPMF Response relates.	8	Shall be interpreted as an unsigned 8 bit integer
Response status	0x00 (meaning successful receipt) or 0x01 (meaning receipt failure)	8	Where a Device receives an IPMF Response containing a response status of 0x00, that Device shall send the next IPMF, or cease transmission, if the response status relates to a final IPMF Where a Device receives an IPMF Response containing a response status of 0x01, that Device may either cease sending IPMFs or may begin re-sending starting from the IPMF with index 0x00

3527 Table 10.5.3.5d: Inter-PAN ZigBee Payload contents when sending an IPMF Response (three
3528 octets total length)

3529 Sub GHz Requirements

3530 In this Section 10.6, 'Duty Cycle' shall mean the percentage of time a Device is transmitting
3531 on Sub GHz frequencies. For clarity, any actions taken in relation to managing 'Duty Cycle',
3532 including the processing of ZSE *Suspend ZCL Messages* commands, shall only relate to
3533 communications with *End Devices* operating on Sub GHz frequencies.

3534 Introduction

3535 This Section 10.6 specifies requirements for Devices which are capable of operating on Sub
3536 GHz for their SMHAN operations. Data items defined in this Section 10.6 shall have their
3537 defined meaning throughout this Section 10.6. For clarity, the requirements of this Section
3538 10.6 shall therefore apply to SAPC which are capable of operating on Sub GHz for their
3539 SMHAN operations.

3540 Communications Hub requirements

3541 Operational data items

3542 A Dual Band CH shall maintain the data items specified in Table 10.6.2.1a and shall make
3543 those data items available through the DLMS COSEM attributes and objects specified in
3544 Table 10.6.2.1a, through the Use Cases specified in Table 20.

Data item	Requirements	OBIS code	Attribute
Operating Sub GHz Channel	The CH shall maintain 'Operating Sub GHz Channel' as a 32 bit bit-string specifying the Sub GHz Channel currently in use on the SMHAN The channel is specified by a channel number within 'page' The bit-string shall follow the same structure as the ZSE Channel Change Attribute. Thus, there shall only be one of bits 0-26 set to 0b1, which identifies the channel number in use, and bits 27-31 shall contain the page number in which the channel number is located (so a binary value equivalent to 28, 29, 30 or 31) Note that, as for 2.4 GHz only CH, there is no requirement on the CH to expose the 2.4 GHz channel	0-0:94.44.10.2	2

Data item	Requirements	OBIS code	Attribute
	in use		
Sub GHz Channel Log	The CH shall maintain 'Sub GHz Channel Log' as a circular log containing date-time stamped entries of the last 100 Operating Sub GHz Channel values used, where each entry in the Log specifies (1) the date-time operation began on the Sub GHz Channel, (2) the Sub GHz Channel (by way of a 32 bit bit-string which has the same structure as Operating Sub GHz Channel) and (3) a flag indicating the trigger for the change In DLMS COSEM terms, the Log shall be exposed as an 'entry_eventLogEntry8' structure as defined in Section 18.2.1.1. The CHF shall create each log entry according to the requirements of 10.6.2.8 For clarity, the first entry created in this Log shall detail the Sub GHz Channel selected by the CH at initial SMHAN network formation	0-0:94.44.10.3	2
CHF Communication Store – Sub GHz link quality.	The CH shall maintain a store holding, for each Device in the CHF Device Log the quality of the any Sub GHz Communications Link established with the CHF by maintaining the 'AverageRSSI' as specified in Section 10.6.2.1.1	0-0:30.3.0.255	Part of attribute 4

3545 Table 10.6.2.1a: Dual Band CH operational data item requirements - external

3546 A Dual Band CH shall maintain the internal data items specified in Table 10.6.2.1b. For clarity,
 3547 there is no requirement to expose these data items either in a Remote Party Response or to
 3548 other Devices on the SMHAN.

Data item	Requirements
Poor Communications Change Channel Matrix	The CH shall maintain, for each Device which is not a GSME, the UTC date-time at which a Sub GHz Channel Scan was undertaken by the CH due to that Device reporting poor radio conditions to the CH At the point where a Device which is not a GSME connects to the CH and when the CHF successfully actions a 'DBCH05 Request CHF Sub GHz Channel Scan' Command, the CH shall set the UTC date-time to the 'Start of Time'

Data item	Requirements
	with its Section 9.1.6 meaning
Unrequited Lost GSME Searches	The CH shall maintain the 'Unrequited Lost GSME Searches' as the number of Sub GHz Channel Changes the CH has undertaken to try to re-establish GSME communications since the last time it communicated with the GSME This value shall be reset to zero whenever communications with the GSME are re-established
MacRxUcastDeltaSum Matrix	The CH shall maintain, for each Device, the number of unicast messages received by the CH's MAC layer from that Device on any Sub GHz Channel over the last Duty Cycle Measurement Period
Non GSME Poor Communications Thirty Minute History Matrix	The CH shall maintain, for each Sub GHz Non-GSME Device, a record of 'Noisy' for each of the preceding 150 thirty minute periods which saw at least one <i>Mgmt_NWK_Unsolicited_Enhanced_Update_notify</i> command being received by the CH from the Device

3549 Table 10.6.2.1b: Dual Band CH operational data item requirements - internal

3550 Calculation of AverageRSSI

3551 The CH shall:

3552 Where communications are received from a Device communicating on Sub GHz frequencies
3553 with the CH, measure RSSI as specified in the ZigBee Specification section D.10.2.2.2;

3554 Where communications are received from a Device not communicating on Sub GHz
3555 frequencies with the CH, set RSSI to zero; and

3556 Use the value of RSSI, to calculate AverageRSSI (part of attribute 4 of DLMS COSEM object
3557 with OBIS code 0-0:30.3.0.255) over the period between 00:00:01 to 00:00:00 UTC in the
3558 UTC day prior to the current day.

3559 For clarity:

3560 AverageRSSI shall be calculated across all packets received from the Device in question
3561 during the relevant 24 hour period; and

3562 The values needed to calculate AverageRSSI shall persist across CH power loss and reboot.

3563 Constant data items

3564 A Dual Band CH shall have the data items in Table 10.6.2.2 set to the values in Table 10.6.2.2
3565 prior to being installed. These values shall not be capable of change without firmware
3566 change.

Data item	Requirements	Value
Suspension Period	This constant shall be the number of minutes for which ZCL communications are to be stopped when Duty Cycle controls are enforced on a Device by the CH. Thus, this shall be the value of the <i>Suspension Period</i> parameter in the <i>Suspend ZCL Messages</i> command, where the CH does not set that parameter to zero	5 (minutes)
Critical-Regulated Duty Cycle Threshold	This Threshold shall be the CH enforced boundary between ‘Critical’ and ‘Regulated’ Duty Cycle where those terms have the ZSE 5.14.6 meaning Thus, the <i>DutyCycleRegulatedThreshold</i> parameter in the NLME-SET-INTERFACE.request shall be 250, meaning 2.5%	2.5%
Duty Cycle Measurement Period	In ZigBee, this is referred to as <i>aDUTYCYCLEMeasurementPeriod</i> , and so <i>aDUTYCYCLEMeasurementPeriod</i> shall be set to this value in the CH	60 (minutes)
ScanDuration	As per the ZigBee Specification and IEEE 802.15.4, <i>ScanDuration</i> determines the period of time that the CH spends scanning each channel as part of a Sub GHz Channel Scan	5

3567 Table 10.6.2.2: Dual Band CH constant data item requirements - internal

3568 Configuration data items

3569 Prior to installation, a Dual Band CH shall have attribute 2 of the object with OBIS Code 0-0:94.44.10.0 populated with the DLMS COSEM
3570 structure specified in Table 10.6.2.3.

3571 CH shall not update attribute 2 of the object with OBIS Code 0-0:94.44.10.0 except as a result of a ‘DBCH04 Set CHF Sub GHz Configuration’.
3572 The CHF shall not process any such Command where the Table 10.6.2.3 requirements for Tag, Length and ‘Device Validation Requirements on
3573 Value’ are not met. Where the CH does not process such a Command as a result of one or more of these checks failing, the CH shall not send a
3574 ‘DBCH04 Set CHF Sub GHz Configuration’ Response but shall issue an Alert and create a Security Log entry in line with the requirements of
3575 Section 6.2.4.2.

3576 Whenever a CHF has updated attribute 2 of object with OBIS Code 0-0:94.44.10.0 as a result of a ‘DBCH04 Set CHF Sub GHz Configuration’
3577 Command, it shall:

- 3578 1. create an entry in the Event Log with Event Code set to 0x8F2A;
- 3579 2. send a ‘DBCH09 Sub GHz Configuration Changed Sub GHz Alert’ with:
3580 the Message Code set to 0x0113;

3581 the Alert Code set to 0x8F2A; and

3582 the Use Case Specific Additional Content set to the updated value of attribute 2 of the object with OBIS Code 0-0:94.44.10.0, so providing full
3583 details of the structure in Table 10.6.2.3.

3584 Whenever the CH sets attribute 2 of the object with OBIS code 0-0:94.44.10.1 as a result of receiving a ‘DBCH05 Request CHF Sub GHz Channel
3585 Scan’ Command, the CH shall send a ‘DBCH05 Request CHF Sub GHz Channel Scan’ Response to confirm that the attribute has been set and
3586 then shall undertake the processing required by Section 10.6.2.7.

Data item	Requirements	DLMS COSEM element	Tag	Length	Device Validation Requirements on Value
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SEC Schedule 8 – GBCS v4.4

Data item	Requirements	DLMS COSEM element	Tag	Length	Device Validation Requirements on Value
Sub GHz Configuration Settings	This shall be a structure containing 15 values	subGhzConfig::= structure {	0x02	0x0F	
Sub GHz Channel Masks	The CHF shall only scan Sub GHz Channels that are allowed to be used by the Sub GHz Channel Masks values Sub GHz Channel Masks shall be a set of four 32 bit bit-strings, one entry for each of the 4 Sub GHz ‘pages’ (each page equates to a set of channels that could be used in a specific frequency range) The bit-strings shall follow the ZigBee bitmap format for these items in the <i>Sub GHz Cluster</i> For clarity: 1. in the channel part, 0b1 shall mean ‘channel can be used by the CH’ and 0b0 shall mean ‘channel cannot be used by the				

SEC Schedule 8 – GBCS v4.4

Data item	Requirements	DLMS COSEM element	Tag	Length	Device Validation Requirements on Value
	CH' 2. bit numbering is in line with GBCS and so is numbered Least Significant Bit is zero (LSB0)				
Page 28 Mask	This shall be the channel mask that the CH applies for <i>Page 28</i> , with its ZSE meaning	page28Mask: bit-string(32),	0x04	0x20	At least two of bits 26-0 must be set to 0b1 AND bits 31-27 must have the value 0b11100
Page 29 Mask	This shall be the channel mask that the CH applies for <i>Page 29</i> , with its ZSE meaning	page29Mask: bit-string(32),	0x04	0x20	At least two of bits 26-0 must be set to 0b1 AND bits 31-27 must have the value 0b11101
Page 30 Mask	This shall be the channel mask that the CH applies for <i>Page 30</i> , with its ZSE meaning	Page30Mask: bit-string(32),	0x04	0x20	At least two of bits 26-0 must be set to 0b1 AND bits 31-27 must have the value 0b11110
Page 31 Mask	This shall be the channel mask that the CH applies for <i>Page 31</i> , with its ZSE meaning	Page31Mask: bit-string(32),	0x04	0x20	bits 31-27 must have the value 0b11111

SEC Schedule 8 – GBCS v4.4

Data item	Requirements	DLMS COSEM element	Tag	Length	Device Validation Requirements on Value
Normal-Limited Duty Cycle Threshold	This Threshold shall define the boundary between 'Normal' and 'Limited' Duty Cycle where those terms have the ZSE 5.14.6 meaning. In line with the ZigBee Specification, the data item shall be a 16 bit unsigned integer where 100 shall mean 1% and all other values shall be interpreted accordingly	normalLimitedDutyCycleThreshold: long-unsigned,	0x12		Must be between 0x0032 and 0x00C8 inclusive (so 50 to 200 meaning 0.5% to 2.0%)
Limited-Critical Duty Cycle Threshold	This Threshold shall define the boundary between 'Limited' and 'Critical' Duty Cycle where those terms have the ZSE 5.14.6 meaning. In line with the ZigBee Specification, the data item shall be a 16 bit unsigned integer where 100 shall mean 1% and all other values shall be interpreted accordingly	limitedCriticalDutyCycleThreshold: long-unsigned,	0x12		Must be between 0x0096 and 0x00FA exclusive (so 150 to 250 meaning 1.5% to 2.5%). Must be greater than normalLimitedDutyCycleThreshold
Maximum Sub GHz Channel Changes Per	The CH shall not undertake more than this number of Channel Changes in any 7*24*60*60 second period except	channelChangesPerWeek: unsigned,	0x11		Must be between 0x01 and 0x07

SEC Schedule 8 – GBCS v4.4

Data item	Requirements	DLMS COSEM element	Tag	Length	Device Validation Requirements on Value
Week	where the one or more additional channel changes results from 'DBCH05 Request CHF Sub GHz Channel Scan' Command(s)				
GSME Curfew	Shall be the numbers of hours without GSME communications before the CH determines the GSME to be a 'Lost GSME' When the CH identifies that there is a 'Lost GSME', it shall take actions as defined in Section 10.6.2.6	gsmeCurfew: long-unsigned,	0x12		Must be greater than 0x0001
Channel Quieter Threshold	Shall be the minimum number of decibels by which an alternative channel needs to be quieter, in the conditions defined in 10.6.2.8	quieterDecibelsLimit: unsigned,	0x11		Must be 0x01 or greater
Channel Noisier Threshold	Shall be the maximum number of decibels by which an alternative channel may be noisier, in the conditions defined in 10.6.2.8	noisierDecibelsLimit: unsigned,	0x11		Must be between 0x00 and 0x14 inclusive
Non GSME Poor	Shall be an integer value between 100	nonGsmePoorCommsPercent	0x12		Must be between 0x0064

Data item	Requirements	DLMS COSEM element	Tag	Length	Device Validation Requirements on Value
Communications Percentage Threshold	and 10,000 where 100 equates to 1%	age: long-unsigned,			and 0x2710 inclusive
Non GSME Poor Communications Thirty Minute Periods Measurement Periods	Shall be the number of 30 minute periods over which assessment of non GSME Device poor communications reports is made by the CH	nonGsmeMeasurementPeriod: unsigned,	0x11		Must be between 0x32 and 0x96 (so between 50 and 150) inclusive
Local CH Noise Measurement Period	The number of trailing minutes over which the CH shall assess its own percentage of retried messages	localChNoiseMeasurementPeriod: long-unsigned,	0x12		Must be greater than 0x003C (so 60 minutes)
Local CH Failure Percentage	Shall be an integer value between 100 and 10,000 where 100 equates to 1%	localChFailurePercentage: long-unsigned,	0x12		Must be between 0x0064 and 0x2710 inclusive
Local CH Retry Percentage	Shall be an integer value between 100 and 10,000 where 100 equates to 1%	localChRetryPercentage: long-unsigned}	0x12		Must be between 0x0064 and 0x2710 inclusive

3587 Table 10.6.2.3: Dual Band CH configuration data item requirements

3588 CHF Sub GHz Alerts and corresponding events

3589 Dual Band CH shall be capable of generating all Sub GHz Alerts and corresponding Log

3590 Entries, and shall generate the triggering events, and corresponding Alerts and Log Entries in

3591 line with the requirements of Table 10.6.2.4 and Table 16.2.

3592 For clarity, in relation to any Sub GHz Alert:

3593 the Business Target ID shall always be the Entity Identifier in the CHF's

3594 {accessControlBroker, keyAgreement, management} Trust Anchor Cell;

3595 the Business Originator ID shall be the CHF's Entity Identifier; and

3596 the Alert Payload shall always, as a DLMS COSEM based payload, be constructed as per Table

3597 7.2.9c.

Event / Alert Code Meaning	Requirements
Limited Duty Cycle Action Taken	This event shall occur when the CH measurement of Duty Cycle rises above the Normal-Limited Duty Cycle Threshold. When this occurs, the CH shall: 1. identify the Device which has the largest value from the MacRxUcastDeltaSum Matrix and set 'Device ID' accordingly; 2. create an entry in the Event Log with Event Code set to 0x8F20 and otherInfo set to 'Device ID'; 3. send a 'DBCH06 Limited Duty Cycle Action Taken Sub GHz Alert' with: 1. the Message Code set to 0x0110; 2. the Alert Code set to 0x8F20; and 3. the Use Case Specific Additional Content set to the concatenation 0x0908 'Device ID'; 4. if 'Device ID' is not that of a GSME, the CH shall send to that Device a <i>Suspend ZCL Messages</i> command with the <i>Suspension Period</i> parameter set to Suspension Period; and 5. if 'Device ID' is that of a GSME, in the <i>Suspend ZCL Messages</i> command response to the next <i>Get Suspend ZCL Messages Status</i> command received by the CH from that GSME, the CH shall set the <i>Suspension Period</i> parameter to Suspension Period.

Event / Alert Code Meaning	Requirements
	For clarity, SMHAN communications with the specified Device will not be possible for Suspension Period
Duty Cycle fallen below Normal-Limited Duty Cycle Threshold	This event shall occur when the CH measurement of Duty Cycle falls back below the Normal-Limited Duty Cycle Threshold
Critical Duty Cycle Action Taken	This event shall occur when the CH takes the action in ZSE 5.14.6 point 6 For clarity, SMHAN communications with any Sub GHz End Device except GSME will not be possible for Suspension Period starting at the date-time in the Alert / CHF Event Log Entry. Further, OTA firmware downloads to GSME will pause for this period. No Remote Party Commands will be sent to Sub GHz End Devices during this period (although Alerts and Responses may be received from GSME)
Duty Cycle fallen below Limited-Critical Duty Cycle Threshold	This event shall occur when the CH measurement of Duty Cycle falls back below the Limited-Critical Duty Cycle Threshold
Regulated Duty Cycle Action Taken	This event shall occur when the CH takes the action in ZSE 5.14.6 point 7
Duty Cycle fallen below Critical- Regulated Duty Cycle Threshold	This event shall occur when the CH measurement of Duty Cycle falls back below the Critical-Regulated Duty Cycle Threshold
Sub GHz Channel Changed	This event shall occur when a CH begins operating on a new Sub GHz Channel. For clarity, this includes the Sub GHz Channel selected by the CH on SMHAN network formation The resulting Alert and Log entry shall be created as per the requirements of Section 10.6.2.8 and Section 16.2. Note these events are recorded in the dedicated Sub GHz Channel Log and not in the CHF Event Log

Event / Alert Code Meaning	Requirements
Sub GHz Channel Scan initiated	This event shall occur whenever the CH undertakes the processing of Section 10.6.2.8.
Sub GHz Channel Scan Request Assessment Outcome	This event shall occur when a Sub GHz Channel Scan trigger has been assessed by the CH, as per the requirements of Section 10.6.2.7. The resulting Alert and Event shall record both the nature of the triggering event and the outcome of the assessment checks, so including whether a resulting Sub GHz Channel Scan was triggered
Sub GHz Channel not changed due to Frequency Agility Parameters	This event occurs when a Sub GHz Channel scan has been undertaken but the CH determines not to change the Sub GHz Channel
Three Lost GSME Searches Failed	When 'Unrequested Lost GSME Searches' reaches three and 24 hours has elapsed since the most recent Sub GHz Channel Change, this event shall occur
Sub GHz Configuration Changed	This event shall occur when the CHF updates attribute 2 of object with OBIS Code 0-0:94.44.10.0
Message Discarded Due to Duty Cycle Management	The event shall occur as specified in Section 10.6.2.5.2
No More Sub GHz Device Capacity	The event shall occur when: 1. a Device is added to the CHF Device Log which is not a GSME, SAPC or HCALCS; 2. there are already four Devices in the CHF Device Log, which are not HCALCS, SAPC or GSME, that joined the SMHAN on a Sub GHz frequency; and 3. the Device added then attempts to join the SMHAN on a Sub GHz Frequency.

Event / Alert Code Meaning	Requirements
	On occurrence of this event, the CH shall: 1. not allow the Device to join the SMHAN on a Sub GHz Frequency; 2. create an entry in the Event Log with Event Code set to 0x8F2D and otherInfo set to 'Device ID' of the Device concerned; and 3. send a 'DBCH11 No More Sub GHz Device Capacity Sub GHz Alert' with: 1. the Message Code set to 0x0115; 2. the Alert Code set to 0x8F2D; and 3. the Use Case Specific Additional Content set to the concatenation 0x0908 'Device ID'

3598 Table 10.6.2.4: CHF Sub GHz Alerts and related Events

3599 CHF functional requirements - general

3600 Dual Band CH shall always form a 'Multi-MAC' network with the meaning of ZSE section
3601 5.14.2.

3602 A Dual Band CH may turn off its Sub GHz HAN interface when either (i) no Device in its CHF
3603 Device Log has joined the SMHAN on a Sub GHz frequency or (ii) it has not communicated
3604 with any Device in its CHF Device Log on a Sub GHz frequency in the prior trailing
3605 7*24*60*60 seconds. For clarity and in line with ZSE, a Dual Band CH shall always have its
3606 Sub GHz HAN interface turned on whenever joining is permitted, with the meaning of ZSE
3607 5.4.1.

3608 Sub GHz GSME message delivery

3609 Whenever a CH is limiting communications to a Sub GHz GSME pursuant to ZSE 5.14.6, the
3610 CH shall buffer any Remote Party Commands or SME.C.PPMID-GSME Messages for the
3611 Device until either they can be delivered to the Device or 24 hours has elapsed since receipt
3612 by the CH.

3613 If 24 hours have elapsed since receipt of a Message by the CH without successful delivery to
3614 the GSME, the CH may discard the Message. If the CHF discards a Remote Party Command
3615 in this way, it shall notify the GPF and the GPF shall log the event in its Event Log and send
3616 an Alert with a GBZ Payload containing an Alert Code 0x819D.

3617 Sub GHz Non GSME Device message delivery

3618 Whenever a CH is limiting communications to a Sub GHz Non GSME Device pursuant to ZSE
 3619 5.14.6, the CH shall on receipt of any Remote Party Command for that Device:

3620 set 'Additional Data' to be the following concatenation, with values taken from the Remote
 3621 Party Message:

3622 0x09 || 0x19 || value Business Originator ID || value of Business Target ID || value
 3623 of CRA Flag || value of Originator Counter;

3624 discard the Remote Party Message;

3625 create an entry in the Event Log with Event Code set to 0x8F2C and otherInfo set to the
 3626 value of Business Target ID; and

3627 send a 'DBCH10 Message Discarded Due to Duty Cycle Management Sub GHz Alert' with:

3628 the Message Code set to 0x0114;

3629 the Alert Code set to 0x8F2C; and

3630 the Use Case Specific Additional Content set to 'Additional Data'.

3631 CHF functional requirements – channel scan triggers

3632 Whenever any of the trigger conditions in Table 10.6.2.6 occurs, the CHF shall set the value
 3633 of scanTrigger according to Table 10.6.2.6 and undertake the processing required by Section
 3634 10.6.2.7.

3635 The scanTrigger value of 0xFF shall be reserved.

Trigger	Value of scanTrigger
The CH sends a Response to a 'DBCH05 Request CHF Sub GHz Channel Scan' Command	0x00 (meaning Remote Party Command)
The CH receives a <i>Mgmt_NWK_Unsolicited_Enhanced_Update_notify</i> command from the GSME	0x01 (meaning GSME request)
With their CHTS meanings, the difference between the 'CH Date and Time' and the date and time in the 'CHF Communications Store' for the GSME rises above 'GSME Curfew' hours	0x02 (meaning GSME missed its curfew)
The time since the last Sub GHz Channel Change rises above 24 hours AND the CH has had no communications with the GSME in the period since that last Sub GHz Channel Change AND 'Unrequested Lost GSME Searches' is less than 3.	0x03 (meaning GSME missing for the last day)

Trigger	Value of scanTrigger
The CH calculated message failure rate averaged over the prior 'Local CH Noise Measurement Period' minutes exceeds 'Local CH Failure Percentage'. The CH shall, in line with the ZSE specification, calculate the message failure rate as the count of the number of unicast messages transmitted at the MAC layer of its Sub GHz interface that fail to be delivered divided by the total number of such unicast messages transmitted	0x04 (meaning CH detected message failure problems)
The CH calculated message retry rate averaged over the prior 'Local CH Noise Measurement Period' minutes exceeds 'Local CH Retry Percentage'. The CH shall, in line with the ZSE specification, calculate the message retry rate as the count of the number of unicast messages transmitted at the MAC layer of its Sub GHz interface that require retry divided by the total number of such unicast messages transmitted	0x05 (meaning CH detected message retry problems)
The 'Non GSME Poor Communications Thirty Minute History Matrix' for a Device records that more than 'Non GSME Poor Communications Percentage Threshold' percent of the entries in the most recent 'Non GSME Poor Communications Thirty Minute Periods Measurement Periods' thirty minute periods are marked as 'Noisy' <u>AND</u> the 'Poor Communications Change Channel Matrix' records, for this Device, that more than 31*24*60*60 seconds have elapsed since a Sub GHz Channel Scan was undertaken for this scanTrigger for this Device	0x06 (meaning Sub GHz Non-GSME Device request)
The CH is forming an SMHAN. For clarity, this shall only happen when a CH is first installed. It shall not happen on any subsequent power cycling or reboot	0x07 (meaning SM HAN formation)

3636 Table 10.6.2.6: Sub GHz Channel Scan triggers

3637 CHF functional requirements – scan trigger assessment

3638 If scanTrigger is not 0x07 (meaning SM HAN formation), the CH shall undertake the checks in
3639 Table 10.6.2.7a, in the sequence specified, prior to undertaking any Sub GHz Channel Scan.
3640 Should a check fail, the CH shall set the statusCode to the value equating to the failed check
3641 in Table 10.6.2.7a and shall not undertake further checks.

3642 If all of the checks are passed or scanTrigger is 0x07 (meaning SM HAN formation), the CH
 3643 shall set statusCode to 0x00 (meaning scanRequestAccepted).

3644 Having set statusCode, the CH shall then:

3645 1. set scanRequestAssessmentOutcomeAndTrigger to the concatenation:
 3646 0x02020901 || statusCode || 0x0901 || scanTrigger;

3647 2. create an entry in the Event Log with Event Code set to 0x8F28 and otherInfo set to
 3648 scanRequestAssessmentOutcomeAndTrigger; and

3649 3. send a 'DBCH08 Sub GHz Channel Scan Request Assessment Outcome Sub GHz Alert'
 3650 with:

3651 1. the Message Code set to 0x0112;

3652 the Alert Code set to 0x8F28; and

3653 the Use Case Specific Additional Content set to scanRequestAssessmentOutcomeAndTrigger.

3654 The CHF shall then undertake the processing specified in Table 10.6.2.7b dependent on the
 3655 values of statusCode and scanTrigger. Where that processing requires another scan trigger
 3656 assessment, the CHF shall set scanTrigger in the subsequent scan to have the same
 3657 scanTrigger value as this scan.

Sequence	Check	statusCode on failure of Check
1	There must not be a Device of type HHT in the CHF Device Log	0x01 (meaning hhtConnected)
2	The CH's current Duty Cycle measurement must be below the Limited-Critical Duty Cycle Threshold	0x02 (meaning dutyCycleUsageIsTooHigh)
3	The CH must not be currently in a state where joining is permitted, with the meaning of ZSE 5.4.1.	0x03 (meaning joiningsIsCurrentlyPermitted)
4	If the Communications Hub has sent at least one <i>Image Block Response</i> command to a GSME relating to a currently stored GSME Firmware image and the CH has communicated with the GSME in the last 24 hours, then the CH must have received a corresponding <i>Upgrade End Request</i> command from that GSME	0x04 (meaning gsmeOTADistributionUnderway)
5	If the trigger for the Channel Scan was not a	0x05 (meaning tooManyScansToday)

Sequence	Check	statusCode on failure of Check
	'DBCH05 Request CHF Sub GHz Channel Scan' Command, there must be no entries in the CHF Event Log that: 1. contain Event Code of 0x8F28 (Sub GHz Channel Scan Request Assessment Outcome); and 2. contain a statusCode of 0x00 (meaning scanRequestAccepted); and 3. are timestamped within the prior 24*60*60 seconds. Note this means that no Channel Scans can occur in the 24 hours following SMHAN formation unless triggered by a 'DBCH05 Request CHF Sub GHz Channel Scan'	
6	If the trigger for the Channel Scan was a 'DBCH05 Request CHF Sub GHz Channel Scan' Command, there must be no entries in the CHF Event Log that: 1. contain an Event Code of 0x8F28 (Sub GHz Channel Scan Request Assessment Outcome); and 2. contain a scanTrigger of 0x00 (meaning Remote Party Command); and 3. contain a statusCode of 0x00 (meaning scanRequestAccepted); and 4. are timestamped within the prior 24*60*60 seconds	0x06 (meaning tooManyCommandsToday)
7	If the trigger for the Channel Scan was not a 'DBCH05 Request CHF Sub GHz Channel Scan' Command, the number of entries in the Sub GHz Channel Log that are timestamped within the prior 7*24*60*60 seconds must be less than the	0x07 (meaning tooManyScansThisWeek)

Sequence	Check	statusCode on failure of Check
	'Maximum Sub GHz Channel Changes Per Week'	

3658 Table 10.6.2.7a: scan trigger assessment checks

3659

statusCode	Required CHF Actions
0x00 (meaning scanRequestAccepted).	The CHF shall undertake the processing specified in Section 10.6.2.8
0x01 (meaning hhtConnected)	When the Device of type of HHT is removed from the CHF Device Log, the CHF shall undertake a scan trigger assessment as specified by this Section 10.6.2.7
0x02 (meaning dutyCycleUsageIsTooHigh)	When the CH's Duty Cycle measurement next falls back below the Limited-Critical Duty Cycle Threshold, the CHF shall undertake a scan trigger assessment as specified by this Section 10.6.2.7
0x03 (meaning joiningIsCurrentlyPermitted)	When the CH changes from the state where joining is permitted to a state where joining is not permitted, with the meaning of ZSE 5.4.1, the CHF shall undertake a scan trigger assessment as specified by this Section 10.6.2.7
0x04 (meaning gsmeOTAUpgradeUnderway)	Whenever the soonest of (1) the CHF sends an <i>Upgrade End Request Response</i> to the GSME or (2) the GSME Firmware image the CHF holds is removed or (3) 24 hours have elapsed, the CHF shall undertake a scan trigger assessment as specified by this Section 10.6.2.7
0x05 (meaning tooManyScansToday)	If scanTrigger relates to a GSME and so has the value 0x01, 0x02 or 0x03 then, after 24 hours have elapsed, the CHF shall undertake a scan trigger assessment as specified by this Section 10.6.2.7 If scanTrigger does not relate to a GSME and so does not have the value 0x01, 0x02 or 0x03 then the CHF shall take no further action
0x06 (meaning tooManyCommandsToday)	The CHF shall take no further action
0x07 (meaning	If scanTrigger relates to a GSME and so has the value 0x01,

statusCode	Required CHF Actions
tooManyScansThisWeek)	0x02 or 0x03 then, after 24 hours have elapsed, the CHF shall undertake a scan trigger assessment as specified by this Section 10.6.2.7 If scanTrigger does not relate to a GSME and so does not have the value 0x01, 0x02 or 0x03 then the CHF shall take no further action

3660 Table 10.6.2.7b: Post scan trigger assessment actions

3661 CHF functional requirements – Sub GHz Channel Scan actions

3662 The CH shall complete a Sub GHz Channel Scan.

3663 The CHF shall then set 'Alternative Sub GHz Channel' to:

3664 The channel which, of the Sub GHz Channels scanned, has the lowest noise, if this lowest
 3665 noise channel is different than the CH Operating Sub GHz Channel or if there is no existing
 3666 channel recorded in the CH Operating Sub GHz Channel; or

3667 The channel which, of the Sub GHz Channels scanned, has the second lowest noise, if the
 3668 lowest noise channel is the CH Operating Sub GHz Channel.

3669 The CHF shall then set 'Relative Noise' to be the SMHAN radio noise difference in decibels
 3670 between the CH Operating Sub GHz Channel and the 'Alternative Sub GHz Channel', or to be
 3671 zero if there is no existing channel recorded in the CH Operating Sub GHz Channel.

3672 The CHF shall then determine whether to change Sub GHz Channel according to Table
 3673 10.6.2.8, the value of scanTrigger and the 'Relative Noise'.

3674 The CHF shall comply with the ZSE requirements whenever undertaking a channel change.
 3675 Additionally, having set the *Channel Change Pending* Functional Notification flag, the CH
 3676 shall wait until the soonest of either:

3677 The GSME has read the Channel Change attribute; or

3678 60 minutes have elapsed

3679 before changing its Operating Sub GHz Channel.

3680 Where the CHF has changed the Operating Sub GHz Channel, the CHF shall:

- 3681 1. create an entry in the Sub GHz Channel Log with Event Code set to 0x8F26 and
- 3682 otherInfo set to the concatenation (value of Operating Sub GHz Channel) || 0x0901
- 3683 || scanTrigger; and

3684 2. send a 'DBCH07 Sub GHz Sub GHz Channel Changed Sub GHz Alert' with:

3685 1. the Message Code set to 0x0111;

3686 the Alert Code set to 0x8F26; and

3687 the Use Case Specific Additional Content set to 0x02020420 || (value of Operating Sub GHz

3688 Channel) || 0x0901 || scanTrigger.

3689

Value of scanTrigger	Actions
0x00 (meaning Remote Party Command)	If 'Relative Noise' is at least 'Channel Quieter Threshold' decibels better, the CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x01 (meaning GSME request)	If 'Relative Noise' is no more than 'Channel Noisier Threshold' decibels worse, the CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x02 (meaning GSME missed its curfew)	The CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x03 (meaning GSME missing for the last day)	The CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x04 (meaning CH detected message failure problems)	If 'Relative Noise' is at least 'Channel Quieter Threshold' decibels better, the CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x05 (meaning CH detected message retry problems)	If 'Relative Noise' is at least 'Channel Quieter Threshold' decibels better, the CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x06 (meaning Sub GHz Non-GSME Device request)	If 'Relative Noise' is no more than 'Channel Noisier Threshold' decibels worse, the CHF shall change its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'
0x07 (meaning SM HAN formation)	The CHF shall set its Operating Sub GHz Channel to the 'Alternative Sub GHz Channel'

3690 Table 10.6.2.8: Sub GHz Channel Scan triggers

3691 Sub GHz End Devices - functional requirements

3692 A Sub GHz End Device shall:

3693 not act as a ZigBee router when operating on a Sub GHz Channel;

3694 where the Device can also support 2.4 GHz operation:

3695 on first connecting to a ZigBee network, attempt to establish network communication in the

3696 2.4GHz band. Only where communications are not of sufficient quality, shall the Device

3697 attempt to establish network communications in the Sub GHz band; and

3698 having connected to a ZigBee network at either 2.4 GHz or at Sub GHz, not attempt to

3699 change to the other of 2.4 GHz or Sub GHz; and

3700 not use the *Mgmt_NWK_Unsolicited_Enhanced_Update_notify* command to notify the CH

3701 of problems with its communications link more frequently than once in any 30-minute

3702 period.

3703 Sub GHz GSME - functional requirements

3704 Sub GHz GSME shall wait at least 2 hours from detecting SMHAN interference before

3705 indicating that the interference is continuing by way of sending a

3706 *Mgmt_NWK_Unsolicited_Enhanced_Update_notify* command to the CH.

3707 When operating on Sub GHz, GSME shall, on each wake up, check the *Functional Notification*

3708 *Flags* for bits 25 (*Energy Scan Pending*) and 26 (*Channel Change Pending*).

3709 If either bit is set (so has a value 0b1) then the GSME shall attempt to retrieve any

3710 Commands buffered for it on the CH before turning off its SMHAN radio. For clarity and in

3711 line with Section 10.6.2.8, the GSME should attempt such retrieval before reading the CHF

3712 *Channel Change* attribute, if it is to maximise the likelihood of Command retrieval before it

3713 turns off its SMHAN radio.

3714 If bit 25 is set, the GSME shall disable the SMETS User Interface Commands '4.5.2.4 Check

3715 for HAN Interface Commands' and '4.5.2.8 Find Smart Metering Home Area Network and Re-

3716 establish Communications Links' and shall not receive and execute '4.5.3.3 Add Credit' and

3717 '4.5.3.1 Activate Emergency Credit' Commands from a PPMID and a Gas Proxy Function

3718 (where those terms have their SMETS meanings), as required by SMETS 4.4.7.2, until it next

3719 turns on its SMHAN radio.

3720 Note that CH may change Sub GHz Channel once every 24 hours to attempt to communicate

3721 with a lost GSME. On each such change, the CH will undertake a Sub GHz Channel Scan,

3722 meaning that it cannot communicate with a GSME for a period of time. GSME should factor

3723 both the 24 hour period and the associated Sub GHz Channel Scan in to their attempts to re-

3724 establish lost communications with the CH.

3725 Sub GHz Available Channels

3726 The Sub GHz Available Channels shall be as defined in the ZigBee Specification at section
 3727 D.10.3.

3728 SMHAN management

3729 Requests to leave the SMHAN

3730 Where a Device is not a ZigBee *end device* or a Communications Hub, the
 3731 *nwkLeaveRequestAllowed* attribute of the *NIB (Network layer information base)* shall be set
 3732 to *FALSE*. This means that, in line with the ZigBee specification, *Leave* commands received
 3733 by such Devices shall be immediately and silently discarded and such Devices shall not leave
 3734 the SMHAN on receipt of such *Leave* commands.

3735 When a Device, which is a ZigBee *end device*, has received and executed a ZigBee *Leave*
 3736 command with a value in the *Rejoin Sub-Field* of 0, the Device shall attempt to re-establish
 3737 SMHAN communications using the ZSE 5.4.2 mechanisms.

3738 Network Key Change

3739 In line with the ‘ZigBee PRO/2007 Layer PICS and Stack Profiles’ requirement, Devices shall
 3740 be capable of maintaining a minimum of two *Network Keys*.

3741 Whenever a CHF adds an Entity Identifier to, or removes an Entity Identifier from, its CHF
 3742 Device Log, where the type of the Device to which that Entity Identifier relates is other than
 3743 HHT or GPF, the CHF shall set a ‘CHF Device Log Changed Today’ flag.

3744 For clarity and in accordance with Section 10.5.3.3, any Device which provided its Entity
 3745 Identifier and *Install Code* in a CCS01 Command to the Communications Hub via *Inter-PAN*
 3746 shall be treated by the CHF as being of type HHT.

3747 Once each day, at a time between 02:00 and 04:00 UTC, the CHF shall check whether the
 3748 ‘CHF Device Log Changed Today’ flag is set and, if it is set, the CHF shall take the following
 3749 steps:

- 3750 1. generate a new *Network Key*;
- 3751 2. set, via the GPF, the *Network Key Active* flag in the *FunctionalNotificationFlags* and
 3752 then wait for the soonest of (1) 60 minutes or (2) the GSME requesting the new
 3753 *Network Key*;
- 3754 3. for each Device in the CHF Device Log which has an associated *TC Link Key*, unicast
 3755 the new *Network Key* to the Device by issuing the *APSME-TRANSPORT-KEY.request*
 3756 primitive with the *DestAddress* parameter set to the address of the Device and the
 3757 communication to the Device being protected by the associated *TC Link Key*. The
 3758 *TransportKeyData* sub-parameters shall be set as follows:

-
- 3759 1. the *KeySeqNumber* sub-parameter shall be set to the sequence count value for the
3760 new *Network Key*;
- 3761 2. The *NetworkKey* sub-parameter shall be set to the new *Network Key*; and
- 3762 3. The *UseParent* sub-parameter shall be set to *FALSE*;
- 3763 4. Send to all Devices in the CHF Device Log the *APSMESWITCH-KEY.request* primitive
3764 with the *KeySeqNumber* parameter set to the sequence count value for the new
3765 *Network Key*; and
- 3766 5. unset the 'CHF Device Log Changed Today' flag.
- 3767 Should any Device not receive the new *Network Key* and the instruction to switch to it, that
3768 Device will discover a communication failure. In such circumstances, the Device shall use
3769 the ZSE 5.4.2 ('Re-joining a Secured Network') mechanisms to re-establish communications.
- 3770 For clarity, the CHF shall not send the new *Network Key* to either:
- 3771 devices which have Entity Identifiers in the CHF Device Log but which do not have a *TC Link*
3772 *Key* established; or
- 3773 devices which do not have an Entity Identifier in the CHF Device Log.
- 3774 Network Key related requirements
- 3775 In this Section, all terms in italics shall have their ZSE or ZigBee Specification meaning and ZS
3776 shall mean the ZigBee Specification.
- 3777 When a Device, which is not a CH, receives a new *Network Key*, the Device shall only store
3778 that *Network Key* where either:
- 3779 1. the Device does not currently hold any *Network Key* (so meaning it is being installed);
3780 or
- 3781 the Device receives the new *Network Key* encrypted with a hash of its *Trust Center Link Key*
3782 (so meaning that a *Trust Center Swapout* is in progress); or
- 3783 the Device receives the new *Network Key* encrypted with its *Trust Center Link Key* (and
3784 potentially with an existing *Network Key*) and either:
- 3785 1. the value of *KeySeqNumber* for the new *Network Key* is greater than the value of the
3786 Device's *nwkActiveKeySeqNumber*; or
- 3787 2. the Device's *nwkActiveKeySeqNumber* is greater than 127 and the value of
3788 *KeySeqNumber* for the new *Network Key* is not greater than
3789 (*nwkActiveKeySeqNumber* + 128) modulo 256.
-

3790 Where a Device, which is not a CH, stores a new *Network Key*, it shall switch to using that
3791 new *Network Key* for outgoing messages where either:

- 3792 1. it does not hold any other *Network Key*;
- 3793 2. it received the new *Network Key* encrypted only with a hash of its *Trust Center Link*
3794 *Key*; or
- 3795 3. it receives a message validly encrypted with the new *Network Key*.

3796 Where a Device stores a new *Network Key* and that storage leads to the Device needing to
3797 remove details related to an old *Network Key*, the Device shall remove the *Network Key* that
3798 it received furthest back in time, and remove the *nwkSecurityMaterialSet* details associated
3799 with that key. Note that, in cases (2.) and (3.b) above, that key would likely not be the one
3800 with the lowest *KeySeqNumber*.

3801 Where a Device, which is not a CH, receives a *switch-key* command requesting that it
3802 switches to using a new *Network Key*, the Device shall only take action in response to that
3803 command where either:

3804 the value of the '*sequence number*' parameter in the *switch-key* command is greater than
3805 the value of the Device's *nwkActiveKeySeqNumber*; or

3806 the Device's *nwkActiveKeySeqNumber* is greater than 127 and the value of the '*sequence*
3807 *number*' parameter in the *switch-key* command is not greater than
3808 (*nwkActiveKeySeqNumber* + 128) modulo 256.

3809 Where a Device, which is not a CH, switches to using a new *Network Key*, the Device shall:

3810 4. in line with ZS 4.3.4 (and contrary to ZS 4.6.3.4.2), only set the associated
3811 *OutgoingFrameCounter* to zero if *OutgoingFrameCounter* is currently greater than
3812 0x80000000; and

3813 5. ensure that, in the *IncomingFrameCounterSet* within the *nwkSecurityMaterialSet* for
3814 this new *Network Key*:

3815 For a Device which is not an End Device, any *SenderAddress* is an identifier for a Device that
3816 is in the Device's *nwkNeighborTable*;

3817 For a Device which is an End Device, the only *SenderAddress* is the identifier for Device's
3818 current parent Device; and

3819 In line with ZS 4.6.3.4.2, all *IncomingFrameCounters* are set to zero.

3820 A Device shall:

-
- 3821 6. only increment the value of the *IncomingFrameCounter* for the sending Device as a
 3822 result of processing incoming messages from the sending Device which are secured
 3823 with the *Network Key* the receiving Device is currently using for outgoing messages;
 3824 and
- 3825 7. whenever it removes a Device from its *nwkNeighborTable*, also remove that Device's
 3826 details from the *IncomingFrameCounterSet* within the *nwkSecurityMaterialSet* for the
 3827 *Network Key* it is currently using to secure outgoing messages.
- 3828 For the purposes of aging out entries from the *nwkNeighborTable*, a Device shall, where it is
 3829 a *Router*:
- 3830 8. only use the ZS table 3-58 specified default values for *nwkRouterAgeLimit* and
 3831 *nwkLinkStatusPeriod*, so 3 and 15 seconds respectively;
- 3832 9. set bit 0 of *nwkParentInformation* to 0b0, and so bit 1 to 0b1, meaning that *End*
 3833 *Devices* need to send *End Device Timeout Request* commands as a unicast to refresh
 3834 the *keepalive timer*;
- 3835 10. Only refresh the *keepalive timer* when the *Network Key* used to secure such *End*
 3836 *Device Timeout Request* commands is that currently in use by the Device for its
 3837 outgoing messages; and
- 3838 11. have the *nwkEndDeviceTimeoutDefault* set to the default 8 (so meaning 256 minutes)
 3839 in line with ZS table 3-58 and not change the value of a Device's *keepalive* timeout
 3840 where it receives an *End Device Timeout Request* command with a *Requested*
 3841 *Timeout Enumeration Value* greater than 10 (so meaning greater than 1,024 minutes).
- 3842 When a Device has chosen a network to join, it shall remove *Neighbor table entries*
 3843 corresponding to Devices that are not members of the chosen network.
- 3844 Where a Device is an *End Device*, the Device shall not send an *End Device Timeout Request*
 3845 command with a *Requested Timeout Enumeration Value* greater than 10 (so meaning
 3846 greater than 1,024 minutes).
- 3847 CH shall not send any new *Network Key* encrypted only with an existing *Network Key*.
- 3848 Where a CH creates a switch-key command, it shall treat the reference in ZS 4.4.6.1.3 to ZS
 3849 4.4.9.6 (which does not exist) as a reference to ZS 4.4.10.5.
- 3850 A CH shall not attempt to remove Devices from the network using the *APSME-REMOVE-*
 3851 *DEVICE* primitives, and so shall not send *Remove Device* commands to remove Devices from
 3852 the network. A CH may send *Remove Device* commands only as required by ZSE 5.4.2.2. For
 3853 clarity, this means that the values in the *ParentAddress* and *TargetAddress* parameters of
 3854 such an *APSME-REMOVE-DEVICE* request shall never be the same.
-

3855 CH routing of Remote Party and SME.C.PPMID-GSME Messages

3856 Defined Terms

3857 The following terms used in this Section 10.8 shall have the meanings defined in this Table
3858 10.8.1.

Defined Term	Meaning
Message Header Structure	Shall, in relation to the Message being processed, be any one of: - a Grouping Header, as specified in Section 7.2.7; - a MAC Header, as specified in Section 7.2.5; - a Message Routing Header, as specified in Section 7.2.11.5; - and - a GBT Header, as specified in Section 7.2.11.6
Non-Validated Components	Shall be: - in a Grouping Header or a Message Router Heading, any of: - The value for Originator Counter in the transaction-id; - The value for Business Originator ID in the originator-system-title; - Any value in the date-time, where such a value is present; - and - Any value in other-information; and - any values in any parts of a Message that are not within a Message Header Structure.

3859 Table 10.8.1: Meanings of Defined Terms

3860 CH Routing of Remote Party Commands and SME.C.PPMID-GSME and Alerts

3861 Whenever a CH receives either:

3862 a Remote Party Message via its WAN Interface; or

3863 a Remote Party Message in the Data parameter payload of a *TransferData* command which
3864 is from an HHT; or

3865 an SME.C.PPMID-GSME Message in the Data parameter payload of a *TransferData* command
3866 from a Device, which is in its CHF Device Log,

3867 the CH shall:

3868 process the Message Header Structure(s) in that Message sufficiently to identify the target

3869 Device's Entity Identifier; and

3870 where the identified Device is in the CHF Device log and is not an HHT, GPF or CHF, attempt

3871 to deliver that Message to the identified Device.

3872 In undertaking such processing, the CH shall not validate any of the 'Non-Validated

3873 Components' as a pre-requisite to attempting to deliver the Message to the identified

3874 Device.

3875 Attempts at delivery of Messages shall be subject to the provisions of Sections 10.2.2,

3876 10.3.4.2 and 10.6.2.5.1.

3877 CH Routing of Remote Party Responses and Alerts

3878 Whenever a CH receives content in the Data parameter payload of a *TransferData* command

3879 which is not from an HHT or VWD, the CH shall process that content including any potential

3880 Message Header Structure(s) in it, so as to decide whether it contains a Remote Party

3881 Message. Where the CH determines that the content contains a Remote Party Message, the

3882 CH shall attempt to deliver that Message via its WAN Interface and, as required by Section

3883 10.5.3.2, by sending a *TransferData* command containing the Message to any HHT in its CHF

3884 Device Log.

3885 In undertaking such processing, the CH shall not validate any of the Non-Validated

3886 Components, or the value for Business Target ID in the recipient-system-title, as a pre-

3887 requisite to attempting to deliver the Message.

3888 VWD interactions

3889 VWD interactions shall only be allowed on a Virtual WAN CH.

3890 Introduction – informative

3891 A VWD allows for delivery of Messages to and from the SMHAN as an alternative delivery

3892 route to the Virtual WAN CH's WAN connection. A VWD maintains a persistent connection

3893 with the Virtual WAN CH.

3894 This Section 10.9 specifies requirements related to:

3895 how a connection is made between a VWD and a Virtual WAN CH; and

3896 how Messages are then to be transferred to and from the VWD.

3897 Establishing a connection between a VWD and a Virtual WAN CH – informative

3898 A VWD may *join* the HAN in two ways:

3899 1. using the Inter-PAN mechanism in the same way as an HHT as per Section 10.5.3.4;
3900 or

3901 2. by a Remote Party adding the Entity Identifier of the VWD to the CHF Device Log
3902 using a CCS01 Command.

3903 If a VWD *joins* the HAN using the Inter-PAN mechanism, it initially behaves in exactly the
3904 same way as an HHT; in this case, the VWD is operating solely as an HHT. A VWD operates in
3905 'Virtual WAN mode' after the CHF sends a *RequestTunnel* command to the VWD and the
3906 VWD sends a *RequestTunnelResponse* command in response with a *TunnelStatus* value of
3907 0x00 ('Success').

3908 If a VWD *joins* the HAN by a Remote Party using a CCS01 Command, the VWD behaves as the
3909 Device type it is added as, i.e. PPMID or Type 2.

3910 WAN proxy operation

3911 Introduction – informative

3912 The Virtual WAN CH maintains a WAN Interface via the WAN Communications Link and the
3913 Virtual WAN Communications Link. The WAN Communications Link is the link to the WAN
3914 Provider. The Virtual WAN Communications Link is a HAN connection to the VWD, ultimately
3915 connected to the Virtual WAN Provider.

3916 Active Communications Link

3917 The WAN Interface of a Virtual WAN CH shall only send and receive Messages on the active
3918 Communications Link, which is either the WAN Communications Link or the Virtual WAN
3919 Communications Link. The method used by the Virtual WAN CH to determine the active
3920 Communications Link is not prescribed in this document. For clarity, Messages that are used
3921 for link maintenance purposes maybe sent on either Communications Link at any time.

3922 VWD and CHF – Device Requirements

3923 A VWD that provides its Entity Identifier and *Install Code* in a CCS01 Command to the
3924 Communications Hub via Inter-PAN shall be treated by the CHF as being of type HHT as per
3925 Section 10.5.3.3.

3926 As per Section 10.2.2, in interactions between a VWD and a Virtual WAN CH over the
3927 SMHAN:

3928 the Virtual WAN CH shall support the *Tunneling Cluster* as a *Client*; and

3929 the VWD shall support the *Tunneling Cluster* as a *Server*.

3930 VWD and CHF – establishing communications via Inter-PAN

3931 The VWD and the Virtual WAN CH shall undertake Steps 1 to 6 as per Section 10.5.3.4. The
3932 VWD and CHF are now operating in 'HHT mode' and shall process any Message as described
3933 in Section 10.5.3.4.

3934 For the VWD and CHF to operate in 'Virtual WAN mode', the VWD forwards an 'Enable
3935 Virtual WAN Device' Command to the CHF using the *TransferData* command with the
3936 *TunnelID* associated with *ProtocolID* 6. The format of this Command and how it is processed
3937 is not described in this document.

3938 If the 'Enable Virtual WAN Device' Command is successfully processed:

3939 1. the CHF shall send a *RequestTunnel* command to the identified VWD using a
3940 *ProtocolID* of 0 (as per Section 10.2.2);

3941 the VWD shall process the *RequestTunnel* command. If successfully processed, the VWD is
3942 operating in 'Virtual WAN mode' and shall send a *RequestTunnelResponse* command in
3943 response;

3944 if the *TunnelStatus* value in the *RequestTunnelResponse* command is 0x00 ('Success'), the
3945 CHF is operating in 'Virtual WAN mode';

3946 if the VWD is a Type 2 Device that has Virtual WAN capability, the VWD shall close the *tunnel*
3947 with the *TunnelID* associated with *ProtocolID* 6;

3948 if the VWD is a PPMID that has Virtual WAN capability, the VWD shall keep the *tunnel* with
3949 the *TunnelID* associated with *ProtocolID* 6 open but shall restrict the payload of a
3950 *TransferData* command to be:

3951 HAN Only Messages sent between PPMID and GSME;

3952 Remote Party Commands destined for the PPMID; and

3953 Remote Party Responses and Alerts originating from the PPMID.

3954 VWD and CHF – establishing communications by Remote Party

3955 The Remote Party shall send a CCS01 Command to the Virtual WAN CH to add the VWD as a
3956 PPMID or Type 2 Device. For clarity, the VWD shall behave as the Device type it is added as.

3957 For the VWD and CHF to operate in 'Virtual WAN mode', an 'Enable Virtual WAN Device'
3958 Command is sent to the CHF. The format of this Command and how it is processed is not
3959 described in this document.

3960 If the 'Enable Virtual WAN Device' Command is successfully processed:

3961 1. the CHF shall send a *RequestTunnel* command to the identified VWD using a
3962 *ProtocolID* of 0 (as per Section 10.2.2);

the VWD shall process the RequestTunnel command. If successfully processed, the VWD is operating in 'Virtual WAN mode' and shall send a RequestTunnelResponse command in response; and

if the TunnelStatus value in the RequestTunnelResponse command is 0x00 ('Success') the CHF is operating in 'Virtual WAN mode'.

VWD and CHF – WAN operation

Whilst the VWD is in the CHF's Device Log and *joined* to the SMHAN, and the CHF is operating in 'Virtual WAN mode', any Responses or Alerts received by the CHF from any SMHAN Device shall be processed according to which WAN Communications Link is active (as defined in Section 10.9.3.2) as follows:

if the WAN Communications Link is active, the Response or Alert shall be sent to the WAN Communications Link; or

if the Virtual WAN Communications Link is active, the Response or Alert shall be sent to the Virtual WAN Communications Link using the *TransferData* command with the *TunnelID* associated with *ProtocolID* 0.

Downloading firmware images to Devices

Requirements and statements in this Section 11, and in Use Cases 'CS05b Distribute Firmware to ESME / GSME' and 'CS06 Activate Firmware', shall apply to an SAPC as if it were an ESME.

Introduction – informative

Compared to other Smart Metering messages, firmware images are large. Further, each image is likely to be applicable to a significant number of Devices. Thus, an end-to-end, unicast Message to each affected Device, with each message containing a copy of the image, is not efficient from a WAN perspective.

This leads to the firmware update process being separated into two stages for ESME, GSME, HCALCS and Communications Hubs:

distribution of the image to end Devices without any activation of that image; and

a separate and subsequent 'activation' Command to each Device.

The Distribute Firmware Command is not a Critical Command (since it does not affect the operating firmware) and does not need to be unicast.

The Activate Firmware Command is a Critical Command and so must be unicast – as it must be digitally signed and be for one, and only one specified Device. Further, the Activate Command must apply to one, and only one, image and that image must have originated

3996 from the same party that signs the Activate Firmware Command (that is, the party
3997 responsible for that Device). To meet these requirements:

3998 the Activate Firmware Command is of type SME.C.C and so the Signature and MAC on the
3999 Command shall have been verified by the Device prior to the Hash validation (see next
4000 bullet); and

4001 a Device receiving an Activate Firmware Command shall calculate a Hash over the
4002 Manufacturer Image it holds and ensure the Hash so calculated matches that in the Activate
4003 Firmware Command, before the Device attempts to activate the corresponding
4004 Manufacturer Image.

4005 For PPMID the firmware image is delivered to the Communications Hub and transferred to
4006 the target Device. Upon successful transfer of the firmware image the Communications Hub
4007 instructs the PPMID to activate the firmware image.

4008 The GBCS does not constrain the mechanisms used by Device manufacturers to ensure that
4009 only valid Manufacturer Images are activated on Devices manufactured by them. The GBCS
4010 does require that the manufacturer information related to a Manufacturer Image is made
4011 available, so that the Upgrade Image and the ZigBee Over-The-Air (OTA) Header can be
4012 provided when requesting distribution of an image.

4013 In common with other Messages, the GBCS shall not constrain the mechanisms by which the
4014 firmware Messages are transported to the Communications Hub. The GBCS constrains HAN
4015 transport mechanisms to those provided by ZSE.

4016 In line with the ZigBee OTA specification at section 5.1, the contents of Manufacturer Images
4017 sent to Devices are manufacturer defined. Thus, a particular Manufacturer Image may
4018 consist of whatever the manufacturer requires to achieve the necessary update which could
4019 be a full image or just a patch to application code or any other manufacturer specified
4020 content.

4021 Therefore the steps taken by a Device when it activates the contents of a particular
4022 Manufacturer Image are manufacturer specific and specified in the release note for that
4023 Manufacturer Image. Thus, activation of a Manufacturer Image means the set of
4024 manufacturer specified steps the Device takes in processing the content of the image and
4025 terms such as ‘activation’, ‘activate’, ‘activated’, ‘installed’ etc. in the GBCS and the Technical
4026 Specifications, are meant in this sense.

4027 Therefore, when a Device reports its ‘installed’ or ‘active’ firmware version it shall respond
4028 with either:

4029 the firmware version for its mostly recently successfully activated Manufacturer Image (with
4030 the value in the corresponding OTA Header and so the value in the Central Products List
4031 (CPL) with its SEC meaning); or

4032 if it has never successfully activated a new Manufacturer Image, the firmware version it held
4033 on installation (which again would correspond to the CPL).

4034 Common Requirements

4035 Transport of firmware images

4036 Italicised terms in this Section 11.2.1 shall have the meanings defined in ZigBee Document
4037 09-5264r23.

4038 For ESME, GSME, HCALCS and PPMID firmware image distribution, the ZigBee Over-The-Air
4039 (OTA) mechanisms shall be used for transport of the image over the HAN. The ESME / GSME
4040 / HCALCS / PPMID firmware image delivered to the Communications Hub shall comply with
4041 ZigBee OTA format requirements.

4042 Communications Hub firmware images shall not be transported over the HAN and so ZigBee
4043 OTA structures shall not be required.

4044 Every Communications Hub shall be configured to act as the single OTA Server on its HAN.

4045 ESME, GSME, HCALCS and PPMID shall be configured to act as an OTA Client. The ESME,
4046 HCALCS and PPMID shall use the '*Image Notify*'²⁴ Command sent by the OTA Server to
4047 inform it that a new firmware image is available. The GSME shall use the notification flags
4048 mechanism whereby a flag shall be set by the OTA Server to inform it that a new firmware
4049 image is available when requested.

4050 The Communications Hub shall:

4051 as required by CHTS, have the capability to store one GSME OTA Upgrade Image and one
4052 ESME OTA Upgrade Image;

4053 overwrite an ESME / GSME image with a subsequently delivered image for the same Device
4054 type unless:

4055 the subsequently delivered image has Force Replace = 0x00; and

24 See section 6.10.3 of ZigBee Document 09-5264r23

4056 the Communications Hub has sent at least one *Image Block Response Command* relating to
4057 the already stored image but has not received a corresponding *Upgrade End Request*
4058 *Command*²⁵.

4059 In such circumstances the Communications Hub shall not overwrite the currently
4060 stored image; and

4061 overwrite an HCALCS image with a subsequently delivered HCALCS image regardless of the
4062 setting of Force Replace.

4063 The Communications Hub shall make the GSME image available for a minimum of fourteen
4064 (14) days and, after this period, replace the GSME image with an image for the PPMID /
4065 HCALCS, if one becomes available. If the transfer of a GSME image to the GSME is in
4066 progress, the Communications Hub shall only replace this GSME image with an image for the
4067 PPMID / HCALCS once the GSME image transfer has completed.

4068 If the Communications Hub has sent at least one Image Block Response Command relating
4069 to the stored GSME image but has not received a corresponding Upgrade End Request
4070 Command within fourteen (14) days, the CH shall discard the GSME image.

4071 The Communications Hub shall make the PPMID / HCALCS image available for fourteen (14)
4072 days unless a new PPMID / HCALCS / GSME image is available.

4073 If an ESME / PPMID / HCALCS / GSME Upgrade Image is discarded or replaced prior to having
4074 been successfully transported over the HAN, the Communications Hub shall send an Alert
4075 for each target Device Entity Identifier associated with the Upgrade Image File with the Alert
4076 Code 0x8F89 as specified in Section 11.7 by setting `firmwareVersion` to the Upgrade
4077 Image File version and `transferResponseCode` to `imageDiscarded`.

4078 Whenever the Communications Hub's OTA Server issues an *Upgrade End Response*
4079 *Command* to a GSME, ESME or HCALCS pursuant to this GBCS, the *UpgradeTime* parameter
4080 shall have the value 0xFFFFFFFF²⁶.

4081 Whenever the Communications Hub's OTA Server issues an *Upgrade End Response*
4082 *Command* to a PPMID pursuant to this GBCS, the *UpgradeTime* parameter shall have the
4083 value 0x00000000 and the PPMID OTA Client shall activate the Firmware.

4084 The OTA Server shall not issue *Image Block Response Commands* with WAIT_FOR_DATA
4085 status except where the OTA Server is communicating over a Sub GHz Channel and it is
4086 issuing such commands to manage Duty Cycle (with its ZSE meaning).

25 As defined in section 6.10 of ZigBee Document 09-5264r23

26 As defined in sections 6.10.10 and 6.8.4 of ZigBee Document 09-5264r23

4087 Contrary to section 6.13 of ZigBee Document 09-5264r23, an ESME, GSME or HCALCS OTA
 4088 Client shall not activate any Manufacturer Image except as specified in Use Case CS06. An
 4089 ESME, GSME or HCALCS OTA Client shall not prevent the replacement of an existing
 4090 Manufacturer Image that it may hold if that Image is not yet activated.

4091 Construction of Upgrade Image

4092 For an ESME, GSME or HCALCS firmware image, the Authorising Remote Party shall be the
 4093 Supplier for the target Device.

4094 For a Communications Hub firmware image, the Authorising Remote Party shall be the WAN
 4095 Provider for the target Device.

4096 For an ESME, GSME, HCALCS or Communications Hub the Upgrade Image shall be the
 4097 concatenation:

4098 Manufacturer Image || Force Replace || 0x40 || Authorising Remote Party Signature
 4099 where:

4100 Manufacturer Image shall contain the firmware image the Device is to apply and any
 4101 manufacturer specific data needed. For clarity, the GBCS shall not constrain the structure or
 4102 contents of Manufacturer Image;

4103 Force Replace shall be a single octet where Force Replace = 0x00 shall mean do not force the
 4104 replacement of the currently stored image; and

4105 Authorising Remote Party Signature shall be calculated across the Manufacturer Image using
 4106 the Authorising Remote Party's Private Digital Signing Key.

4107 For a PPMID firmware image, the Upgrade Image shall be the Manufacturer Image where:

4108 Manufacturer Image shall contain the firmware image the PPMID is to apply and any
 4109 manufacturer specific data needed. For clarity, the GBCS shall not constrain the structure or
 4110 contents of the PPMID Manufacturer Image.

4111

4112 Construction of OTA Upgrade Image

4113 OTA Upgrade Image shall be the concatenation:

4114 OTA Header || Upgrade Image

4115 where OTA Header shall be populated according to Table 11.2.3. For clarity, there shall be
 4116 no other sub-elements present.

OTA Header

ZigBee OTA Message Element	Contents	Length (octets)	Note
OTA upgrade file identifier	0x0BEEF11E	4	Fixed by ZigBee OTA specification
OTA Header version	0x0100	2	Specified by current version of ZigBee OTA specification
OTA Header length	0x003C	2	The length of ZigBee OTA Header which is decimal 60
OTA Header Field control	0x0004	2	Detailing what is / is not present in ZigBee OTA Header
Manufacturer code	ZSE assigned identifier for the Manufacturer of the target Device	2	So this identifies the manufacturer producing the Manufacturer Image
Image type	Manufacturer specific	2	As per the ZigBee OTA specification, this is to differentiate products from the same manufacturer
File version	Manufacturer specific	4	As per the ZigBee OTA specification, this is to differentiate release and build numbers for the product in question
ZigBee Stack version	0x0002	2	ZigBee PRO
OTA Header string	Manufacturer specific	32	May be blank but is not required to be used in Device processing of the firmware image
Total Image size (including header)	The length in octets of OTA Upgrade Image	4	Contents to be interpreted as an unsigned integer
Minimum hardware version	Manufacturer specific	2	
Maximum hardware version	Manufacturer specific	2	

4117 Table 11.2.3: Population of the OTA Header

4118 The OTA Header shall uniquely identify a firmware image.

4119 Construction of Manufacturer Image Hash

4120 Manufacturer Image Hash shall be a Hash calculated across the whole Manufacturer Image
 4121 file that is provided to the Authorising Remote Party or the Remote Party in case of the
 4122 PPMID.

4123 Verification of the authenticity of the Upgrade Image

4124 The Device shall verify Upgrade Image by verifying the Authorising Remote Party Signature
 4125 using Manufacturer Image and the Authorising Remote Party's Public Key. For clarity, this
 4126 shall be the only ECDSA verification required by the GBCS and this is not the ZSE ECDSA
 4127 Signature sub-element.

4128 For an ESME, GSME or HCALCS receiving an Upgrade Image, the Authorising Remote Party's
4129 Public Key shall be that held by the Device in the {supplier, digitalSignature, management}
4130 Trust Anchor Cell.

4131 For a Communications Hub receiving an Upgrade Image, the Authorising Remote Party's
4132 Public Key shall be that held by the Device in the {wanProvider, digitalSignature,
4133 management} Trust Anchor Cell.

4134 For a PPMID receiving an Upgrade Image the verification of the authenticity of the Upgrade
4135 Image in this section shall not apply.

4136

4137 Construction of Firmware Distribution Receipt Alert

4138 If the Device is an ESME or HCALCS, the 'Alert Payload' fields shall be populated according to
4139 Section 7.2.9.

4140 If the Device is a GSME, the 'Alert Payload' fields shall be populated according to Section
4141 7.2.10.

4142 In the above cases, the Device shall:

4143 populate the Use Case Specific Additional Content with the concatenation

4144 0x0920 || the calculated Manufacturer Image Hash

4145 populate the Alert Code field with 0x8F1C (failure), or 0x8F72 (success).

4146 If the Device is a PPMID, no Firmware Distribution Alert shall be sent.

4147

4148 Activation of firmware images

4149 The Activate Firmware Command shall be of type SME.C.C.

4150 A Device other than the PPMID receiving such a Command shall undertake the verifications
4151 required of a SME.C.C Command.

4152 If all such SME.C.C verifications succeed, the Device shall then calculate Manufacturer Image
4153 Hash over the Manufacturer Image it holds and compare that with the Manufacturer Image
4154 Hash specified in the Activate Firmware Command (see Use Case CS06 in Section 11.5 for
4155 details of the Activate Firmware Command Payload construction).

4156 If the two Hashes match, the Device shall attempt to activate the firmware image.

4157 If the two Hashes do not match, the Device shall not attempt to activate the firmware image.

4158 The Device shall issue a relevant Activate Firmware Response detailing the success or failure
 4159 (see Use Case CS06 in Section 11.5 for details of the Activate Firmware Command Payload
 4160 construction).

4161 See Section 11.6 for the activation of Upgrade Images for the PPMID.

4162

4163 CS05a Distribute Firmware to Communications Hub

4164 This Use Case covers the distribution of an Upgrade Image that is intended for a
 4165 Communications Hub to that Communications Hub.

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	None – this is a Variant Message
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	N/A (this Command does not have a Service Reference for CH)
Valid Target Device(s)	Communications Hub
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	WAN Provider
Valid Response Recipient role(s) (only for Messages authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	WAN Provider Specific

4166 Table 11.3: Use Case Cross References for CS05a Distribute Firmware to Communications
 4167 Hub

4168 Pre-conditions

-
- 4169 None.
- 4170 Detailed Steps
- 4171 The Upgrade Image shall be constructed according to Section 11.2.2.
- 4172 The Upgrade Image shall be transported to the Communications Hub.
- 4173 The Communications Hub shall verify the Upgrade Image according to Section 11.2.5, verify
- 4174 the Upgrade Image is suitable for this Communications Hub, and discard the Upgrade Image
- 4175 where it fails to verify.
- 4176 CS05b Distribute Firmware to ESME / GSME / HCALCS
- 4177 This Use Case covers the distribution of an OTA Upgrade Image that is intended for a GSME,
- 4178 ESME or HCALCS to that GSME, ESME or HCALCS.

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	None – this is a Variant Message
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	11.1
Valid Target Device(s)	ESME / GSME / HCALCS
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	WAN Provider Specific to Communications Hub; ZigBee OTA from Communications Hub

Cross Reference	Value
	to ESME / GSME / HCALCS

4179	Table 11.4: Use Case Cross References for CS05b Distribute Firmware to ESME / GSME /
4180	HCALCS
4181	Pre-conditions
4182	None.
4183	Detailed Steps
4184	Italicised terms in this Section 11.4.2 shall have the meaning specified in ZSE.
4185	ESME, GSME and HCALCS shall use the ZCL <i>Image Block Request</i> and <i>Image Block Response</i>
4186	commands to retrieve available OTA images.
4187	ESME, GSME and HCALCS shall not use the ZCL <i>Query Specific File Request</i> and <i>Query Specific</i>
4188	<i>File Response</i> commands.
4189	The OTA Upgrade Image shall be populated according to Section 11.2.3.
4190	The OTA Upgrade Image shall be transported to the Communications Hub through which the
4191	Device communicates.
4192	The Communications Hub shall update its OTA Cluster to reflect availability of the OTA
4193	Upgrade Image, once the image is received by the Communications Hub.
4194	The Communications Hub, as OTA Server, shall indicate availability of an OTA Upgrade Image
4195	differently for ESME, HCALCS and GSME:
4196	for ESME and HCALCS, the Communications Hub shall send a <i>Image Notify</i> command; and
4197	for GSME, the Communications Hub shall set a <i>the New OTA Firmware flag</i> (Bit Number 0) in
4198	<i>FunctionalNotificationFlags</i> .
4199	For each ESME / GSME / HCALCS Device Entity Identifier associated with the OTA Upgrade
4200	Image, the Communications Hub shall
4201	1. verify whether the ESME / GSME / HCALCS <i>Hardware Version</i> is present in the <i>Query</i>
4202	<i>Next Image Request Command</i> ;
4203	1. if the <i>Hardware Version</i> is present and does not fall between the <i>minimum</i> and
4204	<i>maximum Hardware Version</i> in the OTA Header the Communications Hub shall set
4205	<code>transferResponseCode</code> to <code>hardwareVersionMismatch</code> as specified in
4206	Section 11.7 and process from step 5;

-
- 4207 2. if the *Hardware Version* is not present, or is present and falls between the *minimum*
 4208 and *maximum Hardware Version* in the OTA Header, the Communications Hub shall
 4209 make the OTA Upgrade Image available to the ESME / GSME / HCALCS in the *Query*
 4210 *Next Image Response Command*;
- 4211 3. The ESME / GSME / HCALCS shall download an OTA Upgrade Image when it is aware
 4212 of the availability of a suitable OTA Upgrade Image using the *QueryNextImage* and
 4213 *Image Block/Page* commands specified in the *OTA Cluster* specification²⁷;
- 4214 4. upon the reception of the *Upgrade End Request Command* from the ESME / GSME /
 4215 HCALCS, the Communications Hub shall;
- 4216 1. set `transferResponseCode` to `fileTransferSuccess` if the transfer of
 4217 the OTA Upgrade Image has completed successfully and process from step 4;
- 4218 2. set `transferResponseCode` to `fileTransferFailure` if the transfer of
 4219 the OTA Upgrade Image has failed, and process from step 5;
- 4220 3. the Communications Hub shall create an Alert with the Alert Code field set to 0x8F8A
 4221 as specified in Section 11.7, set `firmwareVersion` to the OTA Upgrade Image File
 4222 version, include the `transferResponseCode` from the previous step and
 4223 process from step 6;
- 4224 4. the Communications Hub shall create an Alert with the Alert Code field set to 0x8F89
 4225 as specified in Section 11.7, set `firmwareVersion` to the OTA Upgrade Image File
 4226 version, include the `transferResponseCode` from the previous step and
 4227 process from step 6;
- 4228 5. the Communications Hub shall send the Alert created in the previous step.
- 4229 The ESME / GSME / HCALCS shall verify the Upgrade Image contained within the OTA
 4230 Upgrade Image according to Section 11.2.5, and the ESME / GSME shall update its Event Log
 4231 with the outcome of that verification.
- 4232 If the verification is successful, the ESME / GSME / HCALCS shall construct and send a
 4233 Firmware Distribution Receipt Alert, according to Section 11.2.6, and shall store the
 4234 Manufacturer Image contained within the OTA Upgrade Image.
- 4235 If the verification is not successful, the Device shall discard the OTA Upgrade Image, and
 4236 send a Firmware Distribution Receipt Alert, as detailed in Section 11.2.6.
- 4237 On receipt of a Firmware Distribution Receipt Alert, the Supplier may verify the
 4238 cryptographic protection as specified in Section 6.7.3.
-

4239 Additionally, the Supplier may verify that the Manufacturer Image received by the Device is
 4240 that intended by comparing the Manufacturer Image Hash in the Firmware Distribution
 4241 Receipt Alert, with the Hash which it calculates over the Manufacturer Image provided.

4242

4243 CS06 Activate Firmware

4244 This Use Case covers the activation of a Firmware Image for ESME, GSME, CH and HCALCS.

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command, Response and Alert (if future dated)
Message Type Category	SME.C.C
Capable of future dated invocation?	Yes
Protection Against Replay Required?	No
Service Reference	N/A for Communications Hub (this Command does not have a Service Reference for CH) 11.3 for ESME, GSME and HCALCS
Valid Target Device(s)	ESME / GSME / CH / HCALCS
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier for ESME / GSME / HCALCS WAN Provider for CH
Valid Response Recipient role(s) (only for Messages authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

4245 Table 11.5: Use Case Cross References for CS06 Activate Firmware

- 4246 Pre-conditions
- 4247 None.
- 4248 Detailed Steps
- 4249 Construction of Command
- 4250 Activate Firmware Command Payloads shall be constructed according to the requirements of
- 4251 Section 11.5.2.3 and populated as specified in Table 11.5.2.1.
- 4252 MAC Header, Grouping Header, KRP Signature and ACB-SMD MAC shall be populated as
- 4253 required for a Command of the SME.C.C Message Category.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@ActivateFirmware.CommandPayload	SEQUENCE			
manufacturerImageHash	OCTET STRING	The Manufacturer Image Hash of the image to be activated.	Mandatory	An octet-string of length 32 interpreted as the Manufacturer Image Hash of the Manufacturer Image that is to be activated
originatorCounter	INTEGER (0..18446744073709551615)	The Originator Counter as in the Grouping Header of the Command	Mandatory	
executionDateTime	GeneralizedTime	The date-time at which the Command is to be executed, if future dated	OPTIONAL	

4254 Table 11.5.2.1: @ActivateFirmware.CommandPayload population

- 4255 Device processing of Command and Response handling
- 4256 The Device receiving an Activate Firmware Command shall undertake processing steps in the
- 4257 sequence defined in this Section 11.5.2.2.

-
- 4258 The Device shall:
- 4259 1. undertake Command Authenticity and Integrity Verification as required for a Command
4260 of the SME.C.C Message Category;
- 4261 2. if `executionDateTime` is present then the Device shall:
- 4262 record `manufacturerImageHash`, `originatorCounter` and `executionDateTime`;
- 4263 construct and send a Response where `executionOutcome` is not present. Grouping Header is
4264 constructed and Response Cryptographic Protection is applied as required for a Response of
4265 the SME.C.C Message Categories; and
- 4266 at the date-time specified in `executionDateTime`, or immediately if the `executionDateTime` is
4267 in the past, undertake the processing from step 3.
- 4268 If `executionDateTime` is not present then the Device shall continue processing from step
4269 3 immediately;
- 4270 3. if the Device does not have a stored Manufacturer Image then set
4271 `activateImageResponseCode` to `noImageHeld` and process from step 7;
- 4272 4. calculate Manufacturer Image Hash. If the calculated value does not equal
4273 `manufacturerImageHash` then the Device shall set
4274 `activateImageResponseCode` to `hashMismatch` and process from step 7;
- 4275 5. attempt to activate Manufacturer Image. If the activate fails then the Device shall set
4276 `activateImageResponseCode` to `activationFailure` and process from
4277 step 7;
- 4278 6. set `activateImageResponseCode` to `success`;
- 4279 7. populate the `executionOutcome` according to the requirements of Section 11.5.2.3 using
4280 the `activateImageResponseCode` value produced by the processing in this
4281 Section 11.5.2.2, the value of `originatorCounter` from the Command and the
4282 version of firmware now in operation to populate `firmwareVersion`;
- 4283 8. construct Grouping Header and apply the Response Cryptographic Protection required
4284 for a Response / Alert of the SME.C.C / SME.A.C Message Categories respectively. In
4285 such an Alert, the Message Code shall be 0x00CA. The Response / Alert shall be
4286 addressed to the Business Originator of the Corresponding Command. If
4287 `activateImageResponseCode` is `success` then `alertCode` shall be 0x8F66
4288 else `alertCode` shall be 0x8F67; and
- 4289 9. send the Response if `executionDateTime` was not present in the Command or send
4290 the Alert if `executionDateTime` was present in the Command.
-

4291 On receipt of the Response, the recipient may undertake the 'Response Recipient
4292 Verification' for Responses of type SME.C.C. or for Alerts of type SME.A.C, dependent upon
4293 the Message received.

4294 Activate Firmware Command, Response and Alert Payloads – structure **definition**

4295 Each instance of @ActivateFirmware.CommandPayload and of
4296 @ActivateFirmware.ResponsePayload and of @ActivateFirmware.AlertPayload shall be an
4297 octet string containing the DER encoding of the populated structure defined in this Section
4298 11.5.2.3 which specifies the structure in ASN.1 notation.

4299 ActivateFirmware DEFINITIONS ::= BEGIN

4300

4301 CommandPayload ::= SEQUENCE

4302 {

4303 -- specify the hash of the Manufacturer Image to be activated

4304 manufacturerImageHash OCTET STRING,

4305

4306 -- the Originator Counter as in the Grouping Header of the Command

4307 originatorCounter INTEGER (0..

4308 18446744073709551615),

4309

4310 -- the date-time at which the Command is to execute, if future dated

4311 executionDateTime GeneralizedTime OPTIONAL

4312 }

4313

4314 ResponsePayload ::= CHOICE

4315 {

4316 -- if the Command is future dated, the Response will not have any details of

4317 -- execution (those will be in the subsequent alert)

4318 commandAccepted NULL,

4319

```

4320    -- if the Command is for immediate execution, the Response will detail the
4321    -- outcomes
4322    executionOutcome                               ExecutionOutcome
4323 }
4324
4325 AlertPayload ::=                               SEQUENCE
4326 {
4327     -- specify the Alert Code
4328     alertCode                                     INTEGER(0..4294967295),
4329
4330     -- specify the date-time of execution
4331     executionDateTime                           GeneralizedTime,
4332
4333     -- the Originator Counter as in the Grouping Header of the corresponding Command
4334     originatorCounter                           INTEGER (0..
4335 18446744073709551615),
4336
4337     -- detail what happened when the future dated command was executed
4338     executionOutcome                             ExecutionOutcome
4339 }
4340
4341 ExecutionOutcome ::=                               SEQUENCE
4342 {
4343     -- Specify whether the activation was successful or not
4344     activateImageResponseCode                   ActivateImageResponseCode,
4345

```

4346 -- Specify the Device's now current firmware version. The value shall be four octets in
4347 length and shall correspond to the big endian representation of the unsigned 32-bit integer
4348 File Version field in the ZSE OTA Header structure.

4349	firmwareVersion	OCTET STRING
------	-----------------	--------------

4350 }

4351

4352 ActivateImageResponseCode ::= INTEGER

4353 {

4354 success (0),

4355 nolmageHeld (1),

4356 hashMismatch (2),

4357 activationFailure (3)

4358 }

4359

4360 END

4361 CS05c Distribute Firmware to PPMID

4362 This Use Case covers the distribution of an OTA Upgrade Image that is intended for a PPMID
4363 to the PPMID.

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command
Message Type Category	None – this is a Variant Message
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	11.4
Valid Target Device(s)	PPMID
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages authorised by the	N/A

Cross Reference	Value
Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	WAN Provider Specific to Communications Hub; ZigBee OTA from Communications Hub to PPMID

4364 Table 11.6: Use Case Cross References for CS05c Distribute Firmware to PPMID

4365 Pre-conditions

4366 None.

4367 Detailed Steps

4368 Italicised terms in this Section 11.6.2 shall have the meaning specified in ZSE.

4369 The PPMID shall use the ZCL *Image Block Request* and *Image Block Response* commands to
4370 retrieve available OTA images.

4371 The PPMID shall not use the ZCL *Query Specific File Request* and *Query Specific File Response*
4372 commands.

4373 The OTA Upgrade Image shall be populated according to Section 11.2.3.

4374 The OTA Upgrade Image shall be transported to the Communications Hub through which the
4375 Device communicates:

4376 1. The Communications Hub shall update its OTA Cluster to reflect the availability of the
4377 OTA Upgrade Image;

4378 the Communications Hub, as OTA Server, shall indicate availability of an OTA Upgrade Image
4379 for PPMID by sending a ZSE *Image Notify* command;

4380 the PPMID should include its *Hardware Version* in the *Query Next Image Request Command*;
4381 the Communications Hub shall verify whether the PPMID *Hardware Version* is present in the
4382 *Query Next Image Request Command*;

4383 if the Hardware Version is present and does not fall between the minimum and maximum
4384 Hardware Version in the OTA Header the Communications Hub shall set
4385 transferResponseCode to hardwareVersionMismatch as specified in Section 11.7 and
4386 process from step 7;

4387 if the Hardware Version is not present, or is present and falls between the minimum and
4388 maximum Hardware Version in the OTA Header, the Communications Hub shall make the
4389 OTA Upgrade Image available to the PPMID in the Query Next Image Response Command;

4390 the PPMID shall download an OTA Upgrade Image when it is aware of the availability of a
 4391 suitable OTA Upgrade Image using the *QueryNextImage* and *Image Block/Page* commands
 4392 specified in the *OTA Cluster* specification²⁸;

4393 upon the reception of the *Upgrade End Request Command* from the PPMID, the
 4394 Communications Hub shall:

4395 set `transferResponseCode` to `fileTransferSuccess` if the transfer of the OTA
 4396 Upgrade Image has completed successfully and process from step 6;

4397 set `transferResponseCode` to `fileTransferFailure` if the transfer of the OTA
 4398 Upgrade Image has failed, and process from step 7;

4399 the Communications Hub shall instruct the PPMID to activate the firmware Image as defined
 4400 in Section 11.2.1 and create an Alert with the Alert Code field set to 0x8F8A as specified in
 4401 Section 11.7, set `firmwareVersion` to the OTA Upgrade Image File version, include the
 4402 `transferResponseCode` from the previous step and process from step 8;

4403 the Communications Hub shall create an Alert with the Alert Code field set to 0x8F89 as
 4404 specified in Section 11.7, set `firmwareVersion` to the OTA Upgrade Image File version,
 4405 include the `transferResponseCode` from the previous step and process from step 8;

4406 the Communications Hub shall send the Alert created in the previous step; and

4407 upon receipt of the firmware activation instruction from the Communications Hub the
 4408 PPMID shall activate the firmware and send an Alert with Alert Code 0x8F8B as defined in
 4409 Section 11.9 to this effect.

4410

4411 CCS08 Firmware Transfer Alert

4412 Description

4413 This Section 11.7 covers the creation and processing of Alerts resulting from the
 4414 requirements contained in Section 11.4.2 and Section 11.6.2.

4415 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Alert
Message Type Category	SME.A.C

Cross Reference	Value
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	N/A
Valid Initiating Device(s)	CHF
Valid Business Target role(s) for Alert	Access Control Broker
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

4416 Table 11.7.2: Use Case Cross References for CCS08 Firmware Transfer Alert

4417 Construction of Alert

4418 Grouping Header and SMD Signature shall be populated as required for an Alert of the
4419 SME.A.C Message Category. The Message Code shall be 0x0128.

4420 The Business Originator ID field in the Grouping Header shall contain the CHF's Entity
4421 Identifier.

4422 The Business Target ID field in the Grouping Header shall contain the Entity Identifier in the
4423 CHF's {accessControlBroker, digitalSignature, management} Trust
4424 Anchor Cell.

4425 The Alert Payload shall be constructed according to the requirements of Section 11.7.4 and
4426 the Alerts shall be populated as specified in Table 11.7.3.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@FirmwareTransferAlert.AlertPayload	SEQUENCE			
alertCode	INTEGER	0x8F8A 0x8F89	Mandatory	(1) 0x8F8A: OTA Firmware Image Delivery to the target Devices successful; (2) 0x8F89: OTA Firmware Image Delivery to the target Device failed;

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
executionDateTime	GeneralizedTime	The date-time at which this Alert was created	Mandatory	
otherDeviceEntityIdentifier	OCTET STRING	Entity Identifier	Mandatory	The Entity Identifier of the ESME / GSME / HCALCS / PPMID the firmware upgrade is targeted at
firmwareVersion	OCTET STRING	Firmware version	Mandatory	The big endian representation of the unsigned 32-bit integer firmware version of the transferred or discarded Upgrade
TransferResponseCode	INTEGER	fileTransferSuccess imageDiscarded (1) , hardwareVersionMismatch (2) , fileTransferFailure	Mandatory	(a) 0x8F8A: fileTransferSuccess (0) (b) 0x8F89: imageDiscarded (1); hardwareVersionMismatch (2); fileTransferFailure (3)

4427 Table 11.7.3: @FirmwareTransferAlert.AlertPayload population

4428 Firmware Transfer Alert – structure definition

4429 Each instance of `FirmwareTransferAlert.AlertPayload` shall be an octet string
 4430 containing the DER encoding of the populated structure defined in this Section 11.7.4 which
 4431 specifies the structure in ASN.1 notation.

```

4432 FirmwareTransferAlert DEFINITIONS ::= BEGIN
4433
4434 AlertPayload ::=
4435 {
4436   -- specify the Alert Code
4437   alertCode
4438   INTEGER (0..4294967295) ,
4439   -- specify the date-time of execution
4440   executionDateTime
4441   GeneralizedTime,
4442   -- specify the 64 bit Entity Identifier of the Device the upgrade has been
4443   -- targeted at
4444   otherDeviceEntityIdentifier
4445   OCTET STRING,
4446   -- specify the upgrade image version if replaced or discarded by the
4447   -- Communications Hub or if the transfer to the target device succeeded or
4448   -- failed. The value shall be four octets in length and shall correspond to
4449   -- the big endian representation of the unsigned 32-bit integer FileVersion
4450   -- field in the ZSE OTA Header structure
4451   firmwareVersion
   OCTET STRING,

```

```
4452
4453   -- specify the outcome
4454   transferResponseCode                               TransferResponseCode
4455 }
4456
4457 TransferResponseCode ::=                               INTEGER
4458 {
4459     fileTransferSuccess                               (0),
4460     imageDiscarded                                   (1),
4461     hardwareVersionMismatch                           (2),
4462     fileTransferFailure                               (3)
4463 }
4464
4465 END
4466
```

4467 CS08 Read PPMID/HCALCS Firmware Version

4468 Description

4469 This Section 11.8.1 covers the creation and processing of Commands and Responses related
4470 to reading the Firmware Version. The Message Code shall be 0x0129.

4471 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	SME.C.NC
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	11.2
Valid Target Device(s)	PPMID, HCALCS
Valid Business Target role(s) for Alert	Access Control Broker
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Access Control Broker
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

4472 Table 11.8.2: Use Case Cross References for CS08 Read PPMID/HCALCS Firmware Version

4473 Construction of Command and Response

1. Construction of Commands

'Read PPMID/HCALCS Firmware Version' Command Payloads shall be constructed as specified in Section 11.8.4 and Cryptographic Protection shall be applied as required for a Command of the relevant Message Category.

2. Device processing of Commands and Response handling

The Device receiving a 'Read PPMID/HCALCS Firmware Version' Command shall undertake processing steps in the sequence defined in this Section 11.8.3.2.

In processing a 'Read PPMID/HCALCS Firmware Version' Command, the Device shall:

1. undertake Command Authenticity and Integrity Verification as required for a Command of this Message Category;

if the version of firmware in operation is available, set firmwareVersion to the version of firmware in operation; set firmwareReadCode to firmwareReadSuccess and process from Step 4;

if the version of firmware in operation is not available, set firmwareVersion to 0x00000000; set firmwareReadCode to firmwareReadFailure and process from Step 4; and

create a Response according to the requirements of Section 11.8.4, apply the Response Cryptographic Protection required for a Response of the relevant Message Category, and send the Response.

Read PPMID/HCALCS Firmware Version Command and Response Payloads – structure definition

Each instance of @ReadPpmidHcalcsFirmwareVersion.CommandPayload and ReadPpmidHcalcsFirmwareVersion.ResponsePayload shall be an octet string containing the DER encoding of the populated structure defined in this Section 11.8.4 which specifies the structure in ASN.1 notation.

```

ReadPpmidHcalcsFirmwareVersion          DEFINITIONS ::= BEGIN
CommandPayload ::=
ResponsePayload ::=
{
-- Specify the Device's now current firmware version. The value shall be four
-- octets in length and shall correspond to the big endian representation of the
-- unsigned 32-bit integer File Version field in the ZSE
-- OTA Header structure. Set to 0x00000000 if reading failed.
firmwareVersion          OCTET STRING,

-- specify whether the reading of the firmware was successful.
firmwareReadCode         FirmwareReadCode
}

```



```
4513
4514 FirmwareReadCode ::=
4515 {
4516     firmwareReadSuccess      (0) ,
4517     firmwareReadFailure      (1)
4518 }
4519
4520 END
```

4521 CS09 PPMID Firmware Activation Alert

4522 Description

4523 This Section 11.9 covers the construction of the PPMID Firmware Activation Alert. The
4524 Message Code shall be 0x012B.

4525 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Alert
Message Type Category	SME.A.C
Capable of future dated invocation?	No
Protection Against Replay Required?	Yes
Service Reference	N/A
Valid Target Device(s)	N/A
Valid Business Target role(s) for Alert	Access Control Broker
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

4526 Table 11.9.2: Use Case Cross References for CS09 PPMID Firmware Activation Alert

4527 Construction of PPMID Firmware Activation Alert

4528 Grouping Header and SMD Signature shall be populated as required for an Alert of the
4529 SME.A.C Message Category.

4530 The Business Originator ID field in the Grouping Header shall contain the PPMID’s Entity
4531 Identifier.

4532 The Business Target ID field in the Grouping Header shall contain the Entity Identifier in the
 4533 PPMID's {accessControlBroker, digitalSignature, management} Trust
 4534 Anchor Cell.

4535 The Alert Payload shall be constructed according to the requirements of Section 11.9.4 and
 4536 the Alerts shall be populated as specified in Table 11.9.3.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@PpmidFirmwareActivation.AlertPayload	SEQUENCE			
alertCode	INTEGER	0x8F8B	Mandatory	
executionDateTime	GeneralizedTime	The date-time at which this Alert was created	Mandatory	
firmwareVersion	OCTET STRING	Firmware version	Mandatory	The big endian representation of the unsigned 32-bit integer firmware version in operation or 0x00000000 if reading the firmware version failed
ActivateImageResultCode	INTEGER	activationSuccess (0), activationFailure (1)	Mandatory	

4537 Table 11.9.3: @PpmidFirmwareActivation.AlertPayload population

4538 PPMID Firmware Activation Alert Payloads – structure definition

4539 Each instance of @PpmidFirmwareActivation.AlertPayload shall be an octet
 4540 string containing the DER encoding of the populated structure defined in this Section 11.9.4
 4541 which specifies the structure in ASN.1 notation.

```

4542 PpmidFirmwareActivation                               DEFINITIONS ::= BEGIN
4543
4544
4545 AlertPayload ::=
4546 {
4547   -- specify the Alert Code
4548   alertCode                                           INTEGER(0..4294967295),
4549
4550   -- specify the date-time of execution
4551   executionDateTime                                   GeneralizedTime,
4552
4553   -- Specify the Device's now current firmware version. The value shall be four
4554   -- octets in length and shall correspond to the big endian representation of the
  
```

```

4555 -- unsigned 32-bit integer File Version field in the ZSE
4556 -- OTA Header structure.
4557 firmwareVersion                                OCTET STRING,
4558
4559 -- Specify whether the activation was successful or not
4560 activateImageResultCode                        ActivateImageResultCode
4561 }
4562
4563 ActivateImageResultCode ::= INTEGER
4564 {
4565     activationSuccess                            (0),
4566     activationFailure                            (1)
4567 }
4568
4569 END
4570 Requirements for Certificates

```

4571 This Section 12 lays out requirements as to structure and content to which all valid
 4572 authorised Certificates shall comply, in so far as those requirements affect the processing
 4573 carried out by Devices. All terms in this Section shall, where not defined in the GBCS, have
 4574 the meanings in IETF RFC 575929 and IETF RFC 5280.

4575 Requirements applicable to all Certificates

4576 All Security Credential Documents that are successfully authorised within the APKI for use by
 4577 Devices shall:

4578 be compliant with IETF RFC 5759 and so with IETF RFC 5280. In adherence with the
 4579 requirements of IETF RFC 5759, all Security Credential Documents shall:

4580 contain the authorityKeyIdentifier extension, except where the Security Credential
 4581 Document is self-signed;

4582 contain the keyUsage extension which shall be marked as critical;

4583 be X.509 v3 certificates as defined in IETF RFC 5280, encoded using the ASN.1 Distinguished
 4584 Encoding Rules;

4585 only contain public keys of types that are explicitly allowed within the GBCS. This means all
 4586 public keys shall be elliptic curve public keys on the NIST P-256 curve;

4587 only contain public keys in uncompressed form which shall be elliptic curve points in
 4588 uncompressed form as detailed in section 2.2 of IETF RFC 548030;

29 <http://tools.ietf.org/html/rfc5759>

30 <http://tools.ietf.org/html/rfc5480>

4589 only provide for signature methods that are explicitly allowed within the GBCS. This means
 4590 using P-256 Private Keys with SHA 256 and ECDSA;
 4591 contain a serialNumber of no more than 16 octets in length;
 4592 contain a subjectKeyIdentifier which shall be marked as non-critical;
 4593 contain a certificatePolicies extension containing at least one CertPolicyId which shall be
 4594 marked as critical. A Device shall reject any Certificate where the value in any CertPolicyId is
 4595 not a valid Object Identifier for a Certificate Policy allowed under the Smart Energy Code for
 4596 the Device's operating state. For clarity and in adherence with IETF RFC 5280, Certification
 4597 Path Validation undertaken by Devices shall interpret this extension. Devices shall also
 4598 interpret this extension when processing a Command with Message Code 0x000B (CS02d
 4599 Update Device Certificates on Device);
 4600 contain an authorityKeyIdentifier in the form [0] KeyIdentifier which shall be marked as non-
 4601 critical, except where the Security Credential Document is self-signed. Note this exception
 4602 only applies where RemotePartyRole as specified in the X520OrganizationalUnitName field =
 4603 root;
 4604 only contain KeyIdentifiers generated as per method (2) of section 4.2.1.2 of IETF RFC 5280.
 4605 Thus KeyIdentifiers shall always be 8 octets in length;
 4606 contain an issuer field whose contents are identical to the Security Credential Document's
 4607 signer's subject field in the signer's Security Credential Documents; and
 4608 have a valid notBefore field consisting of the time of issue encoded and a valid notAfter as
 4609 per IETF RFC 5280 section 4.1.2.5.
 4610 Requirements applicable to Organisation Certificates only
 4611 All Organisation Certificates that are Authorised for use by Devices shall:
 4612 have a fixed expiration date in the notAfter field which shall not be GeneralizedTime value of
 4613 99991231235959Z;
 4614 contain a non-empty subject field which shall contain an X520OrganizationalUnitName
 4615 whose value shall be set to the RemotePartyRole that this Certificate allows the subject of
 4616 the Certificate to perform; and
 4617 contain a single Public Key except where the RemotePartyRole = root. Where the
 4618 RemotePartyRole = root, the Certificate shall contain two public keys. The second public key
 4619 shall be referred to as the Contingency Key³¹ and shall be present in the

31 The Contingency Key is a second public key held in the Root Certificate (and protected with an encryption key). Its purpose is to allow the validation of a specific command that

4620 WrappedApexContingencyKey extension with the meaning of IETF RFC 593432. The
 4621 Contingency Key shall be Encrypted as per the requirements of Section 13.3.5.8.1.

4622 Requirements applicable to Certificates where RemotePartyRole = root or
 4623 issuingAuthority

4624 All Remote Parties' Certificates that:
 4625 are Authorised within the APKI for use by Devices; and
 4626 have a X520OrganizationalUnitName whose value is either root or issuingAuthority
 4627 shall:
 4628 have a keyUsage with a value of keyCertSign and cRLSign. For clarity, Devices are not
 4629 required to use the associated Public Keys in relation to the cRLSign keyUsage and shall
 4630 disregard a keyUsage of cRLSign in any processing;
 4631 where X520OrganizationalUnitName = issuingAuthority:
 4632 contain at least one CertPolicyId in the certificatePolicies extension that refers to the OID(s)
 4633 valid for usage in GB Smart Metering;
 4634 contain the basicConstraints extension, with values cA = True, and pathLen = 0. This
 4635 extension shall be marked as critical;
 4636 where X520OrganizationalUnitName = root:
 4637 contain a single CertPolicyId in the certificatePolicies extension that refers to the OID for
 4638 anyPolicy; and
 4639 contain the basicConstraints extension, with the value cA = True and pathLen absent
 4640 (unlimited). This extension shall be marked as critical.

4641 Note that Certificates where RemotePartyRole is root do not have to contain a Subject
 4642 Unique ID. Thus, the Device will not have a value provided to it for this field in the {root,
 4643 keyCertSign, management} Trust Anchor Cell. For consistency, Devices shall record a
 4644 value of 0x0000000000000000 for Subject Unique ID in the {root, keyCertSign,

allows direct replacement of the Root Trust Anchor or the direct replacement of any other Trust Anchor cell. The command (an Apex Trust Anchor Update message) is signed with a private key (used once only, and only to sign this message) that only the second public key (known as the Contingency Key) can verify and therefore authorise action of.

32 Housley, R., Ashmore, S., and C. Wallace, 'Trust Anchor Management Protocol (TAMP)', RFC 5934, August 2010. <https://tools.ietf.org/html/rfc5934>

4645 management} Trust Anchor Cell, and shall use this in responding to Security Credentials
4646 Commands, where values for Subject Unique ID are required from this Trust Anchor Cell.

4647 Requirements applicable to Certificates where RemotePartyRole is neither root nor
4648 issuingAuthority

4649 All Remote Parties' Certificates that:
4650 are Authorised within the APKI for use by Devices; and
4651 have a X520OrganizationalUnitName whose value is not root and is not issuingAuthority
4652 shall:
4653 contain a Subject Unique ID whose value shall be the 8 octet Entity Identifier of the subject
4654 of the Certificate;
4655 have a keyUsage with a value of only one of digitalSignature or keyAgreement; and
4656 contain a single policyIdentifier in the certificatePolicies extension that refers to the OID
4657 applicable to the environment the Certificate has been issued in.

4658 Requirements applicable to Device Certificates

4659 All Device Certificates that are Authorised within the APKI for use by Devices shall:
4660 not have a well-defined expiration date and so the notAfter field shall be assigned the
4661 GeneralizedTime value of 99991231235959Z;
4662 have an empty SubjectName;
4663 have a keyUsage with a value of only one of digitalSignature or keyAgreement;
4664 contain a single policyIdentifier in the certificatePolicies extension that refers to the OID
4665 applicable to the environment the Device Certificate has been issued in;
4666 contain a SubjectAltName extension which shall contain a single GeneralName of type
4667 OtherName that is further sub-typed as a HardwareModuleName (id-on-
4668 HardwareModuleName) as defined in IETF RFC 410833. The hwSerialNum field shall be set
4669 to the Device's Entity Identifier. In adherence to IETF RFC 5280, SubjectAltName shall be
4670 marked as critical; and
4671 contain a single Public Key.

4672 Device processing of Certificates

4673 In relation to Certificates, Devices shall:

33 <http://tools.ietf.org/html/rfc4108>

4674 accept unexpected (not required by the GBCS) certificate extensions and shall ignore silently
4675 non-critical unrecognized certificate extensions;

4676 in adherence with the requirements of IETF RFC 5280, reject any certificate containing
4677 unrecognized critical certificate extensions; and

4678 reject any certificate containing either policy mappings or name constraints.

4679 Managing Security Credentials on Devices

4680 Requirements in:

4681 Section 13.3 and in Use Case CS02b Update Security Credentials Command, Response and
4682 Alert;

4683 Section 13.4 and in Use Case CS02c Issue Security Credentials;

4684 Section 13.5 and in Use Case CS02d Update Device Certificates on Device; and

4685 Section 13.6 and in Use Case CS02e Provide Device Certificates from Device;

4686 shall apply to an SAPC as if it were an ESME.

4687

4688 Introduction – informative

4689 This Section 13 includes the Use Cases related to the management of Security Credentials on
4690 Devices in terms of the relevant Commands, Responses and Alerts:

4691 Section 13.2 - CS02a Provide Security Credential Details Command and Response;

4692 Section 13.3 - CS02b Update Security Credentials Command, Response and Alert;

4693 Section 13.4 - CS02c Issue Security Credentials;

4694 Section 13.5 - CS02d Update Device Certificates on Device;

4695 Section 13.6 - CS02e Provide Device Certificates from Device;

4696 Section 13.7- Pair-wise Authorisation of Devices (covered by various Join / Unjoin Use
4697 Cases);

4698 Section 13.8 - GPF Device Log Backup and Restore (GCS59 and GCS62); and

4699 Section 13.9 - CS02f Provide Security Credential Details Command and Response; and

4700 Section 13.10 - CS02g Update Load Controller Security Credentials Command, Response and
4701 Alert.

4702 Device Security Credentials – informative

4703 In terms of processing relating to a Device's own Security Credentials:

4704 the Command to Devices for issuing Device Certification Requests (and therefore generate
4705 new Public-Private Key Pairs) is covered in Section 13.4;

4706 the Command to Devices for the Device to replace a current Device Certificate with a new
4707 Device Certificate resulting from a Device Certification Request is covered in Section 13.5, as
4708 are the related requirements for the capability to store such Certificates; and

4709 the Command to a Device to provide a copy of its currently held Device Certificates is
4710 covered Section 13.6.

4711 Remote Party Security Credentials – informative

4712 This Section 13.1.2 summarises the GBCS requirements in relation to storing, replacing and
4713 providing details of Remote Party Security Credentials. The use of such credentials to
4714 control access to Device functions is detailed in other Sections of the GBCS and in relevant
4715 Use Cases.

4716 A Remote Party Security Credential is a Public Key Certificate which securely binds together
4717 the Remote Party's identity with a Public Key along with related information, including what
4718 that Public Key can be used for and over what time period it is valid. The corresponding
4719 Private Key should be securely controlled solely by the Remote Party and known only to that
4720 Remote Party.

4721 The purpose of storing each Remote Party Public Key (and related details) on a Device is so
4722 that each Public Key can act as a 'Trust Anchor' for the Device. The Device uses these Trust
4723 Anchors to check cryptographically whether Remote Party Messages can be trusted or not
4724 (and so whether it should act on them or not). Thus, all of a Device's Trust Anchors must be
4725 populated.

4726 Trust Anchors need to be capable of being replaced during a Device's operational life for a
4727 number of reasons including:

4728 the Certificate's expiry (Organisation Certificates will only be valid for a fixed period of time);

4729 the Known Party transferring control to a different organisation (for example on Change of
4730 Supplier);

4731 the cryptographic algorithms, or parameters such as key length, needing to be changed;

4732 the Known Party having lost the use of the corresponding Private Key; or

4733 there being concerns that someone other than the Known Party has use of, or may have use
4734 of, the corresponding Private Key.

4735 Thus, an 'Update Security Credentials Command' must be supported by all Devices that rely
4736 on Remote Party Security Credentials to act as Trust Anchors. Related, all such Devices need
4737 to support a 'Provide Security Credential Details' Command, so that Remote Parties can be
4738 sure which Devices need to have credentials replaced.

4739 However, if these Trust Anchors could be replaced without proper protections, attackers
4740 could take over control of Devices or the Devices could be rendered inoperable. Thus, a
4741 Device needs to do thorough checks before applying an Update Security Credentials
4742 Command. The checks that the Device can and must do vary dependent on the reasons for
4743 the change. Thus, Section 13.2.1 lays out a number of different checks and the
4744 circumstances in which corresponding Commands may be issued. Broadly the following
4745 checks are carried out by the Device:

4746 is the Command properly formed?

4747 is the Command for the Device that it has been delivered to, and is the Command one that it
4748 has not processed previously?

4749 are the Remote Parties apparently authorising the Command allowed to authorise it?

4750 was the Command Authorised by the Remote Parties that it appears to be Authorised by?

4751 were the Certificates in the payload of the Command issued by properly Authorised parties,
4752 specifically by Certification Authorities Authorised (by 'root' under the APKI) to issue GB
4753 Smart Metering Certificates?

4754 Only when a Device has successfully undertaken all five sets of checks should it action the
4755 Update Security Credentials Command.

4756 Other Critical Commands only have to complete the first four categories of check.

4757 Trust Anchor Management Protocol (TAMP) – informative

4758 The GBCS does not specify a fully compliant TAMP solution due to the limited processing and
4759 networking capability of Devices. However, it does incorporate checks that are functionally
4760 derived from relevant checks in IETF RFC 5934.

4761 The GBCS only permits a restricted subset of 'IETF RFC 5934 like' functionality:

4762 replacement of Trust Anchors is required (and specified in this Use Case) but their addition,
4763 change or removal is not allowed;

4764 status queries are supported (and are specified in this Section 13.1.3); and

4765 community related functions are not supported.

4766 CS02a Provide Security Credential Details Command and Response

- 4767 Description
- 4768 This Section covers the creation, validation and processing of (i) CS02a Provide Security
4769 Credential Details Commands and (ii) Responses to such Commands.
- 4770 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	Variant Message and is not a Critical Command
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	6.24
Valid Target Device(s)	GSME / GPF / CHF / HCALCS / PPMID
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier Network Operator Access Control Broker Transitional Change of Supplier WAN Provider Recovery
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

- 4771 Table 13.2.2: Use Case Cross References for Provide Security Credential Details Command
4772 and Response
- 4773 Common Requirements

4774 Summary – informative

4775 Remote Party Security Credentials are provided to Devices as Certificates which are X.509
4776 based, DER encoded ASN.1 structures. Hence, the Command's structure is specified using
4777 ASN.1 with DER encoding to be applied to Command instances. Note that the details
4778 provided in the Response include the related Execution Counter details held on the Device.

4779 The 'CS02a Provide Security Credential Details' Command and Response

4780 This Section 13.2.3.2 summarises the structure of the CS02a Provide Security Credential Details Command.

4781 If protected by an Access Control Broker MAC as per Section 13.2.4.2, a CS02a Provide Security Credential Details Command shall be the
4782 concatenation:

4783 MAC Header || Grouping Header || @ProvideSecurityCredentialDetails.Command || 0x00 || ACB-SMD MAC

4784 If protected by a KRP Signature as per Section 13.2.4.2, a CS02a Provide Security Credential Details Command shall be the concatenation:

4785 Grouping Header || @ProvideSecurityCredentialDetails.Command || 0x40 || KRP Signature

4786 If an SMD Signature is required as per Section 13.2.4.5, a CS02a Provide Security Credential Details Response shall be the concatenation:

4787 Grouping Header || @ProvideSecurityCredentialDetails.Response || 0x40 || SMD Signature

4788 If an SMD Signature is not required as per Section 13.2.4.5, a CS02a Provide Security Credential Details Response shall be the concatenation:

4789 MAC Header || Grouping Header || @ProvideSecurityCredentialDetails.Response || 0x00 || SMD-KRP MAC

4790 Where:

4791 @ProvideSecurityCredentialDetails.Command and Response shall each be an octet string containing the DER encoding of the populated ASN.1
4792 structure (as laid out in Section 13.2.3.3);

4793 0x40 is the length in octets of Signature when a SMD or KRP Signature is present, and 0x00 is the length in octets of Signature when a SMD or
4794 KRP Signature is not present;

4795 KRP Signature and ACB-SMD MAC are as defined in Section 13.2.4.2;

4796 SMD Signature and SMD-KRP MAC are as defined in Section 13.2.4.5; and

4797 MAC Header and Grouping Header are as defined in Section 7.2.

4798 The @ProvideSecurityCredentialDetails.Command and @ProvideSecurityCredentialDetails.Response structure definition

4799 Each instance of @ProvideSecurityCredentialDetails.Command and of @ProvideSecurityCredentialDetails.Response shall be an octet string

4800 containing the DER34 encoding of the populated structure defined in this Section 13.2.3.3 which specifies the structure in ASN.1 notation³⁵.

4801 ProvideSecurityCredentialDetails DEFINITIONS ::= BEGIN

4802

4803 Command ::= SEQUENCE

4804 {

4805 -- Identify which of the Public Keys on the Device is to be used in verifying the Signature or MAC

4806 -- (so defining the nature of the verification by way of the KeyUsage parameter held on the

4807 -- Device for the Public Key so identified).

4808

4809 authorisingRemotePartyTACellIdentifier TrustAnchorCellIdentifier,

4810

4811 -- List the Remote Party Role(s) for which credential details are required

34 <https://www.itu.int/rec/T-REC-X.690/en>

35 <https://www.itu.int/rec/T-REC-X.680/en>

4812

4813 remotePartyRolesCredentialsRequired SEQUENCE OF RemotePartyRole

4814 }

4815

4816 Response ::= SEQUENCE OF RemotePartyDetails

4817

4818 RemotePartyDetails ::= SEQUENCE

4819 {

4820

4821 -- Which Remote Party do these details relate to?

4822 remotePartyRole RemotePartyRole,

4823

4824 -- statusCode shall be success unless the role is not valid on this type of Device or there is a processing failure

4825 statusCode StatusCode,

4826

4827

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4828 -- What is the current Update Security Credentials Execution Counter on the Device for this role, where there is such a number for this role (see
4829 Table 13.2.4.4)?

4830

4831 currentSeqNumber SeqNumber OPTIONAL,

4832

4833 -- What are the details held on the Device for each of the Cells related to this role? The list shall have between one and
4834 -- three entries (e.g. there will be one if role is transitional change of supplier; there may be three if role is supplier)

4835

4836 trustAnchorCellsDetails SEQUENCE OF TrustAnchorCellContents OPTIONAL

4837 }

4838

4839 SeqNumber ::= INTEGER (0.. 18446744073709551615)

4840

4841 TrustAnchorCellContents ::= SEQUENCE

4842 {

4843 -- To what cryptographic use can the Public Key in this Cell be put? Some Remote Party Roles
4844 -- (e.g. supplier) can have more than one Public Key on a Device and each one would only have

4845 -- a single cryptographic use.

4846

4847 trustAnchorCellKeyUsage KeyUsage,

4848

4849 -- trustAnchorCellUsage is to allow for multiple Public Keys of the same keyUsage for the same Remote

4850 -- Party Role. This will be absent except where used to refer to the Supplier Key Agreement Key.

4851 -- This Key is used solely in relation to validating Supplier generated MACs on Prepayment Top Up transactions.

4852

4853 trustAnchorCellUsage CellUsage DEFAULT management,

4854

4855 -- The existingSubjectUniqueID shall be the 64 bit Entity Identifier of the Security Credentials in this Trust Anchor Cell.

4856

4857 existingSubjectUniqueID OCTET STRING,

4858

4859 -- The APKI requirements mean that KeyIdentifier attributes will all be 8 byte SHA-1 Hashes.

4860 -- existingSubjectKeyIdentifier shall be set accordingly based on the contents of the Trust Anchor Cell

4861

4862 existingSubjectKeyIdentifier OCTET STRING

4863 }

4864

4865 TrustAnchorCellIdentifier ::= SEQUENCE

4866 {

4867 -- Which Remote Party Role does this Cell relate to?

4868

4869 trustAnchorCellRemotePartyRole RemotePartyRole,

4870

4871 -- To what cryptographic use can the Public Key in this Cell be put? Some Remote Party Roles

4872 -- (e.g. supplier) can have more than one Public Key on a Device and each one would only have

4873 -- a single cryptographic use.

4874

4875 trustAnchorCellKeyUsage KeyUsage,

4876

4877 -- trustAnchorCellUsage is to allow for multiple Public Keys of the same keyUsage for the same Remote

4878 -- Party Role. This may be absent except where use to refer to the Supplier Key

4879 -- Agreement Key used solely in relation to validating Supplier generated MACs on Prepayment Top Up transactions
4880
4881 trustAnchorCellUsage CellUsage DEFAULT management
4882 }
4883
4884 CellUsage ::= INTEGER {management(0), prePaymentTopUp(1)}
4885
4886 RemotePartyRole ::= INTEGER
4887 {
4888 -- Define the set of Remote Party Roles in relation to which a Device may need to undertake
4889 -- processing. Note that most Devices will only support processing in relation to a subset of these.
4890
4891 root (0),
4892 recovery (1),
4893 supplier (2),
4894 networkOperator (3),
4895 accessControlBroker (4),

4896 transitionalCoS (5),
4897 wanProvider (6),
4898 issuingAuthority (7), -- Devices will receive such Certificates but they do not
4899 -- need to store them over an extended period
4900
4901
4902
4903 -- The 'other' RemotePartyRole is for a party whose role does not allow it to invoke any Device function apart from
4904 -- UpdateSecurityCredentials. This is to allow for Device functionality to be locked out of usage until a valid
4905 -- Remote Party can be identified e.g. where roles cannot be fixed until a Device is bought in to operation
4906 other (127)
4907
4908 }
4909
4910 -- KeyUsage is only repeated here for ease of reference. It is defined in RFC 5912
4911
4912 KeyUsage ::= BIT STRING

4913 {

4914 -- Define valid uses of Public Keys.

4915

4916 digitalSignature (0),

4917 contentCommitment (1), -- not valid for GBCS compliant transactions

4918 keyEncipherment (2), -- not valid for GBCS compliant transactions

4919 dataEncipherment (3),

4920 keyAgreement (4),

4921 keyCertSign (5),

4922 cRLSign (6),

4923 encipherOnly (7),

4924 decipherOnly (8) -- not valid for GBCS compliant transactions

4925 }

4926

4927 -- The GBCS only allows for a constrained set of Trust Anchor Cell operations and so the list of possible outcomes

4928 -- is more limited than in IETF RFC 5934. The list below is that more constrained subset

4929

4930 StatusCode ::= ENUMERATED {
4931
4932 success (0),
4933
4934 -- trustAnchorNotFound indicates that details of a trust anchor were requested, but the referenced trust anchor
4935 -- is not represented on the Device
4936
4937 trustAnchorNotFound (25),
4938
4939 other (127)}
4940
4941
4942 END

4943 CS02a Provide Security Credential Details from a Device – Processing Steps

4944 This Section 13.2.4 lays out the requirements relating to the construction, protection and Authentication of the CS02a Provide Security
4945 Credentials Command, and the construction, protection and Authentication of the corresponding Response.

4946 Command Construction

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- 4947 The Remote Party constructing the Command shall populate Grouping Header according to the requirements of Section 7.2.6.
- 4948 @ProvideSecurityCredentialDetails.Command shall have the structure defined in Section 13.2.3.3, and the Remote Party constructing the
- 4949 Command shall populate with values according to Table 13.2.4.1.
- 4950 The Remote Party constructing the Command shall populate Command Length once it has fully populated
- 4951 @ProvideSecurityCredentialDetails.Command, based on the length of the octet string so constructed.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@ProvideSecurityCredentialDetails.Command ::=	SEQUENCE			
authorisingRemotePartyTACellIdentifier	SEQUENCE		Mandatory	This structure identifies which Public Key on the Device is to be used in checking the Command's cryptographic protection . The key is identified by way of Trust Anchor Cell and so the nature of the check, by way of the KeyUsage parameter, is also identified
trustAnchorCellRemotePartyRole	INTEGER	recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS	Mandatory if authorisingRemotePartyTACellIdentifier present	The role of the Party applying the Command's cryptographic protection

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		(5) , wanProvider (6)		
trustAnchorCellKeyUsage	BIT STRING	digitalSignature (0) , keyAgreement (4)	Mandatory if authorisingRemotePartyTA CellIdentifier present	Where the Command's cryptographic protection is a digital signature (digitalSignature) or a MAC (keyAgreement). The value shall be digitalSignature unless trustAnchorCellRemotePartyRole = accessControlBroker
trustAnchorCellUsage	INTEGER	management (0)	DEFAULT management	Must be absent since the prePaymentTopUp key pair cannot be used in relation to this command
remotePartyRolesCredentialsRequired	SEQUENCE OF			
RemotePartyRole	INTEGER	root (0) , recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS (5) ,	Mandatory to have at least one	List the Remote Party Role(s) for which credential details are required

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		wanProvider (6)		

4952 Table 13.2.4.1: Attribute values for CS02a Provide Security Credentials Command

4953 Command Cryptographic Protection

4954 If the Access Control Broker is undertaking this step to apply a MAC, then the Access Control Broker shall undertake the steps in Section
4955 13.2.4.2.1 otherwise:

4956 the Remote Party originating the Command shall generate a Signature for the Command and set KRP Signature accordingly;

4957 the Signature, for incorporation in the Command, shall only be generated once all fields of the Grouping Header ||

4958 @ProvideSecurityCredentialDetails.Command are populated as per the requirements for the Command Construction stage; and

4959 the Remote Party shall use its Private Digital Signing Key to generate the Signature.

4960 Access Control Broker MAC

4961 If the Access Control Broker is undertaking this step to apply a MAC, then the Access Control Broker shall calculate a MAC using the parameters
4962 in Table 13.2.4.2.1 and set ACB-SMD MAC to the value so calculated.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		

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Private Key Agreement Key	Access Control Broker's	
Public Key Agreement Key	Device's	As identified by the Business Target ID in Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header @ProvideSecurityCredentialDetails.Command 0x00	

4963 Table 13.2.4.2.1: Calculation of Access Control Broker MAC for CS02a Provide Security Credentials command

4964 Command Authenticity and Integrity Verification

4965 The Device shall undertake processing according to the requirements of this Section before undertaking any other processing of the Command.

4966 The checks should be carried out in the order specified. The Device shall cease checking at the point that any one check fails.

4967 The checks required are shown in Table 13.2.4.3.

Check Number	Criteria that shall be tested by the Device	How the Device shall test the Criteria
1.1	The Message is for the Device	The value in the Business Target ID field of the Message Identifier part of the Command instance must be equal to the Device's Entity Identifier

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1.2	The Message Code is for Provide Security Credentials	The value in the Message Code field of the Command instance must be equal to 0x0008
2.1	The Command was protected cryptographically using the Private Key corresponding to the Remote Party Public Key held in the Trust Anchor Cell identified by authorisingRemotePartyTACellIdentifier	As specified in Section 13.2.4.3.1

4968 Table 13.2.4.3: Provide Security Credentials Command authenticity and integrity verification

4969 Should any of the checks detailed in this Section 13.2.4.3 fail then the Device shall:

4970 generate an entry in the Security Log recording failed Authentication;

4971 discard the Command without execution and without sending a Response; and

4972 send an Alert notifying the failed Authentication, constructed as specified in Section 6.2.4.2, populated with the relevant Alert Code from
4973 Section 16 , to the Known Remote Party identified by the Security Credentials it holds in the {supplier, management,
4974 digitalSignature} Trust Anchor Cell.

4975 Where all of the checks detailed in this Section 13.2.4.3 succeed the Device shall process the Command and produce a Response.

4976 Command Authenticity and Integrity Verification

4977 The Device shall undertake the following checks until either all are successful or one has failed.

4978 1. If trustAnchorCellUsage is present this test shall fail.

4979 2. If trustAnchorCellKeyUsage = keyAgreement then

4980 ((trustAnchorCellRemotePartyRole = accessControlBroker) and (the MAC calculated by the Device according to Table 13.2.4.3.1
4981 equates to ACB-SMD MAC)

4982 else

4983 ((trustAnchorCellKeyUsage = digitalSignature) and (the Device shall use the Public Key in the Trust Anchor Cell identified by

4984 authorisingRemotePartyTACellIdentifier to verify that KRP Signature is the Digital Signature across Grouping Header ||

4985 @ProvideSecurityCredentialDetails.Command)

4986 else

4987 3. This test shall fail.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	
Public Key Agreement Key	Access Control Broker's	As held by the Device in Trust Anchor Cell {accessControlBroker, keyAgreement, management}
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header @ProvideSecurityCredentialDetails.Command 0x00	

4988 Table 13.2.4.3.1: Calculation of MAC for Provide Security Credential Details Command

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4989 Response Construction

4990 The Device shall populate Grouping Header according to the requirements of Section 7.2.6.

4991 The @ProvideSecurityCredentialDetails.Response shall have the structure defined in Section 13.2.3.3, and the Device shall populate with values
4992 according to Table 13.2.4.4.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@ProvideSecurityCredentialDetails.Response ::=	SEQUENCE OF			
SEQUENCE				
remotePartyRole	INTEGER	root (0) , recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS (5) , wanProvider (6) ,	Mandatory if SEQUENCE is present	The role to which the credentials in this SEQUENCE relate
statusCode	ENUMERATED	success (0) , trustAnchorNotFound (25) ,	Mandatory if SEQUENCE is present	Whether the Device can supply the details

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		other (127)		
currentSeqNumber	INTEGER	The corresponding Counter value	Present if statusCode=0	The Execution Counter held by the Device for this role's use of the Update Security Credentials Command. Where this role is root, the value of the anyByContingency Execution Counter shall be returned. Where this role is transitionalCoS, the value of the transCoSByTransCos Execution Counter shall be returned. Where this role is supplier, the value of the supplierBySupplier Execution Counter shall be returned.
trustAnchorCellsDetails	SEQUENCE OF		At least one in the SEQUENCE OF must be present if statusCode=0	
SEQUENCE				
trustAnchorCellKeyUsage	BIT STRING	digitalSignature (0) , keyAgreement (4) ,	Mandatory if SEQUENCE is	To what use can the public key in this Cell be put

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		keyCertSign (5)	present	
trustAnchorCellUsage	INTEGER	prePaymentTopUp(1)	DEFAULT management (0)	Only needs to be present for the {supplier, keyAgreement, prePaymentTopUp} Cell
existingSubjectUniqueID	OCTET STRING	Entity Identifier in this Cell	Mandatory if SEQUENCE is present	See Section 12.4 and Section 12.3
existingSubjectKeyIdentifier	OCTET STRING	Key Identifier of the key in this Cell	Mandatory if SEQUENCE is present	

4993 Table 13.2.4.4: Attribute values for Provide Security Credentials Response

4994 Response Cryptographic Protection

4995 If the Command that triggered this Response was protected by a MAC then the Device shall calculate a MAC using the parameters in Table

4996 13.2.4.5 and set SMD-KRP MAC to the value so calculated.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	

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Public Key Agreement Key	Access Control Broker's	As held by the Device in {accessControlBroker, keyAgreement, Management}
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header @ProvideSecurityCredentialDetails.Response 0x00	

4997 Table 13.2.4.5: Calculation of MAC for CS02a Provide Security Credential Details Response

4998 Otherwise:

4999 the Device creating the Response shall generate a Signature for the Response and set SMD Signature to the value calculated;

5000 the Signature, for incorporation in the Response, shall only be generated once all fields of the Grouping Header || Length ||

5001 @ProvideSecurityCredentialDetails.Response are populated, as per requirements for the Response Construction stage; and

5002 the Device shall use its Private Digital Signing Key to generate the Signature.

5003 Response Recipient Cryptographic Verification

5004 If the Response contains a MAC, the Access Control Broker can verify that MAC by calculating a MAC according to the parameters in Table

5005 13.2.4.6 and checking that the MAC so calculated equates to that in the Response.

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Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Access Control Broker's	
Public Key Agreement Key	Device's	As identified by Business Originator ID in the Message Identifier
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Grouping Header @ProvideSecurityCredentialDetails.Response 0x00	

- 5006 Table 13.2.4.6: Calculation of MAC for CS02a Provide Security Credential Details Response Verification
- 5007 CS02b Update Security Credentials Command, Response and Alert
- 5008 Description
- 5009 This Section 13.3 and Section 13.10 cover the creation, validation and processing of:
- 5010 Update Security Credentials Commands;
- 5011 Responses to such Commands; and
- 5012 Alerts resulting from the future dated execution of such Commands.

- 5013 The Update Security Credentials Commands shall be:
- 5014 used solely to replace Remote Party Security Credentials held in Trust Anchor Cells on Devices;
- 5015 supported by any Device that can process Remote Party Messages; and
- 5016 the only Command that Devices are capable of accepting for replacement of Remote Party Security Credentials, once the tamper seal is applied
- 5017 to the Device.
- 5018 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response / Alert
Message Type Category	Variant Message and is Critical
Capable of future dated invocation?	Yes (but see constraint in Table 13.3.5.1, check 1.3)
Protection Against Replay Required?	The Protection Against Replay mechanisms for Update Security Credentials are specified in Section 13.3. The Protection Against Replay mechanisms of other Sections of the GBCS do not apply
Service Reference	6.15, 6.23, 8.5, 6.21
Valid Target Device(s)	ESME / GSME / GPF / CHF / HCALCS / PPMID
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application	Supplier

Cross Reference	Value
Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Network Operator Access Control Broker Transitional Change of Supplier WAN Provider Recovery
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

5019 Table 13.3.2: Use Case Cross References for Update Security Credentials Command

5020 Command, Response and Alert Structure

5021 The Update Security Credentials Command

5022 This Section 13.3.3.1 summarises the structure of the Update Security Credentials Command, which depends on credentialsReplacementMode
5023 and the deviceType of the Device.

5024 If credentialsReplacementMode = anyByContingency or anyExceptAbnormalRootByRecovery then an Update Security Credential Details
5025 Command shall be the concatenation:

-
- 5026 Grouping Header || @UpdateSecurityCredentials.CommandPayload || 0x40 || KRP Signature
- 5027 If `credentialsReplacementMode = accessControlBrokerByACB` and `deviceType` is not
- 5028 `communicationsHubCommunicationsHubFunction` then an Update Security Credentials Command shall be the concatenation:
- 5029 MAC Header || Grouping Header || @UpdateSecurityCredentials.CommandPayload || 0x00 || ACB-SMD MAC
- 5030 In all other cases, the Update Security Credentials Command shall be the concatenation:
- 5031 MAC Header || Grouping Header || @UpdateSecurityCredentials.CommandPayload || 0x40 || KRP Signature || ACB-SMD MAC
- 5032 In these Command structures:
- 5033 @UpdateSecurityCredentials.CommandPayload shall be an octet string containing the DER encoding of the populated ASN.1 structure (as laid
- 5034 out in Section 13.3.5.11);
- 5035 Grouping Header shall be constructed as specified in Section 7.2.7 with Business Originator ID being the Entity Identifier of the Known Remote
- 5036 Party which generated KRP Signature, and with Business Originator Counter being that of the same Known Remote Party;
- 5037 KRP Signature shall be generated as specified in Section 6.3.3;
- 5038 ACB Grouping Header shall be constructed as specified in Section 7.2.7 with Business Originator ID being the Entity Identifier of the Access
- 5039 Control Broker and Business Originator Counter being that of the Access Control Broker;
- 5040 MAC Header shall be constructed as specified in Section 7.2.5; and
- 5041 ACB-SMD MAC shall be calculated as specified in Section 6.2.3.
- 5042 The Update Security Credentials Response
- 5043 An Update Security Credentials Response shall be the concatenation:

5044 Grouping Header || @UpdateSecurityCredentials.ResponsePayload || 0x40 || SMD Signature

5045 where:

5046 @UpdateSecurityCredentials.ResponsePayload shall be an octet string containing the DER encoding of the populated ASN.1 structure (as laid
5047 out in Section 13.3.5.11);

5048 Grouping Header in the Response shall be constructed as specified in Section 7.2.7 with Business Target ID being the Entity Identifier specified
5049 in the corresponding Command's Grouping Header; and

5050 SMD Signature shall be generated as specified in Section 6.3.5.

5051 The Update Security Credentials Alert

5052 An Update Security Credentials Alert shall be the concatenation:

5053 Grouping Header || @UpdateSecurityCredentials.AlertPayload || 0x40 || SMD Signature

5054 where:

5055 @UpdateSecurityCredentials.AlertPayload shall be an octet string containing the DER encoding of the populated ASN.1 structure (as laid out in
5056 Section 13.3.5.11);

5057 Grouping Header in the Alert shall be constructed as specified in Section 7.2.7 with Business Target ID being the Entity Identifier specified in the
5058 corresponding Command's Grouping Header;

5059 the Message Code being 0x00CB; and

5060 2. SMD Signature shall be generated as specified in Section 6.3.5.

5061 The Update Security Credentials Command, Response and Alert – informative

5062 The @UpdateSecurityCredentials.CommandPayload structure has four parts:

5063 authorisingRemotePartyControl: which includes details of what kind of credential replacement this Command is, which Remote Parties are

5064 authorising it and information to support Protection Against Replay protections;

5065 replacements: which is a list of new Certificates the Device is to store details from, along with which Trust Anchor Cell each set of details is to

5066 be stored in on the Device;

5067 certificationPathCertificates: which is a list of Certification Authority Certificates the Device will need to use in checking that the replacement

5068 Certificates were properly issued; and

5069 executionDateTime: which, if present, specifies the date-time at which the certificates in the CommandPayload are to be used to replace the

5070 credentials currently in use on the Device. If this field is not present, the Command shall be executed immediately. If this field has the value

5071 equivalent to 'never' (which is '99991231235959Z') the certificate replacement will never happen. This is to allow cancellation of future dated

5072 Commands. Note that future dating is not supported where certificates are being replaced in exception conditions.

5073 The @UpdateSecurityCredentials.Response structure contains, for immediate execution commands, a list detailing the success of failure of

5074 each of the replacements, including details of the parties affected. For future dated commands, @UpdateSecurityCredentials.AlertPayload

5075 structure contains the list detailing the success, or failure, of each of the replacements, including details of the parties affected.

5076 Section 13.3.5.11 contains narrative for each of the parts of these ASN.1 structures.

5077 Section 18.3.1 provides an illustrative instantiation of @UpdateSecurityCredentials.CommandPayload and its corresponding DER encoding.

5078 Updating Security Credentials on a Device – Processing Steps

5079 This Section lays out the requirements for the construction, protection and Authentication of the Update Security Credentials Command

5080 Payload, the processing required on the Device of the Command, the construction of the corresponding Response Payload and, where required,

5081 the Alert Payload.

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5082 Command Payload construction

5083 The @UpdateSecurityCredentials.CommandPayload shall have the structure defined in Section 13.3.5.11, and the Remote Party constructing
5084 the Command shall populate with values according to Table 13.3.4.1.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@ UpdateSecurityCredentials.Command ::=	SEQUENCE			
authorisingRemotePartyControl	SEQUENCE			This structure provides details to allow the Device to identify the Remote Party Role authorising this Command, check whether the rest of the payload is allowable and allow counters / counter caches on the Device to be reset, if the command changes the Remote Party in control
credentialsReplacementMode	INTEGER	supplierBySupplier (2) , networkOperatorByNetworkOperator (3) , accessControlBrokerByACB (4) , wanProviderByWanProvider (5) ,	Mandatory	Specify the replacement mode so that the Device can check that the Remote Party Role authorising the command is allowed to authorise this type of replacement(s) and that all replacements in the payload are allowed within this replacement mode. The structure of the label is <i>kindOfCertificate(s)BeingReplacedBypartydoingthereplacement</i> . For example,

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		transCoSByTransCoS (6) , supplierByTransCoS (7) , anyExceptAbnormalRootByRecovery (8) , anyByContingency (9)		supplierBySupplier is where a new supplier Certificate is being provided to the Device by its Supplier
plaintextSymmetricKey	[0] IMPLICIT OCTET STRING	The symmetric key that will decrypt the encrypted Contingency Key held on the Device	OPTIONAL	Only to be present if the Contingency Key arrangements are being used (so if credentialsReplacementMode = anyByContingency). The contents provide the symmetric key to decrypt the Contingency Public Key in the (root, keyCertSign, management) Trust Anchor Cell
applyTimeBasedCPVChecks	[1] IMPLICIT INTEGER	disapply(1)	DEFAULT apply	Only to be present if the Remote Party sending the Command is instructing the Device not to apply time based checks as part of Certification Path Validation of replacements. This should only be in exceptional circumstances.

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
authorisingRemotePartyTACellIdentifier	[2] IMPLICIT SEQUENCE		OPTIONAL	This structure identifies which Public Key on the Device is to be used in verifying KRP Signature. The key is identified by way of Trust Anchor Cell and so the nature of the check, by way of the KeyUsage parameter, is also identified. 'authorisingRemotePartyTACellIdentifier' can only be omitted when the Access Control Broker is changing its own Key Agreement credentials
trustAnchorCellRemotePartyRole	INTEGER	root (0) , recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS (5) , wanProvider (6)	Mandatory if authorisingRemotePartyTACellIdentifier present	The role of the Party applying KRP Signature. Note that where root is used, this refers only to the encrypted Contingency key in the root TA Cell, so is only valid if credentialsReplacementMode = anyByContingency and plaintextSymmetricKey is populated with the symmetric key required to decrypt that public key

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
trustAnchorCellKeyUsage	BIT STRING	digitalSignature (0) if credentialsReplacementMode <> anyByContingency, keyCertSign (5) if credentialsReplacementMode = anyByContingency	Mandatory if authorisingRemotePartyTACellIdentifier present	KRP Signature is a digital signature
trustAnchorCellUsage	INTEGER	management (0)	DEFAULT management	Must be absent since the prePaymentTopUp key pair cannot be used in relation to this Command
authorisingRemotePartySeqNumber	[3] IMPLICIT INTEGER	Originator Counter of Remote Party authorising the Command	Mandatory	Specify the Originator Counter for the Remote Party applying KRP Signature, or (for the Access Control Broker changing its credentials) the Access Control Broker's Originator Counter
newRemotePartyFloorSeqNumber	[4] IMPLICIT INTEGER	Originator Counter of Remote Party who will have control of this Remote Party Role if the update is	OPTIONAL	If the Command is to effect a change of control, then newRemotePartyFloorSeqNumber should be included and will be the value

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		successful		used to prevent replay of Update Security Credentials Commands, and other Commands, for the new controlling Remote Party
newRemotePartySpecialistFloorSeqNumber	[5] IMPLICIT SEQUENCE OF		OPTIONAL	Some Commands on the Device may use a different Originator Counter sequence for Protection Against Replay. The only example is the Prepayment Top Up Command on ESME and GSME. The SpecialistSeqNumber structure allows such Counters to also be reset on change of control. Should only be present if this Command changes supplier credentials and the new supplier uses different counters for its Prepayment Top Ups than it does for other Commands
SEQUENCE				
seqNumberUsage	INTEGER	prepaymentTopUp (0)	Mandatory if newRemotePartySpecialistFloorSeqNumber present	Specify the usage of the SeqNumber

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
seqNumber	INTEGER	Relevant Originator Counter	OPTIONAL	Specify the associated SeqNumber
otherRemotePartySeqNumberChanges	[6] IMPLICIT SEQUENCE OF		OPTIONAL	In some cases, one party acting in one Remote Party Role may be replacing certificates for a different Remote Party Role (e.g. transitionalCoS changing Supplier Credentials). In such cases, Execution Counters need also to be reset for that other Remote Party Role
SEQUENCE				
otherRemotePartyRole	INTEGER	supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS (5) , wanProvider (6) ,	Mandatory if otherRemotePartySeqNumberChanges present	The Remote Party Role of the party whose credentials are being placed on the Device but which did not authorise the command directly. Note that this is not valid for root or recovery

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
otherRemotePartyFloorSeqNumber	INTEGER	Relevant Originator Counter	Mandatory if otherRemotePartySeqNumberChanges present	Specify the associated SeqNumber
newRemotePartySpecialistFloorSeqNumber	SEQUENCE OF		OPTIONAL	Should only be present if otherRemotePartyRole = supplier, and that new supplier uses different counters to prevent replay on Prepayment Top Up
SEQUENCE				
seqNumberUsage	INTEGER	prepaymentTopUp (0)	Mandatory if newRemotePartySpecialistFloorSeqNumber present	Specify the usage of the SeqNumber
seqNumber	INTEGER	Relevant Originator Counter	OPTIONAL	Specify the associated SeqNumber
replacements	SEQUENCE OF			Provide a list of replacement Certificates. Each entry in replacements contains a new end entity Certificate and the identity of the Trust Anchor Cell which is to have its contents replaced

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
				using that Certificate.
SEQUENCE			At least one SEQUENCE must be present	One structure is required for each Trust Anchor Cell that is to be updated
replacementCertificate	Certificate	End entity Certificate	Mandatory if SEQUENCE is present	Provide the new end entity certificate
targetTrustAnchorCell	SEQUENCE			Specify where it is to go (specifically which Trust Anchor Cell is to have its details replaced using the new end entity certificate)
trustAnchorCellRemotePartyRole	INTEGER	root (0) , recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS	Mandatory if SEQUENCE is present	To which Remote Party Role does the Trust Anchor Cell relate

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		(5) , wanProvider (6)		
trustAnchorCellKeyUsage	BIT STRING	{digitalSignature (0) , keyAgreement (4) , keyCertSign (5)} ,	Mandatory if SEQUENCE is present	To what use can the public key in this Cell be put
trustAnchorCellUsage	INTEGER	prePaymentTopUp (1) }	DEFAULT management	Should be absent unless: the deviceType is eSME or gSME; and the Supplier operating the Device wishes to use Prepayment Top Up functionality on the Device, and this is a replacement of the corresponding Certificate. Note the Certificate specified for use in the {supplier, keyAgreement, prePaymentTopUp} Trust Anchor Cell may be the same key as that specified for the {supplier, keyAgreement, management} Trust Anchor Cell or may be different.
certificationPathCertificates	SEQUENCE OF Certificate	The list of certificates needed for Certification	At least one Certificate must be	Provide the Certification Authority Certificates needed to undertake

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		Path Validation	present	Certification Path Validation against the root public key held on the Device. The number of these may be less than the number of replacements (e.g. a Supplier may replace all of its Certificates but may only need to supply one Certification Authority Certificate to link them all back to root
executionDateTime	Generalized Time	The date-time at which the replacements are to be used in updating the Device's Security Credentials	OPTIONAL	This field may only be present if credentialsReplacementMode is either supplierBySupplier or supplierByTransCoS

5085 Table 13.3.4.1: Attribute values for Update Security Credentials Command

5086 Command Authenticity and Integrity Verification

5087 The Device shall undertake processing according to the requirements of Section 13.3.5.1.

5088 Should any of the checks detailed in Section 13.3.5.1 fail then the Device shall:

5089 generate an entry in the Security Log recording failed Authentication;

5090 discard the Command without execution and without sending a Response; and

5091 send an Alert notifying the failed Authentication, constructed as specified in Section 6.2.4.2 of the GBCS, populated with the relevant Alert
5092 Code according to Section 16, to the Known Remote Party identified by:

5093 the Trust Anchor Cell {supplier, digitalSignature, management} if the Device's deviceType is not
5094 communicationsHubCommunicationsHubFunction; or

5095 the Trust Anchor Cell {wanProvider, digitalSignature, management} if the Device's deviceType is
5096 communicationsHubCommunicationsHubFunction.

5097 Where all of the checks detailed in Section 13.3.5.1 succeed the Device shall process the Command and produce a Response.

5098 Command Processing

5099 Before undertaking any further processing, the Device shall update Execution Counter to the value of authorisingRemotePartySeqNumber.

5100 If `executionDateTime` is present then the Device shall:

5101 record against the remotePartyRole (as specified in `authorisingRemotePartyControl`), `authorisingRemotePartyControl`, replacements; and
5102 `executionDateTime`;

5103 construct a Response where `executionOutcome` is not present according to the requirements of Section 13.3.4.4; and

5104 at the date-time specified in `executionDateTime`, or immediately if the `executionDateTime` is in the past undertake the processing of Section
5105 13.3.4.3.1 then construct an Alert according to the requirements of Section 13.3.4.5.

5106 If `executionDateTime` is not present then the Device shall:

5107 undertake the processing of Section 13.3.4.3.1; and

5108 construct a Response where `executionOutcome` is present according to the requirements of Section 13.3.4.4.

5109 replacements Processing

-
- 5110 For each of the `targetTrustAnchorCell` in replacements, the Device shall:
- 5111 record the `entityIdentifier` and `subjectKeyIdentifier` currently held in that `targetTrustAnchorCell`;
- 5112 attempt to replace the contents of that `targetTrustAnchorCell` using the corresponding certificate in `TrustAnchorReplacement`; and
- 5113 if the contents of the replacement are successfully applied, undertake the processing required by Section 13.3.5.10 in relation to the
- 5114 `RemotePartyRole` for that `targetTrustAnchorCell`.
- 5115 For clarity, before the contents of a certificate are used to replace the contents of a `targetTrustAnchorCell`, the Device shall have successfully
- 5116 undertaken the checks in Section 13.3.5.9 in relation to that certificate. Should a check in 13.3.5.9 not succeed then, in line with Section
- 5117 4.3.2.8.2, the Device shall undertake no further processing of the Command, except for the issuance of a Message containing an
- 5118 `executionOutcome`.
- 5119 Response Construction
- 5120 The `@UpdateSecurityCredentials.ResponsePayload` shall have the structure defined in Section 13.3.5.11, and the Device shall populate the
- 5121 `executionOutcome`, where present with values according to Table 13.3.4.6.
- 5122 Alert Construction
- 5123 The `@UpdateSecurityCredentials.AlertPayload` shall have the structure defined in Section 13.3.4, and the Device shall populate the
- 5124 `executionOutcome`, with values according to Table 13.3.4.6.
- 5125 `executionOutcome` construction

SEC Schedule 8 – GBCS v4.4

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
executionOutcome	SEQUENCE			
authorisingRemotePartySeqNumber	INTEGER	Originator Counter of Remote Party authorising the Command, as specified in the corresponding Command	Mandatory	This is to allow the Alert to be linked to the Command that caused execution
credentialsReplacementMode	INTEGER	supplierBySupplier (2) , networkOperatorByNetworkOperator (3) , accessControlBrokerByACB (4) , wanProviderByWanProvider (5) , transCoSByTransCoS (6) , supplierByTransCoS (7) , anyExceptAbnormalRootByRecovery (8) , anyByContingency	Mandatory	Provide details of the corresponding Command that are not in the standard GBCS message header. Specifically the mode in which the Command was invoked

SEC Schedule 8 – GBCS v4.4

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		(9) } ,		
remotePartySeqNumberChanges	SEQUENCE OF		Mandatory containing zero, one or many occurrences of the following structure	The resulting changes to any Execution Counter held on the Device
SEQUENCE				
otherRemotePartyRole	INTEGER	root (0) , recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS (5) , wanProvider (6) ,	Mandatory if SEQUENCE is present	The role which has had its Execution Counter values changed on the Device

SEC Schedule 8 – GBCS v4.4

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
otherRemotePartyFloorSeqNumber	INTEGER	The corresponding Execution Counter value	Mandatory if SEQUENCE is present	
newRemotePartySpecialistFloorSeqNumber	SEQUENCE OF		OPTIONAL	Only present where Remote Party Role is supplier
SEQUENCE				
seqNumberUsage	INTEGER	{prepaymentTopUp(0)} ,	Mandatory if newRemotePartySpecialistFloorSeqNumber present	Specify the usage of the SeqNumber
seqNumber	INTEGER		Mandatory if newRemotePartySpecialistFloorSeqNumber present	Specify the associated SeqNumber
replacementOutcomes	SEQUENCE OF		One per replacement in the corresponding Command so at least one	For each replacement in the Command, detail the outcome and impacted parties

SEC Schedule 8 – GBCS v4.4

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
SEQUENCE				
affectedTrustAnchorCell	SEQUENCE		Mandatory if SEQUENCE is present	Specify which Trust Anchor Cell was the target of this replacement
trustAnchorCellRemotePartyRole	INTEGER	root (0) , recovery (1) , supplier (2) , networkOperator (3) , accessControlBroker (4) , transitionalCoS (5) , wanProvider (6)	Mandatory if SEQUENCE is present	Specify the Remote Party Role to which the Trust Anchor Cell relates
trustAnchorCellKeyUsage	BIT STRING	digitalSignature (0) , keyAgreement (4) , keyCertSign (5)	Mandatory if SEQUENCE is present	To what use can the public key in this Cell be put

SEC Schedule 8 – GBCS v4.4

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
trustAnchorCellUsage	INTEGER	{management (0) , prePaymentTopUp (1) }	DEFAULT management	Absent unless: the deviceType is eSME or gSME; and the supplier operating the Device wishes to use Prepayment Top Up functionality on the Device, and this is a replacement of the corresponding certificate. Note the certificate specified for use in the {supplier, keyAgreement, prePaymentTopUp} Trust Anchor Cell may be the same key as that specified for the {supplier, keyAgreement, management} Trust Anchor Cell or may be different.
statusCode	ENUMERATED	success (0) , badCertificate (5) , noTrustAnchor (10) , insufficientMemory (17) , resourcesBusy (30) , other (127)	Mandatory if SEQUENCE is present	Whether the replacement to this Cell was successful or, if it failed, why it failed
existingSubjectUniqueID	OCTET STRING		Mandatory if SEQUENCE is present	The 64 bit Entity Identifier of the Remote Party whose credentials were in this Cell prior to receipt of the corresponding Command

SEC Schedule 8 – GBCS v4.4

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
existingSubjectKeyIdentifier	OCTET STRING		Mandatory if SEQUENCE is present	For the public key in this Cell prior to receipt of the corresponding Command
replacingSubjectUniqueID	OCTET STRING		Mandatory if SEQUENCE is present	The 64 bit Entity Identifier of the Remote Party whose credentials were to be placed in this Cell
replacingSubjectKeyIdentifier	OCTET STRING		Mandatory if SEQUENCE is present	For the public key which was to be placed in this Cell

5126 Table 13.3.4.6: Attribute values for executionOutcome

5127 Common Requirements

5128 Update Security Credentials Command Verification

5129 The Device shall undertake the checks set out in this Section 13.3.5.1 before undertaking any other processing of the Command. The checks
 5130 may be carried out in any order. Checking shall cease at the point that any one check fails. The checks required are shown in Table 13.3.5.1.

Check Number	Criteria that must be tested by the Device	How the Device shall test the Criteria
1.1	The Message is for the Device	The value of the Business Target ID in the Grouping Header

SEC Schedule 8 – GBCS v4.4

Check Number	Criteria that must be tested by the Device	How the Device shall test the Criteria
		in Command instance must be equal to the Device's Entity Identifier
1.2	The Message Code is for Update Security Credentials	The value in the Message Code field of the Grouping Header must be equal to the value specified in Table 13.3.5.2 for the CredentialsReplacementMode specified in CommandPayload
1.3	If executionDateTime is present the Command is to replace Supplier Security Credentials	If executionDateTime is present then credentialsReplacementMode must either supplierBySupplier or supplierByTransCoS
1.4	The Device has not already actioned this Command	As specified in Section 13.3.5.3
2.1	The targetTrustAnchorCells all exist on a Device of this type	As specified in Section 13.3.5.4
2.1	The trustAnchorCellUsage in the authorisingRemotePartyTACellIdentifier must be ABSENT	As specified in Section 13.3.4.1
2.2	The credentialsReplacementMode is one that can be Authorised by the Remote Party / Parties authorising the Command	As specified in Section 13.3.5.5
2.2	The replacements specified are all allowed in this	As specified in Section 13.3.5.6

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Check Number	Criteria that must be tested by the Device	How the Device shall test the Criteria
	credentialsReplacementMode	
2.2	The TrustAnchorCellUsage in the targetTrustAnchorCell in each entry in replacements must be ABSENT unless it has the value prePaymentTopUp as specified in the Notes column of Table 13.3.4.1	As specified in Section 13.3.4.1
2.3	The keyUsage in each of the replacement certificates provided is consistent with the target Trust Anchor Cells identified in replacements	As specified in Section 13.3.5.7
3.1	The Cryptographic Protections are valid	As specified in Section 13.3.5.8

5131 Table 13.3.5.1: Update Security Credentials Command authenticity and integrity verification

5132 Message Code Validation

CredentialsReplacementMode	Message Code
supplierBySupplier	0x0102
networkOperatorByNetworkOperator	0x0103
accessControlBrokerByACB	0x0104
wanProviderByWanProvider	0x0105

CredentialsReplacementMode	Message Code
transCoSByTransCoS	0x0106
supplierByTransCoS	0x0107
anyExceptAbnormalRootByRecovery	0x0108
anyByContingency	0x0109

5133 Table 13.3.5.2: Message Code validation against CredentialsReplacementMode

5134 Preventing Replay of Commands

5135 The Protection Against Replay mechanisms for the Update Security Credentials Command shall be that specified in this Section 13.3.5.3 (which
5136 is different than that for other GBCS Commands).

5137 For each of RemotePartyRole from which the Device can receive a valid Updated Security Credentials Command, the Device shall allocate
5138 storage for an Execution Counter which shall be capable of storing a 64 bit unsigned integer and which shall initially be set to the value zero at
5139 manufacture. For transitionalCoS, the Device may allocate storage for an additional Execution Counter, credentialsReplacementModes.

5140 Before executing any Update Security Credentials Command, a Device shall confirm that, if CredentialsReplacementMode <>
5141 accessControlBrokerByACB, then

5142 (authorisingRemotePartyTACellIdentifier is populated in the Command) and (the authorisingRemotePartySeqNumber is strictly numerically
5143 greater than the Execution Counter the Device has recorded for the RemotePartyRole identified in authorisingRemotePartyTACellIdentifier and,
5144 where relevant for transitionalCoS, the credentialsReplacementMode);

5145 else

5146 (the authorisingRemotePartySeqNumber is strictly numerically greater than the Execution Counter the Device has recorded for the
5147 accessControlBroker).

5148 Required Trust Anchor Cells by Device Type

5149 The Trust Anchor Cells specified in Section 4.3.2.5 are those required on each Device type and so are the only valid targetTrustAnchorCells.

5150 The Device shall ensure that all targetTrustAnchorCells specified in the Command instance are valid for the type of Device it is, according to the
5151 requirements of Section 4.3.2.5. A Command containing any invalid targetTrustAnchorCells shall not be processed any further by the Device.

5152 Valid credentialsReplacementMode by Remote Party Roles authorising the Command

5153 A Command containing a certain credentialsReplacementMode is only Authorised using certain types of Public-Private Key Pairs in certain ways.

5154 The Command identifies the Public Key corresponding to the Private Key used by the authorising Remote Party in the
5155 authorisingRemotePartyTACellIdentifier structure. Table 13.3.5.5 lists the only Authorised combinations. All other combinations represent
5156 Commands not properly Authorised and shall be rejected by a Device.

5157

	authorisingRemotePartyTACellIdentifier		
	RemotePartyRole	KeyUsage	CellUsage
credentialsReplacementMode			
supplierBySupplier	supplier	digitalSignature	management
networkOperatorByNetworkOperator	networkOperator	digitalSignature	management
accessControlBrokerByACB: if authorisingRemotePartyTACellIdentifier is present it must refer to the specified Trust Anchor Cell. If absent and deviceType is not communicationsHubCommunicationsHubFunction, then the mode of	accessControlBroker	digitalSignature	management

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	authorisingRemotePartyTACellIdentifier		
	RemotePartyRole	KeyUsage	CellUsage
credentialsReplacementMode			
accessControlBrokerByACB is valid, otherwise the mode is invalid			
wanProviderByWanProvider	wanProvider	digitalSignature	management
transCoSByTransCoS	transitionalCoS	digitalSignature	management
supplierByTransCoS	transitionalCoS	digitalSignature	management
anyExceptAbnormalRootByRecovery	recovery	digitalSignature	management
anyByContingency	root	keyCertSign	management

5158 Table 13.3.5.5: Valid credentialsReplacementMode by Remote Party Roles authorising the Command

5159 Valid credentialsReplacementMode by the targetTrustAnchorCells specified in the Command

5160 A Command containing a certain credentialsReplacementMode can only validly replace the Security Credentials in a certain subset of Trust
5161 Anchor Cells. The Command identifies the Cells that are to have credentials replaced in each targetTrustAnchorCell within each
5162 TrustAnchorReplacement in replacements.

5163 Table 13.3.5.6 below lists the only valid targetTrustAnchorCell combinations for each credentialsReplacementMode. All other combinations are
5164 invalid. A Command containing any invalid combinations shall not be processed any further by the Device.

	targetTrustAnchorCell		
	RemotePartyRole	KeyUsage	CellUsage
credentialsReplacementMode			

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	targetTrustAnchorCell		
	RemotePartyRole	KeyUsage	CellUsage
credentialsReplacementMode			
supplierBySupplier	supplier	any valid for GBCS	any valid for GBCS
networkOperatorByNetworkOperator	networkOperator	any valid for GBCS	any valid for GBCS
accessControlBrokerByACB	accessControlBroker	any valid for GBCS	any valid for GBCS
wanProviderByWanProvider	wanProvider	any valid for GBCS	any valid for GBCS
transCoSByTransCoS	transitionalCoS	digitalSignature	management
supplierByTransCoS	supplier	any valid for GBCS	any valid for GBCS
anyExceptAbnormalRootByRecovery	any valid for GBCS	any valid for GBCS	any valid for GBCS
anyByContingency	any valid for GBCS	any valid for GBCS	any valid for GBCS

5165 Table 13.3.5.6: Valid credentialsReplacementMode by the targetTrustAnchorCells specified in the Command

5166 Valid usage of Certificates against the targetTrustAnchorCells specified in the Command

5167 [Note: Each of the ‘end entity’ Certificates in the Command must have the same keyUsage as the Trust Anchor Cell it is to be applied to.]

5168 For each instance of the TrustAnchorReplacement structure in the Command, the keyUsage in replacementCertificate shall be equal to
5169 targetTrustAnchorCell.KeyUsage. Where this check fails for any one or more of the TrustAnchorReplacement instances, the Command shall not
5170 be actioned by the Device.

5171 [Note: Save for supplier and network operator roles, each of the ‘end entity’ Certificates in the Command must have the same
5172 RemotePartyRole as the Trust Anchor Cell it is to be applied to.]

5173 For each instance of the TrustAnchorReplacement structure in the Command where (targetTrustAnchorCell.RemotePartyRole <> supplier) and
5174 (targetTrustAnchorCell.RemotePartyRole <> networkOperator), the RemotePartyRole in replacementCertificate shall be equal to
5175 targetTrustAnchorCell.RemotePartyRole. Where this check fails for any one or more of the TrustAnchorReplacement instances, the Command
5176 shall not be actioned by the Device.

5177 Notes:

5178 mismatches between RemotePartyRole in the certificate and the target Trust Anchor Cell are admissible, in relation to CS02b Commands, for
5179 supplier and networkOperator only, and are needed (see Section 4.3.2.5); and

5180 CellUsage is simply a selector to allow a different Key Agreement key pair to be used for Prepayment Top Ups. However, that use of a different
5181 Key Pair is not mandated and so validation is not required; any valid supplier Key Agreement certificate may be used in this Trust Anchor Cell.

5182 Verifying the CryptographicProtections

5183 In verifying Cryptographic Protections pursuant to this Section 13.3.5.8:

5184 KRP Signature shall, where required by Section 13.3.3.1 for the specified Message Code and credentialsReplacementMode, be verified
5185 according to the requirements in this Section 13.3.5.8; and

5186 ACB-SMD MAC, where required by Section 13.3.3.1 for the specified Message Code and credentialsReplacementMode, shall be verified
5187 according to the requirements in Section 6.2.4.1.2.

5188 If credentialsReplacementMode = anyByContingency or Message Code = 0x0109 then KRP Signature shall be verified using the public key
5189 established according to the requirements of Section 13.3.5.8.1.

5190 If credentialsReplacementMode = <> anyByContingency or Message Code =<> 0x0109 then KRP Signature shall be verified using the public key
5191 identified as per Section 4.3.2.7.2.

5192 If `credentialsReplacementMode` = `accessControlBrokerByACB` or Message Code = 0x0104 and deviceType is not
5193 `communicationsHubCommunicationsHubFunction` then ACB-SMD MAC shall be verified as per Section 6.2.4.1.2.

5194 Decrypting the contingency public key and verifying Authorising Remote Party's digital signature against that decrypted key

5195 The Device shall decrypt the Contingency Key that it holds in Trust Anchor Cell {root, keyCertSign, management} by undertaking Decryption
5196 using the following parameters:

5197 setting Ciphertext to be encrypted value of the Contingency Key;

5198 setting Additional Authenticated Data to be 0x31;

5199 setting the Initialization Vector to be 0xFFFFFFFF0000000000000000; and

5200 setting the shared symmetric key to be the value in `plaintextSymmetricKey`.

5201 Where Decryption is successful, the Device shall use the Plaintext produced as the Public Key to verify KRP Signature according to the
5202 requirements at Section 6.3.4.

5203 The Contingency Key shall have been Encrypted accordingly.

5204 Verifying the authenticity of replacement certificates

5205 For each `replacementCertificate` in `replacements`, the Device shall first apply the requirements of Section 12.6 (Device processing
5206 of Certificates). If any of those checks fail, the Section 13.3.5.9 check fails.

5207 Where Certification Path Validation is required by this Section 13.3.5.9, the application of time based checks shall be determined as follows:

5208 if, in the Command, `applyTimeBasedCPVChecks` = `disapply` then time based checks shall NOT be applied by the Device; and

5209 otherwise time based checks shall be applied or not applied in line with the requirements of Section 4.3.2.8.

5210 If (credentialsReplacementMode <> anyByContingency) and (replacements does NOT include a targetTrustAnchorCell of {root, keyCertSign,
5211 management}) then the Device shall, for each replacementCertificate in replacements, undertake Certification Path Validation according to the
5212 requirements of Section 4.3.2.8.

5213 If (credentialsReplacementMode = anyExceptAbnormalRootByRecovery) and (replacements does include a targetTrustAnchorCell of {root,
5214 keyCertSign, management}) then the Device shall:

5215 first undertake the checks at Section 13.3.5.9.1 in relation to the root Certificates; and

5216 then shall, for each of the other replacementCertificate in replacements, undertake Certification Path Validation according to the requirements
5217 of Section 4.3.2.8. In so doing the Device shall use the details from the replacementCertificate in replacements specified for updating {root,
5218 keyCertSign, management} as the root for Certification Path Validation.

5219 If (credentialsReplacementMode = anyByContingency) and (replacements does include a targetTrustAnchorCell of {root, keyCertSign,
5220 management}) then the Device shall, for each of the other replacementCertificate in replacements, undertake Certification Path Validation
5221 according to the requirements of Section 4.3.2.8. In so doing the Device shall use the details from the replacementCertificate in replacements
5222 specified for updating {root, keyCertSign, management} as the root for Certification Path Validation.

5223 If any of the above checks fail, the Section 13.3.5.9 check fails.

5224 Validation of new root Certificate against current root Security Credentials

5225 The Device shall:

5226 identify the Certificate in replacements that corresponds to the targetTrustAnchorCell of {root, keyCertSign, management}. The Certificate shall
5227 be referred to as NewWithNew; then

5228 identify the Certificate in certificationPathCertificates that has the same subjectPublicKey as the NewWithNew Certificate and where
5229 RemotePartyRole in the X520OrganisationalUnitName field = root. The Certificate shall be referred to as NewWithOld. If no such Certificate is
5230 found, the Section 13.3.5.9.1 check fails else:

5231 use the Public Key in the {root, keyCertSign, management} Trust Anchor Cell to verify the digital signature in NewWithOld. If the digital
5232 signature verification fails, the Section 13.3.5.9.1 check fails else:

5233 use the Public Key in NewWithNew to verify the digital signature in NewWithNew. If the digital signature verification fails, the Section
5234 13.3.5.9.1 check fails.

5235 Required Processing on Change of Remote Party Control

5236 If:

5237 the targetTrustAnchorCell is {supplier, digitalSignature, management}; and

5238 the Entity Identifier in the targetTrustAnchorCell is changed by the replacement; and

5239 the Device is an ESME or a GSME,

5240 then the Device shall:

5241 set the Supplier Name which it displays to the Common Name in the subject field of the certificate that was used to populate the
5242 targetTrustAnchorCell; and

5243 add an entry in the Billing Data Log with a snapshot cause of 0x00002000 (Change of Supplier) (with the entry added having the same content
5244 as is required on Set Payment Mode Or Tariff change);

5245 reset the Tariff Block Counter Matrix; and

5246 where the Device is an ESME, end any currently active Boost Period, ALCS [n] Setting Period, APC [n] Limit Period, APC [n] Setting Period and
5247 HCALCS [n] Setting Period, where those terms have their SMETS meanings.

5248 If the targetTrustAnchorCell is {root, keyCertSign, management} and there are any future dated Update Security Credentials or Activate
5249 Firmware Commands held on the Device that have not yet executed, and so the `executionDateTime` is in the future, then the Device shall
5250 set each `executionDateTime` to '99991231235959Z'.

5251 If the Device is an ESME, GSME or HCALCS, the targetTrustAnchorCell is {supplier, digitalSignature, management} and there are any future
5252 dated Update Security Credentials or Activate Firmware Commands held on the Device that have not yet executed, and so their
5253 `executionDateTime` is in the future, then the Device shall set each such `executionDateTime` to '99991231235959Z'.

5254 If the Device is a CHF, the targetTrustAnchorCell is {WANProvider, digitalSignature, management} and there is any future dated Activate
5255 Firmware Command held on the Device that have not yet executed, and so its `executionDateTime` is in the future, then the Device shall
5256 set each such `executionDateTime` to '99991231235959Z'.

5257 If the Device is a GPF, the targetTrustAnchorCell is {supplier, digitalSignature, management} and there are any future dated Update Security
5258 Credentials Commands held on the Device that have not yet executed, and so their `executionDateTime` is in the future, then the Device
5259 shall set each such `executionDateTime` to '99991231235959Z'.

5260 If:

5261 the targetTrustAnchorCell is {supplier, digitalSignature, management}; and

5262 the Entity Identifier in the targetTrustAnchorCell is changed by the replacement

5263 then the Device shall set the execution date-time of any other future dated Commands, that are held on the Device but not yet executed, to
5264 'never', as detailed in Section 9.2. If the `deviceType` of the Device is `gSME` then the Device shall also delete any future dated data items
5265 that are pending activation.

5266 Relevant Execution Counters' shall be those which relate to types of Commands:

5267 which originate from the `remotePartyRole` of this targetTrustAnchorCell; and

5268 whose cryptographic protection is checked using the Public Key in this `targetTrustAnchorCell`.

5269 Where this Update Security Credentials Command contains an `otherRemotePartySeqNumberChange` in which

5270 `otherRemotePartyRole` is the `remotePartyRole` of this `targetTrustAnchorCell`, the Device shall set all Relevant Execution

5271 Counters to be the corresponding value in `otherRemotePartyFloorSeqNumber`. For clarity, this applies to the `loadController`

5272 `remotePartyRole` when executing a `loadControllerBySupplier CS02g Update Security Credentials Command`.

5273 If:

5274 `remotePartyRole` of `targetTrustAnchorCell` and that of `authorisingRemotePartyControl` is `supplier`; and

5275 `keyUsage` of `targetTrustAnchorCell` is `digitalSignature`

5276 then the Device shall set all Relevant Execution Counters to the value in `newRemotePartyFloorSeqNumber`. For clarity, this applies to the

5277 Execution Counter for the `loadControllerBySupplier CS02g Update Security Credentials Command`.

5278 If the `deviceType` is `eSME` or `gSME` and (`targetTrustAnchorCell` is `{supplier, digitalSignature, management}`), the

5279 Device shall:

5280 clear all values from the UTRN Counter Cache; and

5281 place a single value in the UTRN Counter Cache. If `newRemotePartySpecialistFloorSeqNumber` is present and the

5282 `seqNumberUsage` in that `newRemotePartySpecialistFloorSeqNumber` is `prepaymentTopUp` then that value shall be the 32

5283 most significant bits of the `seqNumber` in the `newRemotePartySpecialistFloorSeqNumber`. Otherwise the value shall be the 32

5284 most significant bits of whichever of the `newRemotePartyFloorSeqNumber` or the `otherRemotePartyFloorSeqNumber` fields

5285 present in the Command.

5286 Note, where `SeqNumber` values are to be used to populate Execution Counters on the Device, these `SeqNumber` values shall be processed

5287 as unsigned 64 bit integers, so 8 octets in length. This is because Execution Counters are required to use this representation. Correspondingly,

5288 in relation to values in the UTRN Counter Cache (so UTRN Counters), references to the 32 most significant bits of `SeqNumbers` are references
5289 to the 32 most significant bits when the value is represented as a 64 bit unsigned integer.

5290 The `@UpdateSecurityCredentials.CommandPayload`, `@UpdateSecurityCredentials.ResponsePayload` and
5291 `@UpdateSecurityCredentials.AlertPayload` structure definition

5292 Each instance of `@UpdateSecurityCredentials.CommandPayload`, `@UpdateSecurityCredentials.ResponsePayload` and of
5293 `@UpdateSecurityCredentials.AlertPayload` shall be an octet string containing the DER encoding of the populated structure defined in this
5294 Section 13.3.4, which specifies the structure in ASN.1.

5295 The structure of `Certificate` shall be as defined in ASN.1 in IETF RFC 5912. Note that the `Certificate` structures within IETF RFC 5912 begin
5296 after the phrase ‘Certificate- and CRL-specific structures begin here’.

5297 `UpdateSecurityCredentials DEFINITIONS ::= BEGIN`

5298

5299 `CommandPayload ::= SEQUENCE`

5300 `{`

5301 `-- Provide details to allow the Device to identify the Remote Party Role authorising`

5302 `-- this Command, check whether the rest of the payload is allowable, prevent replay attacks`

5303 `-- and allow counters / counter caches on the Device to be reset, if the Command changes the Remote Party`

5304 `-- in control.`

5305 `-- The Remote Party authorising the Command is that party which generated the KRP Signature (or the Access Control Broker`

5306 `-- if there is no KRP Signature)`

5307

5308 authorisingRemotePartyControl AuthorisingRemotePartyControl,

5309

5310 -- One TrustAnchorReplacement structure is required for each Trust Anchor Cell that is to be updated

5311

5312 replacements SEQUENCE OF TrustAnchorReplacement,

5313

5314 -- Provide the certificates needed to undertake Certification Path Validation of the new

5315 -- end entity certificate against the root public key held on the Device. The number of these may be less

5316 -- than the number of replacement certificates (e.g. a supplier may replace all of its certificates but

5317 -- may only need to supply one Certification Authority Certificate to link them all back to the root public

5318 -- key as currently stored on the Device.

5319

5320 certificationPathCertificates SEQUENCE OF Certificate,

5321

5322 -- If the Command is to be future dated, specify the date-time at which the certificate replacement is to happen

5323

5324 executionDateTime GeneralizedTime OPTIONAL
5325
5326 }
5327
5328 ResponsePayload ::= SEQUENCE
5329 {
5330 -- if the Command is future dated, the Response will not have any details of execution (those will be in the subsequent alert)
5331
5332 commandAccepted NULL,
5333
5334 -- if the Command is for immediate execution, the Response will detail the outcomes
5335
5336 executionOutcome ExecutionOutcome OPTIONAL
5337
5338 }
5339
5340 AlertPayload ::= SEQUENCE

```
5341 {
5342   -- specify the Alert Code
5343   alertCode    INTEGER(0..4294967295),
5344
5345   -- specify the date-time of execution
5346   executionDateTime  GeneralizedTime,
5347
5348
5349   -- detail what happened when the future dated Command was executed
5350
5351   executionOutcome  ExecutionOutcome
5352
5353 }
5354
5355 ExecutionOutcome ::=      SEQUENCE
5356 {
5357   -- Provide details of the corresponding Command that may not be in the standard GBCS message header. Specifically the
```

5358 -- mode in which the Command was invoked, the Originator Counter in the original Command and the resulting changes to any
5359 -- replay counters held on the Device
5360
5361 authorisingRemotePartySeqNumber SeqNumber,
5362 credentialsReplacementMode CredentialsReplacementMode,
5363 remotePartySeqNumberChanges SEQUENCE OF RemotePartySeqNumberChange,
5364
5365 -- For each replacement in the Command, detail the outcome and impacted parties
5366
5367 replacementOutcomes SEQUENCE OF ReplacementOutcome
5368
5369 }
5370
5371 AuthorisingRemotePartyControl ::= SEQUENCE
5372 {
5373 -- Specify the replacement mode so that the Device can check that the Remote Party Role is allowed to
5374 -- authorise this type of replacement and that all replacements in the payload are allowed within this

5375 -- replacement mode

5376

5377 credentialsReplacementMode CredentialsReplacementMode,

5378

5379 -- Only if credentialsReplacementMode = anyByContingency, provide the symmetric key to decrypt

5380 -- the Contingency Public Key in the (root, keyCertSign, management) Trust Anchor Cell

5381

5382 plaintextSymmetricKey [0] IMPLICIT OCTET STRING OPTIONAL,

5383

5384 -- Specify whether the time based checks as part of any Certificate Path Validation should be applied

5385

5386 applyTimeBasedCPVChecks [1] IMPLICIT INTEGER {apply(0), disapply(1)} DEFAULT apply,

5387

5388 -- Identify which of the Public Keys on the Device is to be used in checking KRP Signature

5389 -- 'authorisingRemotePartyTACellIdentifier' may only be omitted when

5390 -- the access control broker is updating its own credentials and the target device is not a CHF.

5391 -- In all other cases it is mandatory.

5392

5393 authorisingRemotePartyTACellIdentifier [2] IMPLICIT TrustAnchorCellIdentifier OPTIONAL,

5394

5395 -- Specify the Originator Counter for the Remote Party Applying KRP Signature, or (for the

5396 -- Access Control Broker changing its credentials) the Access Control Broker's Originator Counter.

5397

5398 authorisingRemotePartySeqNumber [3] IMPLICIT SeqNumber,

5399

5400 -- If the Command is to effect a change of control, then newRemotePartyFloorSeqNumber must be included

5401 -- and will be the value used to prevent replay of Update Security Credentials Commands for the

5402 -- new controlling Remote Party.

5403

5404 newRemotePartyFloorSeqNumber [4] IMPLICIT SeqNumber OPTIONAL,

5405

5406 -- Some Commands on the Device may use a different Originator Counter sequence for Protection Against Replay. At this

5407 -- version of the GBCS, the only example is the Prepayment Top Up Command on ESME and GSME. The

5408 -- SpecialistSeqNumber structure allows such Counters to also be reset on change of control.

5409

5410 newRemotePartySpecialistFloorSeqNumber [5] IMPLICIT SEQUENCE OF SpecialistSeqNumber OPTIONAL,

5411

5412 -- In some cases, one party acting in one Remote Party Role may be replacing certificates for a different Remote Party Role.

5413 -- In some cases, Execution Counters need also to be reset for those other Remote Party Role(s)

5414

5415 otherRemotePartySeqNumberChanges [6] IMPLICIT SEQUENCE OF RemotePartySeqNumberChange OPTIONAL

5416 }

5417

5418 RemotePartySeqNumberChange ::= SEQUENCE

5419 {

5420 otherRemotePartyRole RemotePartyRole,

5421 otherRemotePartyFloorSeqNumber SeqNumber,

5422 newRemotePartySpecialistFloorSeqNumber SEQUENCE OF SpecialistSeqNumber OPTIONAL

5423 }

5424

5425 SpecialistSeqNumber ::= SEQUENCE

```
5426 {
5427 -- Specify the usage of the SeqNumber
5428 seqNumberUsage    SeqNumberUsage,
5429
5430 -- Specify the associated SeqNumber
5431 seqNumber    SeqNumber
5432 }
5433
5434 SeqNumberUsage ::= INTEGER
5435 {
5436 -- Define the full set of discrete usages on a Device. The only specialist
5437 -- counter is for Prepayment Top Up (which is set independently of other counters). This may only be
5438 -- included when changing Supplier Security Credentials on an ESME or GSME.
5439
5440 prepaymentTopUp    (0)
5441 }
5442
```

5443 SeqNumber ::= INTEGER (0.. 18446744073709551615)

5444

5445

5446 TrustAnchorReplacement ::= SEQUENCE

5447 {

5448 -- Provide the new end entity certificate

5449

5450 replacementCertificate Certificate,

5451

5452 -- Specify where it is to go (specifically which Trust Anchor Cell is to have its details replaced using

5453 -- the new end entity certificate)

5454

5455 targetTrustAnchorCell TrustAnchorCellIdentifier

5456 }

5457

5458

5459 ReplacementOutcome ::= SEQUENCE

5460 {
5461 affectedTrustAnchorCell TrustAnchorCellIdentifier,
5462 statusCode StatusCode,
5463
5464 -- The GBCS Certificate requirements mean that the Subject Unique ID attribute in the subject field of a certificate will always
5465 -- contain the 64 bit unique number that equates to Entity Identifier. existingSubjectUniqueID should be set
5466 -- accordingly based on the contents of the Trust Anchor Cell prior to Command processing.
5467
5468 existingSubjectUniqueID OCTET STRING,
5469
5470 -- The GBCS Certificate requirements mean that subjectKeyIdentifier attributes will all be 8 byte SHA-1 Hashes.
5471 -- existingSubjectKeyIdentifier should be set accordingly based on the contents of the Trust Anchor Cell prior to
5472 -- Command processing.
5473
5474 existingSubjectKeyIdentifier OCTET STRING,
5475
5476 -- The Subject Unique ID in the subject field of the certificate in this TrustAnchorReplacement

5477

5478 replacingSubjectUniqueID OCTET STRING,

5479

5480 -- The subjectKeyIdentifier in the certificate in this TrustAnchorReplacement

5481

5482 replacingSubjectKeyIdentifier OCTET STRING

5483 }

5484

5485 TrustAnchorCellIdentifier ::= SEQUENCE

5486 {

5487 -- Which Remote Party Role does this Cell relate to?

5488

5489 trustAnchorCellRemotePartyRole RemotePartyRole,

5490

5491 -- To what cryptographic use can the Public Key in this Cell be put? Some Remote Party Roles

5492 -- (e.g. supplier) can have more than one Public Key on a Device and each one would only have

5493 -- a single cryptographic use.

5494

5495 trustAnchorCellKeyUsage KeyUsage,

5496

5497 -- trustAnchorCellUsage is to allow for multiple Public Keys of the same keyUsage for the same Remote

5498 -- Party Role. It will be absent except where used to refer to the Supplier Key

5499 -- Agreement Key used solely in relation to validating Supplier generated MACs on Prepayment Top Up

5500 -- transactions

5501

5502 trustAnchorCellUsage CellUsage DEFAULT management

5503 }

5504

5505 CellUsage ::= INTEGER {management(0), prePaymentTopUp(1)}

5506

5507 RemotePartyRole ::= INTEGER

5508 {

5509 -- Define the set of Remote Party Roles in relation to which a Device may need to undertake

5510 -- processing. Note that most Devices will only support a subset of these.

5511

5512 root (0),

5513 recovery (1),

5514 supplier (2),

5515 networkOperator (3),

5516 accessControlBroker (4),

5517 transitionalCoS (5),

5518 wanProvider (6),

5519 issuingAuthority (7), -- Devices will receive such Certificates but they do not need to store

5520 -- them over an extended period

5521

5522 -- The 'other' RemotePartyRole is for a party whose role does not allow it to invoke any Device function apart from

5523 -- UpdateSecurityCredentials. This is to allow for Device functionality to be locked out of usage until a valid

5524 -- Remote Party can be identified e.g. where roles cannot be fixed until a Device is brought in to operation

5525

5526 other (127)

5527

```
5528 }
5529
5530 -- KeyUsage is only repeated here for clarity. It is defined in RFC 5912
5531
5532 KeyUsage ::= BIT STRING
5533 {
5534 -- Define valid uses of Public Keys held by Devices in their Trust Anchor Cells.
5535
5536 digitalSignature      (0),
5537 contentCommitment (1), -- not valid for GBCS compliant transactions
5538 keyEncipherment      (2), -- not valid for GBCS compliant transactions
5539 dataEncipherment     (3), -- not valid for GBCS compliant transactions
5540 keyAgreement          (4),
5541 keyCertSign          (5),
5542 cRLSign              (6),
5543 encipherOnly          (7), -- not valid for GBCS compliant transactions
5544 decipherOnly (8) -- not valid for GBCS compliant transactions
```

```
5545 }
5546
5547 CredentialsReplacementMode ::=  INTEGER
5548 {
5549 -- Define the valid combinations as to which Remote Party Roles can replace which kinds of Trust Anchors.
5550
5551 -- Normal operational replacement modes
5552 supplierBySupplier    (2),
5553 networkOperatorByNetworkOperator    (3),
5554 accessControlBrokerByACB    (4),
5555 wanProviderByWanProvider    (5),
5556 transCoSByTransCoS    (6),
5557 supplierByTransCoS    (7),
5558
5559 -- Recovery modes
5560 anyExceptAbnormalRootByRecovery (8),
5561 anyByContingency    (9)
```

5562 }

5563

5564 -- The GBCS only allows for a constrained set of Trust Anchor Cell operations and so the list of possible outcomes

5565 -- is more limited than in RFC 5934. The list below is that more constrained subset

5566

5567 StatusCode ::= ENUMERATED {

5568

5569 success (0),

5570

5571 -- badCertificate is used to indicate that the syntax for one or more certificates is invalid.

5572

5573 badCertificate (5),

5574

5575 -- noTrustAnchor is used to indicate that the authorityKeyIdentifier does not identify the public key of a

5576 -- trust anchor or a certification path that terminates with an installed trust anchor

5577

5578 noTrustAnchor (10),

5579

5580 -- insufficientMemory indicates that the update could not be processed because the Device did not

5581 -- have sufficient memory

5582

5583 insufficientMemory (17),

5584

5585 -- resourcesBusy indicates that the resources necessary to process the replacement are not available at the

5586 -- present time, but the resources might be available at some point in the future.

5587

5588 resourcesBusy (30),

5589

5590 -- other indicates that the update could not be processed, but the reason is not covered by any of the assigned

5591 -- status codes. Use of this status code SHOULD be avoided.

5592

5593 other (127) }

5594

5595 END

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5596 Requirements for AuthorisingRemotePartyControl elements – informative

5597 All bar two parts of the AuthorisingRemotePartyControl structure are optional. This Section summarises when each of the optional elements
5598 needs to be present.

AuthorisingRemotePartyControl element	Notes
credentialsReplacementMode	Always required
plaintextSymmetricKey	Only required if credentialsReplacementMode = anyByContingency (when it is always required)
applyTimeBasedCPVChecks	Only required if the Device is to ignore time when undertaking Certification Path Validation, in which case it needs to have the value 'disapply'
authorisingRemotePartyTACellIdentifier	For a Communications Hub, always present. For all other Devices, always present unless the Access Control Broker is replacing its own Key Agreement credentials (in which case it should be omitted)
authorisingRemotePartySeqNumber	Always required
newRemotePartyFloorSeqNumber	If the Command is to effect a change of control, then newRemotePartyFloorSeqNumber should be included. It can be present in all other situations
newRemotePartySpecialistFloorSeqNumber	Only required on Change of Supplier where the new Supplier has decided to use a different sequence of Originator Counters for Prepayment Top Ups.
otherRemotePartySeqNumberChanges	Should be present if one role (e.g. recovery, transitionalCoS) is changing credentials for

AuthorisingRemotePartyControl element	Notes
	another role or roles (e.g. supplier). In such cases, this should be present to set Execution Counters for that other role or roles

5599 Table 13.3.5.12: Requirements for AuthorisingRemotePartyControl elements

5600 CS02c Issue Security Credentials

5601 Description

5602 This Section covers the creation, validation and processing of (i) Issue Security Credentials
5603 Commands and (ii) Responses to such Commands.

5604 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	SME.C.C
Capable of future dated invocation?	No
Protection Against Replay Required?	Yes
Service Reference	6.17
Valid Target Device(s)	ESME / GSME / GPF / CHF
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier (for Devices other than CHF) WAN Provider (for CHF Devices only)
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	None
Valid initiating Device type(s) [HAN Only Messages]	None
Protocol	ASN.1

5605 Table 13.4.2: Use Case Cross References for Issue Security Credential Details Command and
5606 Response

5607 Construction of Commands

5608 Issue Security Credentials Command Payloads shall be constructed as specified in Section
5609 13.4.7 and Cryptographic Protection I and Cryptographic Protection II shall be applied as
5610 required for a Command of Message Category SME.C.C.

-
- 5611 Device processing of Commands and Response handling
- 5612 The Device receiving an Issue Security Credentials Command shall undertake processing
- 5613 steps in the sequence defined in this Section 13.4.4.
- 5614 In processing an Issue Security Credentials Command, the Device shall:
- 5615 1. undertake Command Authenticity and Integrity Verification as required for a Command
 - 5616 of Message Category SME.C.C. The Security Credentials used to verify Cryptographic
 - 5617 Protection I shall be:
- 5618 those held in the {wANProvider, digitalSignature, management} Trust Anchor Cell, if the
- 5619 target Device's deviceType equals communicationsHubCommunicationsHubFunction;
- 5620 those held in the {supplier, digitalSignature, management} Trust Anchor Cell, if the target
- 5621 Device's deviceType does not equal communicationsHubCommunicationsHubFunction;
- 5622 2. validate that the value of keyUsage in CommandPayload is either digitalSignature only
 - 5623 or keyAgreement only. If this validation fails then the Device shall set
 - 5624 issueCredentialsResponseCode to invalidKeyUsage, and process from step 6;
- 5625 3. generate a Private-Public Key Pair and store the Private Key so generated in the Pending
 - 5626 Private Key Cell determined by the value of keyUsage in CommandPayload . If the step
 - 5627 fails then the Device shall set issueCredentialsResponseCode to keyPairGenerationFailed,
 - 5628 and process from step 6;
- 5629 4. with the ASN.1 terms in this step (that are not defined in this Section 13.4.4) having the
 - 5630 meaning of IETF RFC 298636; generate a CertificationRequest which:
- 5631 complies with the requirements of IETF RFC2986 and IETF RFC 5912;
- 5632 is DER encoded, in line with the recommendation of IETF RFC 596737;
- 5633 has subjectPublicKey set to the bit string representation of the Public Key generated in step
- 5634 3;
- 5635 incorporates an extensionRequest identified by id-ce-keyUsage which shall contain the
- 5636 keyUsage value specified in CommandPayload;
- 5637 incorporates an extensionRequest identified by id-ce-subjectAltName which shall contain a
- 5638 single GeneralName of type OtherName that is further sub-typed as a

36 <https://tools.ietf.org/html/rfc2986>

37 <https://tools.ietf.org/html/rfc5967>

5639 HardwareModuleName (id-on-HardwareModuleName) as defined in IETF RFC 4108. The
5640 hwSerialNum field shall be set to the Device's Entity Identifier; and

5641 has the `signature` generated using the Private Key generated in step 3;

5642 5. if the generation of `CertificationRequest` is not successful then the Device shall set
5643 `issueCredentialsResponseCode` to `cRProductionFailed`; and

5644 6. create a Response according to the requirements of Section 13.4.7, apply the Response
5645 Cryptographic Protection required for a Response of Message Category SME.C.C, and
5646 send the Response.

5647 Response Processing

5648 Response Recipient Verification may be undertaken as specified in this GBCS for a Response
5649 of Message Category SME.C.C. The `issueCredentialsResponseCode` field, where present in
5650 the Response, may be decoded according to the ASN.1 definitions at Section 13.4.6.

```

5651 IssueSecurityCredentials Command and Response payloads – structure definition
5652 Each instance of @IssueSecurityCredentials.CommandPayload and of @IssueSecurityCredentials.ResponsePayload shall be an octet string
5653 containing the DER encoding of the populated structure defined in this Section 13.4.6 which specifies the structure in ASN.1 notation.
5654 IssueSecurityCredentials DEFINITIONS ::= BEGIN
5655
5656 CommandPayload ::=                               SEQUENCE
5657 {
5658     -- specify the keyUsage to which the generated key-pair will be put, if subsequently authorised
5659     keyUsage                               KeyUsage
5660
5661 }
5662
5663 ResponsePayload ::=                               CHOICE
5664
5665 {
5666     -- if the Command was successful, provide the generated Certification Request. CertificationRequest shall
5667     -- be as defined in ASN.1 by IETF RFC 5912. For reference, it is in the section headed ‘ASN.1 Module for RFC 2986’

```

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```
5668      certificationRequest      CertificationRequest,
5669
5670      -- if the Command was unsuccessful, detail the failure
5671
5672      issueCredentialsResponseCode      IssueCredentialsResponseCode
5673  }
5674
5675  -- KeyUsage is only repeated here for ease of reference. It is defined in IETF RFC 5912
5676
5677  KeyUsage ::=  BIT STRING
5678  {
5679      -- Define valid uses of Public Keys held by Devices in their Trust Anchor Cells.
5680
5681      digitalSignature      (0),
5682      contentCommitment (1),      -- not valid for GBCS compliant transactions
5683      keyEncipherment      (2),      -- not valid for GBCS compliant transactions
5684      dataEncipherment      (3),      -- not valid for GBCS compliant transactions
```

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```
5685 keyAgreement      (4),
5686 keyCertSign       (5),  -- not valid for this Use Case
5687 cRLSign            (6),  -- not valid for this Use Case
5688 encipherOnly       (7),  -- not valid for GBCS compliant transactions
5689 decipherOnly       (8)   -- not valid for GBCS compliant transactions
5690 }
5691
5692 IssueCredentialsResponseCode ::=          INTEGER
5693 {
5694     invalidKeyUsage          (1),
5695     keyPairGenerationFailed  (2),
5696     cRProductionFailed       (3)
5697 }
5698
5699
5700 END
5701 Constructing the @IssueSecurityCredentials.CommandPayload and of @IssueSecurityCredentials.ResponsePayload
```

5702 @IssueSecurityCredentials.CommandPayload shall have the structure defined in Section 13.4.6, and the Remote Party constructing the
 5703 Command shall populate with values according to Table 13.4.7a.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@IssueSecurityCredentials.CommandPayload	SEQUENCE			
keyUsage	BIT STRING	Either digitalSignature (0) only, or keyAgreement (4) only	Mandatory	Only one or the other is valid

5704 Table 13.4.7a: @IssueSecurityCredentials.CommandPayload population

5705 @IssueSecurityCredentials.ResponsePayload shall have the structure defined in Section 13.4.6, and the Device constructing the Response shall
 5706 populate with values according to Table 13.4.7b.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@IssueSecurityCredentials.ResponsePayload	CHOICE			
certificationRequest	See IETF RFC	The Certification Request produced according to	Mandatory	Mandatory if certificationRequest

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
	5912	the requirements of Section 13.4.4.		successfully produced
issueCredentialsResponseCode	INTEGER	Shall be populated according to the processing defined in Section 13.4.4	Mandatory	Mandatory if certificationRequest is not successfully produced

5707 Table 13.4.7b: @IssueSecurityCredentials.ResponsePayload population

5708 CS02d Update Device Certificates on Device

5709 Description

5710 This Section 13.5 covers the creation, validation and processing of (i) Update Device
5711 Certificates on Device, Commands and (ii) Responses to such Commands.

5712 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	SME.C.C
Capable of future dated invocation?	No
Protection Against Replay Required?	Yes
Service Reference	6.15
Valid Target Device(s)	ESME / GSME / GPF / CHF
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier (for Devices other than CHF) WAN Provider (for CHF Devices only)
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	None
Valid initiating Device type(s) [HAN Only Messages]	None
Protocol	ASN.1

5713 Table 13.5.2: Use Case Cross References for Update Device Certificate on Device, Command
5714 and Response

5715 Construction of Commands

5716 Update Device Certificate on Device Command Payloads shall be constructed as specified in
5717 Section 13.5.7, and Cryptographic Protection I and Cryptographic Protection II shall be
5718 applied as required for a Command of Message Category SME.C.C.

-
- 5719 Device processing of Commands and Response handling
- 5720 The Device receiving an Update Device Certificate on Device Command shall undertake
5721 processing steps in the sequence defined in this Section 13.5.4.
- 5722 In processing an Update Device Certificate on Device Command, the Device shall:
- 5723 1. undertake Command Authenticity and Integrity Verification as required for a Command
5724 of Message Category SME.C.C, except that check 4 in Section 6.2.4.1.1 may be
5725 undertaken after the checks in Section 6.2.4.1.2. The Security Credentials used to verify
5726 Cryptographic Protection I shall be:

5727 those held in the {wANProvider, digitalSignature, management} Trust Anchor Cell, if the
5728 target Device's deviceType equals communicationsHubCommunicationsHubFunction; or

5729 those held in the {supplier, digitalSignature, management} Trust Anchor Cell, if the target
5730 Device's deviceType does not equal communicationsHubCommunicationsHubFunction.
 - 5731 2. establish the values of keyUsage, subjectPublicKey and hwSerialNum in certificate in the
5732 CommandPayload. If any of the values cannot be established then the Device shall set
5733 updateDeviceCertResponseCode to invalidCertificate, and process from step 10;
 - 5734 3. validate that hwSerialNum established at step 2 is the Device's Entity Identifier. If this
5735 validation fails then the Device shall set updateDeviceCertResponseCode to
5736 wrongDeviceIdentity, and process from step 10;
 - 5737 4. validate that keyUsage established at step 2 is either digitalSignature only or
5738 keyAgreement only. If this validation fails then the Device shall set
5739 updateDeviceCertResponseCode to invalidKeyUsage, and process from step 10;
 - 5740 5. validate that the Device holds a Pending Private Key for the keyUsage as established at
5741 step 2. If this validation fails then the Device shall set updateDeviceCertResponseCode
5742 to noCorrespondingKeyPairGenerated, and process from step 10;
 - 5743 6. validate that subjectPublicKey established at step 2 is the bit string representation of the
5744 Public Key corresponding to the Pending Private Key identified at step 5. If this
5745 validation fails then the Device shall set updateDeviceCertResponseCode to
5746 wrongPublicKey, and process from step 10;
 - 5747 7. store certificate. If this step fails then the Device shall set
5748 updateDeviceCertResponseCode to certificateStorageFailed, and process from step 10;
 - 5749 8. set the Current Private Key to have the value of the Pending Private Key for the
5750 keyUsage established at step 2. If this step fails then the Device shall set
5751 updateDeviceCertResponseCode to privateKeyChangeFailed, and process from step 10;
-

-
- 5752 9. set updateDeviceCertResponseCode to success; and
- 5753 10. create a Response according to the requirements of Section 13.5.7, apply the Response
5754 Cryptographic Protection required for a Response of Message Category SME.C.C, and
5755 send the Response.
- 5756 If all steps were successful and this was a change of digitalSignature certificate, the
5757 Response shall be signed using the private key corresponding to the new certificate. If there
5758 was a failure, the Response shall be signed using the private key corresponding to the pre-
5759 existing key pair.
- 5760 Once the Pending Private Key becomes the Current Private Key, the Device will be using the
5761 new Private Key and this will affect all Remote Parties interacting with the Device;
5762 specifically they will need to use the new Certificate corresponding to the Private Key now in
5763 use.
- 5764 Response Processing
- 5765 Response Recipient Verification may be undertaken as specified in this GBCS for a Response
5766 of the relevant Message Category. The updateDeviceCertResponseCode field may be
5767 decoded according to the ASN.1 definitions at Section 13.5.6.
- 5768 If this was a change of digitalSignature certificate, the public key to be used to verify the
5769 Device's signature is dependent on the value of updateDeviceCertResponseCode. If
5770 updateDeviceCertResponseCode is success then the Private Key used for Signing will
5771 have changed. If updateDeviceCertResponseCode is other than success, the Private
5772 Key used for Signing will not have changed.

5773 Update Device Certificate on Device Command and Response payloads – structure definition

5774 Each instance of @UpdateDeviceCertificateonDevice.CommandPayload and of

5775 @UpdateDeviceCertificateonDevice.ResponsePayload shall be an octet string containing the DER encoding of the populated

5776 structure defined in this Section 13.5.6, which specifies the structure in ASN.1 notation.

5777 UpdateDeviceCertificateonDevice DEFINITIONS ::= BEGIN

5778

5779 CommandPayload ::= Certificate

5780 -- provide the certificate which the Device is to store

5781 -- the ASN.1 specification of certificate shall be as defined in IETF RFC 5912 for IETF RFC 5280

5782

5783 ResponsePayload ::= UpdateDeviceCertResponseCode

5784

5785 -- if the Command was unsuccessful, detail the failure; otherwise respond with success

5786

5787 UpdateDeviceCertResponseCode ::= INTEGER

5788 {

5789 success (0),

5790 invalidCertificate (1),

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5791	wrongDeviceIdentity	(2),
5792	invalidKeyUsage	(3),
5793	noCorrespondingKeyPairGenerated	(4),
5794	wrongPublicKey	(5),
5795	certificateStorageFailed	(6),
5796	privateKeyChangeFailed	(7)
5797	}	
5798		
5799	END	

5800 Constructing the @UpdateDeviceCertificateonDevice.CommandPayload and @UpdateDeviceCertificateonDevice.ResponsePayload

5801 @UpdateDeviceCertificateonDevice.CommandPayload shall have the structure defined in Section 13.5.6, and the Remote Party constructing

5802 the Command shall populate with values according to Table 13.5.7a.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@UpdateDeviceCertificateonDevice.CommandPayload				
Certificate	See IETF RFC 5912	A new Device Certificate that the Device is to process	Mandatory	

5803 Table 13.5.7a: @UpdateDeviceCertificateonDevice.CommandPayload population

5804 @UpdateDeviceCertificateonDevice.ResponsePayload shall have the structure defined in Section 13.5.6, and the Device constructing the
 5805 Response shall populate with values according to Table 13.5.7b.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@UpdateDeviceCertificateonDevice.ResponsePayload				
UpdateDeviceCertResponseCode	INTEGER	Shall be populated according to the processing defined in Section 13.5.4	Mandatory	

5806 Table 13.5.7b: @UpdateDeviceCertificateonDevice.ResponsePayload population

5807 CS02e Provide Device Certificates from Device

5808 Description

5809 This Section covers the creation, validation and processing of (i) Provide Device Certificates
5810 from Device Commands and (ii) Responses to such Commands.

5811 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command & Response
Message Type Category	Variant Message
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	6.24
Valid Target Device(s)	ESME / GSME / GPF / CHF
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier (for Devices other than CHF) WAN Provider (for CHF Devices only)
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	None
Valid initiating Device type(s) [HAN Only Messages]	None
Protocol	ASN.1

5812 Table 13.6.2: Use Case Cross References for Provide Device Certificates from Device
5813 Command and Response

5814 Construction of Commands

5815 Provide Device Certificates from Device Command Payloads shall be constructed as specified
5816 in Section 13.6.7 and Cryptographic Protection I and Cryptographic Protection II shall be
5817 applied as required for a Command of Message Category SME.C.C.

-
- 5818 Device processing of Commands and Response handling
- 5819 The Device receiving a Provide Device Certificates from Device Command shall undertake
5820 processing steps in the sequence defined in this Section 13.6.4.
- 5821 In processing a Provide Device Certificates from Device Command, the Device shall:
- 5822 1. undertake Command Authenticity and Integrity Verification as required for a Command
5823 of Message Category SME.C.C, except that Cryptographic Protection II shall not be
5824 verified. The Security Credentials used to verify Cryptographic Protection I shall be:
- 5825 those held in the {WANProvider, digitalSignature, management} Trust Anchor Cell, if the
5826 target Device's deviceType equals communicationsHubCommunicationsHubFunction; or
- 5827 those held in the {supplier, digitalSignature, management} Trust Anchor Cell, if the target
5828 Device's deviceType does not equal communicationsHubCommunicationsHubFunction;
- 5829 2. validate that keyUsage in CommandPayload is either digitalSignature only or
5830 keyAgreement only. If this validation fails then the Device shall set
5831 provideDeviceCertResponseCode to invalidKeyUsage, and process from step 5;
- 5832 3. confirm that the Device holds a certificate which (1) is for the keyUsage identified at
5833 step 2, (2) contains in hwSerialNum a value equal to the Device's Entity Identifier and (3)
5834 contains in subjectPublicKey the bit string representation of the Public Key
5835 corresponding to the Current Private Key for this keyUsage. If this validation fails then
5836 the Device shall set provideDeviceCertResponseCode to noCertificateHeld, and process
5837 from step 5;
- 5838 4. retrieve the certificate identified in step 3. If this step fails then the Device shall set
5839 provideDeviceCertResponseCode to certificateRetrievalFailure, and process from step 5;
5840 and
- 5841 5. create a Response according to the requirements of Section 13.6.7, apply the Response
5842 Cryptographic Protection required for a Response of Message Category SME.C.C, and
5843 send the Response.
- 5844 Response Processing
- 5845 Response Recipient Verification may be undertaken as specified in this GBCS for a Response
5846 of the Message Category SME.C.C. The provideDeviceCertResponseCode field may be
5847 decoded according to the ASN.1 definitions at Section 13.6.6.

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5848 Provide Device Certificates from Device Command and Response payloads – structure definition

5849 Each instance of @ProvideDeviceCertificateFromDevice.CommandPayload and of @ProvideDeviceCertificateFromDevice.ResponsePayload
5850 shall be an octet string containing the DER encoding of the populated structure defined in this Section 13.6.6 which specifies the structure in
5851 ASN.1 notation.

```
5852 ProvideDeviceCertificateFromDevice DEFINITIONS ::= BEGIN
```

5853

```
5854 CommandPayload ::= SEQUENCE
```

5855 {

```
5856 -- specify the KeyUsage of the Certificate to be returned
```

5857	keyUsage	KeyUsage
------	----------	----------

5858

5859 }

5860

5861 ResponsePayload ::= CHOICE

5862

5863 {

```
5864      -- if the Command was successful, provide the certificate
```

5865	certificate	Certificate,
------	-------------	--------------

5866

5867 -- if the Command was unsuccessful, detail the failure

5868

5869 provideDeviceCertResponseCode ProvideDeviceCertResponseCode

5870 }

5871

5872

5873 -- KeyUsage is only repeated here for ease of reference. It is defined in RFC 5912

5874

5875 KeyUsage ::= BIT STRING

5876 {

5877 -- Define valid uses of Public Keys held by Devices in their Trust Anchor Cells.

5878

5879 digitalSignature (0),

5880 contentCommitment (1), -- not valid for GBCS compliant transactions

5881 keyEncipherment (2), -- not valid for GBCS compliant transactions

5882 dataEncipherment (3), -- not valid for GBCS compliant transactions

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```
5883 keyAgreement      (4),
5884 keyCertSign        (5),  -- not valid for this Use Case
5885 cRLSign             (6),  -- not valid for this Use Case
5886 encipherOnly        (7),  -- not valid for GBCS compliant transactions
5887 decipherOnly        (8)  -- not valid for GBCS compliant transactions
5888 }
5889
5890 ProvideDeviceCertResponseCode ::=      INTEGER
5891 {
5892     invalidKeyUsage      (1),
5893     noCertificateHeld     (2),
5894     certificateRetrievalFailure (3)
5895 }
5896
5897
5898 END
5899 Constructing the @ProvideDeviceCertificateFromDevice.CommandPayload and @ProvideDeviceCertificateFromDevice.ResponsePayload
```

5900 @ProvideDeviceCertificateFromDevice.CommandPayload shall have the structure defined in Section 13.6.6 and the Remote Party constructing
5901 the Command shall populate with values according to Table 13.6.7a.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@ProvideDeviceCertificateFromDevice.CommandPayload	SEQUENCE			
keyUsage	BIT STRING	Either digitalSignature (0) only, or keyAgreement (4) only	Mandatory	Only one or the other is valid

5902 Table 13.6.7a: @ProvideDeviceCertificateFromDevice.CommandPayload population

5903 @ProvideDeviceCertificateFromDevice.ResponsePayload shall have the structure defined in Section 13.6.6, and the Device
5904 constructing the Response shall populate with values according to Table 13.6.7b.

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@ProvideDeviceCertificateFromDevice.ResponsePayload	CHOICE			
Certificate	See IETF RFC 5912	The Device Certificate provided pursuant to Section 13.6.4	Mandatory	Mandatory if certificate successfully produced
provideDeviceCertResponseCode	INTEGER	Shall be populated according to the processing defined in Section 13.6.4	Mandatory	Mandatory if certificate is not successfully produced

5905 Table 13.6.7b: @ProvideDeviceCertificateFromDevice.ResponsePayload population

5906 Pair-wise Authorisation of Devices

5907 Introduction to pair-wise Authorisation of Devices – informative

5908 The role of pair-wise Authorisation – informative

5909 This Section 13.7 includes the Use Cases related to the Authorisation (and the removal of

5910 Authorisation) for pair-wise, secure application layer interaction between two Devices on

5911 the same SMHAN. It also covers the related Use Cases for backing up and restoring the GPF

5912 Device Log.

5913 The process of authorising two Devices to communicate is referred to as ‘Joining’³⁸.

5914 Removal of such authorisation is referred to as ‘Un-joining’. Correspondingly, Remote Party

5915 Commands are specified in this Section 13.7 for instructing Devices that they are to ‘Join’ or

5916 ‘Unjoin’.

5917 In line with the SMETS and CHTS Device Log requirements, two Devices on a HAN must only

5918 be capable of interacting at the application layer if they are currently Joined. They must not

5919 be capable of interacting if (1) they have never been Joined or (2) they have been Unjoined.

5920 The application layer interactions between Devices on the same SMHAN must conform to

5921 the Device Based Access Controls (DBAC) as defined in Section 13.7.3. For example, an

5922 ESME must not be capable of processing an ‘Enable Supply’ Command from an IHD or an

5923 HCALCS.

5924 It is a precondition of Joining that both Devices have been ‘White-listed’ on to the HAN (as

5925 per Use Case ‘CCS01 Add Device to CHF Device log’) so that they are able to communicate

5926 over the HAN at the network layer (and so have network access). The GPF may be

5927 configured to be in the CHF’s Device Log at manufacture. A Device on a white-list can be

5928 removed from the white-list. It must then be unable to communicate over the HAN, and so

5929 unable to interact at the application layer with any Devices to which it was ‘Joined’.

5930 In SMETS terminology:

5931 the CHF’s Device Log holds the list of currently white-listed Devices on the SMHAN; and

5932 the Device Log on an ESME, SAPC, GSME, GPF, HCALCS or PPMID Device holds the Entity

5933 Identifiers, Device Types and related Security Credentials of other Devices on the HAN to

5934 which the Device is currently Joined (and so Authorised to interact with at an application

5935 layer).

5936 The process of white-listing a Device, and its subsequently obtaining network access,

5937 establishes a shared secret key between the Device and the Communications Hub. The Gas

38 This is unrelated to the ZSE meaning of ‘joining’

5938 Proxy Function, which is part of the Communications Hub, uses this shared secret key,
 5939 combined with a Device being entered in to its Device Log, for application layer
 5940 authorisation.

5941 IHDs and other Type 2 Devices are not required to have a Device Log (as defined in SMETS).
 5942 They are required to store security and related details of the Devices to which they are
 5943 Joined as required by ZSE however (otherwise they would be cryptographically unable to
 5944 understand the information being sent to them by the Joined Devices).

5945 IHDs and other Type 2 Devices can only read application layer information from Devices to
 5946 which they are Joined (either by requesting the information from the Device or by receiving
 5947 information published by the Device). When a PPMID is joined to a GPF, the PPMID can only
 5948 read information from the GPF to which it is Joined.

5949 When other types of Device are Joined (e.g. HCALCS, PPMID), they can also exchange
 5950 Commands and Responses at the application layer. For example, an ESME that is Joined to
 5951 an HCALCS can send a Command to the HCALCS to turn its switch on and the HCALCS can
 5952 send a Response saying whether it has done that. A PPMID can send an ‘enable supply’
 5953 Command to an ESME etc.

5954 The joining sequence – informative

5955 Statements in this Section 13.7.1.2 apply to an SAPC as if it were an ESME.

5956 There are three types of Join:

5957 Join Method B: this is a Join involving a Type 2 Device or a GPF;

5958 Join Method C: this is a Join between a GSME and a PPMID; and

5959 Join Method A: this is any Join which is not covered by Method B or C.

5960 Except for Method C, all Joins use the ZSE cryptography which requires exchange of
 5961 messages between the two Devices to establish the shared secret that the two Devices will
 5962 need to use. Method C uses the cryptography of Section 4 of this GBCS.

5963 Only certain combinations of Devices can be validly ‘Joined’. Table 13.7.1.2 summarises valid
 5964 combinations:

Device Name		ESME	GSME	Comms Hub (CHF)	Comms Hub (GPF)	HCALCS	PPMID	Type 2 (IHD or CAD)
	deviceType	1	0	2	3	4	5	6
ESME	1	Not permitted						
GSME	0	Not	Not					

		permitted	permitted					
Comms Hub (CHF)	2	Not permitted	Not permitted	Not permitted				
Comms Hub (GPF)	3	Not permitted	Method B	Not permitted	Not permitted			
HCALCS	4	Method A	Not permitted	Not permitted	Not permitted	Not permitted		
PPMID	5	Method A	Method C	Not permitted	Method B	Not permitted	Not permitted	
Type 2 (IHD or CAD)	6	Method B	Not permitted	Not permitted	Method B	Not permitted	Not permitted	Not permitted

5965 Table 13.7.1.2: Permitted Joins

5966 A Method A Join always involves an ESME and therefore any HAN exchanges required by a
5967 Method A Join shall always be instigated by the ESME involved. In this context the ESME is
5968 referred to as the `methodAInitiator`, since it initiates Method A Joins.

5969 The additional step with a Method A Join is that the other Device must first be sent a Join
5970 Command detailing the ESME with which it is allowed to Join. On receipt, the Device should
5971 add the ESME details to its Device Log and send a Response accordingly. If, subsequently,
5972 the Device is asked to undertake key establishment, it must check that the requesting Device
5973 is in its Device Log.

5974 Only one Device in a Method B Join is remotely instructed. Thus, the HAN exchanges
5975 required by a Method B Join shall always be instigated by the Device receiving such a
5976 Command. From Table 13.7.1.2, this is always a GSME or ESME except where a GPF is to Join
5977 to a PPMID, IHD or CAD. Thus, the sequence of a Method B Join is that the ESME / GSME /
5978 GPF:

5979 is sent a Join Command containing the Entity Identifier of the Device to which it is to Join
5980 and that other Device's `DeviceType`;

5981 verifies the cryptographic protection on the Command and checks to make sure it is well
5982 formed and valid;

5983 updates its Device Log to include details of the new Device;

5984 for an ESME39, undertakes the key establishment process with the specified Device, as per
5985 the ZSE specification; and

5986 creates and sends a Response detailing the success or otherwise of its actions.

5987 A Method C join does not require exchange of Messages between the two Devices for the
5988 establishment of the shared secret. Thus the sequence of a Method C Join is that each of
5989 the GSME and PPMID:

5990 is sent a Join Command containing the Entity Identifier of the Device to which it is to Join,
5991 that other Device's *DeviceType* and *Key Agreement Certificate*;

5992 verifies the cryptographic protection on the Command and does checks to make sure it is
5993 well formed and valid;

5994 updates its Device Log to include details of the new Device;

5995 checks there is a well-formed Device Certificate in the Command;

5996 optionally calculates the shared secret using the Device Certificate of the other Device
5997 (which is provided in the Command); and

5998 creates and sends a Response detailing the success or otherwise of its actions.

5999 The format of Message Payloads – informative

6000 In common with other GBCS Remote Messages related to the management of Security
6001 Credentials, the payloads of Commands and Responses defined in this Section 13.7.1.3 are
6002 specified using ASN.1, with DER encoding to be applied to Command and Response payloads.

6003 Device Requirements

6004 All Devices shall:

6005 support the ZSE Key Establishment Cluster as specified in Annex C of the ZSE cluster;

6006 support 'Crypto Suite 2' as defined in the ZSE specification; and

6007 use 'Crypto Suite 2' when undertaking any associated Key Establishment process.

6008 CHF, ESME, SAPC, HCALCS, PPMID and IHD shall support the mechanism at ZSE 5.4.7.4. A key
6009 that is provided to a pair of Devices by the CHF through that ZSE 5.4.7.4 mechanism shall, in
6010 this Section 13, be referred to as a 'Partner Link Key'.

39 The shared secret between the Communications Hub and the Type 2 Device / GSME established when the Device joined the HAN shall be used by the GPF to authenticate with the Device.

6011 An ESME shall be configured to be a ZSE ‘Router’, as defined in ZSE so that communications
6012 between the ESME and Devices Joined to the ESME, that are not operating on a Sub GHz
6013 Channel, are not reliant on availability of the Communications Hub.

6014 Where an SAPC is not operating on a Sub GHz Channel, the SAPC shall be configured to be a
6015 ZSE ‘Router’, as defined in ZSE so that communications between the SAPC and Devices
6016 Joined to the SAPC, that are not operating on a Sub GHz Channel, are not reliant on
6017 availability of the Communications Hub.

6018 Pursuant to the requirements in the SMETS and the CHTS requirement, Devices shall only
6019 communicate at an application layer with other Devices that are currently in their Device Log
6020 and are permitted by Device Based Access Controls (DBAC) as defined at Section 13.7.3.
6021 Such communications shall always be secured using the shared secrets established pursuant
6022 to Sections 13.7.4.

6023 Application layer communications within the scope of the DBAC requirement are HAN Only
6024 Messages, including provision of information to a PPMID or Type 2 Device. Note that HAN
6025 Only Messages between a PPMID and GSME have a structure that is specified in this GBCS in
6026 the corresponding Use Cases, and those relate only to Add Credit and Activate Emergency
6027 Credit Commands and the corresponding Responses.

6028 Access to the *MeteringDeviceType* attribute in the *Metering* Cluster shall not be within the
6029 scope of the DBAC requirements in Section 13.7.3, so as to facilitate use of the Partner Link
6030 Key mechanism as a pre-requisite to Joins involving an ESME or an SAPC.

6031 Each entry in a non CHF Device Log shall contain the Entity Identifier of the Authorised
6032 Device and its `deviceType`.

6033 The Entity Identifier of a Device with `DeviceType` of
6034 `communicationsHubGasProxyFunction` shall be the EUI 64-bit identifier of the
6035 ZigBee radio interface installed in the Communications Hub.

6036 Device Based Access Control

6037 In relation to information and functionality within SMETS, a Device shall, when it is a
6038 recipient of a Command or request for information from another Device on its SMHAN, only
6039 attempt to action that Command when:

6040 the sending Device’s Entity Identifier is in the recipient Device’s Device Log;

6041 the ZSE cryptographic protection on the Message is authenticated using the Shared Secret /
6042 Shared Secret Key established with the sending Device; and

6043 the Command or request for information is explicitly allowed by a cell in Tables 13.7.3a and
6044 13.7.3b, in terms of the `DeviceType` of the sending (client) and receiving (service) Device.

6045 The receiving Device shall determine the sending Device's DeviceType by reference to its
6046 Device Log entry for that sending Device.

6047 Where a Device is a recipient of a Command or request for information from another Device
6048 on its SMHAN that does not meet the access requirements of this Section 13.7.3, it shall:

6049 generate an entry in the Security Log recording failed Authentication;

6050 discard the Command or request for information without execution and without sending a
6051 Response; and

6052 send an Alert notifying the failed Authentication, constructed as specified in Section 6.2.4.2,
6053 populated with the relevant Alert Code from Section 16, to the Known Remote Party
6054 specified in Table 16.2.

6055 An ESME and an SAPC shall not action any ZSE Local Change Supply command from a PPMID
6056 where Proposed Supply Status has any value other than 0x02 ('Supply ON').

6057 For SMETS section 7.5.4.1 and 7.5.4.2 interactions, the PPMID and GSME shall comply with
6058 Use Cases PCS01 and PCS02 respectively.

Device Name	Server / recipient	ESME / SAPC	GSME	Comms Hub (GPF)	HCALCS	PPMID	Type 2 (IHD or CAD)
Client / sender	deviceType	1	0	3	4	5	6
ESME / SAPC	1	-	-	-	5.6.4.1 / 9.5.3.1	-	-
GSME	0	-	-	-	-	-	-
Comms Hub (GPF)	3	-	Request for Information	-	-	-	-
HCALCS	4	8.5.2.1	-	-	-	-	-
PPMID	5	7.5.5.140 7.5.5.241	7.5.4.1 7.5.4.2	Request for Information	-	-	-

40 For an SAPC, this shall only be allowed by the SAPC where the SAPC supports the corresponding SMETS functionality.

41 For an SAPC, this shall only be allowed by the SAPC where the SAPC supports the corresponding SMETS functionality.

		7.5.5.342 Request for Information					
Type 2 (IHD or CAD)	6	Request for Information	-	Request for Information	-	-	-

6059 Table 13.7.3a: Permitted Access by DeviceType, with Commands shown by SMETS
6060 reference

6061

SMETS Ref	SMETS Command Name	ZSE Ref
5.6.4.1 / 8.5.1.2 / 9.5.3.1	Control HCALCS [n] / Control HCALCS	Load Control Event
5.6.3.21 / 8.5.2.1 / 9.5.2.12	Request Control of HCALCS [n] / Request Control of HCALCS	Get Scheduled Events
7.5.5.1	Request Emergency Credit Activation	Select Available Emergency Credit
7.5.5.2	Request to Add Credit	Consumer Top Up
7.5.5.3	Request to Enable ESME Supply	Local Change Supply
7.5.4.1	Request Emergency Credit Activation	Select Available Emergency Credit
7.5.4.2	Request to Add Credit	Consumer Top Up

6062 Table 13.7.3b: Mapping of Table 13.7.3a command references to SMETS names and ZSE

6063 Use Case Requirements

6064 Requirements:

6065 in this Section 13.7.4 and in all sub Sections of it;

6066 in Use Case CS03A1 Method A Join (Meter);

6067 in Use Case CS03B Method B Join;

6068 in Use Case CS04AC Method A or C Unjoin;

42 For an SAPC, this shall only be allowed by the SAPC where the SAPC supports the corresponding SMETS functionality.

- 6069 in Use Case CS04B Method B Unjoin; and
- 6070 in Use Case CS07 Read Device Join Details;
- 6071 shall apply to an SAPC as if it were an ESME, and as if its `deviceType` were `eSME`.
- 6072 This Section 13.7.4 details requirements which shall be complied with for all Join or Unjoin
- 6073 related Use Cases.
- 6074 Use Cases covered
- 6075 The types of Join Device related Messages, the Grouping names used in this Section 13.7.4,
- 6076 the associated Message Category and the valid recipient `deviceType` for each shall be as
- 6077 specified in Table 13.7.4.143. The Service Reference for all Join Use Cases shall be 8.7, and
- 6078 for Unjoin Use Cases shall be 8.8.

Message Code	Use Case Name	Valid recipient <code>deviceType</code>	Grouping	Message Category	Valid Business Originator role(s) for Command invocation
0x000D	CS03A1 Method A Join (Meter)	eSME	Join Device	SME.C.C	Supplier
0x00AB	CS03A2 Method A Join (non Meter)	type1HANConnectedAuxiliaryLoadControlSwitch type1PrepaymentInterfaceDevice	Join Device	SME.C.C	Access Control Broker where the Command is addressed to a PPMID; Supplier otherwise
0x000E	CS03B Method B Join	gSME eSME communicationsHubGasProxyFunction	Join Device	SME.C.NC	Supplier, Access Control Broker
0x00AF	CS03C Method C Join	gSME type1PrepaymentInterfaceDevice	Join Device	SME.C.C	Access Control Broker where the Command is addressed to a PPMID; Supplier otherwise
0x000F	CS04AC Method A or C Unjoin	gSME eSME type1HANConnectedAuxiliaryLoadControlSwitch type1PrepaymentInterfaceDevice	Unjoin Device	SME.C.C	Access Control Broker where the Command is addressed to a PPMID; Supplier otherwise

43 To avoid duplication of specification, the Use Cases here are grouped together, and the standard Use Case cross reference table is not used.

0x0010	CS04B Method B Unjoin	gSME eSME communicationsHubGasProxyFunction	Unjoin Device	SME.C.NC	Supplier, Access Control Broker
0x0013	CS07 Read Device Join Details	gSME eSME communicationsHubGasProxyFunction type1HANConnectedAuxiliaryLoadControlSwitch type1PrepaymentInterfaceDevice		SME.C.NC	Supplier, Access Control Broker

6079 Table 13.7.4.1: Join Device related Commands, Grouping and Message Categories

6080 Join Device Command and Response Processing

6081 Construction of Commands

6082 ‘Join Device’ Command Payloads shall be constructed as specified in Section 13.7.4.5.2 and
6083 Cryptographic Protection I and Cryptographic Protection II shall be applied as required for a
6084 Command of the relevant Message Category.

6085 For a Command (1) which complies with either Use Case ‘CS03A2 Method A Join (non
6086 Meter)’ or Use Case ‘CS03C Method C Join’ and (2) where the Device to which it is addressed
6087 has a `deviceType` equal to `type1PrepaymentInterfaceDevice`, the Access
6088 Control Broker’s Digital Signing Private Key shall be used in generating the KRP Signature.

6089 Device processing of Commands and Response handling

6090 The Device receiving a ‘Join Device’ Command shall undertake processing steps in the
6091 sequence defined in this Section 13.7.4.2.2. Should a step after step 1 be unsuccessful, the
6092 Device shall create a Response according to the requirements of Section 13.7.4.5.2, apply
6093 the Response Cryptographic Protection required for a Response of the relevant Message
6094 Category, and send the Response and shall not undertake any further steps defined in this
6095 Section 13.7.4.2.2.

6096 In processing a ‘Join Device’ Command, the Device shall:

- 6097 1. undertake Command Authenticity and Integrity Verification as required for a Command
6098 of this Message Category, except that check 4 in Section 6.2.4.1.1 may be undertaken
6099 after the checks in Section 6.2.4.1.2. The Security Credentials used to verify
6100 Cryptographic Protection I shall be:

6101 those held in the {accessControlBroker, digitalSignature, management} Trust Anchor Cell, if
6102 `deviceType` equals `type1PrepaymentInterfaceDevice`; or

6103 those held in the {supplier, digitalSignature, management} Trust Anchor Cell, if `deviceType`
6104 does not equal `type1PrepaymentInterfaceDevice`;

-
- 6105 2. verify the `joinMethodAndRole` as specified in Section 13.7.4.5.3;
- 6106 3. add the `otherDeviceEntityIdentifier` and `otherDeviceType` to its Device
6107 Log as specified in Section 13.7.4.5.4;
- 6108 4. if `deviceType` is `eSME` then undertake Key Establishment with the other Device as
6109 specified in Section 13.7.4.5.5;
- 6110 5. if `joinMethodAndRole` is `methodC`, and so the join is between a `gSME` and a
6111 `type1PrepaymentInterfaceDevice`, check that `otherDeviceCertificate` is
6112 present and validly structured. If the check succeeds the Device shall store, linked to
6113 this Device Log entry, details relating to `otherDeviceCertificate`, such that
6114 the Device is able to use subsequently the Shared Secret derived from
6115 `otherDeviceCertificate` and its own Private Key Agreement Key. If this check
6116 fails the Device shall set `joinResponseCode` to
6117 `invalidOrMissingCertificate` and processing shall be unsuccessful; and
- 6118 6. set `joinResponseCode` to success, create a Response according to the
6119 requirements of Section 13.4.7, apply the Response Cryptographic Protection required
6120 for a Response of the relevant Message Category, and send the Response.
- 6121 Response Processing
- 6122 Response Recipient Verification may be undertaken as specified in this GBCS for a Response
6123 of the relevant Message Category. The `joinResponseCode` field in the Response may be
6124 decoded according to the ASN.1 definitions at Section 13.7.4.5.1.
- 6125 ‘Unjoin Device’ Command and Response Processing
- 6126 Construction of Commands
- 6127 ‘Unjoin Device’ Command Payloads shall be constructed as specified in Section 13.7.4.6.2
6128 and Cryptographic Protection I and Cryptographic Protection II shall be applied as required
6129 for a Command of the relevant Message Category.
- 6130 For a Command where the Device to which it is addressed has a `deviceType` equal to
6131 `type1PrepaymentInterfaceDevice`, the Access Control Broker’s Digital Signing
6132 Private Key shall be used in generating the KRP Signature.
- 6133 Device processing of Commands and Response handling
- 6134 The Device receiving an ‘Unjoin Device’ Command shall undertake processing steps in the
6135 sequence defined in this Section 13.7.4.3.2.
- 6136 In processing an ‘Unjoin Device’ Command, the Device shall:
-

-
- 6137 1. undertake Command Authenticity and Integrity Verification as required for a Command
 6138 of this Message Category. The Security Credentials used to verify Cryptographic
 6139 Protection I shall be:
- 6140 those held in the {accessControlBroker, digitalSignature, management} Trust Anchor Cell, if
 6141 deviceType equals type1PrepaymentInterfaceDevice; or
- 6142 those held in the {supplier, digitalSignature, management} Trust Anchor Cell, if deviceType
 6143 does not equal type1PrepaymentInterfaceDevice;
- 6144 2. set unjoinResponseCode to success;
- 6145 3. verify the otherDeviceEntityIdentifier matches an Entity Identifier currently recorded in
 6146 its Device Log. If it does not then set unjoinResponseCode to
 6147 otherDeviceNotInDeviceLog and process from step 5; otherwise process from step 4;
- 6148 4. delete all information from the entry in its Device Log that has the same Entity Identifier
 6149 as otherDeviceEntityIdentifier along with all shared cryptographic material related to
 6150 that entry. Additionally, If deviceType in that log entry is
 6151 type1HANConnectedAuxiliaryLoadControlSwitch, the Device shall clear the association
 6152 with the corresponding Auxiliary Controller [n], including updating the corresponding
 6153 Auxiliary Controller [n] Type, with its SMETS meaning, to indicate that there is no
 6154 associated Auxiliary Controller. If the deletion does not succeed, set
 6155 unjoinResponseCode to otherFailure; and
- 6156 5. Create a Response according to the requirements of Section 13.7.4.6.2, apply the
 6157 Response Cryptographic Protection required for a Response of the relevant Message
 6158 Category, and send the Response.
- 6159 Response Processing
- 6160 Response Recipient Verification may be undertaken as specified in this GBCS for a Response
 6161 of the relevant Message Category. The unjoinResponseCode field in the Response may
 6162 be decoded according to the ASN.1 definitions at Section 13.7.4.6.1.
- 6163 ‘CS07 Read Device Join Details’ Command and Response Processing
- 6164 Construction of Commands
- 6165 ‘CS07 Read Device Join Details’ Command Payloads shall be constructed as specified in
 6166 Section 13.7.4.7 and Cryptographic Protection II shall be applied as required for a Command
 6167 of the SME.C.NC Message Category.
- 6168 Device processing of Commands and Response handling
-

6169 The Device receiving a 'CS07 Read Device Join Details' Command shall undertake processing
6170 steps in the sequence defined in this Section 13.7.4.4.2.

6171 In processing a 'CS07 Read Device Join Details' Command, the Device shall:

- 6172 1. undertake Command Authenticity and Integrity Verification as required for a Command
6173 of the SME.C.NC Message Category;
- 6174 2. set readLogResponseCode to success;
- 6175 3. attempt to read the Entity Identifier and deviceType for each of the entries in its Device
6176 Log. If the reading does not succeed for all entries, set readLogResponseCode to
6177 readFailure; otherwise populate deviceLogEntries using the data read from its Device
6178 Log; and
- 6179 4. create a Response according to the requirements of Section 13.7.4.7, apply the
6180 Response Cryptographic Protection required for a Response of the SME.C.NC Message
6181 Category, and send the Response.

6182 Response Processing

6183 Response Recipient Verification may be undertaken as specified in this GBCS for a Response
6184 of the SME.C.NC Message Category. The readLogResponseCode and
6185 deviceLogEntries fields in the Response may be decoded according to the ASN.1
6186 definitions at Section 13.7.4.7.

6187 Component Requirements – Join

6188 Join Command and Response payloads – structure definition

6189 Each instance of @JoinDevice.CommandPayload and of @JoinDevice.ResponsePayload shall
6190 be an octet string containing the DER encoding of the populated structure defined in this
6191 Section 13.7.4.5.1 which specifies the structure in ASN.1 notation.

6192 JoinDevice DEFINITIONS ::= BEGIN

6193 CommandPayload ::= SEQUENCE

6194 {

6195 -- specify which type of joining is being authorised and,

6196 -- for Method A Joins, the role the Device is to play

6197

6198 joinMethodAndRole JoinMethodAndRole,

6199

```

6200      -- specify the Entity Identifier of the Device which is to be Joined with
6201
6202      otherDeviceEntityIdentifier          OCTET STRING,
6203
6204      -- specify the DeviceType of that other Device
6205
6206      otherDeviceType                      DeviceType,
6207
6208      -- provide the other Device's Key Agreement certificate, if and only if this
6209      -- is a join between a gSME and a type1PrepaymentInterfaceDevice.
6210      -- Certificate shall be as defined in IETF RFC 5912
6211
6212      otherDeviceCertificate                Certificate OPTIONAL
6213
6214  }
6215
6216  -- detail whether the Command successful executed or, if it did not,
6217  -- what the failure reason was
6218
6219  ResponsePayload ::=                      JoinResponseCode
6220
6221  JoinMethodAndRole ::=                    INTEGER
6222  {
6223      -- methodB is to be used where the other Device is a Type 2 Device or GPF.
6224      -- methodC is used where the Devices involved are a GSME and a PPMID.
6225      -- methodA is used otherwise.
6226      -- methodAInitiator is used where the Device this Command is targeted at

```

```

6227      -- should initiate the Key Agreement process
6228      -- methodAResponder is used where the Device this Command is targeted at
6229      -- should respond in the Key Agreement process, but shall not initiate it
6230
6231      methodAInitiator                (0),
6232      methodAResponder                (1),
6233      methodB                        (2),
6234      methodC                        (3)
6235  }
6236
6237  DeviceType ::=                      INTEGER
6238  {
6239      gSME                            (0),
6240      eSME                            (1),
6241      communicationsHubCommunicationsHubFunction (2),
6242      communicationsHubGasProxyFunction (3),
6243      type1HANConnectedAuxiliaryLoadControlSwitch (4),
6244      type1PrepaymentInterfaceDevice (5),
6245      type2                            (6)
6246  }
6247
6248  JoinResponseCode ::=                INTEGER
6249  {
6250      success                          (0),
6251      invalidMessageCodeForJoinMethodAndRole (1),
6252      invalidJoinMethodAndRole (2),
6253      incompatibleWithExistingEntry (3),

```

6254	deviceLogFull	(4),
6255	writeFailure	(5),
6256	keyAgreementNoResources	(6),
6257	keyAgreementUnknownIssuer	(7),
6258	keyAgreementUnsupportedSuite	(8),
6259	keyAgreementBadMessage	(9),
6260	keyAgreementBadKeyConfirm	(10),
6261	invalidOrMissingCertificate	(11),
6262	noPartnerLinkKeyReceived	(12),
6263	noCBKEResponse	(13)
6264	}	
6265		
6266	END	

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6267 Constructing the @JoinDevice.CommandPayload and of @JoinDevice.ResponsePayload

6268 @JoinDevice.CommandPayload shall have the structure defined in Section 13.7.4.5.1, and the Remote Party constructing the Command
6269 shall populate with values according to Table 13.7.4.5.2a.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@JoinDevice.CommandPayload	SEQUENCE			
joinMethodAndRole	INTEGER	methodAInitiator (0), methodAResponder (1), methodB (2), methodC (3)		See Section 13.7.4.5.3 for valid values
otherDeviceEntityIdentifier	OCTET STRING	Entity Identifier	Mandatory	The Entity Identifier of the Device which is to be entered in this Device's Device Log
otherDeviceType	INTEGER	gSME (0), eSME (1), communicationsHubCommunicationsHubFunction (2), communicationsHubGasProxyFunction (3), type1HANConnectedAuxiliaryLoadControlSwitch (4),	Mandatory	The DeviceType of the Device which is to be entered in this Device's Device Log

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		type1PrepaymentInterfaceDevice (5), type2 (6)		
otherDeviceCertificate	Certificate	The Key Agreement Certificate currently in use by the other Device.	OPTIONAL	The other Device's Key Agreement certificate, which shall only be present if and only if this is a join between a gSME and a type1PrepaymentInterfaceDevice. Certificate shall be as defined in IETF RFC 5912.

6270 Table 13.7.4.5.2a: @JoinDevice.CommandPayload population

6271 @JoinDevice.ResponsePayload shall have the structure defined in Section 13.7.4.5.1, and the Remote Party constructing the

6272 Command shall populate with values according to Table 13.7.4.5.2b.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@JoinDevice.ResponsePayload				

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
JoinResponseCode	INTEGER	Shall be populated according to the processing defined in Section 13.7.4.2.2	Mandatory	

6273 Table 13.7.4.5.2b: @JoinDevice.ResponsePayload population

6274 Verification of joinMethodAndRole

6275 The Device shall first verify the joinMethodAndRole specified in the Command Payload against the Message Code specified in the Grouping

6276 Header of the Command according to Table 13.7.4.5.3a. If the check fails JoinResponseCode in the Response shall be set to the value

6277 invalidMessageCodeForJoinMethodAndRole and no further verification checks in this Section 13.7.4.5.3a shall be undertaken.

Message Code	Use Case Name	Valid joinMethodAndRole
0x000D	CS03A1 Method A Join (Meter)	methodAInitiator
0x00AB	CS03A2 Method A Join (non Meter)	methodAResponder
0x000E	CS03B Method B Join	methodB
0x00AF	CS03C Method C Join	methodC

6278 Table 13.7.4.5.3a: Valid deviceMethod and joinMethodAndRole against Message Code

6279 The Device receiving a Join Device Command shall verify joinMethodAndRole against its own DeviceMethod and the DeviceType specified in

6280 the otherDeviceType parameter of the Command according to the requirements of the remainder of this Section 13.7.4.5.3.

6281 If joinMethodAndRole is methodB then the Device's verification of joinMethodAndRole shall be successful if there is a cell identified by its own
6282 DeviceMethod, and the value of otherDeviceType (as identified in the Command) of a type defined in Table 13.7.4.5.3b, and that cell contains
6283 'success'. Otherwise, the verification shall fail and JoinResponseCode in the Response shall be set to the value invalidJoinMethodAndRole.

	otherDeviceType		
	communicationsHub	type1PrepaymentInterfaceDevice	type2
DeviceType of Device to which the Command is addressed	GasProxyFunction		
gSME	Success	-	-
eSME	-	-	Success
communicationsHubGasProxyFunction	-	Success	Success

6284 Table 13.7.4.5.3b: joinMethodAndRole is methodB

6285 If joinMethodAndRole is methodAInitiator then the Device's verification of joinMethodAndRole shall be successful if there is a cell identified by
 6286 its own DeviceType, and the value of otherDeviceType (as identified in the Command) of a type defined in Table 13.7.4.5.3c, and that cell
 6287 contains 'success'. Otherwise, the verification shall fail and JoinResponseCode in the Response shall be set to the value
 6288 invalidJoinMethodAndRole.

	otherDeviceType	
	type1HANConnected	type1Prepayment
DeviceType of Device to which the Command is addressed	AuxiliaryLoadControlSwitch	InterfaceDevice
eSME	Success	Success

6289 Table 13.7.4.5.3c: joinMethodAndRole is methodAInitiator

6290 If joinMethodAndRole is methodAResponder then the Device's verification of joinMethodAndRole shall be successful if there is a cell identified
6291 by its own DeviceType, and the value of otherDeviceType (as identified in the Command) of a type defined in Table 13.7.4.5.3d, and that cell
6292 contains 'success'. Otherwise, the verification shall fail and JoinResponseCode in the Response shall be set to the value
6293 invalidJoinMethodAndRole.

otherDeviceType	
eSME	
DeviceType of Device to which the Command is addressed	
type1HANConnectedAuxiliaryLoadControlSwitch	Success
type1PrepaymentInterfaceDevice	Success

- 6294
- Table 13.7.4.5.3d: joinMethodAndRole is methodAResponder
- 6295
- If joinMethodAndRole is methodC then the Device’s verification of joinMethodAndRole shall be successful if there is a cell identified by its own
- 6296
- DeviceType and the value of otherDeviceType (as identified in the Command) in Table 13.7.4.5.3e and that cell contains ‘success’. Otherwise,
- 6297
- the verification shall fail and JoinResponseCode in the Response shall be set to the value invalidJoinMethodAndRole.

otherDeviceType		
type1PrepaymentInterfaceDevice		
gSME		
DeviceType of Device to which the Command is addressed		
gSME	Success	-
type1PrepaymentInterfaceDevice	-	Success

- 6298
- Table 13.7.4.5.3e: joinMethodAndRole is methodC
- 6299
- Adding the otherDeviceEntityIdentifier and otherDeviceType to the Device Log
- 6300
- The Device shall undertake the following steps in the sequence specified:

-
- 6301 1. if the `otherDeviceEntityIdentifier` matches an Entity Identifier currently recorded in its Device Log, then the Device shall compare
6302 deviceType in that log entry with `otherDeviceType`. If the Device types match then the addition is successful and processing within this
6303 Section 13.7.4.5.4 shall cease; otherwise the Device shall set `joinResponseCode` to `incompatibleWithExistingEntry` and processing within
6304 this Section 13.7.4.5.4 shall cease;
- 6305 2. the Device shall check if there is capacity for an additional entry in its Device Log, including confirmation that, where `otherDeviceType` is
6306 type1HANConnectedAuxiliaryLoadControlSwitch the Device does not currently have an Auxiliary Controller associated with all five of
6307 Auxiliary Controller [n]. If there is not, the Device shall set `joinResponseCode` to `deviceLogFull` and processing within this Section
6308 13.7.4.5.4 shall cease; and
- 6309 3. the Device shall attempt to create a new Device Log entry using `otherDeviceEntityIdentifier` and `otherDeviceType`. If that entry is not
6310 successfully created, the Device shall set `joinResponseCode` to `writeFailure`. If that entry is successfully created and `otherDeviceType` is
6311 type1HANConnectedAuxiliaryLoadControlSwitch, the Device shall associate one of its Auxiliary Controller [n] with this new HCALCS,
6312 including setting the corresponding Auxiliary Controller [n] Type, with its SMETS meaning, to indicate that this is an HCALCS whose Entity
6313 Identifier is `otherDeviceEntityIdentifier`.
- 6314 Undertaking Key Establishment with the other Device
- 6315 The Device shall undertake the following steps in the sequence specified:
- 6316 1. if the `otherDeviceEntityIdentifier` does not match an Entity Identifier for which the ESME has received a Partner Link Key,
6317 the Device shall set `joinResponseCode` to `noPartnerLinkKeyReceived` and processing within this Section 13.7.4.5.5 shall
6318 cease; and
- 6319 2. the ESME shall initiate, and attempt to complete, Key Establishment according to the ZSE requirements. The ESME shall wait a
6320 minimum of two seconds before timing out any key establishment operation. Should the first attempt to complete Key Establishment
6321 not succeed, the ESME shall wait 120 seconds before making a second attempt. If that second attempt fails, the ESME shall wait a

6322 further 120 seconds before making a final attempt. If that final attempt does not succeed, the Device shall set joinResponseCode to the
 6323 value specified by Table 13.7.4.5.5 or to noCBKEResponse if no response was received from the other Device.

ZSE Response Code ⁴⁴	Value of joinResponseCode
NO_RESOURCES	keyAgreementNoResources
UNKNOWN_ISSUER	keyAgreementUnknownIssuer
UNSUPPORTED_SUITE	keyAgreementUnsupportedSuite
BAD_MESSAGE	keyAgreementBadMessage
BAD KEY_CONFIRM	keyAgreementBadKeyConfirm

6324 Table 13.7.4.5.5: joinResponseCode mapping to ZSE Responses

6325 Component Requirements – Unjoin

6326 Unjoin Command and Response payloads – structure definition

6327 Each instance of @UnjoinDevice.CommandPayload and of @UnjoinDevice.ResponsePayload shall be an octet string containing the DER
 6328 encoding of the populated structure defined in this Section 13.7.4.6.1 which specifies the structure in ASN.1 notation.

6329 UnjoinDevice DEFINITIONS ::= BEGIN

6330

6331 CommandPayload ::= OtherDeviceEntityIdentifier

44 As defined in the ZSE specification

6332 -- specify the Entity Identifier of the Device for which authorisation

6333 -- is to be removed

6334

6335 OtherDeviceEntityIdentifier ::= OCTET STRING

6336

6337 ResponsePayload ::= UnjoinResponseCode

6338

6339 -- detail whether the Command successful executed or, if it did not,

6340 -- what the failure reason was

6341

6342 UnjoinResponseCode ::= INTEGER

6343 {

6344 success (0),

6345 otherDeviceNotInDeviceLog (1),

6346 otherFailure (2)

6347 }

6348

6349 END

6350 Constructing the @UnjoinDevice.CommandPayload and of @UnjoinDevice.ResponsePayload

6351 @UnjoinDevice.CommandPayload shall have the structure defined in Section 13.7.4.6.1, and the Remote Party constructing the Command shall
6352 populate with values according to Table 13.7.4.6.2a.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@UnjoinDevice.CommandPayload				
OtherDeviceEntityIdentifier	OCTET STRING	Entity Identifier	Mandatory	The Entity Identifier of the Device which is to be removed from this Device's Device Log

6353 Table 13.7.4.6.2a: @UnjoinDevice.CommandPayload population

6354 @UnjoinDevice.ResponsePayload shall have the structure defined in Section 13.7.4.6.1, and the Remote Party constructing the Command shall
6355 populate with values according to Table 13.7.4.6.2b.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@UnjoinDevice.ResponsePayload				
unjoinResponseCode	INTEGER	success (0), (0), otherDeviceNotInDeviceLog (1),	Mandatory	Shall be populated according to the processing defined in Section 13.7.4.3

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
		(1), otherFailure (2) (2)		

6356 Table 13.7.4.6.2b: @UnjoinDevice.ResponsePayload population

6357 CS07 Read Device Join Details Command and Response payloads – structure definition

6358 Each instance of @ReadDeviceLog.CommandPayload and of

6359 @ReadDeviceLog.ResponsePayload shall be an octet string containing the DER encoding of

6360 the populated structure defined in this Section 13.7.4.7 which specifies the structure in

6361 ASN.1 notation.

6362 ReadDeviceLog DEFINITIONS ::= BEGIN

6363

6364 CommandPayload ::= NULL

6365

6366 ResponsePayload ::= SEQUENCE

6367 {

6368 -- detail whether the Command successful

6369

6370 readLogResponseCode ReadLogResponseCode,

6371

6372 -- if it was, return the Log Entries

6373 deviceLogEntries SEQUENCE OF DeviceLogEntry

6374 OPTIONAL

6375 }

6376

6377 DeviceLogEntry ::= SEQUENCE

6378 {

6379 deviceIdentifier OCTET STRING,

6380 deviceType DeviceType

6381 }

6382

6383 DeviceType ::= INTEGER

6384 {

6385	gSME	(0),
6386	eSME	(1),
6387	communicationsHubCommunicationsHubFunction	(2),
6388	communicationsHubGasProxyFunction	(3),
6389	type1HANConnectedAuxiliaryLoadControlSwitch	(4),
6390	type1PrepaymentInterfaceDevice	(5),
6391	type2	(6)
6392	}	
6393		
6394		
6395	ReadLogResponseCode::=	INTEGER
6396	{	
6397	success	(0),
6398	readFailure	(1)
6399	}	
6400		
6401	END	
6402		

6403

6404 GCS59 / 62 GPF Device Log Backup and Restore

6405 Introduction to GPF Device Log Backup and Restore – informative

6406 The role of pair-wise authorisation – informative

6407 This Section 13.8 includes the Use Cases related to the backing up and restoring of the GPF's
 6408 Device Log. This is to cater for situation where the existing Communications Hub fails and
 6409 has to be replaced.

6410 In summary:

6411 a GPF sends an Alert whenever its Device Log changes (unless the change is as a result of a
 6412 restore of the Device Log). That Alert contains the contents of the GPF's Device Log after the
 6413 change has been made; and

6414 the Restore GPF Device Log Command shall contain the same structure of Device Log
 6415 contents. If successful, the Command will place those contents in to the GPF's Device Log
 6416 and will have triggered the processing required to authorise the specified Devices
 6417 application layer interaction with the GPF, where required.

6418 The format of Message Payloads – informative

6419 In common with other GBCS Remote Messages related to the management of Security
 6420 Credentials, the Payloads of Alerts, Commands and Responses defined in this Section 13.8
 6421 are specified using ASN.1, with DER encoding to be applied to Command and Response
 6422 payloads.

6423 Each entry in a GPF Device Log shall contain the Entity Identifier of the Authorised Device
 6424 and its deviceType.

6425 GCS62 Backup GPF Device Log

6426 Description

6427 This Section 13.8.2 covers the creation, validation and processing of Alerts resulting from
 6428 changes to the GPF Device Log. One such Alert shall be generated each time that the GPF
 6429 Device Log changes, except where the change arises from a GPF Device Log Restore
 6430 Command.

6431 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message

Cross Reference	Value
Message Type	Alert
Message Type Category	SME.A.NC
Capable of future dated invocation?	N/A
Protection Against Replay Required?	N/A
Service Reference	8.12
Valid Initiating Device(s)	GPF
Valid Business Target role(s) for Alert	Access Control Broker
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

6432 Table 13.8.2.2: Use Case Cross References for GPF Device Log Backup Alert

6433 Construction of Alerts

6434 GPF Device Log Backup Alert Payloads shall be constructed according to the requirements of
6435 Section 13.8.4.1 and populated as specified in Table 13.8.2.3.

6436 MAC Header, Grouping Header and SMD-KRP MAC shall be populated as required for an
6437 Alert of the SME.A.NC Message Category, with the Message Code being 0x00B2. Note that
6438 the Business Target ID in the Grouping Header shall always contain the Entity Identifier of
6439 the Access Control Broker.

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Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@GPFDDeviceLog.BackupAlertPayload	SEQUENCE			
alertCode	INTEGER	0x8071	Mandatory	Fixed value specifying that this is a GPF Device Log Backup Alert
backupDateTime	GeneralizedTime	The date-time at which this Alert was created	Mandatory	This is based on the Device's own clock
deviceLogEntries	SEQUENCE OF		OPTIONAL	There may be 0, 1 or many entries in the Log. The following two fields will be repeated as many times as there are Device Log Entries
deviceEntityIdentifier	OCTET STRING	Entity Identifier	Mandatory	The Entity Identifier of the Device to which this entry relates
deviceType	INTEGER	type1PrepaymentInterfaceDevice (5), type2(6)	Mandatory	The DeviceType of the Device to which this entry relates. These are the only valid entries for the GPF Device Log

6440 Table 13.8.2.3: @GPFDDeviceLog.BackupAlertPayload population

6441 Processing of Alerts

6442 SMD-KRP MAC may be verified by the Access Control Broker as per Section 6.8.4.

6443 GCS59 GPF Device Log Restore

6444 Description

6445 This Section covers the creation, validation and processing of Commands to restore the GPF Device Log, and the creation and validation of the
 6446 corresponding Response.

6447 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	SME.C.NC
Capable of future dated invocation?	No
Protection Against Replay Required?	Yes
Service Reference	8.12
Valid Target Device(s)	GPF
Valid Business Originator role(s) for Command	Access Control Broker
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

6448 Table 13.8.3.2: Use Case Cross References for GPF Device Log Restore

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6449 Construction of Command

6450 GPF Device Log Restore Command Payloads shall be constructed according to the requirements of Section 13.8.4.1 and populated as specified
6451 in Table 13.8.3.3.

6452 MAC Header, Grouping Header and ACB-SMD MAC shall be populated as required for a Command of the SME.C.NC Message Category.

Attribute name	Data Type	Value (blank cells mean the command specific value is derived by the encoding process)	Mandatory, OPTIONAL or DEFAULT value	Notes
@GPFDeviceLog.RestoreCommandPayload	SEQUENCE			
deviceLogEntries	SEQUENCE OF		OPTIONAL	There may be 0, 1 or many entries in the Log. The following two fields will be repeated as many times as there are Device Log Entries. Note that there would be no effect if the Command had no deviceLogEntries.
deviceEntityIdentifier	OCTET STRING	Entity Identifier	Mandatory as part of each entry that is present	The Entity Identifier of the Device to which this entry relates.
deviceType	INTEGER	type1PrepaymentInterfaceDevice (5), type2(6)	Mandatory as part of each entry that is present	The DeviceType of the Device to which this entry relates. These are the only valid entries for the GPF Device Log. Note that the GSME does not need to be in the GPF's Device Log, since the GPF only receives information from the GSME.

6453 Table 13.8.3.3: @GPFDeviceLog.RestoreCommandPayload population

6454 Device processing of Command and Response handling

6455 The GPF receiving a GPF Device Log Restore Command shall undertake processing steps in the sequence defined in this Section 13.8.3.4. The
6456 Device shall undertake Command Authenticity and Integrity Verification as required for a Command of this Message Category, and then, if
6457 successful, for each DeviceLogEntry in deviceLogEntries, shall:

- 6458 1. set deviceLogEntry in the corresponding ResponseOutcome to the values of this DeviceLogEntry in deviceLogEntries;
- 6459 2. set joinResponseCode in the corresponding ResponseOutcome to `success`;
- 6460 3. if the deviceEntityIdentifier matches an Entity Identifier currently recorded in its Device Log, compare deviceType in that log entry with
6461 otherDeviceType. If the Device types match then the addition is successful and processing of this DeviceLogEntry shall cease; otherwise
6462 the Device shall set joinResponseCode to incompatibleWithExistingEntry and processing of this DeviceLogEntry shall cease;
- 6463 4. check if there is capacity for an additional entry in its Device Log. If there is not, the Device shall set joinResponseCode to deviceLogFull
6464 and processing of this DeviceLogEntry shall cease; and
- 6465 5. attempt to create a new Device Log entry using deviceEntityIdentifier and deviceType. If that entry is not successfully created, the Device
6466 shall set joinResponseCode to writeFailure and processing of this DeviceLogEntry shall cease.

6467 Once all DeviceLogEntry in deviceLogEntries have been processed, the GPF shall populate the Response Payload according to the requirements
6468 of Section 13.8.4.1 using the ResponseOutcomes produced by the processing in this Section 13.8.3.4, construct MAC Header, Grouping Header
6469 and apply the Response Cryptographic Protection required for a Response of the SME.C.NC Message Category, and send the Response.

6470 Common Requirements

6471 GPF Device Log Backup Alert, Restore Command and Restore Response Payloads – structure definition

6472 Each instance of @GPFDeviceLog.BackupAlertPayload, @GPFDeviceLog.RestoreCommandPayload and of
6473 @GPFDeviceLog.RestoreResponsePayload shall be an octet string containing the DER encoding of the populated structure defined in this
6474 Section 13.8.4.1 which specifies the structure in ASN.1 notation.

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```
6475  GPFDDeviceLog DEFINITIONS ::= BEGIN
6476
6477  BackupAlertPayload ::=                               SEQUENCE
6478  {
6479      -- specify the Alert Code
6480      alertCode                                         INTEGER(0..4294967295),
6481
6482      -- specify the date-time of the backup
6483      backupDateTime                                    GeneralizedTime,
6484
6485      -- detail the entries in the Device Log now that the change has been made
6486      deviceLogEntries                                  SEQUENCE OF DeviceLogEntry
6487
6488  }
6489
6490  RestoreCommandPayload ::=                             SEQUENCE
6491  {
```

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6492	-- list the Device Log entries that are to be added	
6493	deviceLogEntries	SEQUENCE OF DeviceLogEntry
6494		
6495	}	
6496		
6497	DeviceLogEntry ::=	SEQUENCE
6498	{	
6499		
6500	-- specify the Entity Identifier of the Device	
6501	deviceEntityIdentifier	OCTET STRING,
6502		
6503	-- specify the DeviceType of that Device	
6504		
6505	deviceType	DeviceType
6506	}	
6507		
6508	RestoreResponsePayload ::=	SEQUENCE

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```
6509
6510 {
6511     -- for each DeviceLog Entry, detail whether the Command successfully executed or, if it did not, what the failure reason was
6512
6513     restoreOutcomes                SEQUENCE OF RestoreOutcome
6514 }
6515
6516 RestoreOutcome ::=
6517 {
6518     deviceLogEntry                DeviceLogEntry,
6519     joinResponseCode              JoinResponseCode
6520 }
6521
6522 DeviceType ::=
6523 {
6524     gSME                          (0),
6525     eSME                          (1),
```

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6526	communicationsHubCommunicationsHubFunction	(2),
6527	communicationsHubGasProxyFunction	(3),
6528	type1HANConnectedAuxiliaryLoadControlSwitch	(4),
6529	type1PrepaymentInterfaceDevice	(5),
6530	type2	(6)
6531	}	
6532		
6533	JoinResponseCode::=	INTEGER
6534	{	
6535	success	(0),
6536	invalidMessageCodeForJoinMethodAndRole	(1),
6537	invalidJoinMethodAndRole	(2),
6538	incompatibleWithExistingEntry	(3),
6539	deviceLogFull	(4),
6540	writeFailure	(5),
6541	keyAgreementNoResources	(6),
6542	keyAgreementUnknownIssuer	(7),

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6543	keyAgreementUnsupportedSuite	(8),
6544	keyAgreementBadMessage	(9),
6545	keyAgreementBadKeyConfirm	(10),
6546	invalidOrMissingCertificate	(11)
6547	}	
6548	END	

- 6549 CS02f Provide Security Credential Details Command and Response
- 6550 Description
- 6551 This Section covers the creation, validation and processing of (i) CS02f Provide Security
- 6552 Credential Details Commands and (ii) Responses to such Commands.
- 6553 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	Variant Message and is not a Critical Command
Capable of future dated invocation?	No
Protection Against Replay Required?	No
Service Reference	6.24.1
Valid Target Device(s)	ESME / SAPC
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier Network Operator Access Control Broker Transitional Change of Supplier WAN Provider Recovery Load Controller
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

6554	Table13.9.2: Use Case Cross References for CS02f Provide Security Credential Details
6555	Command and Response
6556	Common Requirements
6557	Summary – informative
6558	The statements in Section 13.2.3.1 apply.

6559 The 'CS02f Provide Security Credential Details' Command and Response

6560 The requirements of Section 13.2.3.2 shall apply as if @ProvideSecurityCredentialDetails
6561 were modified as required by Section 13.2.3.2, and references to sections within
6562 Section 13.2 were references to the corresponding sections within Section 13.9.

6563 The @ProvideSecurityCredentialDetails.Command and
6564 @ProvideSecurityCredentialDetails.Response structure definition

6565 The requirements of Section 13.2.3.3 shall apply as if:

6566 The reference to Section 13.2.4.4 were a reference to Section 13.9.4.4; and

6567 RemotePartyRole were defined as:

6568

```
6569         RemotePartyRole ::= INTEGER
6570         {
6571
6572             root      (0),
6573             recovery   (1),
6574             supplier   (2),
6575             networkOperator (3),
6576             accessControlBroker (4),
6577             transitionalCoS (5),
6578             wanProvider (6),
6579             issuingAuthority (7),
6580             loadController (8),
6581             other      (127)
6582
6583         }
```

6584

6585 CS02f Provide Security Credential Details from a Device – Processing Steps

6586 The requirements of Section 13.2.4, and all sub-sections of it, shall apply as if:

6587 'CS02a Provide Security Credential Details' were 'CS02f Provide Security Credential Details';

6588 RemotePartyRole were defined as required by Section 13.9.3.3;

6589 references to Section 13.2.3.3 were references to Section 13.9.3.3; and

6590 the allowed values for trustAnchorCellRemotePartyRole and

6591 RemotePartyRole in the Command, and remotePartyRole in the Response

6592 included loadController (8).

6593 Command Construction

6594 See Section 13.9.4.

6595 Command Cryptographic Protection

6596 See Section 13.9.4.

6597 Access Control Broker MAC

6598 See Section 13.9.4.

6599 Command Authenticity and Integrity Verification

6600 See Section 13.9.4.

6601 Additionally, the requirements shall apply as if the Message Code required in 'Check Number

6602 1.2' of Table 13.2.4.3 were 0x011B.

6603 Command Authenticity and Integrity Verification

6604 See Section 13.9.4.

6605 Response Construction

6606 See Section 13.9.4.

6607 Response Cryptographic Protection

6608 See Section 13.9.4.

6609 Response Recipient Cryptographic Verification

6610 See Section 13.9.4.

6611 CS02g Update Load Controller Security Credentials Command, Response and Alert

6612 Description

6613 See Section 13.3.1.

6614 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response / Alert
Message Type Category	Variant Message and is Critical
Capable of future dated invocation?	Yes
Protection Against Replay Required?	The Protection Against Replay mechanisms for Update Security Credentials are specified in Section 13.10. The Protection Against Replay mechanisms of other Sections of the GBCS do not apply
Service Reference	6.15.1
Valid Target Device(s)	ESME / SAPC
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier LoadController
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	ASN.1

6615 Table 13.3.2: Use Case Cross References for Update Security Credentials Command

6616 Command, Response and Alert Structure

6617 The Update Load Controller Security Credentials Command

6618 The Update Load Controller Security Credentials Command shall be the concatenation:

6619 MAC Header || Grouping Header ||

6620 @UpdateLoadControllerSecurityCredentials.CommandPayload || 0x40 || KRP Signature ||

6621 ACB-SMD MAC

6622 with the requirements of Section 13.3.3.1, as to the components of that concatenation,

6623 applying as if @UpdateLoadControllerSecurityCredentials were @UpdateSecurityCredentials.

6624 The Update Load Controller Security Credentials Response

6625 An Update Security Credentials Response shall be the concatenation:

6626 Grouping Header || @ UpdateLoadControllerSecurityCredentials.ResponsePayload || 0x40

6627 || SMD Signature

6628 with the requirements of Section 13.3.3.2, as to the components of that concatenation,

6629 applying as if @UpdateLoadControllerSecurityCredentials were @UpdateSecurityCredentials.

6630 Future Dated Update Load Controller Security Credentials Alert

6631 A Future Dated Update Load Controller Security Credentials Alert shall be the concatenation:

6632 Grouping Header || @UpdateLoadControllerSecurityCredentials.AlertPayload || 0x40 ||

6633 SMD Signature

6634 where:

6635 @UpdateLoadControllerSecurityCredentials.AlertPayload shall be an octet string containing

6636 the DER encoding of the populated ASN.1 structure;

6637 Grouping Header in the Alert shall be constructed as specified in Section 7.2.7 with Business

6638 Target ID being the Entity Identifier specified in the corresponding Command's Grouping

6639 Header;

6640 the Message Code being 0x0124; and

6641 SMD Signature shall be generated as specified in Section 6.3.5.

6642 The Update Load Controller Security Credentials Command, Response and Alert –

6643 informative

6644 The statements of Section 13.3.3.4 shall apply as if:

6645 @UpdateLoadControllerSecurityCredentials were @UpdateSecurityCredentials; and

6646 the reference to Section 13.3.5.1 were a reference to Section 13.10.5.1

6647 save for the fact that Section 18.3.1 does not provide an an illustrative instantiation of
 6648 @UpdateLoadControllerSecurityCredentials.CommandPayload.

6649 Updating Load Controller Security Credentials on a Device – Processing Steps

6650 This Section lays out the requirements for the construction, protection and Authentication
 6651 of the Update Load Controller Security Credentials Command Payload, the processing
 6652 required on the Device of the Command, the construction of the corresponding Response
 6653 Payload and, where required, the Alert Payload.

6654 Command Payload construction

6655 The @ UpdateLoadControllerSecurityCredentials.CommandPayload shall have the structure
 6656 defined in Section 13.10.5.11, and the Remote Party constructing the Command shall
 6657 populate with values according to Table 13.3.4.1 as if:

- 6658 1. @UpdateLoadControllerSecurityCredentials were @UpdateSecurityCredentials;
- 6659 2. The allowed values for credentialsReplacementMode were either
 6660 loadControllerByLoadController (10) or
 6661 loadControllerBySupplier (11);
- 6662 3. Within authorisingRemotePartyTACellIdentifier the allowed values
 6663 for trustAnchorCellRemotePartyRole were either supplier (2) or
 6664 loadController (8);
- 6665 4. The otherRemotePartySeqNumberChanges sequence's presence were
 6666 Mandatory in all loadControllerBySupplier (11) Commands, since all
 6667 such Commands effect a change of control;
- 6668 5. When present, the only allowed value for otherRemotePartyRole were
 6669 loadController (8);
- 6670 6. Within targetTrustAnchorCell, the allowed value for
 6671 trustAnchorCellRemotePartyRole were loadController (8); and
- 6672 7. executionDateTime were allowed to be present when
 6673 credentialsReplacementMode were either
 6674 loadControllerByLoadController (10) or
 6675 loadControllerBySupplier (11).

6676 Command Authenticity and Integrity Verification

6677 The Device shall undertake processing according to the requirements of Section 13.10.5.1.

6678 Should any of the checks detailed in Section 13.10.5.1 fail then the Device shall:

6679 generate an entry in the Security Log recording failed Authentication;
6680 discard the Command without execution and without sending a Response; and
6681 send an Alert notifying the failed Authentication, constructed as specified in Section 6.2.4.2
6682 of the GBCS, populated with the relevant Alert Code according to Section 16, to the Known
6683 Remote Party identified by the Trust Anchor Cell {supplier, digitalSignature, management}.
6684 Where all of the checks detailed in Section 13.10.5.1 succeed the Device shall process the
6685 Command and produce a Response.

6686 Command Processing

6687 Before undertaking any further processing, the Device shall update the Execution Counter
6688 for this type of Command to the value of authorisingRemotePartySeqNumber.

6689 If `executionDateTime` is present then the Device shall:

6690 record against the `remotePartyRole` (as specified in `authorisingRemotePartyControl`),
6691 `authorisingRemotePartyControl`, replacements; and `executionDateTime`;

6692 construct a Response where `executionOutcome` is not present according to the
6693 requirements of Section 13.10.4.4; and

6694 at the date-time specified in `executionDateTime`, or immediately if the `executionDateTime` is
6695 in the past undertake the processing of Section 13.10.4.3.1 then construct an Alert according
6696 to the requirements of Section 13.10.4.5.

6697 If `executionDateTime` is not present then the Device shall:

6698 undertake the processing of Section 13.10.4.3.1; and

6699 construct a Response where `executionOutcome` is present according to the requirements of
6700 Section 13.10.4.4.

6701 replacements Processing

6702 For each of the `targetTrustAnchorCell` in replacements, the Device shall:

6703 record the `entityIdentifier` and `subjectKeyIdentifier` currently held in that
6704 `targetTrustAnchorCell`;

6705 attempt to replace the contents of that `targetTrustAnchorCell` using the corresponding
6706 certificate in `TrustAnchorReplacement`; and

6707 if the contents of the replacement are successfully applied, undertake the processing
6708 required by Section 13.10.5.10 in relation to the `RemotePartyRole` for that
6709 `targetTrustAnchorCell`.

For clarity, before the contents of a certificate are used to replace the contents of a `targetTrustAnchorCell`, the Device shall have successfully undertaken the checks in Section 13.10.5.9 in relation to that certificate. Should a check in 13.10.5.9 not succeed then, in line with Section 4.3.2.8.2, the Device shall undertake no further processing of the Command, except for the issuance of a Message containing an `executionOutcome`.

Response Construction

The `@UpdateLoadControllerSecurityCredentials.ResponsePayload` shall have the structure defined in Section 13.10.5.11, and the Device shall populate the `executionOutcome`, where present with values according to Section 13.10.4.6.

Alert Construction

The `@UpdateLoadControllerSecurityCredentials.AlertPayload` shall have the structure defined in Section 13.10.5.11, and the Device shall populate the `executionOutcome`, with values according to Table 13.3.4.6.

`executionOutcome` construction

The requirements of Section 13.3.4.6 shall apply as if:

8. The allowed values for `credentialsReplacementMode` were either `loadControllerByLoadController` (10) or `loadControllerBySupplier` (11);
9. When present, the only allowed value for `otherRemotePartyRole` were `loadController` (8); and
10. Within `affectedTrustAnchorCell`, the allowed value for `trustAnchorCellRemotePartyRole` were `loadController` (8).

Common Requirements

Update Security Credentials Command Verification

The Device shall undertake the checks set out in this Section 13.10.5.1 before undertaking any other processing of the Command. The checks may be carried out in any order. Checking shall cease at the point that any one check fails. The checks required are shown in Table 13.10.5.1.

Check Number	Criteria that must be tested by the Device	How the Device shall test the Criteria
1.1	The Message is for the Device	The value of the Business

Check Number	Criteria that must be tested by the Device	How the Device shall test the Criteria
		Target ID in the Grouping Header in Command instance must be equal to the Device's Entity Identifier
1.2	The Message Code is for Update Security Credentials	The value in the Message Code field of the Grouping Header must be equal to the value specified in Table 13.10.5.2 for the CredentialsReplacementMode specified in CommandPayload
1.4	The Device has not already actioned this Command.	As specified in Section 13.10.5.3
2.1	The <code>targetTrustAnchorCells</code> all exist on a Device of this type	As specified in Section 13.10.5.4
2.1	The <code>trustAnchorCellUsage</code> in the <code>authorisingRemotePartyTACellIdentifier</code> must be ABSENT	As specified in Section 13.10.4.1
2.2	The <code>credentialsReplacementMode</code> is one that can be Authorised by the Remote Party / Parties authorising the Command	As specified in Section 13.10.5.5
2.2	The <code>replacements</code> specified are all allowed in this <code>credentialsReplacementMode</code> .	As specified in Section 13.10.5.6
2.2	The <code>TrustAnchorCellUsage</code> in the <code>targetTrustAnchorCell</code> in each entry in <code>replacements</code> must be ABSENT unless it has the value <code>prePaymentTopUp</code> as specified in the Notes column of Table 13.3.4.1	As specified in Section 13.10.4.1
2.3	The <code>keyUsage</code> in each of the replacement certificates provided is consistent with the target Trust Anchor Cells identified in	As specified in Section 13.10.5.7

Check Number	Criteria that must be tested by the Device	How the Device shall test the Criteria
	replacements	
3.1	The Cryptographic Protections are valid	As specified in Section 13.10.5.8

6738 Table 13.10.5.1: Update Load Control Security Credentials Command authenticity and
6739 integrity verification

6740 Message Code Validation

CredentialsReplacementMode	Message Code
loadControllerByLoadController	0x0125
loadControllerBySupplier	0x0126

6741 Table 13.10.5.2: Message Code validation against CredentialsReplacementMode

6742 Preventing Replay of Commands

6743 The Protection Against Replay mechanisms for the Update Security Credentials Command
6744 shall be that specified in this Section 13.10.5.3 (which is different than that for other GBCS
6745 Commands).

6746 For each of RemotePartyRole from which the Device can receive a valid Updated Load
6747 Controller Security Credentials Command, the Device shall allocate storage for an Execution
6748 Counter which shall be capable of storing a 64 bit unsigned integer and which shall initially
6749 be set to the value zero at manufacture.

6750 Before executing any Update Security Credentials Command, a Device shall confirm that
6751 (authorisingRemotePartyTACellIdentifier is populated in the Command) and (the
6752 authorisingRemotePartySeqNumber is strictly numerically greater than the Execution
6753 Counter the Device has recorded for the RemotePartyRole identified in
6754 authorisingRemotePartyTACellIdentifier).

6755 Required Trust Anchor Cells by Device Type

6756 The Trust Anchor Cells specified in Section 4.3.2.5 for loadController are those
6757 required on each Device type and so are the only valid targetTrustAnchorCells.

6758 The Device shall ensure that all targetTrustAnchorCells specified in the Command instance
6759 are valid for the type of Device it is, according to the requirements of Section 4.3.2.5. A

6760 Command containing any invalid targetTrustAnchorCells shall not be processed any further
6761 by the Device.

6762 Valid credentialsReplacementMode by Remote Party Roles authorising the Command

6763 A Command containing a certain credentialsReplacementMode is only Authorised using
6764 certain types of Public-Private Key Pairs in certain ways. The Command identifies the Public
6765 Key corresponding to the Private Key used by the authorising Remote Party in the
6766 authorisingRemotePartyTACellIdentifier structure. Table 13.10.5.5 lists the only Authorised
6767 combinations. All other combinations represent Commands not properly Authorised and
6768 shall be rejected by a Device.

	authorisingRemotePartyTACellIdentifier		
	RemotePartyRole	KeyUsage	CellUsage
credentialsReplacementMode			
loadControllerBySupplier	supplier	digitalSignature	management
loadControllerByLoadController	loadController	digitalSignature	management

6769 Table 13.10.5.5: Valid credentialsReplacementMode by Remote Party Roles authorising the
6770 Command

6771 Valid credentialsReplacementMode by the targetTrustAnchorCells specified in the Command

6772 A Command containing a certain credentialsReplacementMode can only validly replace the
6773 Security Credentials in a certain subset of Trust Anchor Cells. The Command identifies the
6774 Cells that are to have credentials replaced in each targetTrustAnchorCell within each
6775 TrustAnchorReplacement in replacements.

6776 Table 13.10.5.6 below lists the only valid targetTrustAnchorCell combinations for each
6777 credentialsReplacementMode. All other combinations are invalid. A Command containing
6778 any invalid combinations shall not be processed any further by the Device.

	targetTrustAnchorCell		
	RemotePartyRole	KeyUsage	CellUsage
credentialsReplacementMode			
loadControllerBySupplier	loadController	any valid for GBCS	any valid for GBCS
loadControllerByLoadController	loadController	any valid for GBCS	any valid for GBCS

6779 Table 13.10.5.6: Valid credentialsReplacementMode by the targetTrustAnchorCells specified
6780 in the Command

6781 Valid usage of Certificates against the targetTrustAnchorCells specified in the Command

6782 For each instance of the TrustAnchorReplacement structure in the Command, the keyUsage
6783 in replacementCertificate shall be equal to targetTrustAnchorCell.KeyUsage. Where this
6784 check fails for any one or more of the TrustAnchorReplacement instances, the Command
6785 shall not be actioned by the Device.

6786 Note that mismatches between RemotePartyRole in the certificate and the target Trust
6787 Anchor Cell are admissible for loadController, and are needed (see Section 4.3.2.5).

6788 Verifying the Cryptographic Protections

6789 In verifying Cryptographic Protections:

6790 KRP Signature shall be verified using the public key identified in the Command; and

6791 ACB-SMD MAC shall be verified according to the requirements in Section 6.2.4.1.2.

6792 Verifying the authenticity of replacement certificates

6793 The Device shall first apply the requirements of Section 12.6 (Device processing of
6794 Certificates). If any of those checks fail, the Section 13.10.5.9 check fails.

6795 In relation to Certification Path Validation, the application of time based checks shall be
6796 determined as follows:

6797 if, in the Command, applyTimeBasedCPVChecks = disapply then time based checks shall NOT
6798 be applied by the Device; and

6799 otherwise time based checks shall be applied or not applied in line with the requirements of
6800 Section 4.3.2.8.

6801 The Device shall, for each replacementCertificate in replacements, undertake Certification
6802 Path Validation according to the requirements of Section 4.3.2.8.

6803 Required Processing on Change of Remote Party Control

6804 If the targetTrustAnchorCell is {loadController, digitalSignature, management} and there are
6805 any future dated Update Load Controller Security Credentials held on the Device that have
6806 not yet executed, and so their executionDateTime is in the future, then the Device
6807 shall set each such executionDateTime to '99991231235959Z'.

6808 If:

6809 the targetTrustAnchorCell is {loadController, digitalSignature, management}; and

6810 the Entity Identifier in the `targetTrustAnchorCell` is changed by the replacement

6811 then the Device shall set the execution date-time of any other future dated Commands that

6812 are from the Load Controller and that are held on the Device but not yet executed, to 'never',

6813 as detailed in Section 9.2.

6814 Relevant Execution Counters' shall be those which relate to types of Commands:

6815 which originate from the `remotePartyRole` of this `targetTrustAnchorCell`;

6816 whose cryptographic protection is checked using the Public Key in this

6817 `targetTrustAnchorCell`.

6818 Where this Command contains an `otherRemotePartySeqNumberChange` in which

6819 `otherRemotePartyRole` is the `remotePartyRole` of this

6820 `targetTrustAnchorCell`, the Device shall set all Relevant Execution Counters to be

6821 the corresponding value in `otherRemotePartyFloorSeqNumber`.

6822 If:

6823 `remotePartyRole` of `targetTrustAnchorCell` and that of

6824 `authorisingRemotePartyControl` is `loadController`; and

6825 `keyUsage` of `targetTrustAnchorCell` is `digitalSignature`,

6826 then the Device shall set all Relevant Execution Counters to the value in

6827 `newRemotePartyFloorSeqNumber`.

6828 Note, where `SeqNumber` values are to be used to populate Execution Counters on the

6829 Device, these `SeqNumber` values shall be processed as unsigned 64 bit integers, so 8 octets

6830 in length.

6831 The

6832 `@UpdateLoadControllerSecurityCredentials.CommandPayload`, `@UpdateLoadControllerSecurityCredentials.ResponsePayload` and

6833 `@UpdateLoadControllerSecurityCredentials.AlertPayload` structure definition

6834

6835 The requirements of Section 13.3.5.11 shall apply as if:

6836 11. `@UpdateLoadControllerSecurityCredentials` were `@UpdateSecurityCredentials`;

6837 12. `RemotePartyRole` were defined as:

6838

6839 `RemotePartyRole ::= INTEGER`

6840 {

```

6841
6842         root    (0),
6843         recovery  (1),
6844         supplier   (2),
6845         networkOperator (3),
6846         accessControlBroker (4),
6847         transitionalCoS    (5),
6848         wanProvider  (6),
6849         issuingAuthority (7),
6850         loadController (8),
6851         other (127)
6852
6853     }
6854 13.    and CredentialsReplacementMode were defined as:
6855
6856         CredentialsReplacementMode ::=  INTEGER
6857         {
6858         supplierBySupplier    (2),
6859         networkOperatorByNetworkOperator    (3),
6860         accessControlBrokerByACB    (4),
6861         wanProviderByWanProvider    (5),
6862         transCoSByTransCoS    (6),
6863         supplierByTransCoS    (7),
6864         anyExceptAbnormalRootByRecovery (8),
6865         anyByContingency    (9),
6866         loadControllerByLoadController    (10),
6867         loadControllerBySupplier    (11)

```

6868 }

6869

6870 Apply Prepayment Top Up to an ESME or GSME

6871 Requirements in this Section 14, and sub-Sections of it, shall, where the SAPC supports the
6872 corresponding SMETS functionality, apply to an SAPC as if it were an ESME.

6873 Defined Terms

6874 The following terms used in this Section 14 shall have the meanings defined in this Table
6875 14.1.

Defined Term	Meaning
Currency Unit	Shall be either GB Pound or European Central Bank Euro
Maximum Credit Threshold	Shall be the maximum value of any single Prepayment Top Up with a resolution of five decimal places. Its value shall be interpreted by the Device in whole Currency Units
Maximum Meter Balance Threshold	Shall be the maximum total credit value recorded on the ESME / GSME with a resolution of five decimal places. Its value shall be interpreted by the Device in whole Currency Units
Highest UTRN Counter	The highest numerical value of any UTRN Counter in the UTRN Counter Cache
Prepayment Token Decimal (PPTD)	Shall have the meaning specified in Section 14.3.1
Prepayment Top Up Token (PTUT)	Shall have the meaning specified in Section 14.3.2
Unique Transaction Reference Number (UTRN)	Shall have the meaning specified in Section 14.3.3
UTRN Check Digit	Shall be the 20 th digit of the UTRN
UTRN Counter Cache	Shall be an array capable of containing up to 100 entries, each entry containing an unsigned integer of 32 bits in length. At manufacture, the UTRN Counter Cache shall contain at least one zero value and shall contain no values

	which are non-zero. The array shall be arranged as a circular buffer such that, when full, further writes shall cause the lowest numerical value entry to be overwritten
UTRN Counter	The 32 most significant bits of the Originator Counter

6876 Table 14.1: Meanings of Defined Terms

6877 Maximum Credit Threshold and Maximum Meter Balance Threshold are represented using a
6878 value and associated scaler. The scaler determines the fractional part of the value, for
6879 example, to represent £123.25478 requiring a resolution of 5 decimal places would use a
6880 value of 12325478 and an associated scaler of -5 (i.e. multiply by 10⁻⁵). To interpret a value
6881 in whole Currency Units, the fractional part would be ignored, so for example, 12325478
6882 would become £123.

6883

6884 Description – informative

6885 This Section covers the application of a Prepayment Top Up, that has been purchased for a
6886 particular ESME or GSME, to that ESME or GSME.

6887 It covers four options:

6888 applying a Prepayment Top Up to an ESME without consumer intervention;

6889 applying a Prepayment Top Up to a GSME without consumer intervention;

6890 applying a Prepayment Top Up to an ESME or GSME with consumer entry of a numeric code
6891 on the ESME or GSME; and

6892 applying a Prepayment Top Up to an ESME or GSME with consumer entry of a numeric code
6893 on a PPMID.

6894 Some requirements are common to all four options. Accordingly, this Section 14 is split in to
6895 five subsections:

6896 an initial subsection covering requirements common to all four options; and

6897 four subsequent subsections covering one option in each subsection.

6898 By way of context:

6899 any Prepayment Top Up Message is a Remote Party Command in GBCS terms (because it is
6900 from a Remote Party to a GSME or ESME). The means of delivery (typing in on meter, typing
6901 in on PPMID, sending over WAN, etc.) does not affect this classification;

as a Remote Party Command, it must result in the GSME or ESME generating a Response back to the Remote Party who issued it (so the Supplier), unless there is an Authentication failure (in which case the Supplier has to be sent an Alert), as per the Technical Specifications;

because the ranges are exclusive, the Originator Counter in Prepayment Top Up transactions cannot collide with the Originator Counter in any other transaction; and

there is no requirement to include the Device's ID explicitly in the locally entered transaction, so a PPMID joined to more than one Smart Meter will need to allow the Consumer to pick which Smart Meter the Prepayment Top Up is for.

Common Requirements

Construction of the PPTD

The PPTD shall be a 19 decimal digit integer. The most significant two digits of the PPTD shall always be between 73 and 96, which shall be constructed and represented according to the requirements of this Section 14.3.1.

The decimal representation of the PPTD shall be the result of the addition of 7,394,156,990,786,306,048 to the decimal representation of the PTUT.

Construction of the PTUT

The PTUT shall be an unsigned 64 bit integer (so 8 octets), which shall be constructed and represented according to the requirements of this Section 14.3.2.

The bits within the PTUT shall be numbered from 63 for the most significant bit through to 0 for the least significant bit.

The bits of the PTUT shall be set to the values in Table 14.3.2.

PTUT component	Value	Bits	Note
PTUT Lead	0b000	63-61	Fixed Value
PTUT Sub Class	0b0000	60-57	Fixed value
PTUT Truncated Originator Counter	Bits 41-32 of the Originator Counter	56-47	Used for Protection Against Replay purposes when the transaction is entered locally
PTUT Value Class	0b00 if PTUT Value is to be interpreted as multiples of 1/100 of Currency Unit;	46-45	If Currency Unit is set to GB Pounds on the ESME or GSME, 0b00 means PTUT Value will be interpreted as

PTUT component	Value	Bits	Note
	OR 0b01 if PTUT Value is to be interpreted as multiples of Currency Unit.		GB Pennies; and 0b01 means PTUT Value will be interpreted as GB Pounds
PTUT Value	The quantum of the PTUT expressed as an unsigned binary number of 13 bits in length, so with leading binary zeros where required	44-32	Thus, the maximum value is either: £81.91 if PTUT Value Class =0b00; or £8,191.00 if PTUT Value Class =0b01.
PTUT Supplier MAC	See Section 14.3.4	31-0	

6924 Table 14.3.2: Values of PTUT bits

6925 Construction of the Unique Transaction Reference Number (UTRN)

6926 The Unique Transaction Reference Number (UTRN) shall be a 20 decimal digit which shall be
 6927 the 19 decimal digits of the PPTD with a 20th decimal digit which shall be appended after the
 6928 least significant digit of the 19 decimal digit representation of PPTD. This 20th decimal digit
 6929 shall be the UTRN Check Digit. The UTRN Check Digit shall be calculated according to the
 6930 requirements of Section 14.8.

6931 Construction of the PTUT Supplier MAC

6932 The PTUT Supplier MAC shall only be calculated once the 32 most significant bits of PTUT
 6933 (bits 63-32 of the PTUT) have been populated as per the requirements of Section 14.3.2.

6934 The Remote Party, whose Security Credentials are stored against the Supplier role of the
 6935 target Device, shall calculate a MAC using the parameters in Table 14.3.4 then setting the
 6936 PTUT Supplier MAC to be the 32 least significant bits of the 128 bit MAC produced by the
 6937 MAC calculation.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Supplier's Prepayment Top Up Key	

Input Parameter	Value	Note
	Agreement Key [which the Supplier may elect to be different than the Key Agreement Key they use for other interactions with the Device]	
Public Key Agreement Key	Device's	
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Supplier's prePaymentTopUp Entity Identifier Device's Entity Identifier 0x01 Originator Counter 32 most significant bits of the PTUT	

6938 Table 14.3.4: Calculation of the PTUT Supplier MAC

6939 Validating the PTUT Supplier MAC

6940 To validate the PTUT Supplier MAC, the Device shall calculate the MAC using the parameters
 6941 in Table 14.3.5, then ensure the 32 least significant bits of the 128 bit MAC produced by the
 6942 MAC calculation has the same value as the PTUT Supplier MAC.

Input Parameter	Value	Note
To calculate the Shared Secret ('Z') input to the KDF:		
Private Key Agreement Key	Device's	
Public Key Agreement Key	Supplier's Prepayment Top Up Key Agreement Key	
The other input to the KDF ('OtherInfo') shall be calculated according to the requirements of Section 4.3.3.3.		

Input Parameter	Value	Note
As input to the GMAC function, the IV shall be constructed according to the requirements of Section 4.3.3.4, the Plaintext shall be empty and:		
Additional Authenticated Data shall be the concatenation:	0x110000000000 Supplier's prePaymentTopUp Entity Identifier Device's Entity Identifier 0x01 Originator Counter 32 most significant bits of the PTUT	

6943 Table 14.3.5: Validation of the PTUT Supplier MAC

6944 Checking the UTRN Counter against the UTRN Counter Cache

6945 The Device shall set the UTRN Counter to be the 32 most significant bits of the Originator
6946 Counter.

6947 The Device shall check that the UTRN Counter is strictly numerically greater than the
6948 numerically lowest value in the UTRN Counter Cache, and is not equal to any value in the
6949 UTRN Counter Cache.

6950 Updating the UTRN Counter Cache

6951 Where the Prepayment Top Up is successfully applied and prior to sending any Response,
6952 the Device shall add a new entry to the UTRN Counter Cache whose UTRN Counter value
6953 shall be set to the 32 most significant bits of Originator Counter.

6954 Validating the Maximum Credit Values

6955 Maximum Credit Threshold

6956 The Device shall ensure that the Prepayment Top Up's value specified by PTUT Value Class
6957 and PTUT Value does not exceed the Device's Maximum Credit Threshold parameter.

6958 Maximum Meter Balance Threshold

6959 The Device shall ensure that the Prepayment Top Up's value specified by PTUT Value Class
6960 and PTUT Value when added to the Device's Credit Balance does not exceed the Device's
6961 Maximum Meter Balance Threshold parameter.

6962 Validating the PTUT Sub-Class

6963 The Device shall ensure that the value specified by PTUT Sub-Class is of value 0b0000.

6964 Applying Prepayment Top Ups

- 6965 For clarity, ESME and GSME shall apply the value of Prepayment Top Ups as specified in
6966 SMETS.
- 6967 CS01a Applying a Prepayment Top Up to an ESME without consumer intervention
- 6968 Description
- 6969 This Section covers the application of a Prepayment Top Up that has been bought for an
6970 ESME to that ESME, in the case where the consumer does not enter any details on Devices in
6971 their premises.
- 6972 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	SME.C.NC but with additional cryptographic processing specified in Sections 14.3.4 and 14.3.5
Capable of future dated invocation?	No
Protection Against Replay Required?	The Protection Against Replay mechanisms for Prepayment Top Ups are specified in Section 14.3.6. The Protection Against Replay mechanisms specified elsewhere in the GBCS do not apply
Service Reference	2.2
Valid Target Device(s)	ESME
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the ESME) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A

Cross Reference	Value
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	DLMS COSEM

6973 Table 14.4.2: Use Case Cross References for Prepayment Top Up to an ESME without
6974 consumer intervention

6975 Pre-conditions

6976 None.

6977 Detailed Steps

6978 The Device shall undertake the checks set out in this Section 14.4.4 in the sequence laid out:
6979 only once all checks in Section 6.2.4.1.1 have been successfully completed; and
6980 before undertaking any other processing of the Command.

6981 If any of the checks specified in this Section 14.4.4 fail, the Device shall not carry out further
6982 checks, and the requirements of Section 6.2.4.2 shall apply. Otherwise, processing shall
6983 continue as per the requirements of Section 6.2.4.1.2. Where that check is successful,
6984 processing shall continue as below.

6985 Verifying against the maximum credit values

6986 The Device shall carry out the checks specified in Section 14.3.8.1 and Section 14.3.8.2.

6987 Verifying the Originator Counter

6988 The Device shall verify the Originator Counter against the UTRN Counter Cache according to
6989 Section 14.3.6.

6990 Validating the PTUT Supplier MAC

6991 The Device shall validate the PTUT Supplier MAC according to Section 14.3.5.

6992 Response Construction

6993 At the Response Construction stage, the Device shall first update the UTRN Counter Cache
6994 according to Section 14.3.7, and shall then populate the Response according to the
6995 requirements of the Message Template for CS01a.

6996 CS01b Applying a Prepayment Top Up to a GSME without consumer intervention

6997 Description

6998 This Section covers the application of a Prepayment Top Up that has been bought for a
 6999 GSME to that GSME, in the case where the consumer does not enter any details on Devices
 7000 in their premises, except for the additional processing defined in this Section.

7001 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	See Table 14.4.2
Capable of future dated invocation?	No
Protection Against Replay Required?	See Table 14.4.2
Service Reference	2.2
Valid Target Device(s)	GSME
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the ESME) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	GBZ

7002 Table 14.5.2: Use Case Cross References for Prepayment Top Up to a GSME without
 7003 consumer intervention

7004 Pre-conditions

7005 None.

7006 Detailed Steps

7007 The Device shall undertake the checks set out in this Section 14.5.4 in the sequence laid out:

- 7008 only once all checks in Section 6.2.4.1.1 have been successfully completed; and
- 7009 before undertaking any other processing of the Command.
- 7010 Verifying against the maximum credit values
- 7011 The Device shall carry out the checks specified in Section 14.3.8.1 and Section 14.3.8.2.
- 7012 Verifying the Originator Counter
- 7013 The Device shall verify the Originator Counter against the UTRN Counter Cache according to
- 7014 Section 14.3.6.
- 7015 Validating the PTUT Supplier MAC
- 7016 The Device shall validate the PTUT Supplier MAC according to Section 14.3.5.
- 7017 If any of the checks specified in this Section 14.5.4 fail, the requirements of Section 6.2.4.2
- 7018 shall apply. Otherwise, processing shall continue as per the requirements of Section
- 7019 6.2.4.1.2.
- 7020 Response Construction
- 7021 At the Response Construction stage, the Device shall first update the UTRN Counter Cache
- 7022 according to Section 14.3.7 and shall then populate the Response according to the
- 7023 requirements of Use Case CS01b.
- 7024 Applying a Prepayment Top Up to an ESME or GSME with consumer entry of a numeric code
- 7025 on the ESME or GSME
- 7026 Description
- 7027 This Section covers the application of a Prepayment Top Up that has been bought for a
- 7028 GSME or ESME to that GSME or ESME in the case where the consumer enters the
- 7029 corresponding UTRN on the GSME or ESME.
- 7030 The Use Case covering the Response is referenced in Section 14.6.5.
- 7031 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Response
Message Type Category	This is a Variant Message type. The Command shall be the UTRN constructed in accordance with Section 14.3.3. The

Cross Reference	Value
	Command includes cryptographic protections as specified in Sections 14.3.4 and 14.3.5. The Response shall be of Message Type Category SME.C.NC.
Capable of future dated invocation?	No
Protection Against Replay Required?	See Table 14.4.2
Service Reference	N/A
Valid Target Device(s)	ESME or GSME
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [HAN Only Messages]	N/A
Protocol	Outside of protocols since entered via User Interface

7032 Table 14.6.2: Use Case Cross References for Prepayment Top Up through consumer UTRN
7033 entry

7034 Pre-conditions

7035 None.

7036 Detailed Steps

7037 Detailed Steps/Sequence

7038 The Device shall undertake the validation checks set out in this Section 14.6.4.1 before
7039 undertaking any other processing of the Command. The validation checks shall be
7040 undertaken in the sequence laid out. Should a validation check fail, subsequent validation
7041 checks shall not be undertaken by the Device.

-
- 7042 Should any of the checks fail (save for the optional UTRN Check Digit verification), the
 7043 requirements of Section 6.2.4.2 shall apply.
- 7044 Verifying the UTRN Check Digit
- 7045 The Device:
- 7046 may validate the 20th digit (the UTRN Check Digit) as specified at Section 14.8 (Calculating
 7047 and Verifying the UTRN Check Digit); and
- 7048 shall disregard the 20th decimal digit to determine PPTD prior to undertaking any
 7049 subsequent checks.
- 7050 Using the PPTD to calculate the PTUT
- 7051 PTUT shall take the value of PPTD minus 7,394,156,990,786,306,048.
- 7052 The Device shall interpret the resulting unsigned integer according to Table 14.3.2.
- 7053 Verifying PTUT subclass category
- 7054 The Device shall carry out the checks specified in Section 14.3.9.
- 7055 Verifying against the maximum credit values
- 7056 The Device shall carry out the checks specified in Section 14.3.8.1 and Section 14.3.8.2.
- 7057 Deriving the Originator Counter⁴⁵
- 7058 The Originator Counter shall be derived by:
- 7059 1. creating four 32 bit signed integer variables p, q, r and s;
 - 7060 2. setting p = the numeric value of the 10 least significant bits of Highest UTRN Counter;
 - 7061 3. setting q = (the numeric value Highest UTRN Counter) – p;
 - 7062 4. setting r = the numeric value of PTUT Truncated Originator Counter;
 - 7063 5. if $r < (p - 2^9)$ then setting $s = (r + 2^{10})$ else if $r > (p + 2^9)$ then setting $s = (r - 2^{10})$ else
 7064 setting $s = r$;
 - 7065 6. setting Originator Counter equal to $((q + s) * 2^{32})$
- 7066 Verifying the Originator Counter

45 This derivation places a practical limit on the maximum increment between issued sequentially UTRN Counters. An increment of greater than $(2^9 - 1)$ between a UTRN Counter and the next one issued will cause this derivation to be inaccurate

7067 The Device shall verify the Originator Counter against the UTRN Counter Cache according to
7068 Section 14.3.6.

7069 Deriving the Message Identifier

7070 The Device shall derive the Message Identifier by:

7071 setting the Business Target ID to the Entity Identifier in the Key Agreement Security
7072 Credentials it holds for the Trust Anchor Cell with Remote Party Role as supplier and cell
7073 usage of prePaymentTopUp;

7074 setting the Business Originator ID to its own Entity Identifier; and

7075 setting Message Identifier to the concatenation Business Originator ID || Business Target ID
7076 || 0x02 || Originator Counter.

7077 Validating the PTUT Supplier MAC

7078 The Device shall validate the PTUT Supplier MAC according to Section 14.3.5.

7079 Response Construction

7080 The Device shall first update the UTRN Counter Cache according to Section 14.3.7.

7081 Where the Device is an ESME, the Device shall construct, and send via its HAN interface, a
7082 Response message complying with the requirements of Use Case CS01a, using a Message
7083 Identifier as specified in Section 14.6.4.1.7, and where the Originator Counter is as derived
7084 by the calculations in Section 14.6.4.1.5.

7085 Where the Device is a GSME, the Device shall construct, and send via its HAN interface, a
7086 Response message complying with the requirements of Use Case CS01b, using Message
7087 Identifier as specified in Section 14.6.4.1.7, and where the Originator Counter is as derived
7088 by the calculations in Section 14.6.4.1.5.

7089 Applying a Prepayment Top Up to an ESME or GSME with consumer entry of a numeric code
7090 on a PPMID

7091 Description

7092 This Section covers the application of a Prepayment Top Up that has been bought for a
7093 specific GSME or ESME to that GSME or ESME in the case where the consumer enters the
7094 corresponding UTRN on a PPMID on the same SMHAN.

7095 The Use Case covering the Command is referenced in Section 14.7.4.1.2, The Use Case
7096 covering the Response is referenced in Section 14.7.4.1.4.

7097 Use Case Cross References

Cross Reference	Value
Grouping	Remote Party Message
Message Type	Command and Responses
Message Type Category	The Command and Response requirements are specifically as detailed in this Section 14.7
Capable of future dated invocation?	No
Protection Against Replay Required?	See Table 14.4.2
Service Reference	N/A
Valid Target Device(s)	ESME or GSME
Valid Business Originator role(s) for Command invocation (and so, for DLMS COSEM Commands, which Application Association is to be used for delivery of the Command to the Device) [Remote Party Messages Only]	Supplier
Valid Response Recipient role(s) (only for Messages Authorised by the Access Control Broker on behalf of parties not known to the Device) [Remote Party Messages Only]	N/A
Valid initiating Device type(s) [SMHAN Only Messages]	N/A
Protocol	See this Section 14.7

7098 Table 14.7.2: Use Case Cross References for Prepayment Top Up through PPMID entry

7099 Pre-conditions

7100 None.

7101 Detailed Steps

7102 Detailed Steps / Sequence

7103 Verifying the UTRN check digit

7104 The PPMID may validate the 20th digit (the UTRN Check Digit) as specified at Section 14.8

7105 (Calculating and Verifying the UTRN Check Digit). Where this check fails, the PPMID shall

7106 cease processing the Command and shall inform the consumer of the failure of the check

7107 digit.

7108 Command Construction by the PPMID

7109 Where the target Device is a GSME, the PPMID shall construct the Command according to
7110 the requirements of Use Case PCS01.

7111 Where the target Device is an ESME, the PPMID shall construct a ZSE Consumer Top Up
7112 command.

7113 In all cases:

7114 the value of the Top Up Code, with its ZSE meaning, shall be set to be a VisibleString whose
7115 value is the 20 digit UTRN; and

7116 the value of the Originating Device, with its ZSE meaning, shall be 0x02 (IHD).

7117 HAN Only Command Validation by the ESME / GSME

7118 If the ESME / GSME has no PPMID in its Device Log, the ESME / GSME shall apply the
7119 requirements of Section 6.2.4.2 to reflect the failed check, and undertake no additional
7120 processing.

7121 If the ESME / GSME has a PPMID in its Device Log:

7122 if the receiving Device is an ESME, the ESME shall use ZSE cryptographic processes to
7123 establish whether the Command was authentically issued by a PPMID that is in its Device
7124 Log; or

7125 if the receiving Device is a GSME, the GSME shall undertake Command Authenticity and
7126 Integrity Verification, as required for a Command of Message Category SME.C.PPMID-GSME
7127 to establish whether the Command was authentically issued by the PPMID that is in its
7128 Device Log. For clarity, the 'PPMID entity ID' and the 'Meter entity ID' from PCS01 and
7129 PCS02 Commands are the Business Originator ID and Business Target ID respectively used in
7130 the Section 6.2.4.1.1 checks.

7131 If the Command was not authentically issued by a PPMID within the Device Log, the ESME /
7132 GSME shall apply the requirements of Section 6.2.4.2 to reflect the failed check.

7133 If the Command was authentically issued by a PPMID within the Device Log, the ESME /
7134 GSME shall comply with the requirements of Section 14.6.4 (but excluding requirements in
7135 Sections 14.6.4.1.1, save that the ESME / GSME shall disregard the 20th digit before
7136 undertaking any further steps), and so process the contents of the Command accordingly.

7137 HAN Only Response Construction and Issue

7138 Where the ESME / GSME successfully creates a Remote Party Response to its Supplier, as per
7139 the requirements in Section 14.6.5, the ESME / GSME shall also:

7140 where the Device is a GSME, construct the HAN Only Response according to the
 7141 requirements of Use Case PCS01 and send it to the PPMID; or

7142 where the Device is an ESME, construct a ZSE Consumer Top Up Response command, and
 7143 send it to the PPMID.

7144 In all cases the value of the Source of Top up, with its ZSE meaning, shall be 0x02 (IHD).

7145 Calculating and Verifying the UTRN Check Digit

7146 The UTRN Check Digit shall be calculated from the the PPTD by a process equivalent to the
 7147 following (Verhoeff's) Algorithm⁴⁶:

7148 setting an interim digit (referred to as *IntDig*) to have a value of zero;

7149 setting an index (referred to as *K*) to have a value of four;

7150 repeating the following steps with another index (referred to as *J*) taking the nineteen values
 7151 of the integers from 1 to 19 in succession;

7152 setting *CurDig* to the value of the *J*th digit of the 19 decimal digits of the PPTD, where the
 7153 first digit is the most significant (leftmost as written) and the nineteenth digit the least
 7154 significant;

7155 setting a third index (referred to as *L*) to the value in Table 14.8a using *K* as the Row Index
 7156 and *CurDig* as the Column Index;

7157 if the value of *K* is less than 7, setting *K* to the value of *K*+1, otherwise setting *K* to zero;

7158 setting *IntDig* to the value in Table 14.8b using *IntDig* as the Row Index and *L* as the Column
 7159 Index;

7160 setting *IntDig* to the value in row 1 of Table 14.8c using the value of *IntDig* as the Column
 7161 Index; and

7162 setting the UTRN Check Digit to the value of *IntDig*.

7163 The UTRN Check Digit may be verified by undertaking exactly the same calculation on the 19
 7164 most significant digits of the UTRN, and comparing the result (the final value of *IntDig*, which

46 See: (1) Verhoeff, J. (1969). Error Detecting Decimal Codes (Tract 29). The Mathematical Centre, Amsterdam. doi:10.1002/zamm.19710510323., (2) Kirtland, Joseph (2001). Identification Numbers and Check Digit Schemes. Mathematical Association of America. p. 153. ISBN 0-88385-720-0. Retrieved August 26, 2011. (3) Salomon, David (2005). Coding for Data and Computer Communications. Springer. p. 56. ISBN 0-387-21245-0. Retrieved August 26, 2011

7165 would be used to set the UTRN Check Digit) to the 20th decimal digit which is the UTRN
7166 Check Digit.

		Column Index									
		0	1	2	3	4	5	6	7	8	9
Row Index	0	0	1	2	3	4	5	6	7	8	9
	1	1	5	7	6	2	8	3	0	9	4
	2	5	8	0	3	7	9	6	1	4	2
	3	8	9	1	6	0	4	3	5	2	7
	4	9	4	5	3	1	2	6	8	7	0
	5	4	2	8	6	5	7	3	9	0	1
	6	2	7	9	3	8	0	6	4	1	5
	7	7	0	4	6	9	1	3	2	5	8

7167 Table 14.8a: Setting a third index

7168

		Column Index									
		0	1	2	3	4	5	6	7	8	9
Row Index	0	0	1	2	3	4	5	6	7	8	9
	1	1	2	3	4	0	6	7	8	9	5
	2	2	3	4	0	1	7	8	9	5	6
	3	3	4	0	1	2	8	9	5	6	7
	4	4	0	1	2	3	9	5	6	7	8
	5	5	9	8	7	6	0	4	3	2	1
	6	6	5	9	8	7	1	0	4	3	2
	7	7	6	5	9	8	2	1	0	4	3
	8	8	7	6	5	9	3	2	1	0	4
	9	9	8	7	6	5	4	3	2	1	0

7169 Table 14.8b: Setting IntDig using IntDig as a Row Index

7170

Column Index											
Row		0	1	2	3	4	5	6	7	8	9
Index	1	1	2	6	7	5	8	3	0	9	4

7171 Table 14.8c: Setting IntDig using IntDig as a Column Index

7172 Message Codes

7173 Message Codes shall be 2 octets in length and shall take the values specified in the 'Use Case
7174 reference' tab in the Mapping Table.

7175 For Messages specified by this GBCS, the most significant bit of the Message Code shall be
7176 0b0.

7177 Event / Alert Codes and related requirements

7178 Italicised terms in this Section 16 shall have their meaning in the ZCL / ZSE specifications.

7179 Introduction – informative

7180 This Section 16 sets out how Events and Alerts are handled. The Technical Specifications
7181 define when Events occur and whether these Events are logged (in an Event Log) and
7182 whether sent as an Alert via the HAN / WAN.

7183 Table 16.2 defines Event Codes for events defined in the Technical Specifications. It also
7184 indicates whether, as per the Technical Specifications, there is a corresponding Alert issued
7185 over the Device's network interface (containing the relevant Event Code). It is important to
7186 note that not all Event Codes have a corresponding Alert. Where Alert Code is used
7187 elsewhere in this document, it equates to Event Code in Table 16.2.

7188 Alerts sent over the SMHAN are not subject to the same message categorisation as those
7189 sent over the WAN. An Alert sent over the SMHAN is a native ZSE message.

7190 Types of Alert

7191 There are two Alert types. All have the same Grouping Header but different payloads as set
7192 out below:

7193 Alert type 1 - Payload comprises Alert Code and Timestamp only (two sub-types: DLMS and
7194 ZigBee). These are labelled 'Y(1)' in the 'Alert WAN (Alert type)' column in Table 16.2; and

7195 Alert type 2 - Payload comprises Alert Code, Timestamp and Use Case specific data as
7196 defined in Table 16.2 or main body of document (three sub-types: ASN.1, DLMS and ZigBee).
7197 These are labelled 'Y(2)' in the 'Alert WAN (Alert type)' column in Table 16.2.

7198 Table 16.2 sets out the Alert type for each Alert Code. Examples of Use Case specific data
7199 include Billing Data Logs and content relating to future dated Commands (e.g. Message ID).

7200 Table 16.2 sets out whether Alerts are mandated, mandatory conditional or non-mandated:

7201 Mandated - Alerts that Devices of the applicable Device Type must support;

7202 Mandated conditional – Devices of the applicable Device Type must support at least one
7203 from the specified group (e.g. there are seven Alerts in ‘mandated – conditional group 1’,
7204 Devices must support at least one of these seven); and

7205 Non-mandatory – no requirement for Devices to support, but where implemented Alert
7206 Codes shall have the meaning shown in Table 16.2.

7207 An SAPC shall support the Alerts, and support corresponding logging requirements, for rows
7208 in Table 16.2 marked ‘mandatory’ in the column titled ‘SAPC support’. For clarity, ‘ECS100
7209 Command not supported by Device’ is marked as ‘mandatory’ and must be supported by an
7210 SAPC unless the SAPC supports all Use Cases required of an ESME which is a Single Element
7211 Electricity Metering Equipment, with its SMETS meaning.

7212 An SAPC, where it has a Boost Function, with its SMETS meaning, shall support the Alerts,
7213 and support corresponding logging requirements, for rows in Table 16.2 marked ‘mandatory
7214 where the SAPC has a Boost Function, with its SMETS meaning’ in the column titled ‘SAPC
7215 support’.

7216 An SAPC shall support, for each group, at least one of the Alerts, and support corresponding
7217 logging requirements, for rows in Table 16.2 marked ‘mandatory – conditional group N’ in
7218 the ‘SAPC support’ column, where ‘N’ is a number.

7219 An SAPC shall additionally support the Alerts, and corresponding logging requirements, for
7220 rows in Table 16.2 marked ‘optional’ in the ‘SAPC support’ column, where the SAPC supports
7221 the corresponding SMETS functionality.

7222 Alert Construction

7223 Alert construction is described in the GBCS in a number of places, including:

7224 Section 7.2.3 details common Message construction for all Alert types;

7225 Section 7.2.9 details Message construction for Alerts with DLMS COSEM Payloads. Table
7226 7.2.9c details the required components of the Alert;

7227 Section 7.2.10 details Message construction for Alerts with ZSE Payloads. Table 7.2.10c
7228 details the required components of the Alert;

7229 Sections 11.2 and 13.3 detail the Message construction for the Alerts with ASN.1 Payload;
7230 and

7231 Section 9.2.2 details the Message construction for future dated Alerts.

7232 Event Behaviour

7233 Detail on Event behaviour can be found in the Technical Specifications using the relevant
7234 SMETS and CHTS reference in Table 16.2.

7235 Event and Alert Codes

7236 Table 16.2 lists the valid Event and Alert Codes, and sets out their requirements.



GBCS v4.4 Table
16.2.xlsx

7237

7238 Table 16.2: Event and Alert Codes

7239 Event Logs

7240 Only GSME, ESME, SAPC, CHF and GPF have Event Logs. The requirement set out in Table
7241 16.2 to log entries into Event Logs only applies to GSME, SAPC, ESME, CHF and GPF as
7242 follows:

7243 Event Log (GSME, SAPC, ESME, CHF and GPF);

7244 Security Event Log (GSME, SAPC, ESME, CHF and GPF);

7245 Power Event Log (ESME and SAPC where the SAPC supports the corresponding SMETS
7246 functionality); and

7247 Auxiliary Controller Event Log (SAPC and ESME).

7248 Use Cases to read logs (all) and clear logs (event logs only) are detailed in the Mapping Table.

7249 Requirements

7250 Event / Alert codes shall be 2 octets in length and shall take the values specified in Table 16.2.

7251 As per the Technical Specifications, all Alerts, Event Log entries, Security Log entries, Power

7252 Event Log entries and Auxiliary Controller Event Log entries shall contain a UTC date time

7253 stamp, in addition to the Event / Alert code. For Event / Alert Codes beginning 0x80 or 0x81

7254 in Table 16.2 which relate to a GSME or ESME, the requirements in Table 16.2 in relation to:

7255 sending Alerts;

7256 notifying events to other Devices to which it is connected by means of a Communications

7257 Link; and

7258 logging events in the Event Log or Power Event Log,

7259 shall be those configured in ESME, SAPC and GSME prior to installation.

7260 In relation to Event / Alert Codes 0x8161 and 0x8162 and the notification of those events to
7261 other Devices to which the Device is connected by means of a Communication Link, the
7262 Device shall be configured prior to installation so as to not send such notifications unless:

7263 the Device is an ESME which includes an APC; or

7264 the Device is an SAPC which includes a User Interface and that User Interface provides
7265 access to User Interface Commands detailed in Table 16.4;

7266 in which cases:

7267 the Device shall be configured prior to installation so as to send such notifications; and

7268 when sending such notifications, the Device shall, in relation to any ZigBee Publish Event
7269 command used to send such a notification:

7270 populate the Event Data parameter with the value 0x02 || (Table 16.4 'User
7271 Interface Command Code' corresponding to the Command invoked); and

7272 populate the Event ID parameter with either 0x8161 or 0x8162 to reflect the
7273 Outcome of that Command.

7274 For clarity, the requirement to populate the Event Data parameter applies only to ESME with
7275 APC and SAPC with a User Interface. Other Devices, if configured after installation to notify
7276 events Event / Alert Codes 0x8161 and 0x8162, would not be required to populate that field
7277 with the Table 16.4 value, and so that value may not be present. Alternatively, for these
7278 other Devices a value may be present but may not follow the requirements of this Section
7279 16.4.

7280 Prior to installation, all such Event / Alert Codes shall be configured not to sound an Alarm.
7281 For clarity, Event / Alert Codes beginning 0x8F and any applying on Devices other than ESME
7282 / GSME / SAPC are not configurable and Table 16.2 specifies the fixed settings. Once
7283 installed, the configurations relating to sending Alerts, notifying other Devices, logging and
7284 sounding Alarms may be modified using the event and Alert configuration Use Cases.

7285 GSME shall reject any ZSE *SetEventConfiguration* command containing an *Event ID* in the
7286 *Event Configuration Payload* with 0x8F in the most significant octet, to ensure Critical Alerts
7287 are always configured on. For clarity, the ESME Alert Configuration Use Cases, which may
7288 also be supported by SAPC, do not allow for Alert Codes starting 0x8F.

7289 SAPC is not required to support the Event / Alert configuration Use Cases. Where SAPC does
7290 not support such Use Cases, the settings required by Table 16.2 shall remain unchanged,
7291 including for any Event / Alert Codes beginning 0x80 or 0x81.

7292 As specified in Table 16.2 by way of 'x' in a cell, deviceType (and for ESME, variant of ESME)
7293 shall determine which Alerts a device shall issue and which Event Log and Security Log
7294 entries it shall record. Where deviceType = 0x04 (HCALCS) or 0x05 (PPMID), this Section 16
7295 only requires the sending of Alerts, since neither Device type is required to have either an
7296 Event Log or a Security Log.

7297 Where an Alert and a Log entry have the same trigger in a Device, the Device shall record
7298 the same UTC date time stamp and the same Event / Alert code in both.

7299 The Remote Party to which an Alert containing a specific Event Code is addressed shall be
7300 determined by the Remote Party Role as specified in Table 16.2. Where the Remote Party
7301 Role is stated as 'Supplier (not CHF) or WAN Provider (CHF only)', the Alert shall be
7302 addressed:

7303 to the WAN Provider if deviceType = 0x02 (CHF); or
7304 to the Supplier for all other deviceType values.

7305 Where the Remote Party Role is stated as 'Transitional Change of Supplier or Supplier (not
7306 CHF) or WAN Provider (CHF only)', the Alert shall be addressed to the Transitional Change of
7307 Supplier if the Alert is as a result of a `supplierByTransCoS` CS02b Update Security
7308 Credentials Command; otherwise is shall be addressed:

7309 to the WAN Provider if deviceType = 0x02 (CHF); or
7310 to the Supplier for all other deviceType values.

7311 Where a Use Case is specified in Table 16.2 the corresponding Alert shall be constructed
7312 according to the specified Use Case. Where no Use Case is specified the Alert shall be
7313 constructed according to Section 7.

7314 Where an Alert has two recipient roles identified, the Device shall place the Entity ID of the
7315 Supplier in the Business Target ID field and the Entity ID of the other recipient in the
7316 Supplementary Remote Party ID field.

7317 For any Event Log entries relating to Event Codes 0x8161 and 0x8162, the Device shall record
7318 the commands input on the User Interface by including the User Interface Command Code in
7319 the Event Log entry as defined in Table 16.4.

7320 Where an SAPC has a User Interface to support SMETS requirements, for any Event Log
7321 entries relating to Event Codes 0x8161 and 0x8162, the SAPC shall record the Commands
7322 input on the User Interface by including the User Interface Command Code in the Event Log
7323 entry as defined in Table 16.4, as if the SAPC were an ESME.

User Interface Command Code	User Interface Command (from SMETS)	GSME	ESME	Additionally for ESME with ALCS	Additionally for ESME with Boost Function
0x0001	Activate Boost Period				x
0x0002	Activate Emergency Credit [PIN]	x	x		
0x0005	Add Credit	x	x		
0x0008	Allow Access to User Interface	x	x		
0x000A	Cancel Boost Period				x
0x000B	Check for HAN Interface Commands	x			
0x000C	Disable Privacy PIN Protection [PIN]	x	x		
0x000E	Enable Supply [PIN]	x	x		
0x000F	Extend Boost Period				x
0x0012	Set Privacy PIN [PIN]	x	x		
0x0013	Test ALCS [1]			x	
0x0014	Test ALCS [2]			x	
0x0015	Test ALCS [3]			x	
0x0016	Test ALCS [4]			x	
0x0017	Test ALCS [5]			x	
0x0018	Test Valve	x			
0x0019	Reset Remaining Battery Capacity	x			
0x001A	Find and Join SMHAN	x	x	x	x

7324 Table 16.4: User Interface Command Codes by Device

7325 For any Event Log entries relating to Event Codes 0x8154 and 0x8155, the Device shall record
7326 the Commands received on the Network Interface by including the Message Code in the
7327 Event Log. For clarity, such entries are only required where the command received has a
7328 Message Code specified in Table 20.

7329 Where a log entry is required to have data additional to the Alert Code and date-time stamp,
7330 that additional data shall be recorded in the 'otherInformation' field of that log entry.

7331 For Alerts with Alert Code 0x81A0, the Device shall structure and populate the Message
7332 according to the requirements of Table 16.2 and:

7333 an ESME shall create a DLMS COSEM Alert Payload, which shall be as defined in Table 7.2.9c,
7334 where the Use Case specific additional content contains the concatenation 0x09 || 0x02 ||
7335 otherInformation; or

7336 a GSME shall create a GBZ Alert Payload, which shall be as defined in Table 7.2.10c, where
7337 the Use Case specific additional content is otherInformation.

7338 The Device shall populate the 'otherInformation' field with one of the following two octet
7339 long octet string values:

7340 0x0000 (other);

7341 0x0001 (Error Non Volatile Memory);

7342 0x0002 (Error Program Execution);

7343 0x0003 (Error Program Storage);

7344 0x0004 (Error RAM);

7345 0x0005 (Error Unexpected Hardware Reset);

7346 0x0006 (Error Watchdog);

7347 0x0007 (Error Metrology Firmware Verification Failure);

7348 0x0008 (Error Measurement Fault); or

7349 0x0009 (Unspecified Smart Meter Operational Integrity Error).

7350 Remote Party Usage Rights

7351 Remote Party Access Rights to Attributes and Methods

7352 Access rights to attributes and methods shall be enforced by the Device as per the
7353 requirements in the 'SMETS required objects' tab in the Mapping Table. 'R' shall mean that
7354 the Remote Party Role shall have read access to the attribute. 'W' shall mean that the
7355 Remote Party Role shall have write access to the attribute. 'A' shall mean that the Remote
7356 Party Role shall be able to invoke the method. There shall be no other access to these
7357 attributes and methods allowed by the Device.

7358 Encryption of attributes whenever transiting the HAN Interface shall be enforced by the
7359 Device as per the requirements in the 'SMETS required objects' tab in the Mapping Table. 'Y'
7360 in the column headed 'Encrypted' shall mean that the Encryption shall always be applied to
7361 the corresponding attribute as it crosses the HAN Interface.

7362 Remote Party Usage Rights to Use Cases

7363 Access rights to Use Cases shall be enforced by the Device as per the requirements in the
7364 Use Case Access Permissions table in each Use Case (see Table 19.3). A Remote Party Role in
7365 that table shall have access to the Use Case. There shall be no other access allowed by the
7366 Device. Remote Party roles align to the Trust Anchor Cells in Section 4.3.2.5. The Access
7367 Control Broker controls access for Unknown Remote Parties.

7368 Message Templates

7369 GBZ and ZSE Message Templates

7370 Message Templates for GBZ Use Cases are detailed in the embedded Use Cases, Section 19.3.
7371 These Message Templates are derived from the Mapping Table, and shall be complied with
7372 in the construction and population of all such Messages.

7373 Message Templates for ZSE commands between ESME and HCALCS

7374 Requirements in this Section 18.1.1, and sub-Sections of it, shall apply to an SAPC as if it
7375 were an ESME.

7376 ZSE Load Control Event command

7377 The ZSE Load Control Event command shall be sent by an ESME:

7378 following successful authentication of a Command with Message Code 0x011E (so 'ECS47a
7379 Set Auxiliary Controller [n] State'), where that command requires an HCALCS to change its
7380 state;

7381 to control an HCALCS according to the Auxiliary Controller Calendar; or

7382 as required by Section 18.1.1.3.

7383 In executing this command, the ESME shall send the ZCL Load Control Event command to the
 7384 HCALCS identified in that Command / in that Get Scheduled Events command / in that entry
 7385 of the Auxiliary Controller Calendar (as determined by the requirements of Section 7.3.6.1)
 7386 with:

7387 the values of each field populated in the ZCL Load Control Event command as specified in
 7388 Table 18.1.1.1;

7389 the 'Duration in Minutes' field set according to the respective triggers above calculated as
 7390 per SMETS;

7391 the 'Duty Cycle' field set to 0x00 or to 0x64, determined according to the requirements of
 7392 Section 7.3.6.1.

7393 The recipient HCALCS shall interpret the value in Duty Cycle accordingly.

7394 Whenever it sends a *Load Control Event* command, the ESME shall add an entry to the
 7395 Auxiliary Controller Event Log where:

7396 hanCommandID is set to the value of the *Issuer Event ID* in the command;

7397 outcome is set to 0x00 ('Outcome not known'); and

7398 switchNumberAndAction is set to the concatenation 0x1'n' (where 'n' is the value of
 7399 Auxiliary Controller [n] for the HCALCS in question) || Duty Cycle in the command.

7400 On successful authentication of such a ZCL command, the recipient HCALCS shall respond
 7401 with a *Report Event Status* ZCL command populated as per Table 18.1.1.4, with Event Status
 7402 set to:

7403 0x02 ('Event started'), if the command was successfully executed; or

7404 0xFE ('Load Control Event command Rejected'), if the command was not successfully
 7405 executed.

7406 Whenever it receives a *Report Event Status* command with an *Event Status* of either 0x02 or
 7407 0xFE, the ESME shall add an entry to the Auxiliary Controller Event Log where:

7408 hanCommandID is set to the value of the *Issuer Event ID* in the command;

7409 outcome is set to:

7410 1. 0x01 ('Success') if Event Status is 0x02 ('Event started') in the
 7411 command; or

7412 2. 0x02 ('Failure') if Event Status is 0xFE ('Load Control Event command
7413 Rejected'), in the command; and

7414 switchNumberAndAction is set to the concatenation 0x1'n' (where 'n' is the value of
7415 Auxiliary Controller [n] for the HCALCS in question) || Duty Cycle Applied in the command.

7416 After the 'Duration In Minutes' specified in such a Load Control Event command has elapsed
7417 according to the HCALCS timer, the HCALCS shall send to the ESME a Get Scheduled Events
7418 command in accordance with Table 18.1.1.3. For clarity, an HCALCS may additionally send a
7419 Get Scheduled Events command to the ESME at any time.

Element	Meaning	Value	Octets
ZCL header			
Frame control	Cluster-specific; not manufacturer specific; server-client; allow default response;	0b00001001	1
Transaction sequence number		0x00	1
Command identifier	Load Control Event	0x00	1
ZCL payload			
Issuer Event ID (UINT32)	Set to the ESME's current UTC time	See 'Meaning' column	4
Device Class (BITMAP16)	All device types	0xFFFF	2
Utility Enrollment Group (UINT8)	All groups	0x00	1
Start Time (UTCTime)	Start immediately	0x00000000	4
Duration In Minutes (UINT16)	A value between 1 and 1440 minutes	See 'Meaning' column	2
Criticality Level (UINT8)	Voluntary	0x01	1
Cooling Temperature Offset (UINT8)	Not used	0xFF	1
Heating Temperature Offset (UINT8)	Not used	0xFF	1
Cooling Temperature Set	Not used	0x8000	2

Element	Meaning	Value	Octets
Point (INT16)			
Heating Temperature Set Point (INT16)	Not used	0x8000	2
Average Load Adjustment Percentage (INT8)	Not used	0x80	1
Duty Cycle (UINT8)	0x00 (0) = switch OFF; 0x64 (100) = switch ON	See 'Meaning' column	1
Event Control (BITMAP8)	Do not randomise	0x00	1

7420 Table 18.1.1.1: ZSE Load Control Event command

7421 Intentionally blank

7422 ZSE Get Scheduled Events command

7423 When sending a ZSE Get Scheduled Event command pursuant to section 8.5.2.1 of SMETS,
7424 an HCALCS shall populate that ZCL Command according to Table 18.1.1.3.

7425 On authenticated receipt of a ZSE Get Scheduled Events command, the ESME shall, if the
7426 Supply State is Enabled (where those terms have their SMETS meaning) send a ZSE Load
7427 Control Event command instructing the HCALCS whether it is to be open or closed, and for
7428 how long it is to be in that state, as required by SMETS.

7429 Otherwise, the ESME shall send a ZSE Default Response command with Status =
7430 NOT_FOUND.

Element	Meaning	Value	Octets
ZCL header			
Frame control	Cluster-specific; not manufacturer specific; client-server; disable default response;	0b00010001	1
Transaction sequence number		0x00	1
Command identifier	Get Scheduled Events	0x01	1
ZCL payload			
Start Time (UTCtime)	Retrieve active event	0x00000000	4

Number of Events (UINT8)	Device can only accept 1 event	0x01	1
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7431 Table 18.1.1.3: ZSE Get Scheduled Events command

7432 ZSE Report Event Status command

Element	Meaning	Value	Octets
ZCL header			
Frame control	Cluster-specific; not manufacturer specific; client-server; allow default response;	0b00000001	1
Transaction sequence number		0x00	1
Command identifier	Report Event Status	0x00	1
ZCL payload			
Issuer Event ID (UINT32)	Set to the Issuer Event ID from the corresponding ZSE Load Control Event command received from the ESME	See 'Meaning' column	4
Event Status (UINT8)	Refer to ZigBee standard	As per the requirements of this Section 18.1.1	1
Event Status Time (UTCTime)	Fixed value	0x00000001	4
Criticality Level Applied (UINT8)	0x01 = Voluntary	0x01	1
Cooling Temperature Set Point Applied (UINT16)	Not used	0x8000	2
Heating Temperature Set Point Applied (UINT16)	Not used	0x8000	2
Average Load Adjustment Percentage Applied (INT8)	Not used	0x80	1
Duty Cycle Applied (UINT8)	0x00 (0) = switched OFF; 0x64 (100) = switched ON	See 'Meaning' column	1
Event Control (BITMAP8)	Do not randomise	0x00	1
Signature Type (UINT8)	No signature	0x00	1

7433	Table 18.1.1.4: ZSE Report Event Status command
7434	Message Template for ZSE commands between SAPC / ESME with APC and PPMID / Type 2
7435	Device
7436	ZSE Load Control Event command
7437	In this section:
7438	Source Device shall mean either (1) an ESME with at least one APC or (2) an SAPC; and
7439	Requesting Device shall mean a Device which is (1) in the Source Device's Device Log and (2)
7440	is a Type 2 Device or a PPMID, according to the Source Device's Device Log.
7441	Where the Source Device has received a <i>ZDO Bind Request</i> in relation to the <i>Demand</i>
7442	<i>Response and Load Control Cluster</i> from a Requesting Device (regardless of whether the
7443	Source Device supports binding with its ZigBee meaning), the Source Device shall:
7444	whenever, in relation to any of its Auxiliary Controller [n], the Source Device has changed the
7445	current output level; or
7446	when its Auxiliary Controller [n] is an APC, the Source Device has changed the current input
7447	level, or input limit level or output limit level;
7448	Send a <i>ZSE Load Control Event</i> command to the Requesting Device, populated as specified in
7449	Table 18.1.2.1 for the Auxiliary Controller [n] whose level has changed. If:
7450	the Load Control Event Command issued has a value of 1440 minutes for the 'Duration In
7451	Minutes' parameter; and
7452	1440 minutes then elapse without the Source Device sending a subsequent Load Control
7453	Event Command to the Requesting Device in relation to Auxiliary Controller [n];
7454	then the Source Device shall send a Load Control Event Command to the Requesting Device
7455	in relation to Auxiliary Controller [n]'s now current levels.
7456	For clarity:
7457	the above requirements for the Service Device pushing <i>ZSE Load Control Event</i> commands
7458	are additional to any other ZigBee or GBCS requirements to push such commands;
7459	the Requesting Device shall not be required to undertake any ZigBee specified client
7460	processing specific to the Load Control Event Command; and
7461	the Requesting Device shall not respond to the Load Control Event with a Report Event
7462	Status Command (the use of a reserved bit in the Device Class bitmap of the Load Control
7463	Event ensures that this behaviour aligns with the ZSE requirements).

7464 Where the Source Device receives an authenticated Get Scheduled Events Command from a
 7465 Requesting Device (so a Device which is not an HCALCS according to the Source Device's
 7466 Device Log), the Source Device shall, for each of Auxiliary Controller [n] (where n = 1 to 5)
 7467 which has an associated ALCS or APC:

7468 where Auxiliary Controller [n] is an ALCS, send a ZSE Load Control Event command to the
 7469 Requesting Device, populated as specified in Table 18.1.2.1 detailing the current state;

7470 where Auxiliary Controller [n] is an APC, send two ZSE Load Control Event commands to the
 7471 Requesting Device, populated as specified in Table 18.1.2.1, one detailing the current input
 7472 levels and the other detailing the current output levels; and

7473 where Auxiliary Controller [n] is an APC or an ALCS, send, for each of the next 25 state
 7474 changes which are required by the currently configured Auxiliary Controller Calendar and
 7475 which relate to Auxiliary Controller [n], a ZSE Load Control Event command to the
 7476 Requesting Device, populated as specified in Table 18.1.2.1. Where there are fewer than 25
 7477 such state changes, the Source Device shall send such lower number of commands as is
 7478 required to detail all such state changes, as are specified by the configured Auxiliary
 7479 Controller Calendar.

7480 For clarity, this means that 1 authenticated Get Scheduled Events Command from a
 7481 Requesting Device can result in a maximum of 135 ZSE Load Control Event commands from
 7482 the Source Device (specifically in the case where the Source Device has five APCs installed =
 7483 $5 \times (2 \text{ (current states)} + 25 \text{ (future states)})$).

7484 The Source Device shall not apply any ZSE specified requirements specific to the processing
 7485 of Get Scheduled Events Commands, and so shall discard the values of any parameters in the
 7486 ZCL payload of such Commands.

Element	Meaning	Value	Octets
ZCL header			
Frame control	Cluster-specific; not manufacturer specific; server-client; allow default response;	0b00001001	1
Transaction sequence number		See ZigBee specifications	1
Command identifier	Load Control Event	0x00	1
ZCL payload			
Issuer Event ID (UINT32)	Set to the Source Device's current UTC time	See 'Meaning' column	4

Element	Meaning	Value	Octets
Device Class (BITMAP16)	See value	Bit 0: 0b1 if Auxiliary Controller [n] is an ALCS; 0b0 if it is an APC. Note that, as per SMETS 5.5.11, n is between 1 and 5 inclusive. Bit 15: 0b1 All other bits shall have the value 0b0 except: If Auxiliary Controller [n] is an APC and this command relates to its output or is an ALCS, bit 2n-1 shall have the value 0b1; and If Auxiliary Controller [n] is an APC and this command relates to its input, bit 2n shall have the value 0b1	2
Utility Enrollment Group (UINT8)	Not used	0x00	1
Start Time (UTCTime)	Where this command is to report an expected, future Auxiliary Controller state change, the UTCTime and Duration in Minutes values shall be calculated by the Source Device according to Remote Party Commands successfully processed by the Source Device at the time this command is created. Where this command is to report the Auxiliary Controller's current state, the Source Device shall report the UTCTime at which, according to the Source Device's Clock, this Auxiliary Controller was set to this state. For clarity, this means that where a state change is not caused by the Auxiliary Controller Calendar, the Source Device shall additionally: record the UTCTime of that state change ('State Change Time'); and record, at the time of the state change, the expected time in minutes that the state is to be maintained, if that can be established, or a value of 1440 if it cannot (for example, where it is a state change caused by SMETS required defaulting behaviour). The lesser of this expected time in minutes value, or the time in	See 'Meaning' column	4

Element	Meaning	Value	Octets
	minutes between the 'State Change Time' recorded and the now expected next change to state (in light of subsequent Remote Party Command processing leading to a duration change but not yet a state change) shall be reported in any associated 'Duration in Minutes' parameter. Where, at the time of a state change, the Source Device has a Time Status of Invalid, the Source Device shall record, and so subsequently report, a value of 0xFFFFFFFF for the UTCTime (meaning, as per ZCL 2.5.2.21, invalid value of this type) and a 'Duration in Minutes' of 0 minutes, to indicate the error state. Where the Auxiliary Controller has been in the reported state throughout its life, it shall report a UTCTime of 0x00000000 and a 'Duration in Minutes' of the lesser of 1440 minutes and any now expected next change to state (in light of Remote Party Command processing)		
Duration In Minutes (UINT16)	See prior 'UTCTime' row	See 'Meaning' column	2
Criticality Level (UINT8)	The value specifies the cause of the Device's current Duty Cycle value in this command, which, in line with SMETS, will be one of: Auxiliary Controller Calendar; Remote Party Command ('ECS47a Set Auxiliary Controller [n] State'); Boost Function; or Default Device behaviour required by SMETS when no other causes apply	0x0A: Default Device behaviour 0x0B: Auxiliary Controller Calendar 0x0C: Remote Party Command 0x0D: Boost Function	1
Cooling Temperature Offset (UINT8)	Not used	0xFF	1
Heating Temperature Offset (UINT8)	Not used	0xFF	1
Cooling Temperature Set Point (INT16)	Not used	0x8000	2
Heating Temperature Set	Not used	0x8000	2

Element	Meaning	Value	Octets
Point (INT16)			
Average Load Adjustment Percentage (INT8)	See value	If Auxiliary Controller [n] is an APC and a limit is currently being applied by the Source Device in the corresponding direction (so input or output as specified in Device Class), as a result of an 'ECS47e Limit APC [n] Level' Command, the value between 0x00 (0) and 0x64 (100) reflecting that limit as required by section 7.3.6.1. Otherwise, 0x64(100)	1
Duty Cycle (UINT8)	See value	If Auxiliary Controller [n] is an APC and this command relates to its input, the value of 'q' required by section 7.3.6.1. Otherwise, the value of 'p' required by section 7.3.6.1	1
Event Control (BITMAP8)	Do not randomise	0x00	1

7487 Table 18.1.2.1: ZSE Load Control Event command providing current state of Auxiliary
7488 Controller [n]

7489 DLMS COSEM Message Templates

7490 Table 18.2 contains Message Templates for all Use Case with DLMS COSEM payloads. These
7491 Message Templates are derived from the Mapping Table, and shall be complied with in the
7492 construction and population of all such Messages.



GBCS v4.4 Table
18.2 DLMS COSEM M

7493

7494 Table 18.2: DLMS COSEM Message Templates

7495 Encoding

7496 Italicised terms in this Section 18.2.1 shall have their DLMS COSEM meaning.

7497 Requirements in this Section 18.2.1, and sub-Sections of it, shall, where the SAPC supports
7498 the corresponding SMETS functionality, apply to an SAPC as if it were an ESME.

7499 Compact array encoding

7500 The Blue Book definition of *attribute 2* of *Profile Generic* objects may be interpreted as
7501 requiring ‘*entry*’ to be a *structure* containing a single choice from the DLMS data types. The
7502 GBCS interprets it as meaning that ‘*entry*’ is a *structure* that can contain multiple choices of
7503 DLMS data types. These choices vary between instances of *Profile Generic* object. To
7504 identify these different structures, the naming convention ‘*entry_nameOfStructure*’ is used.

7505 The GBCS uses the *compact-array* data type in attribute 2 of *Profile Generic* objects. Table
7506 18.2.1.1 details the derivation of the *contents-description* element within the *compact-array*
7507 *structure* for the structures used in the *Profile Generic* objects required by this GBCS. These
7508 encodings are reflected in the DLMS COSEM Message Templates.

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
entry_dIValueLogEntry ::= structure {	0x02	0x02		0x1302020606
timestamp: double-long-unsigned,	0x06			
dIValue: double-long-unsigned	0x06			
}				
entry_enumValueLogEntry ::= structure {	0x02	0x02		0x1302020616
timestamp: double-long-unsigned,	0x06			
enumValue: enum	0x16			
}				
entry_eventLogEntry12 ::= structure {	0x02	0x03		0x130203061209
timestamp: double-long-unsigned,	0x06			
logCode: long-unsigned,	0x12			
otherInformation: octet-string(12)	0x09			
}				

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
entry_powerLogEntry ::= structure {	0x02	0x03		0x130203061206
timestamp: double-long-unsigned,	0x06			
logCode: long-unsigned,	0x12			
otherInformation: double-long-unsigned	0x06			
}				
entry_eventLogEntry8 ::= structure {	0x02	0x03		0x130203061209
timestamp: double-long-unsigned,	0x06			
logCode: long-unsigned,	0x12			
otherInformation: octet-string(8)	0x09			
}				
entry_securityLogEntry ::= structure {	0x02	0x02		0x1302020612
timestamp: double-long-unsigned,	0x06			
logCode: long-unsigned	0x12			

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
}				
entry_billingCalendarLogEntry::= structure{	0x02	0x07 or 0x09		0x13020706060100300601000806010008060100080601000806 (single element) or 0x1302090606060100300601000406010008060100080601000806 (twin element)
timestamp: double-long-unsigned,	0x06			
activeImportRegisterValue: double-long-unsigned,	0x06			
secondaryActiveImportRegisterValue: double-long-unsigned, [[MAY NOT BE PRESENT]]	0x06			
tariffTOURegisterValues: array double-long-unsigned,	0x01	0x0030	0x06	
secondaryTariffTOURegisterValues: array double-long-unsigned, [[MAY NOT BE PRESENT]]	0x01	0x0004	0x06	
tariffTOUBlock1RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
tariffTOUBlock2RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock3RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock4RegisterValues: array double-long-unsigned	0x01	0x0008	0x06	
}				
entry_billingCalendarOnSetModeOrTariffLogEntry::= structure{	0x02	0x0D or 0x0F		0x13020D060601003006010008060100080601000806010008060505050505 (single element) or 0x13020F0606060100300601000406010008060100080601000806010008060505050505 (twin element)
timestamp: double-long-unsigned,	0x06			
activeImportRegisterValue: double-long-unsigned,	0x06			
secondaryActiveImportRegisterValue: double-long-unsigned, [[MAY NOT BE PRESENT]]	0x06			

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
tariffTOURegisterValues: array double-long-unsigned,	0x01	0x0030	0x06	
secondaryTariffTOURegisterValues: array double-long-unsigned, [[MAY NOT BE PRESENT]]	0x01	0x0004	0x06	
tariffTOUBlock1RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock2RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock3RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock4RegisterValues: array double-long-unsigned	0x01	0x0008	0x06	
emergencyCreditBalanceValue: double-long,	0x05			
meterBalanceValue: double-long,	0x05			
paymentDebtRegisterValue: double-long,	0x05			
timeDebtRegisters1Value: double-long,	0x05			
timeDebtRegisters2Value: double-long,	0x05			
accumulatedDebtRegisterValue: double-long	0x05			

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
}				
entry_boostFunctionLogEntry ::= structure {	0x02	0x02		0x1302020606
boost_start: double-long-unsigned,	0x06			
boost_end: double-long-unsigned	0x06			
}				
entry_prepaymentReadLogEntry ::= structure {	0x02	0x07		0x130207060505050505
timestamp: double-long-unsigned,	0x06			
emergencyCreditBalanceValue: double-long,	0x05			
meterBalanceValue: double-long,	0x05			
paymentDebtRegisterValue: double-long,	0x05			
timeDebtRegisters1Value: double-long,	0x05			
timeDebtRegisters2Value: double-long,	0x05			
accumulatedDebtRegisterValue: double-long	0x05			

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
}				
entry_registerReadLogEntry::= structure{	0x02	0x07 or 0x09		0x13020706060100300601000806010008060100080601000806 (single element) or 0x1302090606060100300601000406010008060100080601000806 (twin element)
timestamp: double-long-unsigned,	0x06			
activeImportRegisterValue: double-long-unsigned,	0x06			
secondaryActiveImportRegisterValue: double-long-unsigned, [[MAY NOT BE PRESENT]]	0x06			
tariffTOURegisterValues: array double-long-unsigned,	0x01	0x0030	0x06	
secondaryTariffTOURegisterValues: array double-long-unsigned, [[MAY NOT BE PRESENT]]	0x01	0x0004	0x06	
tariffTOUBlock1RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
tariffTOUBlock2RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock3RegisterValues: array double-long-unsigned,	0x01	0x0008	0x06	
tariffTOUBlock4RegisterValues: array double-long-unsigned	0x01	0x0008	0x06	
}				
entry_activeImportLogEntry ::= structure {	0x02	0x03 or 0x02		0x130203060606 (twin element) or 0x1302020606 (single element)
timestamp: double-long-unsigned,	0x06			
primaryValue: double-long-unsigned,	0x06			
secondaryValue: double-long-unsigned [[MAY NOT BE PRESENT]]	0x06			
}				
entry_twoDlValueLogEntry ::= structure {	0x02	0x03		0x130203060606
timestamp: double-long-unsigned,	0x06			
dIValue: double-long-unsigned,	0x06			

<i>Structure definition</i>	<i>Tag</i>	<i>Number of entries (structures and arrays only)</i>	<i>Tag of entries in array</i>	<i>contents-description for compact-array</i>
dlValue2: double-long-unsigned	0x06			
}				
entry_auxiliaryControllerLogEntry::= structure {	0x02	0x04		0x13020406121606
timestamp: double-long-unsigned,	0x06			
switchNumberAndAction: long-unsigned,	0x12			
outcome: enum,	0x16			
hANCommandID: double-long-unsigned	0x06			
}				

7509 Table18.2.1.1: derivation of the *contents-description* element within the *compact-array structure*

7510 Values of the credit_charge_configuration attribute of Account (Class ID 111) objects

7511 There are three SMETS parameters required for all Set Payment Mode Use Cases:

7512 Payment Mode, being Credit or Prepayment;

7513 Suspend Debt Emergency, being True or False and only being relevant when Payment Mode = Prepayment; and

- 7514 Suspend Debt Disabled, being True or False and only being relevant when Payment Mode = Prepayment.
- 7515 Note that Disablement Threshold can also be set through the 'Set Payment Mode to Prepayment' Use Case.
- 7516 The combination of these values determines, and is reflected in, the five possible values in the *credit_charge_configuration* attribute of the
- 7517 Account objects.
- 7518 On an ESME that is not a Twin Element variant, the ESME shall accept only the five values for the *credit_charge_configuration* attribute set out
- 7519 in Table 18.2.1.2a.

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
Credit	Not relevant	Not relevant	0x0102 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0	44
Prepayment	True	True	0x010D 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403C0 020309060000130A00FF09060000131402FF0403C0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403C0 020309060000130A01FF09060000131402FF0403C0 020309060000130A02FF09060000131404FF0403E0	275

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
			020309060000130A02FF09060000131401FF0403E0 020309060000130A02FF09060000131402FF0403E0	
Prepayment	True	False	0x010D 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403E0 020309060000130A00FF09060000131402FF0403E0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403E0 020309060000130A01FF09060000131402FF0403E0 020309060000130A02FF09060000131404FF0403E0 020309060000130A02FF09060000131401FF0403E0 020309060000130A02FF09060000131402FF0403E0	275
Prepayment	False	True	0x010A 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403C0 020309060000130A00FF09060000131402FF0403C0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0	212

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
			020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403C0 020309060000130A01FF09060000131402FF0403C0	
Prepayment	False	False	0x010A 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403E0 020309060000130A00FF09060000131402FF0403E0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403E0 020309060000130A01FF09060000131402FF0403E0	212

7520 Table 18.2.1.2a: allowable values for the *credit_charge_configuration* attribute for all ESME (except Twin Element variant)

7521 On an ESME that is a Twin Element variant, the ESME shall accept only the five values for the *credit_charge_configuration* attribute in Table
7522 18.2.1.2b.

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
Credit	Not relevant	Not relevant	0x0103 020309060000130A00FF09060000131400FF0403E0	65

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
			020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131405FF0403E0	
Prepayment	True	True	0x010F 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403C0 020309060000130A00FF09060000131402FF0403C0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403C0 020309060000130A01FF09060000131402FF0403C0 020309060000130A02FF09060000131404FF0403E0 020309060000130A02FF09060000131401FF0403E0 020309060000130A02FF09060000131402FF0403E0 020309060000130A00FF09060000131405FF0403E0 020309060000130A01FF09060000131405FF0403E0	317
Prepayment	True	False	0x010F 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403E0 020309060000130A00FF09060000131402FF0403E0	317

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
			020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403E0 020309060000130A01FF09060000131402FF0403E0 020309060000130A02FF09060000131404FF0403E0 020309060000130A02FF09060000131401FF0403E0 020309060000130A02FF09060000131402FF0403E0 020309060000130A00FF09060000131405FF0403E0 020309060000130A01FF09060000131405FF0403E0	
Prepayment	False	True	0x010C 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403C0 020309060000130A00FF09060000131402FF0403C0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403C0 020309060000130A01FF09060000131402FF0403C0 020309060000130A00FF09060000131405FF0403E0	254

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>	<i>Value of credit_charge_configuration attribute</i>	<i>Length in octets</i>
			020309060000130A01FF09060000131405FF0403E0	
Prepayment	False	False	0x010C 020309060000130A00FF09060000131400FF0403E0 020309060000130A00FF09060000131404FF0403E0 020309060000130A00FF09060000131401FF0403E0 020309060000130A00FF09060000131402FF0403E0 020309060000130A00FF09060000131403FF0403E0 020309060000130A01FF09060000131403FF0403E0 020309060000130A01FF09060000131400FF0403E0 020309060000130A01FF09060000131404FF0403E0 020309060000130A01FF09060000131401FF0403E0 020309060000130A01FF09060000131402FF0403E0 020309060000130A00FF09060000131405FF0403E0 020309060000130A01FF09060000131405FF0403E0	254

7523 Table 18.2.1.2b: allowable values for the *credit_charge_configuration* attribute for all ESME (Twin Element variant)

7524 Deriving the values of the *credit_charge_configuration* attribute of Account (Class ID 111) objects – informative

7525 This Section explains the derivation of the five values of this attribute that an ESME can accept.

7526 The *credit_charge_configuration* attribute encoding is shown in Table 18.2.1.3a.

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
credit_charge_configuration			

Component	Hex value	Length in octets	Notes
Tag	0x01	1	tag for array
Length	Variable	1	entries in array
credit_charge_configuration_element			
Tag	0x02	1	
Length	0x03	1	3 elements in this structure
credit_reference			
Tag	0x09	1	tag for octet-string
Length	0x06	1	logical_name is 6 octets
Value	Variable	6	OBIS code for this class 112 object
charge_reference			
Tag	0x09	1	tag for octet-string
Length	0x06	1	logical_name is 6 octets
Value	Variable	6	OBIS code for this class 113 object
collection_configuration			
Tag	0x04	1	tag for bit-string
Length	0x03	1	3 as per the Blue Book
Value	0b11Z	1	Where Z is the variable Bit 0;
trailing_bits	0b00000	1	

7527 Table 18.2.1.3a: *credit_charge_configuration* attribute encoding

7528 So the value of the *credit_charge_configuration_element* attribute is a 21 octet long concatenation:

7529 0x02030906 || credit object OBIS code || 0x0906 || charge object OBIS code || 0x0403 || collection bit string || 0b00000

7530 The meaning of each *credit_charge_configuration_element* is that this charge can be collected from this credit object, except in possible meter
7531 states specified by the *collection_configuration* bit string.

7532 On an ESME, there shall be three class 112 Credit objects, as shown in Table 18.2.1.3b. Two are not relevant in Credit Mode.

<i>SMET Reference Component</i>	<i>OBIS Code (decimal)</i>	<i>OBIS Code (hexadecimal)</i>	<i>Payment Mode</i>
MeterBalance	0-0:19.10.0.255	0x0000130A00FF	Prepayment and Credit
AccumulatedDebt	0-0:19.10.2.255	0x0000130A02FF	Prepayment
EmergencyCreditBalance	0-0:19.10.1.255	0x0000130A01FF	Prepayment

7533 Table 18.2.1.3b: Class 112 Credit objects

7534 There shall be five class 113 Charge objects on an ESME (or six on a Twin Element ESME), as shown in Table 18.2.1.3c. Three are not relevant in
7535 Credit Mode.

<i>SMET Reference Component</i>	<i>OBIS Code (decimal)</i>	<i>OBIS Code (hexadecimal)</i>	<i>Payment Mode</i>
DebtRecoveryRates[1]	0-0:19.20.1.255	0x0000131401FF	Prepayment
DebtRecoveryRates[2]	0-0:19.20.2.255	0x0000131402FF	Prepayment
DebtRecoveryPerPayment	0-0:19.20.3.255	0x0000131403FF	Prepayment
SecondaryTariffTOUPriceMatrix (Twin element ESME only)	0-0:19.20.5.255	0x0000131405FF	Prepayment and Credit
StandingCharge	0-0:19.20.4.255	0x0000131404FF	Prepayment and Credit
TariffBlockPriceMatrixTOU	0-0:19.20.0.255	0x0000131400FF	Prepayment and Credit

7536 Table 18.2.1.3c: Class 113 Charge objects

- 7537 As defined in the Blue Book, the *collection_configuration* bit string determines whether a charge is collected from a credit dependent on ESME
7538 state.
- 7539 Bit 1 affects charging in load limiting periods. There is no such requirement in SMETS, so this value is always 0b1 (charges are applied in load
7540 limiting periods).
- 7541 In Credit Mode, collection continues in all states, so the value of all three bits is always 0b1.
- 7542 In Prepayment Mode, *collection_configuration* is set according to Suspend Debt Disabled (affects Bit 0) values, and the pairing of charge and
7543 credit object.
- 7544 Suspend Debt Emergency being True means that DebtRecoveryRates[1..2] and StandingCharge are collected from AccumulatedDebt rather
7545 than EmergencyCreditBalance, when Emergency Credit is in use, so Suspend Debt Emergency is specified by way of pairing charge and credit
7546 objects accordingly. Note that Bit 2 of *collection_configuration* shall always be fixed at 0b1.
- 7547 Suspend Debt Disabled being True means that DebtRecoveryRates[1..2] are no longer collected when the supply is disabled due to lack of
7548 credit.
- 7549 Table 18.2.1.3d sets out the *credit_charge_configuration_element* array entries in Credit Mode.

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131400FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131404FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131405FF0403E0

7550 Table 18.2.1.3d: Class 113 Charge objects

7551 When the mode is set as in Table 18.2.1.3e:

<i>Payment Mode</i>	<i>Suspend Debt Emergency</i>	<i>Suspend Debt Disabled</i>
Prepayment	False	False

7552 Table 18.2.1.3e: Prepayment states

7553 the *credit_charge_configuration_element* array entries are as per Table 18.2.1.3f.

<i>Tag & Length</i>	<i>Credit object</i>	<i>Tag & Length</i>	<i>Charge object</i>	<i>Tag & Length</i>	<i>Collection bit string</i>	<i>trailing_bits</i>	<i>Array entry</i>
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131400FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131404FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131401FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131402FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131400FF0403E0

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131404FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131401FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131402FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131405FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131405FF0403E0

7554 Table 18.2.1.3f: *credit_charge_configuration_element* array entries

7555 Note that, as per SMETS, the value of MeterBalance determines whether charges are collected from EmergencyCreditBalance or MeterBalance.

7556 When the mode is set as in Table 18.2.1.3g:

Payment Mode	Suspend Debt Emergency	Suspend Debt Disabled
Prepayment	False	True

7557 Table 18.2.1.3g: Prepayment states

7558 the *credit_charge_configuration_element* array entries are as per Table 18.2.1.3h.

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF	0x0906	0x0000131400FF	0x0403	0b111	0b00000	0x020309060000130A00FF09060000131400FF0403E0

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Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
	(MeterBalance)		(TariffBlockPriceMatrixTOU)		(collectable in all circumstances)		
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131404FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A00FF09060000131401FF0403C0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A00FF09060000131402FF0403C0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131400FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131404FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A01FF09060000131401FF0403C0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A01FF09060000131402FF0403C0

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131405FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131405FF0403E0

7559 Table 18.2.1.3h: *credit_charge_configuration_element* array entries

7560 When the mode is set as in Table 18.2.1.3i:

Payment Mode	Suspend Debt Emergency	Suspend Debt Disabled
Prepayment	True	False

7561 Table 18.2.1.3i: Prepayment states

7562 the *credit_charge_configuration_element* array entries are as per Table 18.2.1.3j.

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131400FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131404FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131401FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131402FF0403E0

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Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b1110 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131400FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131404FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131401FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131402FF0403E0
0x02030906	0x0000130A02FF (AccumulatedDebt)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collect in Emergency Credit period – see note at bottom of table)	0b00000	0x020309060000130A02FF09060000131404FF0403E0
0x02030906	0x0000130A02FF (AccumulatedDebt)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b111 (collect in Emergency Credit period – see note at bottom of table)	0b00000	0x020309060000130A02FF09060000131401FF0403E0
0x02030906	0x0000130A02FF (AccumulatedDebt)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b111 (collect in Emergency Credit period – see note at bottom of table)	0b00000	0x020309060000130A02FF09060000131402FF0403E0

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131405FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131405FF0403E0

7563 Table 18.2.1.3j: *credit_charge_configuration_element* array entries

7564 Note that, as per SMETS, charges shall only accrue to AccumulatedDebt in Emergency Credit periods.

7565 When the mode is set as in Table 18.2.1.3k:

Payment Mode	Suspend Debt Emergency	Suspend Debt Disabled
Prepayment	True	True

7566 Table 18.2.1.3k Prepayment states

7567 the *credit_charge_configuration_element* array entries are as per Table 18.2.1.3l.

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131400FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131404FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A00FF09060000131401FF0403C0

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Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A00FF09060000131402FF0403C0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131403FF (DebtRecoveryPerPayment)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131403FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131400FF (TariffBlockPriceMatrixTOU)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131400FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131404FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A01FF09060000131401FF0403C0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b110 (do not collect when supply is disabled due to no credit)	0b00000	0x020309060000130A01FF09060000131402FF0403C0
0x02030906	0x0000130A02FF (AccumulatedDebt)	0x0906	0x0000131404FF (StandingCharge)	0x0403	0b111 (collect in Emergency Credit period – see note at bottom of table)	0b00000	0x020309060000130A02FF09060000131404FF0403E0
0x02030906	0x0000130A02FF (AccumulatedDebt)	0x0906	0x0000131401FF (DebtRecoveryRates[1])	0x0403	0b111 (collect in Emergency Credit period – see note at bottom of table)	0b00000	0x020309060000130A02FF09060000131401FF0403E0

Tag & Length	Credit object	Tag & Length	Charge object	Tag & Length	Collection bit string	trailing_bits	Array entry
0x02030906	0x0000130A02FF (AccumulatedDebt)	0x0906	0x0000131402FF (DebtRecoveryRates[2])	0x0403	0b111 (collect in Emergency Credit period – see note at bottom of table)	0b00000	0x020309060000130A02FF09060000131402FF0403E0
0x02030906	0x0000130A00FF (MeterBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A00FF09060000131405FF0403E0
0x02030906	0x0000130A01FF (EmergencyCreditBalance)	0x0906	0x0000131405FF (SecondaryTariffTOUPriceMatrix (Twin element ESME only))	0x0403	0b111 (collectable in all circumstances)	0b00000	0x020309060000130A01FF09060000131405FF0403E0

7568 Table 18.2.1.3l: *credit_charge_configuration_element* array entries

7569 Note that, as per SMETS, charges shall only accrue to AccumulatedDebt in Emergency Credit periods.

7570 Encoding of Billing Calendar start date-time and periodicity where periodicity is monthly or shorter

7571 Table 18.2.1.4 sets out how the components of the Billing Calendar start date-time and periodicity should be encoded where periodicity is
7572 monthly or shorter.

Component	Hex value	Length in octets	Notes
execution_time			
Tag	0x01	1	Tag for array
Length	0x01	1	1 entry in array
execution_time_date			
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	0xFF	1	0xFF means not specified
day of month,	0xFF unless periodicity is monthly. If periodicity is monthly, this shall be the day of the month of the start date-time.	1	0xFF means not specified.
day of week	0xFF unless periodicity is weekly If periodicity is weekly, this shall be the day of the week of the start date-time.	1	0xFF means not specified

7573 Table 18.2.1.4: Encoding of Billing Calendar start date-time and periodicity

7574 Encoding of Billing Calendar start date-time and periodicity where periodicity is quarterly or longer

- 7575 Where the periodicity is quarterly or longer, this Section sets out how the components of the Billing Calendar start date-time and periodicity
7576 shall be encoded in attribute 4 (execution_time) of the DLMS COSEM class 22 (OBIS 0-0:15.0.0.255 in Use Case ECS30a).
- 7577 If the required periodicity is quarterly (so once every three months), the execution_time shall be encoded as per Table 18.2.1.5a.
- 7578 If the required periodicity is six monthly, the execution_time shall be encoded as per Table 18.2.1.5b.
- 7579 If the required periodicity is yearly, the execution_time shall be encoded as per Table 18.2.1.5c.
- 7580 Note that month values are to be calculated using modulo twelve addition. For example, if quarterly billing is required and the start date-time
7581 is in October (so month 10), the value of months in the following three quarters is calculated by adding three modulo 12, so giving 1 (January),
7582 4 (April) and 7 (July).

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
execution_time			
Tag	0x01	1	Tag for array
Length	0x04	1	4 entries in the array, one for each of the four quarters in each year
execution_time_date			First entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified
execution_time_date			Second entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time plus three modulo twelve, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified
execution_time_date			Third entry in the array

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time plus six modulo twelve, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified
execution_time_date			Fourth entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time plus nine modulo twelve, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified

7583 Table 18.2.1.5a: Encoding of Billing Calendar start date-time and periodicity where periodicity is quarterly

7584

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
execution_time			
Tag	0x01	1	Tag for array
Length	0x02	1	2 entries in the array, one for each of the six months in each year
execution_time_date			First entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified
execution_time_date			Second entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time plus six modulo twelve, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified

7585 Table 18.2.1.5b: Encoding of Billing Calendar start date-time and periodicity where periodicity is six monthly

7586

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
execution_time			
Tag	0x01	1	Tag for array
Length	0x01	1	1 entry in the array
execution_time_date			First entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time

Component	Hex value	Length in octets	Notes
Value	See note	4	Time part of the start date-time, as per section 4.1.6.1 of the Blue Book
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	See note	4	Month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of month,	See note	4	Day of month part of the start date-time, as per section 4.1.6.1 of the Blue Book
day of week	0xFF	1	0xFF means not specified

7587 Table 18.2.1.5c: Encoding of Billing Calendar start date-time and periodicity where periodicity is yearly

7588 To detail the fixed value required in the ESME for attribute 3 of the DLMS COSEM class 22 object ((OBIS 0-0:15.0.0.255 in Use Cases ECS30a), so
7589 as to allow for a variable size of the execution_time array.

7590 Encoding to turn daily Block Counter resetting on and off

7591 This Section sets out how attribute 4 (execution_time) of the DLMS COSEM class 22 object (OBIS 0-0:15.1.0.255 in Use Case ECS48) is to be set,
7592 so as to either turn on or off daily resetting of Block Counters.

7593 If daily resetting of Block Counters is required, the execution_time shall be encoded as per Table 18.2.1.6a.

7594 If daily resetting of Block Counters is not required, the execution_time shall be encoded as per Table 18.2.1.6b. Note this uses the same
 7595 31/12/3000 date as DCC User Interface Specification to mean never execute.

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
execution_time			
Tag	0x01	1	Tag for array
Length	0x01	1	1 entry in the array
execution_time_date			First entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	0x00000000	4	Midnight UTC
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0xFF	1	0xFF means not specified
year lowbyte,	0xFF	1	0xFF means not specified
month,	0xFF	1	0xFF means not specified
day of month,	0xFF	1	0xFF means not specified
day of week	0xFF	1	0xFF means not specified

7596 Table 18.2.1.6a: Encoding of execution_time where daily resetting of Block Counters is required.

7597

<i>Component</i>	<i>Hex value</i>	<i>Length in octets</i>	<i>Notes</i>
execution_time			
Tag	0x01	1	Tag for array
Length	0x01	1	1 entry in the array
execution_time_date			First entry in the array
Tag	0x02	1	Tag for structure
Length	0x02	1	2 elements in structure
Time			
Tag	0x09	1	Tag for structure
Length	0x04	1	4 octets in DLMS encoded time
Value	0x00000000	4	Midnight UTC
Date			
Tag	0x09	1	5 octets in DLMS encoded date
Length	0x05	1	2 elements in structure
Value			
year highbyte,	0x0B	1	Year 3000 highbyte
year lowbyte,	0xB8	1	Year 3000 lowbyte
month,	0x0C	1	Meaning December
day of month,	0x1F	1	31st
day of week	0xFF	1	0xFF means not specified

7598 Table 18.2.1.6b: Encoding of execution_time where daily resetting of Block Counters is not required.

7599

- 7600 Illustrative command and response instantiation and DER encoding
- 7601 Illustrative @UpdateSecurityCredentials.CommandPayload instantiation and its DER encoding – informative
- 7602 supplierUpdatingAllSupplierCertificates in Table 18.3.1a is an ASN.1 structured value assignment. This specific example is where a Device's
- 7603 Supplier is instructing the Device to replace both the Supplier Digital Signing and Key Agreement credentials on the Device, and resetting
- 7604 Execution Counters. In business terms, an example of this would be at Change of Supplier.
- 7605 The black text specifies the parts of the ASN.1 structure, the blue text specifies the value it is set to and the comments explain each of the
- 7606 values.

ASN.1	Notes
<pre>supplierUpdatingAllSupplierCertificates CommandPayload ::= {authorisingRemotePartyControl {credentialsReplacementMode supplierBySupplier, authorisingRemotePartyTACellIdentifier {trustAnchorCellRemotePartyRole supplier, trustAnchorCellKeyUsage { digitalSignature}}, authorisingRemotePartySeqNumber 123456789,</pre>	This message is for the supplier replacing supplier credentials The public key to be used to check the signature on this message is the supplier digital signing key currently held by the Device. This is the existing supplier's counter, so greater than any this supplier has used This is the new supplier's counter, which the Device

ASN.1	Notes
<pre> newRemotePartyFloorSeqNumber 987654321} replacements {{replacementCertificate '0A7C8E9F123456789ABCDEF01234'H, targetTrustAnchorCell {trustAnchorCellRemotePartyRole supplier, trustAnchorCellKeyUsage { digitalSignature}}}} {replacementCertificate '0B34269F123456789ABCDEF01234'H, targetTrustAnchorCell {trustAnchorCellRemotePartyRole supplier, trustAnchorCellKeyUsage {keyAgreement}}}} certificationPathCertificates { 'FFAABB9F123456789ABCDEF01234'H } } </pre>	should use if the Command is successful The new supplier's digital signing certificate which is to be placed in the Device's supplier, digital signature Trust Anchor Cell The new supplier's key agreement certificate... which is to be placed in the Device's supplier, key agreement Trust Anchor Cell The Certificate for the CA which issued the new supplier's certificates. The Device will use this to check that the new supplier certificates were properly issued.

7607 Table 18.3.1a: Illustrative @UpdateSecurityCredentials.CommandPayload instantiation – ASN.1 structure

7608 The message sent to the Device would contain the DER encoding of the above ASN.1 value assignment. This DER encoding is laid out and
7609 explained in Table 18.3.1b. For these purposes, the Certificate is simply shown as an OCTET STRING.

Component	Value	Notes
CommandPayload SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x64	100 octet length follows
contents =:		
authorisingRemotePartyControl AuthorisingRemotePartyControl SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x18	Length of authorisingRemotePartyControl
contents =:		
credentialsReplacementMode CredentialsReplacementMode INTEGER:		
tag = [UNIVERSAL 2] primitive;	0x02	
length =	0x01	
contents =:	0x02	Representing supplierBySupplier
authorisingRemotePartyTACellIdentifier TrustAnchorCellIdentifier SEQUENCE:		

Component	Value	Notes
tag = [2] constructed;	0xA2	Tag for authorisingRemotePartyTACellIdentifier
length =	0x07	Length of authorisingRemotePartyTACellIdentifier
contents =:		
trustAnchorCellRemotePartyRole RemotePartyRole INTEGER:		
tag = [UNIVERSAL 2] primitive;	0x02	Tag for INTEGER
length =	0x01	1 octet length INTEGER
contents =:	0x02	Representing supplier RemotePartyRole
trustAnchorCellKeyUsage KeyUsage BIT STRING:		
tag = [UNIVERSAL 3] primitive;	0x03	Tag for BIT STRING
length =	0x02	2 octet length BIT STRING
contents =:	0x0780	Representing digitalSignature
authorisingRemotePartySeqNumber SeqNumber INTEGER:		
tag = [3] primitive;	0x83	Tag for INTEGER

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Component	Value	Notes
length =	0x04	4 octet length INTEGER
contents =:	0x075bcd15	The old supplier's Execution Counter in hex
newRemotePartyFloorSeqNumber SeqNumber INTEGER:		
tag = [4] primitive;	0x84	Tag for INTEGER
length =	0x04	4 octet length INTEGER
contents =:	0x3ade68b1	The new supplier's Execution Counter in hex
replacements SEQUENCE OF:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x36	Length of replacements
contents =:		
TrustAnchorReplacement SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x19	Length of first TrustAnchorReplacement
contents =:		

Component	Value	Notes
replacementCertificate Certificate OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0x04	Tag for OCTET STRING
length =	0x0e	Length of certificate
contents =:	0x0a7c8e9f123456789abcdef0 1234	New supplier's digitalSignature certificate
targetTrustAnchorCell TrustAnchorCellIdentifier SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x07	Length of targetTrustAnchorCell
contents =:		
trustAnchorCellRemotePartyRole RemotePartyRole INTEGER:		
tag = [UNIVERSAL 2] primitive;	0x02	Tag for INTEGER
length =	0x01	1 octet length INTEGER
contents =:	0x02	Representing supplier RemotePartyRole
trustAnchorCellKeyUsage KeyUsage BIT STRING:		

Component	Value	Notes
tag = [UNIVERSAL 3] primitive;	0x03	Tag for BIT STRING
length =	0x02	2 octet length BIT STRING
contents =:	0x0780	Representing digitalSignature
TrustAnchorReplacement SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x19	Length of second TrustAnchorReplacement
contents =:		
replacementCertificate Certificate OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0x04	Tag for OCTET STRING
length =	0x0e	Length of certificate
contents =:	0x0b34269f123456789abcdef01234	New supplier's keyAgreement certificate
targetTrustAnchorCell TrustAnchorCellIdentifier SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x07	Length of targetTrustAnchorCell

Component	Value	Notes
contents =:		
trustAnchorCellRemotePartyRole RemotePartyRole INTEGER:		
tag = [UNIVERSAL 2] primitive;	0x02	Tag for INTEGER
length =	0x01	1 octet length INTEGER
contents =:	0x02	Representing supplier RemotePartyRole
trustAnchorCellKeyUsage KeyUsage BIT STRING:		
tag = [UNIVERSAL 3] primitive;	0x03	Tag for BIT STRING
length =	0x02	2 octet length BIT STRING
contents =:	0x0308	Representing keyAgreement
certificationPathCertificates SEQUENCE OF:		
tag = [UNIVERSAL 16] constructed;	0x30	Tag for SEQUENCE
length =	0x10	Length of certificationPathCertificates
contents =:		
Certificate OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0x04	Tag for OCTET STRING

Component	Value	Notes
length =	0x0e	Length of certificate
contents =:	0xffaabb9f123456789abcdef01234	CA certificate for new supplier

7610 Table 18.3.1b: Illustrative @UpdateSecurityCredentials.Command instantiation – DER encoding

7611 Illustrative @UpdateSecurityCredentials.ResponsePayload instantiation and its DER encoding – informative

7612 supplierUpdatingAllSupplierCertificatesResponse in Table 18.3.2a is an ASN.1 structured value assignment. This specific
7613 example is where a Device is responding successfully to a Command.

7614 The black text specifies the parts of the ASN.1 structure, the *blue text* specifies the value it is set to by the Device and the comments explain
7615 each of the values.

ASN.1	Notes
<pre>supplierUpdatingAllSupplierCertificatesResponse ResponsePayload ::= { commandAccepted NULL, executionOutcome {authorisingRemotePartySeqNumber 123456789, credentialsReplacementMode <i>supplierBySupplier</i>, remotePartySeqNumberChanges {{otherRemotePartyRole <i>supplier</i>, otherRemotePartyFloorSeqNumber <i>987654321</i>}} },</pre>	The corresponding Command was for the Supplier replacing supplier credentials This is the new supplier's counter, which the Device will now use for Protection Against Replay in relation to the supplier role

ASN.1	Notes
<pre> replacementOutcomes {{ {affectedTrustAnchorCell { trustAnchorCellRemotePartyRole <i>supplier,</i> trustAnchorCellKeyUsage { <i>digitalSignature}}</i>}, statusCode <i>success,</i> existingSubjectUniqueID <i>'123456789ABCDEF0'</i>H, existingSubjectKeyIdentifier <i>'1234567890123456'</i>H, replacingSubjectUniqueID <i>'FEDCBA9876543210'</i>H, replacingSubjectKeyIdentifier <i>'ABCDEABCDEABCDEA'</i>H}, {affectedTrustAnchorCell {trustAnchorCellRemotePartyRole <i>supplier,</i> trustAnchorCellKeyUsage { <i>keyAgreement}}</i>}, statusCode <i>success,</i> existingSubjectUniqueID <i>'123456789ABCDEF0'</i>H, existingSubjectKeyIdentifier <i>'0987654321098765'</i>H, replacingSubjectUniqueID <i>'FEDCBA9876543210'</i>H, replacingSubjectKeyIdentifier <i>'FEDCBFEDCBFEDCBF'</i>H}}}} </pre>	This outcome is for the supplier digital signing store The old supplier's Entity Identifier The KeyIdentifier for the old supplier's digital signing key The new supplier's Entity Identifier The KeyIdentifier for the old supplier's digital signing key This outcome is for the supplier key agreement store

7616 Table 18.3.2a: Illustrative @UpdateSecurityCredentials.Response instantiation – ASN.1 structure

7617 The message sent by the Device would contain the DER encoding of the above ASN.1 value assignment. This DER encoding is laid out and
 7618 explained in Table 18.3.2b.

Component	Value	Notes
ResponsePayload SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X8189	Length 137
content =		
commandAccepted NULL		
tag = [UNIVERSAL 5] primitive		0X05
length =	0X00	
executionOutcome ExecutionOutcome SEQUENCE		
tag = [UNIVERSAL 16] constructed	0X30	Tag for SEQUENCE
length =	0X8184	Length 132
content =		
authorisingRemotePartySeqNumber SeqNumber INTEGER:		
tag = [UNIVERSAL 2] primitive	0x02	Tag for INTEGER
length =	0x04	4 octet length INTEGER

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Component	Value	Notes
contents =	0X075BCD15	The old supplier's Execution Counter in hex
credentialsReplacementMode CredentialsReplacementMode INTEGER:		
tag = [UNIVERSAL 2] primitive;	0X02	Tag for INTEGER
length =	0X01	
content =	0X02	Value for supplierBySupplier
remotePartySeqNumberChanges SEQUENCE OF:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X0B	
content =		
RemotePartySeqNumberChange SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X09	
content =		
otherRemotePartyRole RemotePartyRole INTEGER:		
tag = [UNIVERSAL 2] primitive;	0X02	Tag for INTEGER

SEC Schedule 8 – GBCS v4.4

Component	Value	Notes
length =	0X01	
content =	0X02	Value for supplier
otherRemotePartyFloorSeqNumber SeqNumber INTEGER:		
tag = [UNIVERSAL 2] primitive;	0X02	Tag for INTEGER
length =	0X04	
content =	0X3ADE68B1	The new supplier's Execution Counter in hexadecimal
replacementOutcomes SEQUENCE OF:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X6C	Length of 108
content =		
ReplacementOutcome SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X34	Length of 52
content =		
affectedTrustAnchorCell TrustAnchorCellIdentifier SEQUENCE:		

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Component	Value	Notes
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X07	
content =		
trustAnchorCellRemotePartyRole RemotePartyRole INTEGER:		
tag = [UNIVERSAL 2] primitive;	0X02	Tag for INTEGER
length =	0X01	
content =	0X02	Value for supplier
trustAnchorCellKeyUsage KeyUsage BIT STRING:		
tag = [UNIVERSAL 3] primitive;	0x03	Tag for BIT STRING
length =	0X02	
content =	0X0780	Tag for digitalSignature
statusCode StatusCode ENUMERATED:		
tag = [UNIVERSAL 10] primitive;	0X0A	Tag for ENUMERATED
length =	0X01	
content =	0X00	Value for success

SEC Schedule 8 – GBCS v4.4

Component	Value	Notes
existingSubjectUniqueID OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	8 octet length of Entity Identifier
content =	0X123456789ABCDEFO	
existingSubjectKeyIdentifier OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	length of KeyIdentifier
content =	0X1234567890123456	KeyIdentifier
replacingSubjectUniqueID OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	8 octet length of Entity Identifier
content =	0XFEDCBA9876543210	
replacingSubjectKeyIdentifier OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	length of KeyIdentifier
content =	0XABCDEABCDEABCDEA	KeyIdentifier

SEC Schedule 8 – GBCS v4.4

Component	Value	Notes
ReplacementOutcome SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X34	
content =		
affectedTrustAnchorCell TrustAnchorCellIdentifier SEQUENCE:		
tag = [UNIVERSAL 16] constructed;	0X30	Tag for SEQUENCE
length =	0X07	
content =		
trustAnchorCellRemotePartyRole RemotePartyRole INTEGER:		
tag = [UNIVERSAL 2] primitive;	0X02	Tag for INTEGER
length =	0X01	
content =	0X02	Value for supplier
trustAnchorCellKeyUsage KeyUsage BIT STRING:		
tag = [UNIVERSAL 3] primitive;	0x03	Tag for BIT STRING
length =	0X02	

SEC Schedule 8 – GBCS v4.4

Component	Value	Notes
content =	0X0308	
statusCode StatusCode ENUMERATED:		
tag = [UNIVERSAL 10] primitive;	0X0A	Tag for ENUMERATED
length =	0X01	
content =	0X00	Value for success
existingSubjectUniqueID OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	8 octet length of Entity Identifier
content =	0X123456789ABCDEFO	
existingSubjectKeyIdentifier OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	length of KeyIdentifier
content =	0X0987654321098765	KeyIdentifier
replacingSubjectUniqueID OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	8 octet length of Entity Identifier

Component	Value	Notes
content =	0XFEDCBA9876543210	
replacingSubjectKeyIdentifier OCTET STRING:		
tag = [UNIVERSAL 4] primitive;	0X04	Tag for OCTET STRING
length =	0X08	length of KeyIdentifier
content =	0XFEDCBFEDCBFEDCBF	KeyIdentifier

7619 Table 18.3.2b: Illustrative @UpdateSecurityCredentials.ResponsePayload instantiation – DER encoding

7620 Cryptographic Test Vectors

7621 This Section 18.4 provides cryptographic calculations in relation to a number of sample messages. The sample messages' contents align with
 7622 the corresponding Message Templates in Section 18.2. To undertake cryptographic calculations, a number of details about the Smart Metering
 7623 Entities involved are also required, not least Key Pairs, Entity Identifiers and Originator Counters. This Section specifies and uses sample values
 7624 of such attributes.

7625 Cryptographic Calculations

7626 Create details for three Smart Metering Entities with associated Keys and shared secrets:

7627

7628 An Entity called SupplierA:

--With an Entity ID:	0x12:34:56:78:9A:BC:DE:F0
--With a current Originator Counter:	0x00:00:00:00:00:00:00:01

--Digital Signing Private key :	0x3A:6B:2E:AA:0D:9F:25:A9:E4:55:98:3F:EB:5B:B9:47:52:81:21:91:1B:F3:B7:6B:E5:66:1C:89:DB:F2:4B:26
--Digital Signing Public key :	0x76:62:8E:1C:84:EF:79:35:54:8A:E5:D6:2C:7B:B3:AD:28:96:4C:F7:94:F0:38:7A:69:7E:EC:19:CD:D9:8F:46:0A:4D:5E:19:08:7E:F7:21:6E:D8:9C:29:83:1A:6E:E8:38:C8:DE:88:EF:34:F1:1D:3F:41:F3:6D:80:B2:A5:D5
--Key Agreement Private key :	0x3D:9D:FB:33:2E:B4:D6:D6:06:D7:47:18:55:3E:5E:61:B3:92:B0:FC:4C:90:CE:6A:A4:CE:DA:81:7E:80:11:B1
--Key Agreement Public key :	0xEF:F2:1D:5D:D6:74:EE:C6:E0:87:40:70:3B:52:25:52:CB:B7:4F:FC:A1:15:36:C5:37:C3:C8:06:E4:14:3C:8F:B2:E7:CA:3E:73:06:CB:46:DB:E4:BD:59:9C:C4:A3:1F:78:8C:2F:B7:A9:B9:BC:97:BE:98:C8:1E:F1:82:1A:30
--The shared secret calculated with DeviceA is :	0x15:45:AD:F2:75:DC:8E:57:AB:E4:71:E9:F0:C1:20:C2:FA:DD:5B:12:51:AF:B7:BD:AB:25:3C:80:1B:41:11:CE

7629

7630 An Entity called Access Control Broker:

--With an Entity ID:	0xAB:AB:AB:AB:AB:AB:AB:AB
--With a current Originator Counter:	0x10:00:00:00:00:00:00:01
--Key Agreement Private key :	0xE4:A6:CF:B4:31:47:1C:FC:AE:49:1F:D5:66:D1:9C:87:08:2C:F9:FA:77:22:D7:FA:24:B2:B3:F5:66:9D:BE:FB
--Key Agreement Public key :	0x29:2F:97:FE:C1:B3:0C:38:49:B8:06:D9:04:46:E4:A0:37:D6:D1:78:01:97:96:E7:6E:52:55:BD:C3:A0:8E:34:6F:9F:6E:6E:7E:8F:6A:4D:55:96:2D:2F:2D:0E:16:CF:F2:7B:F3:F9:25:FA:7D:BA:FD:15:A8:B1:DC:69:58:94
--The shared secret calculated with DeviceA is :	0x9A:AC:F2:E6:D5:1B:D5:FF:8F:37:BF:36:80:19:A6:91:CB:5B:2F:CB:7B:5F:03:0A:00:06:36:47:B2:0E:13:FE

7631

7632 An Entity called DeviceA:

--With an Entity ID:	0xFF:FF:FF:FF:FF:FF:FF:FE
--With a current Originator Counter:	0x20:00:00:00:00:00:00:01
--Digital Signing Private key :	0xFC:9B:B7:73:E6:C8:35:0A:DB:40:51:AC:91:3C:A4:70:CF:42:2D:8A:53:DE:8C:88:1D:BF:FE:B4:0B:A4:70:51
--Digital Signing Public key :	0x86:FB:5E:B3:CA:05:07:22:6B:E7:19:70:58:B9:EC:04:1D:3A:37:58:D9:D9:C9:19:02:AC:A3:39:1F:4E:58:AE:F1:3A:FF:63:CC:4E:F6:89:42:B9:B9:49:04:DC:1B:89:0E:DB:EA:BD:16:B9:92:11:06:24:96:8E:89:4E:56:0E
--Key Agreement Private key :	0xFB:9F:4C:02:B7:AB:F8:B0:DA:BA:02:7E:0B:C8:1B:8D:D2:09:68:3B:1C:88:93:EE:45:3F:AD:F3:A8:0F:73:E5
--Key Agreement Public key :	0x2D:B4:5A:3F:21:88:94:38:B4:2C:8F:46:4C:75:29:2B:AC:F5:FD:DB:5D:A0:B4:92:50:1B:29:9C:BF:E9:2D:8F:DB:90:FC:8F:F4:02:61:29:83:8B:1B:CA:D1:40:2C:AE:47:FE:7D:80:84:E4:09:A4:1A:FC:E1:6D:63:57:9C:5F
--The shared secret calculated with AccessControlBroker is :	0x9A:AC:F2:E6:D5:1B:D5:FF:8F:37:BF:36:80:19:A6:91:CB:5B:2F:CB:7B:5F:03:0A:00:06:36:47:B2:0E:13:FE
--The shared secret calculated with SupplierA is :	0x15:45:AD:F2:75:DC:8E:57:AB:E4:71:E9:F0:C1:20:C2:FA:DD:5B:12:51:AF:B7:BD:AB:25:3C:80:1B:41:11:CE

7633

7634 Create a Critical Command from SupplierA to Device A: ECS04b Reset Meter Balance on the ESME:

--GBCS Message Category:	SME.C.C
--------------------------	---------

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--GBCS Message Type:	Command
--CRA Flag:	0x01
--Originator Counter:	0x00:00:00:00:00:00:00:01
--Business Originator ID:	0x12:34:56:78:9A:BC:DE:F0
--Business Target ID:	0xFF:FF:FF:FF:FF:FF:FF:FE
--Date Time:	0x
--Other Info:	0x00:B3
--Message Content:	0xD9:20:00:00:01:00:03:03:00:70:00:00:13:0A:00:FF:02:03:00:70:00:00:13:0A:01:FF:02:03:00:70:00:00:13:0A:02:FF:02:03:05:00:00:00:00:05:00:00:00:00:05:00:00:00:00
--The originator's Private Signing Key:	0x3A:6B:2E:AA:0D:9F:25:A9:E4:55:98:3F:EB:5B:B9:47:52:81:21:91:1B:F3:B7:6B:E5:66:1C:89:DB:F2:4B:26
--The Message parts used in Signing:	0x09:01:00:00:00:00:00:00:01:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:B3:35:D9:20:00:00:01:00:03:03:00:70:00:00:13:0A:00:FF:02:03:00:70:00:00:13:0A:01:FF:02:03:00:70:00:00:13:0A:02:FF:02:03:05:00:00:00:00:05:00:00:00:00:05:00:00:00:00
--The per message secret number:	60336962327539050191752715802083620799857562569756173778554746475842619617775
--The resulting Signature in Plain Format:	0x59:04:5A:B0:F5:54:62:2B:03:60:34:0B:9D:87:B5:A2:3E:D5:72:3B:41:DE:3F:20:6E:58:CD:D1:0F:91:5B:9F:E2:E1:2E:2D:A3:63:24:78:A8:DF:67:8E:41:88:95:86:9A:C1:E5:53:18:CC:E0:4D:12:0D:2D:6B:44:DC:16:7B
--The Grouping Header:	0xDF:09:01:00:00:00:00:00:00:01:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:B3:35

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--All of the Message parts covered by the general-signing structure	0xDF:09:01:00:00:00:00:00:00:01:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:B3:35:D9:20:00:00:01:00:03:03:00:70:00:00:13:0A:00:FF:02:03:00:70:00:00:13:0A:01:FF:02:03:00:70:00:00:13:0A:02:FF:02:03:05:00:00:00:00:00:05:00:00:00:00:40:59:04:5A:B0:F5:54:62:2B:03:60:34:0B:9D:87:B5:A2:3E:D5:72:3B:41:DE:3F:20:6E:58:CD:D1:0F:91:5B:9F:E2:E1:2E:2D:A3:63:24:78:A8:DF:67:8E:41:88:95:86:9A:C1:E5:53:18:CC:E0:4D:12:0D:2D:6B:44:DC:16:7B
--The KDF OtherInfo:	0x60:85:74:06:08:03:00:12:34:56:78:9A:BC:DE:F0:09:01:00:00:00:00:00:00:01:FF:FF:FF:FF:FF:FF:FF:FE
--The per message secret symmetric key:	177594815140134193685548970760141301611
--The Initialization Vector:	0x12:34:56:78:9A:BC:DE:F0:00:00:00:00
--The Additional Authenticated Data:	0x11:00:00:00:00:00:DF:09:01:00:00:00:00:00:00:00:01:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:B3:35:D9:20:00:00:01:00:03:03:00:70:00:00:13:0A:00:FF:02:03:00:70:00:00:13:0A:01:FF:02:03:00:70:00:00:13:0A:02:FF:02:03:05:00:00:00:00:00:05:00:00:00:00:05:00:00:00:00:40:59:04:5A:B0:F5:54:62:2B:03:60:34:0B:9D:87:B5:A2:3E:D5:72:3B:41:DE:3F:20:6E:58:CD:D1:0F:91:5B:9F:E2:E1:2E:2D:A3:63:24:78:A8:DF:67:8E:41:88:95:86:9A:C1:E5:53:18:CC:E0:4D:12:0D:2D:6B:44:DC:16:7B
--The resulting MAC:	0x5D:83:2D:15:B5:7A:56:D6:20:F1:98:B3
--The MAC Header excluding the Security Header	0xDD:00:00:00:00:00:00:81:A9
--The Security Header fields:	0x11:00:00:00:00
--The resulting Message:	0xDD:00:00:00:00:00:00:81:A9:11:00:00:00:00:DF:09:01:00:00:00:00:00:00:00:01:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:B3:35:D9:20:00:00:01:00:03:03:00:70:00:00:13:0A:00:FF:02:03:00:70:00:00:13:0A:01:FF:02:03:00:70:00:00:13:0A:02:FF:02:03:05:00:00:00:00:00:05:00:00:00:00:05:00:00:00:00:05

7635

	:00:00:00:00:40:59:04:5A:B0:F5:54:62:2B:03:60:34:0B:9D:87:B5:A2:3E:D5:72:3B:41:DE:3F:20:6E:58:CD:D1:0F:91:5B:9F:E2:E1:2E:2D:A3:63:24:78:A8:DF:67:8E:41:88:95:86:9A:C1:E5:53:18:CC:E0:4D:12:0D:2D:6B:44:DC:16:7B:5D:83:2D:15:B5:7A:56:D6:20:F1:98:B3
--	---

7636 And get a Critical Response to SupplierA from Device A: ECS04b Reset Meter Balance on the ESME:

--GBCS Message Category:	SME.C.C
--GBCS Message Type:	Response
--CRA Flag:	0x02
--Originator Counter:	0x00:00:00:00:00:00:00:01
--Business Originator ID:	0xFF:FF:FF:FF:FF:FF:FF:FE
--Business Target ID:	0x12:34:56:78:9A:BC:DE:F0
--Date Time:	0x
--Other Info:	0x00:B3
--Message Content:	0xDA:20:00:00:01:00:00:03:00:00:00:03:03:00:03:00:03:00
--The originator's Private Signing Key:	0xFC:9B:B7:73:E6:C8:35:0A:DB:40:51:AC:91:3C:A4:70:CF:42:2D:8A:53:DE:8C:88:1D:BF:FE:B4:0B:A4:70:51
--The Message parts used in Signing:	0x09:02:00:00:00:00:00:00:01:08:FF:FF:FF:FF:FF:FF:FF:FE:08:12:34:56:78:9A:BC:DE:F0:00:02:00:B3:12:DA:20:00:00:01:00:00:03:00:00:00:03:03:00:03:00:03:00
--The per message secret number:	7051630462891053624226802980820342349692853825315679846585611221854040699

	8811
--The resulting Signature in Plain Format:	0xA0:1A:B6:9E:D8:A6:56:6A:B1:16:43:C7:35:82:60:A8:8A:8B:60:97:85:93:E7:6A:4E:93:19:35:85:D9:8D:9B:AD:84:38:78:F2:2E:79:4B:53:F6:F2:80:F9:F1:C8:48:D9:D3:8F:C0:50:CD:DD:58:82:75:63:E2:B0:FA:19:E2
--The Grouping Header:	0xDF:09:02:00:00:00:00:00:00:01:08:FF:FF:FF:FF:FF:FF:FF:FE:08:12:34:56:78:9A:BC:DE:F0:00:02:00:B3:12
--All of the Message parts covered by the general-signing structure	0xDF:09:02:00:00:00:00:00:00:01:08:FF:FF:FF:FF:FF:FF:FF:FE:08:12:34:56:78:9A:BC:DE:F0:00:02:00:B3:12:DA:20:00:00:01:00:00:03:00:00:00:03:03:00:03:00:03:00:40:A0:1A:B6:9E:D8:A6:56:6A:B1:16:43:C7:35:82:60:A8:8A:8B:60:97:85:93:E7:6A:4E:93:19:35:85:D9:8D:9B:AD:84:38:78:F2:2E:79:4B:53:F6:F2:80:F9:F1:C8:48:D9:D3:8F:C0:50:CD:DD:58:82:75:63:E2:B0:FA:19:E2
--The resulting Message:	0xDF:09:02:00:00:00:00:00:00:01:08:FF:FF:FF:FF:FF:FF:FF:FE:08:12:34:56:78:9A:BC:DE:F0:00:02:00:B3:12:DA:20:00:00:01:00:00:03:00:00:00:03:03:00:03:00:03:00:40:A0:1A:B6:9E:D8:A6:56:6A:B1:16:43:C7:35:82:60:A8:8A:8B:60:97:85:93:E7:6A:4E:93:19:35:85:D9:8D:9B:AD:84:38:78:F2:2E:79:4B:53:F6:F2:80:F9:F1:C8:48:D9:D3:8F:C0:50:CD:DD:58:82:75:63:E2:B0:FA:19:E2

7637

7638 Supplier A has now increased its Originator Counter by 1.

7639

7640 Create a non-Critical Command from SupplierA to Device A: ECS12 Set Change of Tenancy date on ESME:

--GBCS Message Category:	SME.C.NC
--GBCS Message Type:	Command
--CRA Flag:	0x01

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--Originator Counter:	0x00:00:00:00:00:00:00:02
--Business Originator ID:	0x12:34:56:78:9A:BC:DE:F0
--Business Target ID:	0xFF:FF:FF:FF:FF:FF:FF:FE
--Date Time:	0x
--Other Info:	0x00:22
--Message Content:	0xD9:20:00:00:02:00:01:02:00:01:00:00:5E:2C:03:02:02:01:09:0C:07:DF:01:05:FF:00:00:00:00:80:00:FF
--The Grouping Header:	0xDF:09:01:00:00:00:00:00:00:00:02:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:22:20
--The KDF OtherInfo:	0x60:85:74:06:08:03:00:12:34:56:78:9A:BC:DE:F0:09:01:00:00:00:00:00:00:00:00:02:FF:FF:FF:FF:FF:FF:FF:FE
--The per message secret symmetric key:	323267885984686097664772256155520506945
--The Initialization Vector:	0x12:34:56:78:9A:BC:DE:F0:00:00:00:00
--The Additional Authenticated Data:	0x11:00:00:00:00:00:DF:09:01:00:00:00:00:00:00:00:02:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FF:FE:00:02:00:22:20:D9:20:00:00:02:00:01:02:00:01:00:00:5E:2C:03:02:02:01:09:0C:07:DF:01:05:FF:00:00:00:00:80:00:FF:00
--The resulting MAC:	0x0F:1D:D0:0D:67:45:EB:D8:E0:A6:63:A4
--The MAC Header excluding the Security Header	0xDD:00:00:00:00:00:00:54
--The Security Header fields:	0x11:00:00:00:00
--The resulting Message:	0xDD:00:00:00:00:00:00:54:11:00:00:00:00:DF:09:01:00:00:00:00:00:00:00:00:02

7641

	:08:12:34:56:78:9A:BC:DE:F0:08:FF:FF:FF:FF:FF:FF:FE:00:02:00:22:20:D9: 20:00:00:02:00:01:02:00:01:00:00:5E:2C:03:02:02:01:09:0C:07:DF:01:05:FF:0 0:00:00:00:80:00:FF:00:0F:1D:D0:0D:67:45:EB:D8:E0:A6:63:A4
--	---

7642 And get a non-Critical Response to SupplierA from Device A: ECS12 Set Change of Tenancy date on ESME:

--GBCS Message Category:	SME.C.NC
--GBCS Message Type:	Response
--CRA Flag:	0x02
--Originator Counter:	0x00:00:00:00:00:00:00:02
--Business Originator ID:	0xFF:FF:FF:FF:FF:FF:FF:FE
--Business Target ID:	0x12:34:56:78:9A:BC:DE:F0
--Date Time:	0x
--Other Info:	0x00:22
--Message Content:	0xDA:20:00:00:02:00:00:01:00:01:02:00
--The Grouping Header:	0xDF:09:02:00:00:00:00:00:00:02:08:FF:FF:FF:FF:FF:FF:FE:08:12:34:56: :78:9A:BC:DE:F0:00:02:00:22:0C
--The KDF OtherInfo:	0x60:85:74:06:08:03:00:FF:FF:FF:FF:FF:FF:FE:09:02:00:00:00:00:00:00: :02:12:34:56:78:9A:BC:DE:F0
--The per message secret symmetric key:	102613665902023293907968102748610736248
--The Initialization Vector:	0xFF:FF:FF:FF:FF:FF:FF:FE:00:00:00:00

--The Additional Authenticated Data:	0x11:00:00:00:00:00:DF:09:02:00:00:00:00:00:02:08:FF:FF:FF:FF:FF:FF:FE:08:12:34:56:78:9A:BC:DE:F0:00:02:00:22:0C:DA:20:00:00:02:00:00:01:00:01:02:00:00
--The resulting MAC:	0x0B:3C:1B:31:2C:EA:E9:C1:30:06:0E:29
--The MAC Header excluding the Security Header	0xDD:00:00:00:00:00:00:40
--The Security Header fields:	0x11:00:00:00:00
--The resulting Message:	0xDD:00:00:00:00:00:00:40:11:00:00:00:00:DF:09:02:00:00:00:00:00:00:00:02:08:FF:FF:FF:FF:FF:FF:FE:08:12:34:56:78:9A:BC:DE:F0:00:02:00:22:0C:DA:20:00:00:02:00:00:01:00:01:02:00:00:0B:3C:1B:31:2C:EA:E9:C1:30:06:0E:29

7643 Example Messages Produced

7644

7645 ECS04b Reset Meter Balance on the ESME (Message Category: SME.C.C)

Command Message Structure		
Name	Encoded Content	Encoded Length
MAC Header (general-ciphering)		
____tag	0xDD	1
____contents	0x00000000000000	6
____ciphered-service		
____length	0x8200A9	3
____security header		
____security control byte (SC)	0x11	1

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_____ invocation counter (IC)	0x00000000	4
Grouping Header (general-signing)		
____ tag	0xDF	1
____ transaction-id		
_____ length	0x09	1
_____ value (CRA FLAG)	0x01	1
_____ value (Originator Counter)	0x000000000000000001	8
____ originator-system-title		
_____ length	0x08	1
_____ value	0x123456789ABCDEF0	8
____ recipient-system-title		
_____ length	0x08	1
_____ value	0xFFFFFFFFFFFFFFFFFE	8
____ date-time		
_____ length	0x00	1
____ other-information		
_____ Length	0x02	1
_____ Message Code	0x00B3	2
____ content		
_____ length	0x35	1
access-request		
____ tag	0xD9	1
____ long-invoke-id-and-priority		
_____ configuration	0x20	1
_____ invoke-id	0x000001	3
____ date-time	0x00	1
____ access-request-body		

__access-request-specification		
__SEQUENCE OF	0x03	1
____Request number 1		
____access-request-action	0x03	1
____cosem-method-descriptor		
____class-id	0x0070	2
____instance-id	0x0000130A00FF	6
____method-id	0x02	1
____Request number 2		
____access-request-action	0x03	1
____cosem-method-descriptor		
____class-id	0x0070	2
____instance-id	0x0000130A01FF	6
____method-id	0x02	1
____Request number 3		
____access-request-action	0x03	1
____cosem-method-descriptor		
____class-id	0x0070	2
____instance-id	0x0000130A02FF	6
____method-id	0x02	1
__access-request-list-of-data		
__SEQUENCE OF	0x03	1
____Parameter for request number 1		
____Names		
____Tag	0x05	1
____Value	0x00000000	4
____Parameter for request number 2		

_____Names		
_____Tag	0x05	1
_____Value	0x00000000	4
_____Parameter for request number 3		
_____Names		
_____Tag	0x05	1
_____Value	0x00000000	4
_____signature-length	0x40	1
_____signature-content	0x59045AB0F554622B0360340B9D87B5A23ED5723B41DE3F2 06E58CDD10F915B9FE2E12E2DA3632478A8DF678E41889586 9AC1E55318CCE04D120D2D6B44DC167B	64
_____mac-content	0x5D832D15B57A56D620F198B3	12

7646

7647 Response Message Structure

Name	Encoded Content	Encoded Length
Grouping Header (general-signing)		
_____tag	0xDF	1
_____transaction-id		
_____length	0x09	1
_____value (CRA FLAG)	0x02	1

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_____value (Originator Counter)	0x0000000000000001	8
_____originator-system-title		
_____length	0x08	1
_____value	0xFFFFFFFFFFFFFFFFFE	8
_____recipient-system-title		
_____length	0x08	1
_____value	0x123456789ABCDEF0	8
_____date-time		
_____length	0x00	1
_____other-information		
_____Length	0x02	1
_____Message Code	0x00B3	2
_____content		
_____length	0x12	1
access-response		
_____tag	0xDA	1
_____long-invoke-id-and-priority		
_____configuration	0x20	1
_____invoke-id	0x000001	3
_____date-time	0x00	1
_____access-request-specification	0x00	1
_____access-response-list-of-data		
_____SEQUENCE OF	0x03	1
_____Response for request number 1		
_____Tag	0x00	1
_____Response for request number 2		
_____Tag	0x00	1

_____Response for request number 3		
_____Tag	0x00	1
_____access-response-specification		
_____SEQUENCE OF	0x03	1
_____Result for request number 1		
_____access-response-action	0x03	1
_____result	0x00	1
_____Result for request number 2		
_____access-response-action	0x03	1
_____result	0x00	1
_____Result for request number 3		
_____access-response-action	0x03	1
_____result	0x00	1
_____signature-length	0x40	1
_____signature-content	0xA01AB69ED8A6566AB11643C7358260A88A8B60978593E76 A4E93193585D98D9BAD843878F22E794B53F6F280F9F1C848 D9D38FC050CDDD58827563E2B0FA19E2	64

7648

7649 ECS12 Set Change of Tenancy date on ESME

7650

7651 Command Message Structure

SEC Schedule 8 – GBCS v4.4

Name	Encoded Content	Encoded Length
MAC Header (general-ciphering)		
____tag	0xDD	1
____contents	0x00000000000000	6
____ciphered-service		
____length	0x54	1
____security header		
____security control byte (SC)	0x11	1
____invocation counter (IC)	0x00000000	4
Grouping Header (general-signing)		
____tag	0xDF	1
____transaction-id		
____length	0x09	1
____value (CRA FLAG)	0x01	1
____value (Originator Counter)	0x00000000000000002	8
____originator-system-title		
____length	0x08	1
____value	0x123456789ABCDEF0	8
____recipient-system-title		
____length	0x08	1
____value	0xFFFFFFFFFFFFFFFFFE	8
____date-time		
____length	0x00	1
____other-information		
____Length	0x02	1
____Message Code	0x0022	2

____content		
_____length	0x20	1
access-request		
____tag	0xD9	1
____long-invoke-id-and-priority		
_____configuration	0x20	1
_____invoke-id	0x000002	3
____date-time	0x00	1
____access-request-body		
____access-request-specification		
____SEQUENCE OF	0x01	1
_____Request number 1		
_____access-request-set	0x02	1
_____cosem-attribute-descriptor		
_____class-id	0x0001	2
_____instance-id	0x00005E2C0302	6
_____attribute-id	0x02	1
____access-request-list-of-data		
____SEQUENCE OF	0x01	1
_____Parameter for request number 1		
_____Names		
_____Tag	0x09	1
_____Length	0x0C	1
_____Value	0x07DF0105FF000000008000FF	12
____signature-length	0x00	1

7652

7653 Response Message Structure

____mac-content	0x0F1DD00D6745EBD8E0A663A4	12
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Name	Encoded Content	Encoded Length
MAC Header (general-ciphering)		
____tag	0xDD	1
____contents	0x000000000000	6
____ciphered-service		
____length	0x40	1
____security header		
____security control byte (SC)	0x11	1
____invocation counter (IC)	0x00000000	4
Grouping Header (general-signing)		
____tag	0xDF	1
____transaction-id		
____length	0x09	1
____value (CRA FLAG)	0x02	1
____value (Originator Counter)	0x0000000000000002	8
____originator-system-title		
____length	0x08	1
____value	0xFFFFFFFFFFFFFFFE	8
____recipient-system-title		
____length	0x08	1

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_____value	0x123456789ABCDEF0	8
_____date-time		
_____length	0x00	1
_____other-information		
_____Length	0x02	1
_____Message Code	0x0022	2
_____content		
_____length	0x0C	1
access-response		
_____tag	0xDA	1
_____long-invoke-id-and-priority		
_____configuration	0x20	1
_____invoke-id	0x000002	3
_____date-time	0x00	1
_____access-request-specification	0x00	1
_____access-response-list-of-data		
_____SEQUENCE OF	0x01	1
_____Response for request number 1		
_____Tag	0x00	1
_____access-response-specification		
_____SEQUENCE OF	0x01	1
_____Result for request number 1		
_____access-response-set	0x02	1
_____result	0x00	1
_____signature-length	0x00	1

____mac-content	0x0B3C1B312CEAE9C130060E29	12
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7654 Use Cases

7655 The Use Cases are contained in the embedded HTML document at Table 19.3. Each Use
 7656 Case represents one or more interactions with a Device that make up a GBCS Command,
 7657 Response and / or Alert. This Section 19 provides an overview of the repeatable content
 7658 within these Use Cases.

7659 By definition, logs can contain zero entries (e.g. because the Device is newly installed or,
 7660 where allowed by SMETS/CHTS, a log has just been cleared). Any lower size limits for logs in
 7661 Use Cases, Message Templates and the Mapping Table shall be interpreted accordingly.

7662 Use Case Title

7663 Each Use Case Title section in Table 19.3 provides common information regarding the Use
 7664 Cases that follow. Each section and its purpose is outlined in Table 19.1.

Section	Content
Description	A textual summary of the purpose and scope of the Use Cases encompassed by the Use Case Title
Use Case	Details the Unique Use Case reference, the Use Case name and the Use Case's Message Code (see Section 15)
Use Case Cross References	See Section 19.1.1
Use Case Access Permissions	A summary of User Roles that can perform the Use Case. See Section 17 for Remote Party Usage Rights and Section 4.3.2.6 for Trust Anchor Cells applicable. Note that Use Cases from Unknown Remote Parties are performed using the Remote Party Role of Access Control Broker
SMETS / CHTS Objects applicable to Use Case	A list of SMETS / CHTS attributes and associated methods that are applicable to the Use Case. This confirms the properties required by SMETS / CHTS for the attribute/method. This also provides information on the Service Reference invoked This table is sorted alphabetically by the entry in the column 'name' concatenated with the entry in the column 'attribute / method'

7665 Table 19.1: SMETS / CHTS content of Use Cases

7666 Use Case Cross Reference Section

7667 Table 19.3 provides an overview of important information relevant to the Use Case. It has a
 7668 structured table as summarised in Table 19.1.1.

Cross Reference	Possible Values	Notes
Remote Party or HAN message	HAN Only Message / Remote Party Message	Needed to identify which GBCS requirements apply. See Section 6
Message Type	Command and Response / Alert with reference to the message categories in Section 6	Needed to identify which GBCS requirements apply
Capable of future dated invocation?	Yes / No	Needed to identify which GBCS requirements apply. See Section 9.2
Requires protection against replay?	Yes / No	Needed to identify which GBCS requirements apply. See Section 4.3.1.5
Service Reference	[e.g. 6.20 SetDeviceConfiguration(MPxN)]	Traceability to other Smart Energy Code documents
Read Or Update	Read, Update	Identifies whether the purpose of the Use Case is 'Read' or 'Update'
Response Recipient different from Command Sender	Yes or Blank	Identifies where a Response is sent to a different Remote Party than the originator of the associated Service Request
Use Case Access Permissions	Supplier (C) Supplier (NC) Supplier prepay top up Network Operator (C) Network Operator (NC) Access Control Broker (NC)	Lists which Remote Party Roles may originate the Command within the Use Case. This separates (C) critical and (NC) non critical See Section 17 for more details

Cross Reference	Possible Values	Notes
	WAN Provider (C) Access Control Broker (C) Load Controller (C) Load Controller (NC)	

7669 Table 19.1.1: Allowable values for SMETS / CHTS Use Case Cross References

7670 Objects Applicable to Use Case Section

7671 This section in Table 19.3 contains a 'SMETS objects applicable to Use Case' table to provide
7672 traceability between SMETS functions and methods and the Use Case.

7673 The table contains the values set out in Table 19.1.2.

Row Name	Meaning
Mapping Table row #	Identifier of the SMETS / CHTS object's row in the Mapping Table
Ref	SMETS/CHTS document location of the Attribute (prefixed by the document)
Name	The attribute name as specified in SMETS or CHTS
Attribute / Method	The attribute or method being applied to the SMETS/CHTS
Notes	Describes the Method being applied to the SMETS/CHTS attribute or method in the Use Case
Sub Category	Specifies whether an attribute or method
Data Type	Details of the data type for the attribute as specified in SMETS or CHTS

7674 Table 19.1.2: SMETS objects applicable to Use Case

7675 Beneath that 'SMETS objects applicable to Use Case' table, there is, in each Use Case, one or
7676 two sets of detail relating to either (1) the Use Case's DLMS COSEM implementation or (2)
7677 the Use Case's GBZ implementation or (3) both.

7678 Each such section begins with a statement as to:

7679 the 'Target' Device Types that must support the Use Case. Where the 'Target' includes ESME,
7680 an SAPC may or shall support the Use Case, as determined by whether the 'SAPC support'
7681 column in the 'Use Case reference' tab of Table 20 states 'optional' or 'mandatory'
7682 respectively; and

7683 the required Message Code.

7684 Pre-conditions

7685 Pre-conditions represent conditions for which Device logic is required to ensure correct
7686 operation of commands contained within a message, on the Device. Exception conditions
7687 (such as failures) that are managed by the Protocol are not captured as Pre-conditions.
7688 Manufacturers of Devices must only enforce Pre-conditions that are stated in the Use Cases.
7689 Note that the use of Pre-conditions is minimised in favour of controls being implemented on
7690 Service User systems.

7691 Actions

7692 Actions stipulate additional Device actions that must be performed together with successful
7693 execution of the Use Case.

7694 Use Case-specific content

7695 Each Use Case is given a unique reference and a title.

7696 DLMS COSEM specific content

7697 Table 19.2.1 sets out the Use Case specific attributes and methods and the DLMS specific
7698 mapping.

7699 Within any DLMS COSEM Payload, cosem-attribute-descriptors and cosem-method-
7700 descriptors, and associated fields, shall be ordered based on the contents of columns in the
7701 Mapping Table. The sort order, described by columns headings in the Mapping Table, shall
7702 be:

7703 first by DLMS Sequence;

7704 then by DLMS: Class;

7705 then by OBIS Code;

7706 then by Attribute (A) / Method (M); and

7707 then by Attribute / Method Number.

7708 The sort order shall be ascending in all cases.

7709 For structures, the sequence of elements within a structure shall be as defined in the Blue
7710 Book, where it is defined in the Blue Book, or as per the Mapping Table, where it is not
7711 defined in the Blue Book.

7712 For clarity, the SMETS / CHTS table in each Use Case is sorted in this same order.

Row Name	Meaning
Mapping table row #	Identifier of the SMETS / CHTS object's row in the Mapping Table
SMETS / CHTS Ref	The section(s) with SMETS/CHTS that refer to the SMETS / CHTS name
SMETS / CHTS Name	The attribute name as specified in SMETS / CHTS
Class	Denotes the DLMS Interface Class
OBIS Code	defines identification codes for all data in DLMS / COSEM compliant metering equipment
Attribute or Method	Denotes whether the row relates to an (A)tttribute or (M)ethod
Attribute / Method Number	Forms part of the attribute identity.
Attribute / Method Name and Blue Book reference	The name given to the DLMS object
DLMS COSEM Data type	The data type assigned to the DLMS object
Constant Value	Where this field is present, this is a fixed value for the life of the Device
Notes	Additional useful information

7713 Table 19.2.1: DLMS mapping for Use Case specific attributes / methods

7714 ZSE specific content

7715 Table 19.3 provides information on the ZSE commands required successfully to complete the
 7716 Use Case. These must be processed in the order listed in Table 19.2.2.

7717 Table 19.2.2 is grouped by ZSE command.

Row Name	Meaning
Mapping table row #	Identifier of the row in the Mapping Table
SMETS / CHTS Ref	Identifies the SMETS / CHTS section that describes the attribute
SMETS / CHTS Name	The attribute name as specified in SMETS / CHTS The method being applied to the SMETS / CHTS attribute

Data Type	Identifies the ZSE data type for the attribute
Range	The allowable value range for the attribute
Attribute / Value / Parameter	For ZSE read operations – the attribute or a value returned For ZSE update operations – the attribute or parameter updated
Cluster :ID	Identifies the ZSE cluster that supports the required functionality
Command :ID	Identifies the command and its unique identifier within the ZSE cluster that is used to read or manipulate the attribute for the purpose of the Use Case. Its ZSE identifier is included
Response :ID	Where specified, this identifies the Response and its unique identifier to the read or update command

7718 Table 19.2.2: ZSE specific content

7719 Embedded Use Cases

7720 Table 19.3 contains the Use Cases that fulfil the interface requirements to cover Commands
7721 (and their Responses) and Alerts (where applicable). In addition, it includes ZSE Message
7722 Templates.

7723 Note: DLMS COSEM methods that have values which have an impact on the execution of
7724 the method (that is, methods with input values that are not integer(0)), the DLMS part of
7725 the Mapping Table and the Use Case include two or more rows. One row contains the
7726 method, and the subsequent row(s) contain the value(s) to be sent with the method.

7727 A number of Use Cases are also covered in GBCS main body. These are identifiable from the
7728 Table of Contents.



GBCS v4.4 Table
19.3 Use Cases.html

7729

7730 Table 19.3: Use Cases

7731 Mapping Table

7732 Table 20 contains the Mapping Table from which the Use Cases and Message Templates
7733 were generated. These tables map between SMETS attributes and methods, SEC Service
7734 Requests, Use Cases, DLMS COSEM attributes and methods and ZSE clusters, attributes and
7735 commands.

7736 In addition to the Use Cases, certain columns in the Mapping Table are directly referenced
7737 from this document.

7738 Please note that in the SMETS required objects tab only rows marked 'E' (External to HAN) in
7739 column F are fully specified, since those rows relate to Remote Party Messages. Other rows
7740 are only specified to the extent that these elements of Remote Party Messages rely on them.



V4.4 Table 20.xlsx

7741

7742 Table 20: Mapping Table

7743 Glossary

7744 ||

7745 X || Y shall mean the concatenation of the two octet strings X and Y.

7746 Abstract Syntax Notation One (ASN.1)

7747 ASN.1 is a standard notation for the definition of data types and values. A data type (or type
7748 for short) is a category of information (for example, numeric, textual, still image or video
7749 information). A data value (or value for short) is an instance of such a type. ASN.1 defines
7750 several basic types and their corresponding values, and rules for combining them into more
7751 complex types and values. In some protocol architectures, each message is specified as the
7752 binary value of a sequence of octets. However, standards-writers need to define quite
7753 complex data types to carry their messages, without concern for their binary representation.
7754 In order to specify these data types, they require a notation that does not necessarily
7755 determine the representation of each value. ASN.1 is such a notation.

7756 Access Control Broker (ACB)

7757 In the context of a specific Device, the Known Remote Party whose Security Credentials are
7758 stored in the {accessControlBroker, digitalSignature, management} Trust Anchor Cell where
7759 present, and stored in the {accessControlBroker, keyAgreement, management} Trust Anchor
7760 Cell otherwise.

7761 The ACB applies Cryptographic Protections to all Commands addressed to the Device in
7762 question, except potentially for certain recovery scenarios catered for by the Security
7763 Credentials Commands.

7764 Access Control Broker to Device MAC (ACB-SMD MAC)

7765 A MAC generated by the Access Control Broker in relation to a Command which can only be
7766 verified by the Device which is the target of the Command.

7767	Activate Emergency Credit
7768	A Command described in SMETS.
7769	Active Energy
7770	The integral with respect to time of the Active Power in units of watt-hours (Wh) or standard
7771	multiples thereof (for example, kWh).
7772	Additional Authenticated Data (AAD)
7773	One of the inputs to the calculation of a MAC. The AAD is protected by the MAC but is not
7774	encrypted. AAD has the same meaning as in <i>NIST Special Publication 800-38D</i> :
7775	http://csrc.nist.gov/publications/nistpubs/800-38D/SP-800-38D.pdf .
7776	Alert
7777	A Message generated by a Device including in response to a problem or the risk of a
7778	potential problem.
7779	Alert Code
7780	A 16 bit unsigned integer taking the values specified in Section 16. The Alert Code and Event
7781	Code are the same for a given Event.
7782	Application Association
7783	Shall have the meaning specified in the DLMS COSEM standards.
7784	Application Layer Protocol Data Unit (APDU)
7785	Information delivered as a unit among peer entities of networks.
7786	Association LN Object
7787	A DLMS Component specified in the Blue Book which provides role based access control.
7788	Authenticated Decryption
7789	Has the same meaning as specified in <i>NIST Special Publication 800-38D</i> :
7790	http://csrc.nist.gov/publications/nistpubs/800-38D/SP-800-38D.pdf
7791	Authenticated Encryption (AE)
7792	Has the same meaning as specified in <i>NIST Special Publication 800-38D</i> :
7793	http://csrc.nist.gov/publications/nistpubs/800-38D/SP-800-38D.pdf
7794	Authentication
7795	The method used to confirm the identity of entities or Devices wishing to communicate and
7796	‘Authenticated’ and ‘Authenticity’ shall be construed accordingly.

7797	Authentication Key
7798	Shall be as defined in the Green Book.
7799	Authorisation
7800	The process of granting access to a resource and 'Authorised' shall be construed accordingly.
7801	Authorised Public Key Infrastructure (APKI)
7802	A key infrastructure that is compliant with the Certificate related requirements of this GBCS.
7803	Auxiliary Controller
7804	Shall have the meaning defined in SMETS.
7805	Auxiliary Controller Calendar
7806	Shall have the meaning defined in SMETS.
7807	Auxiliary Load Control Switch
7808	Shall have the meaning defined in SMETS.
7809	Blue Book
7810	The DLMS Blue Book Version <i>DLMS UA 1000-1 Ed. 12.0</i> . This document can be obtained
7811	from the DLMS User Association: http://www.dlms.com .
7812	Boost Function
7813	ESME and SAPC functionality described in SMETS.
7814	Break On Error
7815	Shall have the Green Book meaning of Break On Error used in relation to 'Processing_Option'.
7816	Business Originator
7817	The Smart Metering Entity sending the first Message in a Use Case.
7818	Business Target
7819	The Smart Metering Entity receiving the first Message in a Use Case.
7820	Certificate
7821	An electronic document that binds an identity, and possibly other information, to a Public
7822	Key.
7823	Certification Request
7824	A message requesting the issue of a Certificate by a Certification Authority.

7825	Certification Authority (CA)
7826	A trusted entity which issues Certificates.
7827	Certification Authority Certificate
7828	A Certificate issued to a Certification Authority that allows Certification Path Validation in
7829	relation to Remote Party's Certificates.
7830	Certification Path Validation
7831	Shall have the meaning defined in Section 4.3.2.8.
7832	Certification Revocation List (CRL) Validation
7833	Shall have the meaning defined in Section 4.3.2.8.
7834	Ciphered Information
7835	Shall have the meaning defined in Section 8.4.
7836	Ciphertext
7837	An output of the Authenticated Encryption function or an input of the Authenticated
7838	Decryption function defined in <i>NIST Special Publication 800-38D</i> . The unencrypted form of
7839	the Ciphertext is the Plaintext.
7840	Clock
7841	A timing mechanism that has a minimum resolution of 1 second.
7842	Command
7843	An instruction to perform a function received or sent via any interface.
7844	Command Response Alert (CRA) Flag
7845	An element within a Message Header that enumerates whether the Message is a Command
7846	or a Response or an Alert.
7847	Commercial Product Assurance Security Characteristic
7848	The security characteristics for the relevant Device as indicated in Section 1.0.
7849	Common Name
7850	The Subject X520 Common Name within an Organisation Certificate, where those terms
7851	have their SEC Appendix B Organisation Certificate Policy meaning.
7852	Communications Hub
7853	A device complying with the CHTS.

7854	Communications Hub Function (CHF)
7855	A Communications Hub Function as defined in the Communications Hub Technical
7856	Specifications.
7857	Communications Hub Technical Specifications (CHTS)
7858	Communications Hub Technical Specifications set out in Schedule 10 of the Smart Energy
7859	Code.
7860	Communications Link
7861	Shall have the meaning defined in CHTS.
7862	Confidentiality
7863	The state of information, in transit or at rest, where there is assurance that it is not
7864	accessible by Unauthorised parties through either unintentional means or otherwise.
7865	Consumer
7866	A person who lawfully resides at the Premises that is being Supplied.
7867	Consumer Access Device (CAD)
7868	A Device which, in terms of this GBCS, is permitted to use the same Messages as an IHD.
7869	Consumption
7870	In the context of GSME, Gas Consumption or in the context of ESME, Electricity
7871	Consumption information.
7872	Contingency Key
7873	A feature of Trust Anchor Management Protocol (RFC 5934), and only ever used in a
7874	recovery scenario when the root Certificate (Apex Trust Anchor) needs to be replaced.
7875	Critical Message
7876	A Remote Party Message which may relate to supply being affected, financial fraud or the
7877	compromise of Device security. Critical, Critical Commands, Critical Alerts and Critical
7878	Responses shall be construed accordingly.
7879	Cryptographic Algorithm
7880	An algorithm for performing one or more cryptographic functions which may include
7881	Encryption; Decryption; Digitally Signing or Hashing of information, data, or messages; or
7882	exchange of Security Credentials.
7883	Cryptographic Protection

7884	A part of a Message constructed to provide assurance to the Message recipient in terms of
7885	one or more of integrity, authenticity, non-repudiation and Confidentiality.
7886	Currency Units
7887	The units of monetary value in major and minor units.
7888	Current Private Key
7889	A Device's Private Key for which the Device has most recently successfully received and
7890	processed a Certificate for the corresponding Public Key as defined in Section 13.5.
7891	Data and Communications Company (DCC)
7892	The holder of the licence for the provision of a smart meter communication service granted
7893	pursuant to section 6(1)(f) or 6(1A) of the Electricity Act 1989 or section 7AB of the Gas Act
7894	1986.
7895	Data Store
7896	An area of a Device capable of storing information for future retrieval.
7897	Decryption
7898	The process of converting Encrypted information by an Authorised party to recover the
7899	original information. Like terms shall be construed accordingly.
7900	Device
7901	A Device that is one of ESME, GSME, Gas Proxy Function, Communications Hub Function,
7902	HICALCS, PPMID, SAPC or a Type 2 Device.
7903	Device Based Access Control (DBAC)
7904	Shall have the meaning defined in Section 13.7.3.
7905	Device Certificate
7906	Shall have the meaning set out in Section 12.
7907	Device Log
7908	A data item having the meaning set out in SMETS or CHTS (as relevant).
7909	Digital Signature
7910	The information appended to a Message which is created using the sender's Private Key,
7911	that can be verified using the corresponding Public Key contained in the sender's Certificate,
7912	and provides the receiver with assurance that the sender is who they claim to be, the

7913	message has not been altered in transit and that the holder of the the sender's Private Key
7914	created the Message.
7915	Digital Signing
7916	The creation of a Digital Signature.
7917	Digital Signing Certificate
7918	A Certificate which states that the Public Key contained within, and its associated Private Key,
7919	may be used for Digital Signing purposes.
7920	Distinguished Encoding Rules
7921	Shall have the meaning defined in https://www.itu.int/rec/T-REC-X.690/en
7922	DLMS COSEM
7923	Device Language Message Specification / Companion Specification for Energy Metering - an
7924	Application Layer protocol.
7925	Dual Band Communications Hub (Dual Band CH)
7926	A Communications Hub that is capable of operating at 2.4 GHz and Sub GHz frequencies
7927	simultaneously.
7928	Electricity Consumption
7929	The Active Energy Imported into the Premises and 'Consumed' shall be construed
7930	accordingly.
7931	Elliptic Curve DSA (ECDSA)
7932	The Elliptic Curve Digital Signature Algorithm (see
7933	http://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.186-4.pdf) as specified in Section 4.3.3.
7934	Encoding(X)
7935	The encoding of a variable length integer X, as specified in Section 3.3.
7936	Encryption
7937	The process of converting information in order to make it unintelligible other than to
7938	Authorised parties. Like terms shall be construed accordingly.
7939	Encryption Originator Counter
7940	Shall have the meaning defined in Table 23.
7941	Encryption Remote Party

7942	The Remote Party that can decrypt Encrypted data items.
7943	Entity Identifier
7944	A 64 bit unsigned integer uniquely identifying a Smart Metering Entity.
7945	ESME
7946	Electricity Smart Metering Equipment, as described in SMETS.
7947	ESMETS
7948	The Electricity Smart Metering Equipment (ESME) Technical Specifications in SMETS.
7949	Event
7950	A change in state generated by a Device in response to an internal or external trigger.
7951	Event Code
7952	A 16 bit unsigned integer taking the values specified in Section 16, <u>which equates to the</u>
7953	<u>ZigBee Event ID parameter</u> . The Alert Code and Event Code are the same for a given Event.
7954	Event Log
7955	Data item described in the Technical Specifications.
7956	Execution Counter
7957	Shall have the meaning defined in Section 4.3.1.5 and in Section 13.3.5.3.
7958	Firmware
7959	The embedded software programmes and / or data structures that control Devices.
7960	Firmware Distribution Receipt Alert
7961	Shall have the meaning set out in Section 11.2.6.
7962	Force Replace
7963	The means to instruct a Communications Hub to replace an ESME, SAPC or GSME Firmware
7964	image that it holds, e.g. when the image has only been partially downloaded to the ESME,
7965	SAPC or GSME. This enables recovery from failures.
7966	Gas Consumption
7967	The volume of gas in cubic metres (m3) Supplied to the Premises and 'Consumed' shall be
7968	construed accordingly.
7969	Gas Proxy Function (GPF)

7970	A Gas Proxy Function as defined in the Communications Hub Technical Specifications.
7971	Galois Counter Mode (GCM)
7972	The mode of operation specified in <i>NIST Special Publication 800-38D</i> .
7973	GBZ
7974	A set of structures in the GBCS which carry ZCL / ZSE commands.
7975	General Block Transfer (GBT) / GBT Message
7976	General Block Transfer is a DLMS COSEM mechanism for decomposing APDUs above
7977	maximum sizes that can be transported in to a number of smaller APDUs, which are no
7978	larger than the maximum sizes. A GBT Message is one of these smaller APDUs.
7979	GMAC
7980	Variant of GCM that is used to generate Message Authentication Code from non-
7981	Confidential data, as specified in <i>NIST Special Publication 800-38D</i> .
7982	Green Book
7983	The DLMS Green Book Version <i>DLMS UA 1000-2 Ed.8</i> . This document can be obtained from
7984	the DLMS User Association: http://www.dlms.com .
7985	GSME
7986	Gas Smart Metering Equipment, as described in SMETS.
7987	GSMETS
7988	The Gas Smart Metering Equipment (GSME) Technical Specifications in SMETS.
7989	HAN Only Message
7990	A Message where both the sender and recipient are Devices on the same Smart Metering
7991	Home Area Network.
7992	HAN Connected Auxiliary Load Control Switch (HCALCS)
7993	HAN Connected Auxiliary Load Control Switch, as described in SMETS..
7994	Hashing
7995	A repeatable process to create a fixed size condensed representation of a Message or any
7996	arbitrary data, as further set out in Section 4.3.3. Hash and like terms shall be construed
7997	accordingly.
7998	HCALCSTS

7999	The HAN Connected Auxiliary Load Control Switches (HCALCS) Technical Specifications in
8000	SMETS.
8001	Home Area Network Interface (HAN Interface)
8002	A component of GSME, ESME, IHD or other Device that is capable of sending and receiving
8003	information to and from other Devices.
8004	IHDTs
8005	The In Home Display Technical Specifications in SMETS.
8006	IHD
8007	In Home Display with its SMETS meaning.
8008	Initialization Vector (IV)
8009	An input to the Authenticated Encryption and Authenticated Decryption functions defined in
8010	<i>NIST Special Publication 800-38D</i> . Where the GBCS applies, it shall have the values as
8011	specified at Section 4.3.3.4.
8012	Inter-PAN
8013	Shall have the meaning defined in CHTS.
8014	Join
8015	The process of authorising two Devices to communicate at the application layer.
8016	Key
8017	Data used to determine the output of a cryptographic operation.
8018	Key Agreement
8019	A means to calculate a shared Key between two parties.
8020	Key Agreement Certificate
8021	A Certificate which states that the Public Key contained within, and its associated Private Key,
8022	may be used for Key Agreement purposes.
8023	Key Derivation Function (KDF)
8024	A function to generate derived keying material from a Shared Secret and other information.
8025	Known Remote Party (KRP)
8026	In the context of a specific Device, a Remote Party whose Security Credentials are stored on
8027	that Device in at least one Trust Anchor Cell.

8028	KRP Signature
8029	A Digital Signature generated by a Known Remote Party.
8030	Len(X)
8031	The number of octets in the variable length octet string X.
8032	Load Controller
8033	In the context of a specific Device, the Known Remote Party whose Security Credentials are
8034	stored in the { loadController, digitalSignature, management} Trust Anchor Cell.
8035	MAC Header
8036	As defined in Section 6, a part of a message which is only present when the Message
8037	contains a MAC but which is additional to the MAC.
8038	Manufacturer Image Hash
8039	Shall have the meaning defined in Section 11.2.4.
8040	Mapping Table
8041	The spreadsheet detailing Use Cases and associated protocol requirements as embedded in
8042	Section 20.
8043	Maximum Credit Threshold
8044	Shall have the meaning defined in SMETS.
8045	Maximum Meter Balance Threshold
8046	Shall have the meaning defined in SMETS.
8047	Message
8048	A Command, Response or Alert.
8049	Message Authentication
8050	The process by which the receiver of a Message is provided with assurance that the sender
8051	is who they claim to be and that the Message is in the form originally sent.
8052	Message Authentication Code (MAC)
8053	The number incorporated in a Message to provide Message Authentication, as set out in
8054	Section 4.3.3.
8055	Message Category
8056	A grouping of Remote Party Messages.

8057	Message Code
8058	A 16 bit unsigned integer identifying the Use Case that the Message in question must
8059	conform to. Message Codes have the values specified in Section 15.
8060	Message Identifier
8061	Message Identifier shall be the concatenation of:
8062	Business Originator ID;
8063	Business Target ID;
8064	CRA Flag; and
8065	Originator Counter.
8066	Message Series
8067	Shall have the meaning defined in Section 7.2.11.1.
8068	Message Template
8069	A protocol-specific table defining the encoding of a Message.
8070	Message Type
8071	The Message Types are Command, Response or Alert.
8072	Network Interface
8073	A WAN Interface or HAN Interface.
8074	Network Operator
8075	In the context of a specific Device, the Known Remote Party whose Security Credentials are
8076	stored in the {networkOperator, digitalSignature, management} Trust Anchor Cell.
8077	Object Identifier (OID)
8078	An identifier used to name an object. Structurally, an OID consists of a node in a
8079	hierarchically-assigned namespace, formally defined using the ASN.1 standard.
8080	Operating Sub GHz Channel
8081	The Sub GHz Channel on which a Device is currently operating.
8082	Organisation Certificate
8083	Shall have the meaning set out in Section 12.
8084	Originator Counter

8085	Shall have the meaning defined in Section 4.3.1.2.
8086	OTA Header
8087	Shall have the meaning defined in Section 11.2.3.
8088	OTA Upgrade Image
8089	Shall have the meaning defined in Section 11.2.3.
8090	OtherInfo
8091	An input to the KDF with the meaning as specified in section 5.8.1 of <i>NIST Special Publication</i>
8092	<i>800-56Ar2</i> .
8093	Other User
8094	A Remote Party which is not a Known Remote Party in relation to any Device, and so is
8095	always an Unknown Remote Party in any communication with a Device.
8096	Outcome
8097	The result of executing a Command, expressed as success or failure.
8098	Payload
8099	Part of the Message that provides the message-specific content.
8100	Payment Mode
8101	The information held on Smart Metering Equipment as described at sections 4 and 5 in
8102	SMETS.
8103	Pending Private Key
8104	A Private Key held on a Device for which a Device has not successfully received and
8105	processed a Device Certificate for the corresponding Public Key as defined in Section 13.5.
8106	Plain Format
8107	A Signature is a pair of integers, <i>r</i> and <i>s</i> . For the Elliptic Curve required by the GBCS, each
8108	can be represented as a 256 bit (or 32 octet) string. The Plain Format of a GBCS signature is
8109	the concatenation <i>R S</i> where <i>R</i> is the 32 octet string representing <i>r</i> and <i>S</i> is the 32 octet
8110	string representing <i>s</i> . Thus, a GBCS Signature is an octet string of length 64.
8111	Plaintext
8112	An input to the Authenticated Encryption function and an output from the Authenticated
8113	Decryption function defined in <i>NIST Special Publication 800-38D</i> . Plaintext is the data whose

8114	Confidentiality is to be protected by Encryption. The encrypted form of the Plaintext is the
8115	Ciphertext.
8116	PPMIDTS
8117	The Prepayment Interface Device (PPMID) Technical Specifications in SMETS.
8118	Polyphase
8119	ESME containing three measuring elements suitable for a polyphase supply with up to three
8120	phases and neutral.
8121	Premise(s)
8122	The premise(s) which is / are being Supplied.
8123	Prepayment Daily Read Log
8124	Shall have the meaning defined in SMETS.
8125	Prepayment Interface Device (PPMID)
8126	A Device that provides a User Interface for Prepayment Mode related information and
8127	Commands.
8128	Prepayment Token Decimal (PPTD)
8129	Shall have the meaning defined in Section 14.1.
8130	Prepayment Top Up
8131	The addition of credit to an ESME (or, where it supports it, an SAPC) or GSME operating in
8132	prepayment mode.
8133	Prepayment Top Up Token
8134	Shall have the meaning defined in Section 14.1.
8135	Private Digital Signing Key
8136	A Private Key used for Digital Signing only.
8137	Private Key
8138	The key in a Public-Private Key Pair which must be kept secure by the entity to which it
8139	relates.
8140	Private Key Cell
8141	Shall have the meaning defined in Section 4.3.2.3. A Private Key Cell may be Current or
8142	Pending.

8143	Private Key Agreement Key
8144	A Private Key used for Key Agreement only.
8145	Protection Against Replay
8146	An attribute defined in a Use Case specifying whether a recipient Device is required to
8147	implement the Protection Against Replay mechanisms, as defined in Section 4.3.1.5, for the
8148	Command covered by the Use Case.
8149	Protocol Data Unit (PDU)
8150	Information delivered as a unit among peer entities of networks containing control
8151	information, address information or data.
8152	Provide Security Credentials Command
8153	A Command of a type constructed according to either Section 13.2 or Section 13.9
8154	Public Digital Signing Key
8155	A Public Key used for Digital Signing only.
8156	Public Key
8157	The key in a Public-Private Key Pair which can be distributed to other parties.
8158	Public Key Agreement Key
8159	A Public Key used for Key Agreement only.
8160	Public Key Security Credentials
8161	Security Credentials which include a Public Key.
8162	Public-Private Key Pair
8163	Two mathematically related numbers that are used in Cryptographic Algorithms.
8164	Recovery
8165	In the context of a specific Device, the Known Remote Party whose Security Credentials are
8166	stored in the {recovery, digitalSignature, management} Trust Anchor Cell.
8167	Reliable Time
8168	The state of the Device clock such that is within 10 seconds of UTC, synchronised with the
8169	HAN time server and confirmed by Set Clock Command from the Remote Party whose
8170	security Credentials are stored in the {supplier, digitalSignature, management} Trust Anchor
8171	Cell.

8172	Remote Party
8173	An entity which is remote from a Device and is able to either send Messages to or receive
8174	Messages from a Device, whether directly or via a third party.
8175	Remote Party Alert
8176	Shall have the meaning defined in Section 7.2.3.
8177	Remote Party Command
8178	Shall have the meaning defined in Section 7.2.1.
8179	Remote Party Message
8180	A Message where either the sender(s) or recipient(s) are not Devices.
8181	Remote Party Role
8182	The role of a Remote Party in relation to which one or more Devices is capable of storing
8183	Security Credentials.
8184	Replay Attack
8185	A form of attack on a Communications Link in which a valid information transmission is
8186	repeated through interception and retransmission.
8187	Response
8188	A response to a Command received or sent over any interface.
8189	Response Payload
8190	The parts of a Response that are not related to Cryptographic Protections for integrity,
8191	authenticity or non-repudiation, as defined in Section 7.2.2.
8192	Role
8193	The entitlement of a party to execute one or more Commands.
8194	Root
8195	In the context of a specific Device, the entity whose Security Credentials are stored in the
8196	{root, keyCertSign, management} Trust Anchor Cell.
8197	SAPC
8198	Standalone Auxiliary Proportional Controller, as described in SMETS.
8199	SAPCTS
8200	The Standalone Auxiliary Proportional Controller (SAPC) Technical Specifications in SMETS.

8201	Security Credential Document
8202	Either a:
8203	Device's Certificate; or a
8204	Remote Party's Certificate; or a
8205	Certification Authority Certificate
8206	Security Credentials
8207	Information used to Authenticate a Device, Party or system.
8208	Security Log
8209	Data item described in the Technical Specifications.
8210	Service Reference
8211	Shall have the meaning as set out in Appendix AD of the Smart Energy Code.
8212	SHA-256
8213	The Hashing algorithm of that name approved by the NIST (see
8214	http://csrc.nist.gov/groups/ST/toolkit/secure_hashing.html).
8215	Shared Secret
8216	A number which is established by two parties through the Key Agreement technique
8217	specified in this GBCS and which can be used as input to a KDF.
8218	Shared Secret Key
8219	A number which is derived using the KDF specified in this GBCS.
8220	Smart Energy Code (SEC)
8221	The document of that name, as designated by the Secretary of State under Condition 22 of
8222	the DCC Licence.
8223	Smart Metering Device to Known Remote Party MAC (SMD-KRP MAC)
8224	A MAC generated by a Device in relation to a Response or Alert which can only be verified by
8225	the Known Remote Party which is the target of the Response or Alert.
8226	Smart Metering Entity
8227	An entity that is either a Device or a Remote Party.
8228	Smart Metering Equipment Technical Specifications (SMETS)

8229	A version of the Smart Metering Equipment Technical Specifications set out in Schedule 9 of
8230	the Smart Energy Code, including the ESMETS, GMSETS, HCALCSTS, PPMIDTS and SAPCTS.
8231	Smart Metering Home Area Network (SMHAN)
8232	The network enabling communications between the Devices recorded within a
8233	Communications Hub Function's Device Log (as defined in CHTS).
8234	SMD Signature
8235	A Digital Signature generated by a Device.
8236	Sub GHz
8237	The range of radio frequencies catered for by the ZSE 'Sub-GHz Cluster'.
8238	Sub GHz Alert
8239	An Alert which, in the column headed '0x02 CHF' in Table 16.2, is required to be issued by a
8240	'Dual Band CH Only'. All such Alerts are generated by the CHF.
8241	Sub GHz Channel
8242	One of the radio frequencies catered for by the ZSE 'Sub-GHz Cluster'.
8243	Sub GHz Channel Masks
8244	Configuration settings to limit the Sub GHz frequencies that a Dual Band CH can switch to.
8245	Sub GHz Available Channels
8246	The subset of Sub GHz frequencies that could be used by Dual Band CH and so could be set
8247	in the Sub GHz Channel Masks, so as to comply with UK telecommunications regulations.
8248	This subset of Sub GHz frequencies is defined in Section 10.6.5.
8249	Sub GHz Channel Change
8250	The Dual Band CH action of changing its Operating Sub GHz Channel.
8251	Sub GHz Channel Scan
8252	A process undertaken by a Dual Band Communications Hub to identify the level of radio
8253	noise on each of the Sub GHz Channels that it is allowed to use by the Sub GHz Channel
8254	Masks. This is also referred to in the IEEE 802.15.4a 2007 standard as 'active channel scan'
8255	and in ZSE as 'Energy Scan'.
8256	Sub GHz End Device
8257	A Device which is capable of operating on Sub GHz frequencies and which is not a
8258	Communications Hub.

8259	Sub GHz GSME
8260	A GSME that is capable of operating on Sub GHz frequencies.
8261	Sub GHz Non-GSME Device
8262	A Sub GHz End Device that is not a GSME.
8263	Subject Unique ID
8264	In the context of a Certificate, shall mean the value field of the
8265	<code>AttributeTypeAndValue</code> structure within the <code>subject</code> field whose type is <code>id-at-</code>
8266	<code>uniqueIdentifier</code> .
8267	Supplementary Originator Counter
8268	Shall have the meaning defined in Section 4.3.1.4.
8269	Supplementary Remote Party
8270	The Remote Party identified by a Supplementary Remote Party ID.
8271	Supplementary Remote Party ID
8272	In relation to an Alert which has two recipient roles identified as required, the Entity
8273	Identifier of the recipient which is not the Supplier shall be the Supplementary Remote Party
8274	ID. In relation to a Command and the resulting Response, the Supplementary Remote Party
8275	ID shall be the Entity Identifier of the Remote Party requesting the creation of the Command
8276	by the Access Control Broker.
8277	Supplementary Remote Party Counter
8278	In relation to a Command and the resulting Response, a 64 bit integer specified by the
8279	Remote Party requesting the creation of the Command by the Access Control Broker.
8280	Supply
8281	The supply of gas to Premises for GSME and the supply of electricity to Premises for ESME
8282	and 'Supplied' shall be construed accordingly.
8283	Supplier
8284	A person authorised by licence to Supply gas to Premises for GSME and a person authorised
8285	by licence to Supply electricity to Premises for ESME. In the context of a specific Device, the
8286	Known Remote Party whose Security Credentials are stored in the {supplier, digitalSignature,
8287	management} Trust Anchor Cell.
8288	Supplier Name

8289	The name set in the name part of Contact Details, where that term has its SMETS meaning,
8290	and so the name of the Supplier.
8291	Tag
8292	The first element within a Message Header or part of a Message that provides identification
8293	of the Message or part of Message that follows.
8294	Tapping Off Mechanism (TOM)
8295	Shall have the meaning defined in Section 10.3.4.
8296	Tariff
8297	The structure of prices and other charges relating to a Supply.
8298	Tariff Block Counter Matrix
8299	Data item described in SMETS.
8300	Technical Specifications
8301	The document set comprising SMETS (incorporating the ESMETS, GSMETS, IHDTs, HCalcSTS,
8302	PPMIDTS and SAPCTS), and CHTS.
8303	TOU
8304	Time of Use.
8305	Transitional Change of Supplier
8306	In the context of a specific Device, the Known Remote Party whose Security Credentials are
8307	stored in the {transitionalCoS, digitalSignature, management} Trust
8308	Anchor Cell.
8309	Trust Anchor (TA)
8310	A Trust Anchor represents a Remote Party via a Public Key and associated data stored on a
8311	Device. A Trust Anchor is used by the Device in specified cryptographic operations to
8312	determine whether it should act on Remote Party Commands received.
8313	Trust Anchor Cell
8314	A data store on a Device capable of storing one Trust Anchor. Each Trust Anchor Cell is for a
8315	fixed and pre-specified KeyUsage, CellUsage and RemotePartyRole.
8316	Trust Anchor Management Protocol (TAMP)
8317	A range of IETF RFCs relate to Trust Anchor Management, including:

8318	[RFC4210] Adams, C., Farrell, S., Kause, T., and T. Mononen, 'Internet X.509 Public Key
8319	Infrastructure Certificate Management Protocol (CMP)', RFC 4210, September 2005.
8320	[RFC5280] Cooper, D., Santesson, S., Farrell, S., Boeyen, S., Housley, R., and W. Polk, 'Internet
8321	X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile', RFC
8322	5280, May 2008.
8323	[RFC5914] Housley, R., Ashmore, S., and C. Wallace, 'Trust Anchor Format', RFC 5914, June
8324	2010.
8325	[RFC5934] Housley, R., Ashmore, S., and C. Wallace, 'Trust Anchor Management Protocol
8326	(TAMP)', RFC 5934, August 2010.
8327	[RFC6024] Reddy, R. and C. Wallace, 'Trust Anchor Management Requirements', RFC 6024,
8328	October 2010.
8329	Twin Element
8330	ESME containing two measuring elements.
8331	Type 2 Device
8332	A Device that is not required to have a Device Log, with its SMETS meaning.
8333	Unauthorised
8334	Not Authorised.
8335	Unique Transaction Reference Number (UTRN)
8336	A 20 decimal digit number that is used to convey a Prepayment Top Up Remote Party
8337	Command to an ESME / GSME / SAPC (where it supports such functionality).
8338	Unknown Remote Party (URP)
8339	In the context of a specific Device, a Remote Party whose Security Credentials are not stored
8340	on that Device.
8341	Update Security Credentials Command
8342	A Command of a type constructed according to either Section 13.3 or Section 13.10.
8343	Upgrade Image
8344	Shall have the meaning defined in Section 11.2.2.
8345	Use Case
8346	The structure, format and processing of a Message.
8347	User

8348	Shall have the meaning as set out in Section A of the Smart Energy Code
8349	User Interface
8350	An interface for providing local human interaction with Devices which supports input and
8351	visual output.
8352	User Interface Command
8353	A Remote Party Command that is entered through the User Interface.
8354	UTC
8355	Coordinated Universal Time.
8356	UTRN Check Digit
8357	Shall have the meaning defined in Section 14.1.
8358	UTRN Counter Cache
8359	Shall have the meaning defined in Section 14.1.
8360	Variant Message
8361	A Message that does not fall in to any of the Message Categories defined in Section 6.
8362	Virtual WAN Communications Hub (Virtual WAN CH)
8363	A Communications Hub that is capable of operating using a Virtual WAN Communications
8364	Link in Virtual WAN Mode.
8365	Virtual WAN Communications Link
8366	A component of a Communications Hub that is capable of sending and receiving information
8367	usually sent over the WAN Communications Link via a HAN-connected device. A Virtual
8368	WAN CH is capable of operating the Virtual WAN Communications Link in Virtual WAN Mode.
8369	Virtual WAN Device (VWD)
8370	A PPMID or Type 2 Device that has Virtual WAN capability.
8371	Virtual WAN Mode
8372	A mode of operation of a Virtual WAN CH that allows sending of Messages via the Virtual
8373	WAN Communications Link.
8374	Virtual WAN Provider (VWP)
8375	The organisation providing communications over a secure channel via the Virtual WAN
8376	Communications Link.

-
- 8377 Wide Area Network (WAN) Communications Link
- 8378 A component of a Communications Hub that is capable of sending and receiving information
8379 via the Wide Area Network Provider.
- 8380 Wide Area Network (WAN) Interface
- 8381 A component of a Communications Hub that is capable of sending and receiving information
8382 via the WAN Communications Link and the Virtual WAN Communications Link.
- 8383 Wide Area Network (WAN) Provider
- 8384 The organisation providing communications via the WAN Communications Link of the
8385 Communications Hub. Consequently, in the context of a specific Communications Hub, the
8386 Known Remote Party whose Security Credentials are stored in the {wanProvider,
8387 digitalSignature, management} Trust Anchor Cell.
- 8388 ZigBee Cluster Library (ZCL)
- 8389 The ZigBee Cluster Library Specification reference document as defined in the
8390 'Documentation Alignment' Section of this GBCS.
- 8391 ZigBee PRO/2007 Layer PICS and Stack Profiles
- 8392 The ZigBee PRO/2007 Layer PICS and Stack Profiles reference document as defined in the
8393 'Documentation Alignment' Section of this GBCS.
- 8394 ZigBee Specification
- 8395 The ZigBee Specification reference document as defined in the 'Documentation Alignment'
8396 Section of this GBCS.
- 8397 ZigBee Smart Energy (ZSE) Profile (SEP)
- 8398 The ZigBee Smart Energy (ZSE) Profile Specification reference document as defined in the
8399 'Documentation Alignment' Section of this GBCS.
- 8400 Annex 1 – Additional DLMS Class
- 8401 The class described below shall be supported by ESME, and shall be supported by SAPC
8402 where SAPC supports Use Cases that require this class.
- 8403 Extended Data (class_id: 9000 version: 0)

Attribute(s)			Data type	Min.	Max.	Def.
1.	logical_name	(static)	octet-string[6]			
2.	value_active	(dyn.)	CHOICE			

3.	scaler_unit_active	(dyn.)	scal_unit_type			
4.	value_passive	(static)	CHOICE			
5.	scaler_unit_passive	(static)	scal_unit_type			
6.	activate_passive_value_time	(static)	octet-string			
Methods(s)			Data type			
1.	reset(data)		Integer			
2.	activate_passive_value(data)		integer			

8404

Attribute description

logical_name	Identifies the 'Data' object instance
	Contains the data.
	<i>CHOICE</i> <pre> { -- simple data types null-data [0], Boolean [3], bit-string [4], double-long [5], double-long-unsigned [6], octet-string [9], visible-string [10], utf8-string [12], Bcd [13], integer [15], long [16], unsigned [17], long-unsigned [18], long64 [20], long64-unsigned [21], enum [22], float32 [23], float64 [24], date-time [25], date [26], time [27], -- complex data types array [1], structure [2], compact-array [19] } </pre> The data type depends on the instantiation defined by the 'logical name'. It has to be chosen so, that together with the logical name, an unambiguous interpretation is possible.
value_active	
	Provides information on the unit and the scalar for the value.
scaler_unit_active	scal_unit_type: structure <pre> { scalar, </pre>

	<pre> unit } scalar: integer This is the exponent (to the base of 10) of the multiplication factor unit: enum Enumeration defining the physical unit; for more information check the Blue Book </pre>
value_passive	<pre> Contains the data. CHOICE { -- simple data types null-data [0], Boolean [3], bit-string [4], double-long [5], double-long-unsigned [6], octet-string [9], visible-string [10], utf8-string [12], Bcd [13], integer [15], long [16], unsigned [17], long-unsigned [18], long64 [20], long64-unsigned [21], enum [22], float32 [23], float64 [24], date-time [25], date [26], time [27], -- complex data types array [1], structure [2], compact-array [19] } </pre> The data type depends on the instantiation defined by the 'logical name'. It has to be chosen so, that together with the logical name, an unambiguous interpretation is possible.
scaler_unit_passive	<pre> Provides information on the unit and the scalar for the value. scal_unit_type: structure { scalar, unit } scalar: integer This is the exponent (to the base of 10) of the multiplication factor unit: enum Enumeration defining the physical unit; for more information check the Blue Book </pre>
activate_passive_value_time	Defines the time when the object itself calls the specific method activate_passive_value. A definition with 'not specified' notation in all fields of the attribute will deactivate this automatic activation. Partial 'not specified' notation in just some fields of date and time is

not allowed.

octet-string, formatted as set in 4.1.6.1 for date_time of the Blue DLMS Book

8405 **Method description**

Reset

This method forces a reset of the object. By invoking this method, the value is set to the default value. The default value is an instance specific constant.

data ::= integer(0)

activate_passive_value

This method copies all attributes called ..._passive to the corresponding attributes called ..._active.

data ::= integer(0)

8406 Annex 2 – This Section is intentionally left blank

8407 Annex 3 – This Section is intentionally left blank

8408 Annex 4 – Use of ZigBee in GBCS – informative

8409 Purpose

8410 This annex briefly summarises where the GBCS:

8411 requires the use of ZigBee, specifically where it uses parts of the ZigBee specifications, or
8412 takes an approach which aligns to the ZigBee specifications; and

8413 does not allow the use of ZigBee / requires its use to be modified, specifically where it:

8414 mandates a solution that is not ZigBee derived but where there is ZigBee equivalent in the
8415 specifications;

8416 specifies an approach that is derived from ZigBee but the approach is not part of the ZigBee
8417 specifications; and

8418 specifies an approach that uses parts of the ZigBee but varies from it on specific points.

8419 The document is based on the content of ZigBee specifications referenced in the GBCS.

8420 GBCS requirements to use ZigBee

8421 For all Smart Metering Equipment, the GBCS requires the implementation of functionality
8422 equivalent to a subset of the ZigBee standards, including all mandatory components
8423 required to achieve ZSE certification.

8424 GBCS and the ZigBee standards specify all items that need to be certified.

8425 GBCS requirements not to use ZigBee / vary from it

8426 This Section summarises areas where GBCS requires that parts of the ZigBee standards are
8427 not used or used in a different way. It is an informative summary. The normative
8428 requirements are as stated in the normative Sections of this document.

8429 For some communications between Devices, GBCS requires a solution other than solely
8430 ZigBee. For example, Commands from a PPMID to a GSME (and the corresponding
8431 Responses) are GBCS specified and carried in ZSE TransferData commands.

8432 GBCS requires certain ZigBee features to have a GB-specific interpretation. For example,
8433 Section 10.4.2.11 specifies how the ZSE AccumulatedDebt attribute shall be populated by
8434 the GSME and interpreted by all other devices.

8435 GBCS requires specific population of certain ZigBee commands for GB usage. For example,
8436 Section 10.4.2.11 specifies how the ZSE Get Event Log and Clear Event Log commands can be
8437 used to access GSME Proxy Log copies of the GSME Event or Security Log on the GPF.

8438 GBCS requires a different interpretation of certain ZigBee parameters to support GB usage.
8439 For example, Section 10.4.2.11 specifies that when using the ZSE GetSampledData and
8440 GetSampledDataResponse commands, a SampleRequestInterval field shall contain 0xFFFF
8441 whenever the SampleID field is 0x0001. GBCS specifies that a SampleID of 0x0001
8442 corresponds to the Daily Consumption Log, and SMETS specifies that this log contains
8443 reading data taken at midnight UTC, so once every 24 hours. This is a longer period than can
8444 be specified in the SampleRequestInterval, given that parameter's size and hence the
8445 deviation from the standard.

8446 GBCS lays out specific requirements regarding the use of Inter-PAN communications
8447 between a Communications Hub and an HHT. These are stated in Section 10.

8448 GBCS bars or modifies some internal Device behaviour that is specified in ZigBee standards,
8449 for example:

8450 The ZSE capability for the OTA Client to self-activate any Firmware is not allowed, as stated in
8451 Section 11.2.1.

8452 The ZSE constraint requiring Trust Center involvement in CBKE shall not be applied, as stated
8453 in Section 13.7.2.

8454 GBCS does not require ZSE certification either for non-standard ZSE features or for non-ZSE
8455 features in GBCS.

8456 Annex 5 – Use of DLMS COSEM in GBCS – informative

8457 Statements in the Section regarding an ESME apply to an SAPC as if it were an ESME, where
8458 the SAPC supports the functionality to which the statement relates.

8459 Statements that do not apply to an ESME also do not apply to an SAPC.

8460 Purpose

8461 This annex briefly summarises where the GBCS:

8462 requires the use of DLMS COSEM: specifically where it uses parts of the DLMS COSEM
8463 specification, or takes an approach which aligns to the DLMS COSEM specification; and
8464 does not allow the use of DLMS COSEM / requires its use to be modified: specifically where
8465 it:
8466 Mandates a solution that is not DLMS COSEM derived but where there are DLMS COSEM
8467 equivalents;
8468 Specifies an approach that is derived from DLMS COSEM but the approach is not part of the
8469 DLMS COSEM specification; or
8470 Specifies an approach that uses parts of the DLMS COSEM but varies from it on specific
8471 points.
8472 GBCS requirements to use DLMS COSEM
8473 For ESME and CHF, the GBCS requires the implementation of functionality equivalent to a
8474 subset of the Blue Book Classes. It does not require functionality equivalent to other Blue
8475 Book classes not identified in the GBCS.
8476 For all Devices, GBCS requires a set of cryptographic primitives that align to DLMS Security
8477 Suite 1, and so all Devices will need functionality which is in line with the cryptography
8478 related parts of the Green Book (for both GBCS and DLMS COSEM, these requirements are
8479 NIST derived).
8480 GBCS requires that all Devices use X.509 Certificates and Certification Requests with a
8481 number of optional elements being used / barred. These requirements align with the Green
8482 Book requirements (which are X.509 derived).
8483 For ESME and CHF, the GBCS requires functionality equivalent to Green Book access and
8484 data notification services.
8485 For all Devices, the GBCS requires functionality equivalent to the Green Book's general
8486 ciphering and general signing services.
8487 For all Devices, the GBCS requires functionality equivalent to the Green Book's authenticated
8488 encryption and decryption.
8489 For all Devices, the GBCS requires corresponding alignment with DLMS COSEM's ASN.1
8490 schema and its A-XDR encoding.
8491 GBCS requirements not to use DLMS COSEM / vary from it
8492 Mandates a solution that is not DLMS COSEM derived but where there are DLMS COSEM
8493 equivalent in the specification

8494 For Devices other than ESME and CHF, the GBCS requires functionality equivalent to DLMS
8495 COSEM classes but does not use DLMS COSEM classes (rather GBZ / ASN.1 is used).

8496 For Devices other than ESME and CHF, the GBCS requires support for equivalents of the
8497 Green Book's access and data notification services, but uses GBZ or ASN.1 specific structures.

8498 For all Devices, the GBCS requires that the management of X.509 certificates and Device's
8499 key pairs is undertaken using ASN.1 messages derived from the IETF's TAMP RFCs.

8500 Over the HAN, the GBCS mandates, for all Devices, the use of ZigBee for the communication
8501 layers below the DLMS/COSEM Application Layer and so does not allow the use of the
8502 equivalent Green Book communication profiles. (WAN transport is outside GBCS scope).

8503 For ESME and GSME, distribution of firmware is through the ZSE OTA mechanism.

8504 Specifies an approach that is derived from DLMS COSEM but the approach is not part of the
8505 specification

8506 GBCS specifies the use of a Class 9000 object, not specified in the Blue Book.

8507 For large Remote Party Messages, the GBCS uses DLMS type structures but in a way not
8508 specified in the DLMS COSEM specification.

8509 Messages using pairwise key agreement between GSME and PPMID uses a structure similar
8510 to DLMS COSEM's message structure, but that structure is not part of the DLMS COSEM
8511 specification.

8512 Specifies an approach that uses parts of the DLMS COSEM but varies from it on specific
8513 points

8514 For all bar Type 2 Devices, the DLMS general-signing structure is used in all Remote Party
8515 Messages but the signature field is not populated in Messages that do not require a
8516 signature (i.e. those that are not critical).

8517 For all bar Type 2 Devices, the GBCS uses the general-ciphering structure for Remote Party
8518 Messages that require a MAC. The GBCS leaves most values empty in the header part of the
8519 structure (these values are either in the general-signing structure or are already known to
8520 the meter). Correspondingly, the values used in the OtherInput field of the KDF at section
8521 9.2.3.4.6.5 of the Green Book are those taken from the general-signing structure, rather
8522 than the corresponding fields in the general-ciphering structure.

8523 For ESME and GSME, the GBCS specifies particular, additional interpretation of parameters
8524 within the DLMS COSEM Class 8 object (Clock).

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8525 Annex 6 – Deducing the UTRN Counter from the Truncated UTRN Counter – informative

8526 This annex provides a worked example of the calculation described in Section 14.6.4.1.5. The calculation uses the 10-bit Truncated UTRN
8527 Counter received with the Prepayment Top Up command is received via Consumer Entry to the Device, either directly or via a PPMID. The
8528 calculation uses the highest UTRN Counter value held in the Device's UTRN Counter cache, and a window of 512 either side of this value in
8529 making the deduction.

8530 In this case, the UTRN Counter being entered into the Device is 5 greater than the highest thus far received by the Device.

Parameter	Value (Binary)	Decimal Representation
Vended by supplier		
Originator Counter (64 bits)	100100101000111111000111001011000000000000000000000000000000000 000	10,560,878,642,999,590,912
UTRN Counter (32 bits)	10010010100011111100011100101100	2,458,896,172
PTUT Truncated UTRN Counter (10 bits)	1100101100	812
Recorded on Device		
Highest entry in UTRN Counter Cache (32 bits) = V	10010010100011111100011100100111	2,458,896,167

8531

Step	Description	Example
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		Binary Representation	Decimal Representation
1	The method requires 4 signed 32 bit integers, p , q , r and s		
2	Set p = the numeric value of the least significant 10 bits of the highest UTRN Counter value in the UTRN Counter cache (V)	1100100111	807
3	Set $q = V - p$ $q = 2,458,896,167 - 807$	10010010100011111100010000 000000	2,458,895,360
4	Set r = PTUT Truncated Originator Counter	1100101100	812
5	Calculate $p - 2^9$ (Call this variable, x) ^(See footnote 47) $x = 807 - 512$	100100111	295
6	Calculate $p + 2^9$ (Call this variable, y) $y = 807 + 512$	10100100111	1319
7	Test r against x and y and set s accordingly If $r < x$ then $s = r + 2^{10}$ If $r > y$ then $s = r - 2^{10}$ Else $s = r$	1100101100	812

47 In some cases where $p < 512$, this result may be negative. How negative binary numbers are represented in the calculation is an implementation decision, and not a matter for the GBCS since there is no impact on interoperability.

Step	Description	Example	
		Binary Representation	Decimal Representation
	295 < 812 < 1319, therefore s = r		
8	Set deduced Originator Counter = (q + s) * 2 ³²	10010010100011111100011100 10110000000000000000000000 000000000000	10,560,878,642,999,590,912
9	Set deduced UTRN Counter as most significant 32 bits of Deduced Originator Counter	10010010100011111100011100 101100	2,458,896,172

8532 Table 27: Derivation of the UTRN Counter from the PTUT Truncated UTRN Counter – a worked example

8533

- 8534 Annex 7 – Data Item Values to be set prior to installation of Devices
- 8535 Tables 28a and 28b lists data items and values that shall be configured in ESME prior to
8536 installation.
- 8537 ESME shall also be configured prior to installation to give effect to the settings in Table 28d.
- 8538 The values in Tables 28a and 28b, that relate to ‘Randomised Offset Limit’, shall be configured in SAPC prior to
8539 installation.
- 8540 Where an SAPC supports the functionality associated with a row in Tables 28a, 28b or 28d, the SAPC shall be
8541 configured according to those rows prior to installation.

Device	Data Item	Reference	Value	Notes
ESME (all variants)	Maximum Meter Balance Threshold	SMETS 5.7.4.27	300,000,000 millipence	
ESME (all variants)	Randomised Offset Limit	SMETS 5.7.4.33	600 seconds	The Randomised Offset Is the product of the Randomised Offset Limit(5.7.4.33) and the Randomised Offset Number(5.7.1.5) rounded to the nearest second. This value is used to delay the Tariff Switching Table times and the Auxiliary Controller switching times.
ESME (all variants)	RMS Extreme Over Voltage Threshold	SMETS 5.7.4.35	265.0 volts	GBCS Use Cases specify a resolution to tenths of volts
ESME (all variants)	RMS Extreme Over Voltage Measurement Period	SMETS 5.7.4.34	180 seconds	
ESME (all variants)	RMS Extreme Under Voltage Threshold	SMETS 5.7.4.37	190.0 volts	
ESME (all variants)	RMS Extreme Under Voltage Measurement Period	SMETS 5.7.4.36	180 seconds	
ESME (all variants)	RMS Voltage Sag Threshold	SMETS 5.7.4.40	190.0 volts	
ESME (all variants)	RMS Voltage Sag Measurement Period	SMETS 5.7.4.38	180 seconds	
ESME (all variants)	RMS Voltage Swell	SMETS 5.7.4.41	265.0 volts	

Device	Data Item	Reference	Value	Notes
	Threshold			
ESME (all variants)	RMS Voltage Swell Measurement Period	SMETS 5.7.4.39	180 seconds	
ESME (all variants)	(Phase[1]) Average RMS Voltage Measurement Period	SMETS 5.7.4.6 (5.19.1.3)	1800 seconds	
ESME (all variants)	(Phase[1]) Average RMS Under Voltage Threshold	SMETS 5.7.4.5 (5.19.1.2)	212.0 volts	
ESME (all variants)	(Phase[1]) Average RMS Over Voltage Threshold	SMETS 5.7.4.4 (5.19.1.1)	258.0 volts	
Polyphase ESME	Phase[2] Average RMS Voltage Measurement Period	SMETS 5.19.1.3	1800 seconds	
Polyphase ESME	Phase[2] Average RMS Under Voltage Threshold	SMETS 5.19.1.2	212.0 volts	
Polyphase ESME	Phase[2] Average RMS Over Voltage Threshold	SMETS 5.19.1.1	258.0 volts	
Polyphase ESME	Phase[3] Average RMS Voltage Measurement Period	SMETS 5.19.1.3	1800 seconds	
Polyphase ESME	Phase[3] Average RMS Under Voltage Threshold	SMETS 5.19.1.2	212.0 volts	
Polyphase ESME	Phase[3] Average RMS Over Voltage Threshold	SMETS 5.19.1.1	258.0 volts	
ESME (all variants)	Maximum Demand Configurable Time Period: - start time	SMETS 5.7.4.26	16:00 in hh:mm	
ESME (all variants)	Maximum Demand Configurable Time Period:- end time	SMETS 5.7.4.26	20:00 in hh:mm	

8542 Table 28a: Data items and values to be configured prior to installation of Devices

Data Item	Reference	COSEM class ID	OBIS Code	Attribute ID	Attribute Name	COSEM datatype	Encoded value	Decoded value
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Data Item	Reference	COSEM class ID	OBIS Code	Attribute ID	Attribute Name	COSEM datatype	Encoded value	Decoded value
Maximum Meter Balance Threshold	SMETS 5.7.4.27	9000	0-0:94.44.2.20	4	value_passive	double-long	0x11E1A300	300,000,000
Randomised Offset Limit	SMETS 5.7.4.33	1	0-0:94.44.0.1	2	value	long-unsigned	0x0258	600
RMS Extreme Over Voltage Threshold	SMETS 5.7.4.35	71	0-0:17.0.1.255	4	threshold_normal	double-long-unsigned	0x00000A5A	2650
RMS Extreme Over Voltage Measurement Period	SMETS 5.7.4.34	71	0-0:17.0.1.255	6	min_over_threshold_duration	double-long-unsigned	0x0000000B4	180
RMS Extreme Under Voltage Threshold	SMETS 5.7.4.37	71	0-0:17.0.2.255	4	threshold_normal	double-long-unsigned	0x00000076C	1900
RMS Extreme Under Voltage Measurement Period	SMETS 5.7.4.36	71	0-0:17.0.2.255	6	min_over_threshold_duration	double-long-unsigned	0x0000000B4	180
RMS Voltage Sag Threshold	SMETS 5.7.4.40	71	0-0:17.0.3.255	4	threshold_normal	double-long-unsigned	0x00000076C	1900

Data Item	Reference	COSEM class ID	OBIS Code	Attribute ID	Attribute Name	COSEM datatype	Encoded value	Decoded value
RMS Voltage Sag Measurement Period	SMETS 5.7.4.38	71	0-0:17.0.3.255	6	min_over_threshold_duration	double-long-unsigned	0x000000B4	180
RMS Voltage Swell Threshold	SMETS 5.7.4.41	71	0-0:17.0.4.255	4	threshold_normal	double-long-unsigned	0x000000A5A	2650
RMS Voltage Swell Measurement Period	SMETS 5.7.4.39	71	0-0:17.0.4.255	6	min_over_threshold_duration	double-long-unsigned	0x000000B4	180
(Phase[1]) Average RMS Voltage Measurement Period	SMETS 5.7.4.6 (5.19.1.3)	7	1-0:32.24.0.255	4	capture_period	double-long-unsigned	0x00000008	1800
(Phase[1]) Average RMS Under Voltage Threshold	SMETS 5.7.4.5 (5.19.1.2)	1	1-0:32.31.0.4	2	value	double-long-unsigned	0x0000000848	2120
(Phase[1]) Average RMS Over Voltage Threshold	SMETS 5.7.4.4 (5.19.1.1)	1	1-0:32.35.0.4	2	value	double-long-unsigned	0x000000A14	2580
Phase[2] Average RMS Voltage	SMETS 5.19.1.3	7	1-0:52.24.0.255	4	capture_period	double-long-unsigned	0x00000008	1800

Data Item	Reference	COSEM class ID	OBIS Code	Attribute ID	Attribute Name	COSEM datatype	Encoded value	Decoded value
Measurement Period								
Phase[2] Average RMS Under Voltage Threshold	SMETS 5.19.1.2	1	1-0:52.31.0.4	2	value	double-long-unsigned	0x00000848	2120
Phase[2] Average RMS Over Voltage Threshold	SMETS 5.19.1.1	1	1-0:52.35.0.4	2	value	double-long-unsigned	0x00000A14	2580
Phase[3] Average RMS Voltage Measurement Period	SMETS 5.19.1.3	7	1-0:72.24.0.255	4	capture_period	double-long-unsigned	0x00000708	1800
Phase[3] Average RMS Under Voltage Threshold	SMETS 5.19.1.2	1	1-0:72.31.0.4	2	value	double-long-unsigned	0x00000848	2120
Phase[3] Average RMS Over Voltage Threshold	SMETS 5.19.1.1	1	1-0:72.35.0.4	2	value	double-long-unsigned	0x00000A14	2580

8543 Table 28b: DLMS COSEM values to be configured in Devices at manufacture.

8544 Table 28c: This table is not used.

Attribute	COSEM datatype	Tag	Length	Value	Meaning
entries:entries[1..2]	array	0x01	0x02		An array with two entries, the first turns on monitoring and the second

Attribute	COSEM datatype	Tag	Length	Value	Meaning
					turns it off
entries:entries[1]. schedule_table_entry. Index	long-unsigned	0x12		0x0001	The first entry which turns monitoring on
entries:entries[1]. schedule_table_entry. Enable	boolean	0x03		0x01	True, so the entry always executes
entries:entries[1]. schedule_table_entry. script_logical_name	octet-string(6)	0x09	0x06	0x00000A8064FF	0-0:10.128.100.255 which, as per Table 7.3.8, is the script table controlling monitoring
entries:entries[1]. schedule_table_entry. script_selector	long-unsigned	0x12		0x0001	Meaning start monitoring at the time in this entry
entries:entries[1]. schedule_table_entry. switch_time	octet-string(4)	0x09	0x04	0x10000000	16:00:00:00 - the time at which monitoring is to turn on
entries:entries[1]. schedule_table_entry. validity_window	long-unsigned	0x12		0xFFFF	The script is processed at any time after power failure
entries:entries[1]. schedule_table_entry. exec_weekdays	bit-string(7)	0x04	0x07	0xFE	0xFE = 0b11111110, which means execute this script every day
entries:entries[1]. schedule_table_entry. exec_specdays	bit-string(0)	0x04	0x00		No special day processing
entries:entries[1]. schedule_table_entry. begin_date	octet-string(5)	0x09	0x05	0x000001FFFFFF	From the start of time
entries:entries[1]. schedule_table_entry. end_date	octet-string(5)	0x09	0x05	0xFFFFFFFF	For all time
entries:entries[2]. schedule_table_entry. Index	long-unsigned	0x12		0x0002	The second entry which turns monitoring off
entries:entries[2]. schedule_table_entry. Enable	boolean	0x03		0x01	True, so the entry always executes
entries:entries[2]. schedule_table_entry. script_logical_name	octet-string(6)	0x09	0x06	0x00000A8064FF	0-0:10.128.100.255 which, as per Table 7.3.8, is the script table controlling monitoring

Attribute	COSEM datatype	Tag	Length	Value	Meaning
entries:entries[2]. schedule_table_entry. script_selector	long-unsigned	0x12		0x0002	Meaning stop monitoring at the time in this entry
entries:entries[2]. schedule_table_entry. switch_time	octet-string(4)	0x09	0x04	0x14000000	20:00:00:00 - the time at which monitoring is to turn off
entries:entries[2]. schedule_table_entry. validity_window	long-unsigned	0x12		0xFFFF	The script is processed at any time after power failure
entries:entries[2]. schedule_table_entry. exec_weekdays	bit-string(7)	0x04	0x07	0xFE	0xFE = 0b11111110, which means execute this script every day
entries:entries[2]. schedule_table_entry. exec_specdays	bit-string(0)	0x04	0x00		No special day processing
entries:entries[2]. schedule_table_entry. begin_date	octet-string(5)	0x09	0x05	0x000001FFFF	From the start of time
entries:entries[2]. schedule_table_entry. end_date	octet-string(5)	0x09	0x05	0xFFFFFFFF	For all time

8545 Table 28d: Tag, length and values to be populated in attribute 2 of OBIS code 0-0:12.0.0.255
8546 (which relates to the SMETS 'Maximum Demand Configurable Time Period ') to be
8547 configured prior to installation of ESME.

